

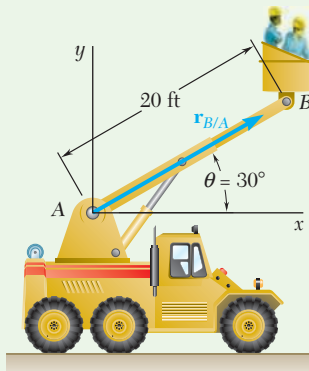
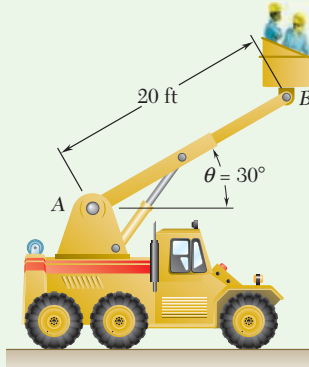


Fig. 1 The rotating coordinate system is attached to the truck at A.

Sample Problem 15.17

At the instant shown, the truck is moving forward with a speed of 2 ft/s and is slowing down at a rate of 0.25 ft/s^2 . The length of the boom AB is decreasing at a constant rate of 0.5 ft/s , the angular velocity of the boom is 0.1 rad/s , and the angular acceleration of the boom is 0.02 rad/s^2 , both clockwise. Determine the velocity and acceleration of point B .

STRATEGY: Since you are not given any forces and are asked to find the velocity and acceleration of a point, use rigid-body kinematics. The boom is moving with respect to the truck, so use a rotating reference frame.

MODELING and ANALYSIS: Attach a rotating coordinate system to the boom housing with its origin at A (Fig. 1).

Velocity of B . From Eq. (15.32'), you know

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{B/A} + \mathbf{v}_{\text{rel}} \quad (1)$$

where $\mathbf{v}_A = (2 \text{ ft/s})\mathbf{i}$, $\mathbf{r}_{B/A} = (20 \cos 30^\circ \text{ ft})\mathbf{i} + (20 \sin 30^\circ \text{ ft})\mathbf{j}$, and $\boldsymbol{\Omega} = (-0.1 \text{ rad/s}^2)\mathbf{k}$. To find the relative velocity, ask yourself what the velocity of B would be, assuming that the rotating coordinate system is not moving. In this case, $\mathbf{v}_{\text{rel}} = -(0.5 \cos 30^\circ \text{ ft/s})\mathbf{i} - (0.5 \sin 30^\circ \text{ ft/s})\mathbf{j}$. Substituting into Eq. (1) gives

$$\mathbf{v}_B = 2\mathbf{i} + (-0.1\mathbf{k}) \times (17.32\mathbf{i} + 10\mathbf{j}) - (0.433\mathbf{i} + 0.25\mathbf{j})$$

$$\mathbf{v}_B = (2.57 \text{ ft/s})\mathbf{i} - (1.982 \text{ ft/s})\mathbf{j} \quad \blacktriangleleft$$

Acceleration of B . From Eq. (15.35'), you know

$$\mathbf{a}_B = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A} - \Omega^2 \mathbf{r}_{B/A} + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}} \quad (2)$$

where $\mathbf{a}_A = -(0.25 \text{ ft/s}^2)\mathbf{i}$, $\dot{\boldsymbol{\Omega}} = -(0.02 \text{ rad/s}^2)\mathbf{k}$, and $\mathbf{a}_{\text{rel}} = 0$. Substituting into Eq. (2) gives

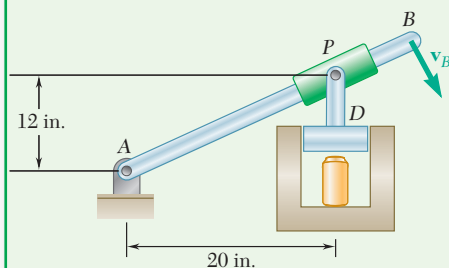
$$\begin{aligned} \mathbf{a}_B &= -0.25\mathbf{i} + (-0.02\mathbf{k}) \times (17.32\mathbf{i} + 10\mathbf{j}) - 0.1^2(17.32\mathbf{i} + 10\mathbf{j}) \\ &\quad + 2(-0.1\mathbf{k}) \times (-0.433\mathbf{i} - 0.25\mathbf{j}) + 0 \end{aligned}$$

$$= -0.25\mathbf{i} + (-0.3464\mathbf{j} + 0.2\mathbf{i}) - (0.1732\mathbf{i} + 0.10\mathbf{j}) + (0.0866\mathbf{j} - 0.05\mathbf{i}) + 0$$

$$\mathbf{a}_B = (-0.273 \text{ ft/s}^2)\mathbf{i} - (0.360 \text{ ft/s}^2)\mathbf{j} \quad \blacktriangleleft$$

REFLECT and THINK: The biggest challenge with this problem is interpreting what you are given in the problem statement. After that, it is straightforward to substitute into the governing equations. The last four terms in Eq. (2) are analogous to the polar coordinate expressions we used in Chapter 11. The following terms represent the same physical quantities: $\dot{\boldsymbol{\Omega}} \times \mathbf{r}_{B/A} \rightarrow r\ddot{\theta}$, $-\Omega^2 \mathbf{r}_{B/A} \rightarrow -r\dot{\theta}^2$, $2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} \rightarrow 2\dot{r}\dot{\theta}$, and $\mathbf{a}_{\text{rel}} \rightarrow \ddot{r}$.

Sample Problem 15.18



In a can crusher, bar AB has a length of 30 in. and slides inside a collar located at point P . This collar is attached to plunger DP , which is constrained to move vertically. At the instant shown, the velocity of point B is a constant 4 ft/s perpendicular to the bar. Determine the velocity and acceleration of the plunger D .

STRATEGY: You are not given any forces and are asked to find the velocity and acceleration of a point, so use rigid-body kinematics. Since the collar is moving with respect to the bar, use a rotating reference frame.

MODELING and ANALYSIS: Attach a rotating coordinate system to the bar with its origin at A (Fig. 1).

Angular Velocity of AB . Rod AB is undergoing fixed-axis rotation, so

$$\omega_{AB} = \frac{v_B}{r_{B/A}} = \frac{48 \text{ in./s}}{30 \text{ in.}} = 1.60 \text{ rad/s} \downarrow$$

Velocity of P . Points D and P have the same velocity and acceleration because the plunger is constrained to translate only. From Eq. (15.32') you know

$$\mathbf{v}_P = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{P/A} + \mathbf{v}_{\text{rel}} \quad (1)$$

where $\mathbf{v}_A = 0$, $\mathbf{r}_{P/A} = (20 \text{ in.})\mathbf{i} + (12 \text{ in.})\mathbf{j}$, and $\boldsymbol{\Omega} = -(1.6 \text{ rad/s})\mathbf{k}$. To find the relative velocity, ask yourself what the velocity of P would be assuming that the rotating coordinate system is not moving. In this case, $\mathbf{v}_{\text{rel}} = v_{\text{rel}} \cos \theta \mathbf{i} + v_{\text{rel}} \sin \theta \mathbf{j}$ where $\theta = \tan^{-1}(12/20) = 30.96^\circ$. Substituting into Eq. (1) gives

$$\begin{aligned} -v_P \mathbf{j} &= 0 + (-1.6\mathbf{k}) \times (20\mathbf{i} + 12\mathbf{j}) + (v_{\text{rel}} \cos \theta \mathbf{i} + v_{\text{rel}} \sin \theta \mathbf{j}) \\ &= -32\mathbf{j} + 19.2\mathbf{i} + 0.8575v_{\text{rel}}\mathbf{i} + 0.5145v_{\text{rel}}\mathbf{j} \end{aligned}$$

Equating components allows you to solve for the unknown velocities:

$$\begin{aligned} \mathbf{i}: \quad 0 &= 19.2 + 0.8575v_{\text{rel}} & \longrightarrow & \quad v_{\text{rel}} = -22.39 \text{ in./s} \\ \mathbf{j}: \quad -v_P &= -32 + 0.5145v_{\text{rel}} & \longrightarrow & \quad v_P = 43.53 \text{ in./s} \end{aligned}$$

$$\mathbf{v}_P = 43.53 \text{ in./s} \downarrow \quad \blacktriangleleft$$

Acceleration of P . From Eq. (15.35'), you know

$$\mathbf{a}_P = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{P/A} - \Omega^2 \mathbf{r}_{P/A} + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}} \quad (2)$$

where $\mathbf{a}_A = 0$, $\dot{\boldsymbol{\Omega}} = 0$, $\mathbf{a}_{\text{rel}} = a_{\text{rel}} \cos \theta \mathbf{i} + a_{\text{rel}} \sin \theta \mathbf{j}$. Substituting into Eq. (2) gives

$$\begin{aligned} -a_P \mathbf{j} &= 0 + 0 - 1.6^2(20\mathbf{i} + 12\mathbf{j}) + 2(-1.6\mathbf{k}) \times \\ &\quad (-22.39 \cos \theta \mathbf{i} - 22.39 \sin \theta \mathbf{j}) + (a_{\text{rel}} \cos \theta \mathbf{i} + a_{\text{rel}} \sin \theta \mathbf{j}) \\ &= (-51.2\mathbf{i} - 30.72\mathbf{j}) + (61.44\mathbf{j} - 36.86\mathbf{i}) + (0.8575a_{\text{rel}}\mathbf{i} + 0.5145a_{\text{rel}}\mathbf{j}) \end{aligned}$$

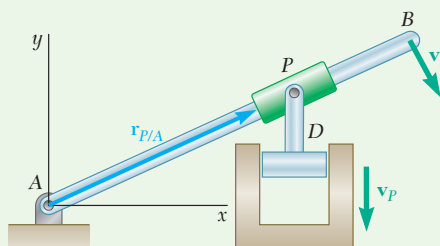


Fig. 1 The rotating coordinate system is attached to arm AB .

Equating components allows you to solve for the unknown accelerations:

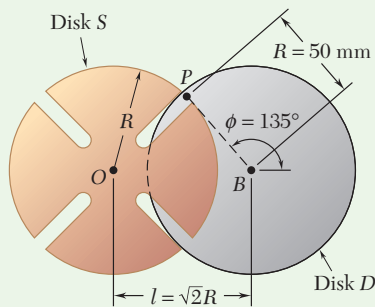
$$\mathbf{i}: 0 = -51.2 - 36.86 + 0.8575a_{\text{rel}} \longrightarrow a_{\text{rel}} = 102.7 \text{ in./s}^2$$

$$\mathbf{j}: -a_P = -30.72 + 61.44 + 0.5145a_{\text{rel}} \longrightarrow a_P = -83.56 \text{ in./s}^2$$

$$a_P = -83.6 \text{ in./s}^2 \downarrow \quad \blacktriangleleft$$

REFLECT and THINK: You used the same strategy for the telescoping boom in Sample Prob. 15.17 as you did for the sliding collar in this problem. For each case, the point of interest was moving with respect to a coordinate frame attached to a rigid body. The same strategy is used in problems where pins move within slotted bodies (such as the Geneva mechanism in Sample Prob. 15.19).

Sample Problem 15.19



The Geneva mechanism shown is used in many counting instruments and in other applications where an intermittent rotary motion is required. Disk D rotates with a constant counterclockwise angular velocity ω_D of 10 rad/s. A pin P is attached to disk D and slides along one of several slots cut in disk S . It is desirable that the angular velocity of disk S be zero as the pin enters and leaves each slot; in the case of four slots, this occurs if the distance between the centers of the disks is $l = \sqrt{2}R$.

At the instant when $\phi = 150^\circ$, determine (a) the angular velocity of disk S , (b) the velocity of pin P relative to disk S .

STRATEGY: You have two rigid bodies whose motions are related; therefore use rigid-body kinematics. Since point P is moving in a slot, use a rotating reference frame.

MODELING and ANALYSIS:

Using geometry, you can solve triangle OPB , which corresponds to the position $\phi = 150^\circ$ (Fig. 1). Using the law of cosines, you have

$$r^2 = R^2 + l^2 - 2Rl \cos 30^\circ = 0.551R^2 \quad r = 0.742R = 37.1 \text{ mm}$$

Then, from the law of sines, you have

$$\frac{\sin \beta}{R} = \frac{\sin 30^\circ}{r} \quad \sin \beta = \frac{\sin 30^\circ}{0.742} \quad \beta = 42.4^\circ$$

Since pin P is attached to disk D and disk D rotates about point B , the magnitude of the absolute velocity of P is

$$v_P = R\omega_D = (50 \text{ mm})(10 \text{ rad/s}) = 500 \text{ mm/s}$$

$$\mathbf{v}_P = 500 \text{ mm/s} \nearrow 60^\circ$$

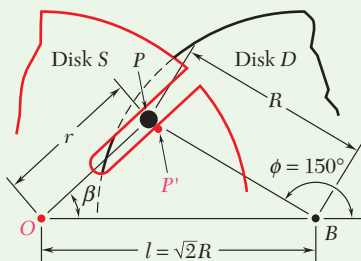


Fig. 1 Distances and angles relating points O , P , and B .

(continued)

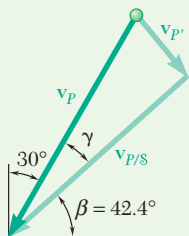


Fig. 2 Vector diagram for the velocity of point P .

Now consider the motion of pin P along the slot in disk S . Denote the point of disk S that coincides with P by P' at the instant considered and select a rotating frame $\mathcal{S}$ attached to disk S . Then from Eq. (15.33), you have

$$\mathbf{v}_P = \mathbf{v}_{P'} + \mathbf{v}_{P/\mathcal{S}} \quad (1)$$

In Eq. (1), $\mathbf{v}_{P'}$ is perpendicular to the radius OP , and $\mathbf{v}_{P/\mathcal{S}}$ is directed along the slot. Draw the velocity triangle corresponding to Eq. (1) (see Fig. 2). From the triangle, you can compute

$$\begin{aligned} \gamma &= 90^\circ - 42.4^\circ - 30^\circ = 17.6^\circ \\ v_{P'} &= v_P \sin \gamma = (500 \text{ mm/s}) \sin 17.6^\circ \\ \mathbf{v}_{P'} &= 151.2 \text{ mm/s} \searrow 42.4^\circ \\ v_{P/\mathcal{S}} &= v_P \cos \gamma = (500 \text{ mm/s}) \cos 17.6^\circ \\ \mathbf{v}_{P/\mathcal{S}} &= \mathbf{v}_{P/\mathcal{S}} = 477 \text{ mm/s} \nearrow 42.4^\circ \quad \blacktriangleleft \end{aligned}$$

Since $\mathbf{v}_{P'}$ is perpendicular to the radius OP , you have

$$\begin{aligned} v_{P'} &= r\omega_{\mathcal{S}} & 151.2 \text{ mm/s} &= (37.1 \text{ mm})\omega_{\mathcal{S}} \\ \omega_{\mathcal{S}} &= \omega_{\mathcal{S}} = 4.08 \text{ rad/s} \downarrow \quad \blacktriangleleft \end{aligned}$$

REFLECT and THINK: The result of the Geneva mechanism is that disk S rotates $1/4$ turn each time pin P engages, then it remains motionless while pin P rotates around before entering the next slot. Disk D rotates continuously, but disk S rotates intermittently. An alternative approach to drawing the vector triangle is to use vector algebra, as was done in Sample Prob. 15.18.

Sample Problem 15.20

In the Geneva mechanism of Sample Prob. 15.19, disk D rotates with a constant counterclockwise angular velocity ω_D of 10 rad/s. At the instant when $\phi = 150^\circ$, determine the angular acceleration of disk S .

STRATEGY: You have two rigid bodies whose motions are related; therefore use rigid-body kinematics. Since point P is moving in a slot, use a rotating reference frame.

MODELING and ANALYSIS: Since you are computing accelerations instead of velocities, you need to use Eq. (15.36), which includes the Coriolis acceleration. You found the angular velocity of the frame $\mathcal{S}$ attached to disk S and the velocity of the pin relative to $\mathcal{S}$ in Sample Prob. 15.19:

$$\begin{aligned} \omega_{\mathcal{S}} &= 4.08 \text{ rad/s} \downarrow \\ \beta &= 42.4^\circ & \mathbf{v}_{P/\mathcal{S}} &= 477 \text{ mm/s} \nearrow 42.4^\circ \end{aligned}$$

Since pin P moves with respect to the rotating frame $\mathcal{S}$, you have

$$\mathbf{a}_P = \mathbf{a}_{P'} + \mathbf{a}_{P/\mathcal{S}} + \mathbf{a}_c \quad (1)$$

Investigate each term of this vector equation separately.

Absolute Acceleration $\mathbf{a}_P$. Since disk D rotates with a constant angular velocity, the absolute acceleration $\mathbf{a}_P$ is directed toward B . This gives

$$a_P = R\omega_D^2 = (500 \text{ mm})(10 \text{ rad/s})^2 = 5000 \text{ mm/s}^2$$

$$\mathbf{a}_P = 5000 \text{ mm/s}^2 \searrow 30^\circ$$

Acceleration $\mathbf{a}_{P'}$ of the Coinciding Point P' . Resolve into normal and tangential components the acceleration $\mathbf{a}_{P'}$ of the point P' of the frame $\mathcal{S}$ that coincides with P at the given instant. (Recall from Sample Prob. 15.19 that $r = 37.1 \text{ mm}$.)

$$(a_{P'})_n = r\omega_S^2 = (37.1 \text{ mm})(4.08 \text{ rad/s})^2 = 618 \text{ mm/s}^2$$

$$(\mathbf{a}_{P'})_n = 618 \text{ mm/s}^2 \nearrow 42.4^\circ$$

$$(a_{P'})_t = r\alpha_S = 37.1\alpha_S \quad (\mathbf{a}_{P'})_t = 37.1\alpha_S \searrow 42.4^\circ$$

Relative Acceleration $\mathbf{a}_{P/\mathcal{S}}$. Since the pin P moves in a straight slot cut in disk S , the relative acceleration $\mathbf{a}_{P/\mathcal{S}}$ must be parallel to the slot; i.e., its direction must be $\nearrow 42.4^\circ$.

Coriolis Acceleration $\mathbf{a}_C$. Rotating the relative velocity $\mathbf{v}_{P/\mathcal{S}}$ through 90° in the sense of $\boldsymbol{\omega}_S$, you obtain the direction of the Coriolis component of the acceleration: $\nwarrow 42.4^\circ$. You have

$$a_C = 2\omega_S v_{P/\mathcal{S}} = 2(4.08 \text{ rad/s})(477 \text{ mm/s}) = 3890 \text{ mm/s}^2$$

$$\mathbf{a}_C = 3890 \text{ mm/s}^2 \nwarrow 42.4^\circ$$

Rewrite Eq. (1) and substitute the accelerations found (Fig. 1):

$$\mathbf{a}_P = (\mathbf{a}_{P'})_n + (\mathbf{a}_{P'})_t + \mathbf{a}_{P/\mathcal{S}} + \mathbf{a}_C$$

$$[5000 \searrow 30^\circ] = [618 \nearrow 42.4^\circ] + [37.1\alpha_S \searrow 42.4^\circ]$$

$$+ [a_{P/\mathcal{S}} \nearrow 42.4^\circ] + [3890 \nwarrow 42.4^\circ]$$

Equating components in a direction perpendicular to the slot,

$$5000 \cos 17.6^\circ = 37.1\alpha_S - 3890$$

$$\alpha_S = \alpha_S = 233 \text{ rad/s}^2 \quad \blacktriangleleft$$

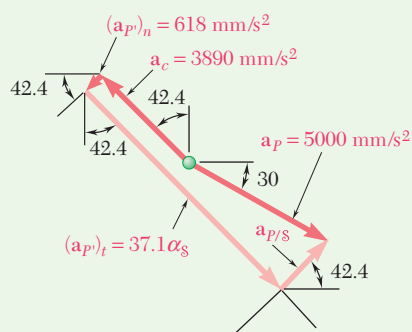


Fig. 1 Vector polygon for the acceleration of point P .

REFLECT and THINK: It seems reasonable that, since disk S starts and stops over the very short time intervals when pin P is engaged in the slots, the disk must have a very large angular acceleration. An alternative approach would have been to use the vector algebra.

SOLVING PROBLEMS ON YOUR OWN

In this section you studied the rate of change of a vector with respect to a rotating frame and then applied that idea to the analysis of the plane motion of a particle relative to a rotating frame.

1. Rate of change of a vector with respect to a fixed frame and with respect to a rotating frame. Denoting the rate of change of a vector $\mathbf{Q}$ with respect to a fixed frame $OXYZ$ by $(\dot{\mathbf{Q}})_{OXYZ}$ and its rate of change with respect to a rotating frame $Oxyz$ by $(\dot{\mathbf{Q}})_{Oxyz}$, we obtained the fundamental relation

$$(\dot{\mathbf{Q}})_{OXYZ} = (\dot{\mathbf{Q}})_{Oxyz} + \boldsymbol{\Omega} \times \mathbf{Q} \quad (15.31)$$

where $\boldsymbol{\Omega}$ is the angular velocity of the rotating frame.

You can now apply this fundamental relation to the solution of two-dimensional problems.

2. Plane motion of a particle relative to a rotating frame. Using Eq. (15.31) and designating the rotating frame by $\mathcal{F}$, we obtained the following expressions for the velocity and the acceleration of a particle P :

$$\mathbf{v}_P = \mathbf{v}_{P'} + \mathbf{v}_{P/\mathcal{F}} \quad (15.33)$$

or

$$\mathbf{v}_P = \mathbf{v}_O + \boldsymbol{\Omega} \times \mathbf{r} + \mathbf{v}_{\text{rel}} \quad (15.32')$$

and

$$\mathbf{a}_P = \mathbf{a}_{P'} + \mathbf{a}_{P/\mathcal{F}} + \mathbf{a}_C \quad (15.36)$$

or

$$\mathbf{a}_P = \mathbf{a}_O + \boldsymbol{\alpha} \times \mathbf{r} - \Omega^2 \mathbf{r} + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}} \quad (15.35')$$

The notation in Eqs. (15.33) and (15.36) is as follows.

a. The subscript P refers to the absolute motion of the particle P ; that is, to its motion with respect to a fixed or newtonian frame of reference OXY .

b. The subscript P' refers to the motion of the point P' of the rotating frame $\mathcal{F}$ that coincides with P at the instant considered.

c. The subscript $P/\mathcal{F}$ refers to the motion of the particle P relative to the rotating frame $\mathcal{F}$.

d. The term $\mathbf{a}_C$ represents the Coriolis acceleration of point P . Its magnitude is $2\Omega v_{P/\mathcal{F}}$, and its direction is found by rotating $\mathbf{v}_{P/\mathcal{F}}$ through 90° in the sense of rotation of the frame $\mathcal{F}$.

You should keep in mind that you need to take the Coriolis acceleration into account whenever a point has a relative velocity in a rotating frame. The problems you will encounter in this section involve collars that slide on rotating rods, booms that extend from cranes rotating in a vertical plane, etc.

When solving a problem involving a rotating frame, you can either *a)* draw vector diagrams representing Eqs. (15.33) and (15.36), respectively, and use these diagrams to obtain either an analytical or a graphical solution, or *b)* use vector algebra.

Problems

CONCEPT QUESTIONS

15.CQ8 A person walks radially inward on a platform that is rotating counterclockwise about its center. Knowing that the platform has a constant angular velocity ω and the person walks with a constant speed u relative to the platform, what is the direction of the acceleration of the person at the instant shown?

- Negative x
- Negative y
- Negative x and positive y
- Positive x and positive y
- Negative x and negative y

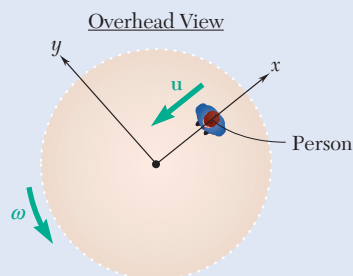


Fig. P15.CQ8

END-OF-SECTION PROBLEMS

15.150 and 15.151 Pin P is attached to the collar shown; the motion of the pin is guided by a slot cut in rod BD and by the collar that slides on rod AE . Knowing that at the instant considered the rods rotate clockwise with constant angular velocities, determine for the given data the velocity of pin P .

15.150 $\omega_{AE} = 8 \text{ rad/s}$, $\omega_{BD} = 3 \text{ rad/s}$

15.151 $\omega_{AE} = 7 \text{ rad/s}$, $\omega_{BD} = 4.8 \text{ rad/s}$

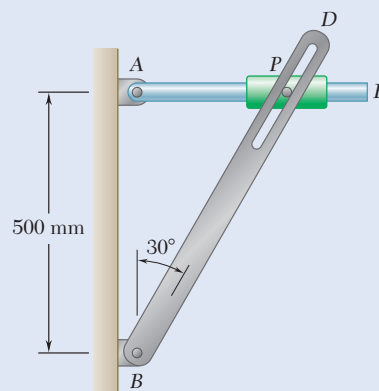


Fig. P15.150 and P15.151

15.152 and 15.153 Two rotating rods are connected by slider block P . The rod attached at A rotates with a constant clockwise angular velocity ω_A . For the given data, determine for the position shown (a) the angular velocity of the rod attached at B , (b) the relative velocity of slider block P with respect to the rod on which it slides.

15.152 $b = 8 \text{ in.}$, $\omega_A = 6 \text{ rad/s}$

15.153 $b = 300 \text{ mm}$, $\omega_A = 10 \text{ rad/s}$

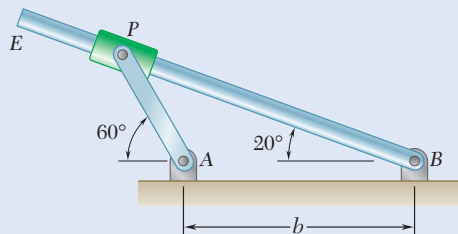


Fig. P15.152

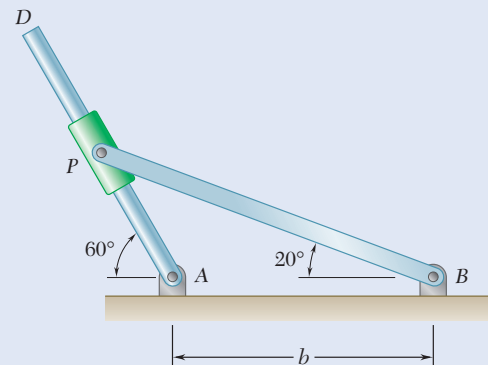


Fig. P15.153

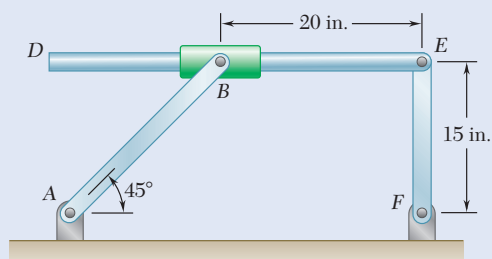


Fig. P15.155 and P15.156

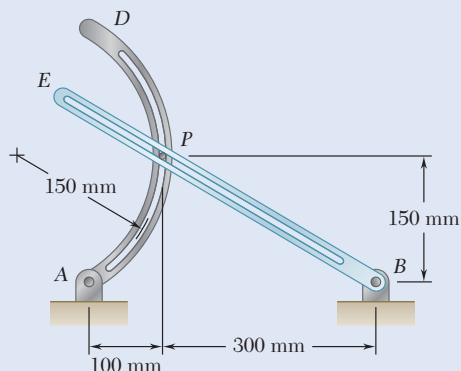


Fig. P15.157

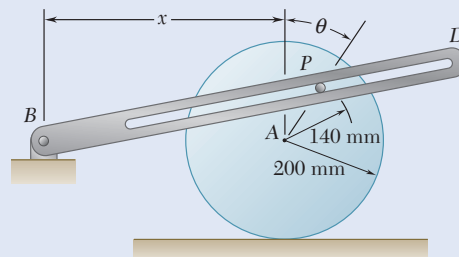


Fig. P15.154

- 15.154** Pin P is attached to the wheel shown and slides in a slot cut in bar BD . The wheel rolls to the right without slipping with a constant angular velocity of 20 rad/s . Knowing that $x = 480 \text{ mm}$ when $\theta = 0$, determine the angular velocity of the bar and the relative velocity of pin P with respect to the rod when (a) $\theta = 0$, (b) $\theta = 90^\circ$.

- 15.155** Knowing that at the instant shown the angular velocity of bar AB is 15 rad/s clockwise and the angular velocity of bar EF is 10 rad/s clockwise, determine (a) the angular velocity of rod DE , (b) the relative velocity of collar B with respect to rod DE .

- 15.156** Knowing that at the instant shown the angular velocity of rod DE is 10 rad/s clockwise and the angular velocity of bar EF is 15 rad/s counterclockwise, determine (a) the angular velocity of bar AB , (b) the relative velocity of collar B with respect to rod DE .

- 15.157** The motion of pin P is guided by slots cut in rods AD and BE . Knowing that bar AD has a constant angular velocity of 4 rad/s clockwise and bar BE has an angular velocity of 5 rad/s counterclockwise and is slowing down at a rate of 2 rad/s^2 , determine the velocity of P for the position shown.

- 15.158** Four pins slide in four separate slots cut in a circular plate as shown. When the plate is at rest, each pin has a velocity directed as shown and of the same constant magnitude u . If each pin maintains the same velocity relative to the plate when the plate rotates about O with a constant counterclockwise angular velocity ω , determine the acceleration of each pin.

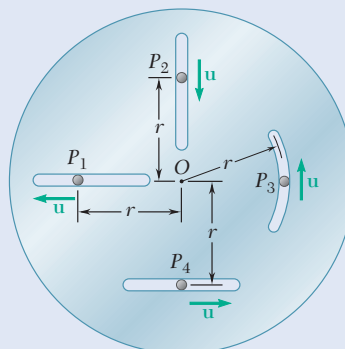


Fig. P15.158

- 15.159** Solve Prob. 15.158, assuming that the plate rotates about O with a constant clockwise angular velocity ω .

15.160 The cage of a mine elevator moves downward at a constant speed of 12.2 m/s. Determine the magnitude and direction of the Coriolis acceleration of the cage if the elevator is located (a) at the equator, (b) at latitude 40° north, (c) at latitude 40° south.

15.161 Pin P is attached to the collar shown; the motion of the pin is guided by a slot cut in bar BD and by the collar that slides on rod AE . Rod AE rotates with a constant angular velocity of 6 rad/s clockwise and the distance from A to P increases at a constant rate of 8 ft/s. Determine at the instant shown (a) the angular acceleration of bar BD , (b) the relative acceleration of pin P with respect to bar BD .

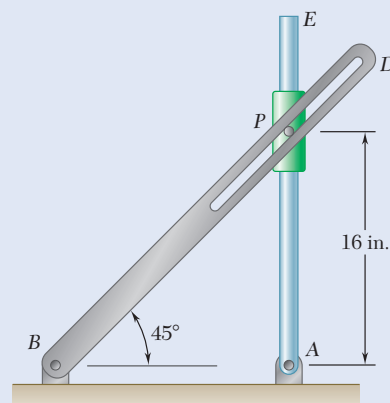


Fig. P15.161

15.162 A rocket sled is tested on a straight track that is built along a meridian. Knowing that the track is located at latitude 40° north, determine the Coriolis acceleration of the sled when it is moving north at a speed of 900 km/h.

15.163 Solve the Geneva mechanism of Sample Prob. 15.20 using vector algebra.

15.164 At the instant shown the length of the boom AB is being *decreased* at the constant rate of 0.2 m/s and the boom is being lowered at the constant rate of 0.08 rad/s. Determine (a) the velocity of point B , (b) the acceleration of point B .

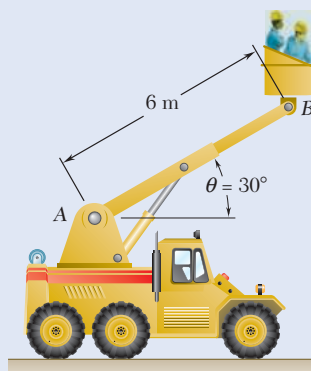


Fig. P15.164 and P15.165

15.165 At the instant shown the length of the boom AB is being *increased* at the constant rate of 0.2 m/s and the boom is being lowered at the constant rate of 0.08 rad/s. Determine (a) the velocity of point B , (b) the acceleration of point B .

15.166 In the automated welding setup shown, the position of the two welding tips G and H is controlled by the hydraulic cylinder D and rod BC . The cylinder is bolted to the vertical plate that at the instant shown rotates counterclockwise about A with a constant angular velocity of 1.6 rad/s. Knowing that at the same instant the length EF of the welding assembly is increasing at the constant rate of 300 mm/s, determine (a) the velocity of tip H , (b) the acceleration of tip H .

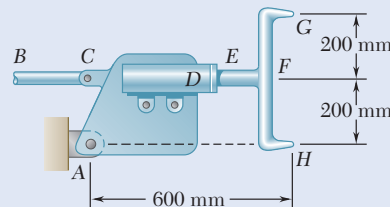


Fig. P15.166 and P15.167

15.167 In the automated welding setup shown, the position of the two welding tips G and H is controlled by the hydraulic cylinder D and rod BC . The cylinder is bolted to the vertical plate that at the instant shown rotates counterclockwise about A with a constant angular velocity of 1.6 rad/s. Knowing that at the same instant the length EF of the welding assembly is increasing at the constant rate of 300 mm/s, determine (a) the velocity of tip G , (b) the acceleration of tip G .

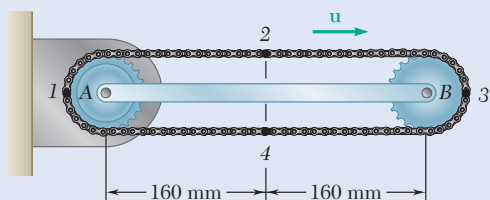


Fig. P15.168 and P15.169

15.168 and 15.169 A chain is looped around two gears of radius 40 mm that can rotate freely with respect to the 320-mm arm AB . The chain moves about arm AB in a clockwise direction at the constant rate of 80 mm/s relative to the arm. Knowing that in the position shown arm AB rotates clockwise about A at the constant rate $\omega = 0.75$ rad/s, determine the acceleration of each of the chain links indicated.

15.168 Links 1 and 2

15.169 Links 3 and 4

15.170 A basketball player shoots a free throw in such a way that his shoulder can be considered a pin joint at the moment of release as shown. Knowing that at the instant shown the upper arm SE has a constant angular velocity of 2 rad/s counterclockwise and the forearm EW has a constant clockwise angular velocity of 4 rad/s with respect to SE , determine the velocity and acceleration of the wrist W .

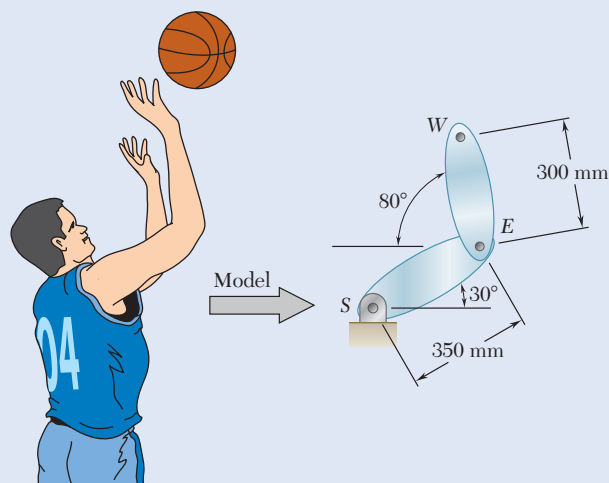


Fig. P15.170

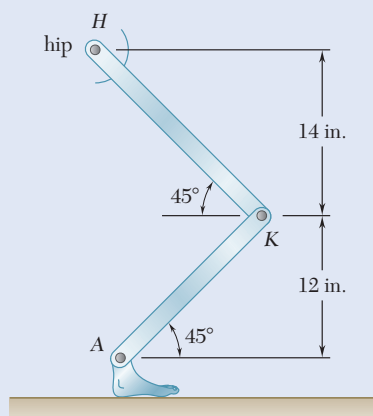


Fig. P15.171

15.171 The human leg can be crudely approximated as two rigid bars (the femur and the tibia) connected with a pin joint. At the instant shown, the velocity of the ankle A is zero, the tibia AK has an angular velocity of 1.5 rad/s counterclockwise and an angular acceleration of 1 rad/s² counterclockwise. Determine the relative angular velocity and relative angular acceleration of the femur KH with respect to AK so that the velocity and acceleration of H are both straight up at this instant.

15.172 The collar P slides outward at a constant relative speed u along rod AB , which rotates counterclockwise with a constant angular velocity of 20 rpm. Knowing that $r = 250$ mm when $\theta = 0$ and that the collar reaches B when $\theta = 90^\circ$, determine the magnitude of the acceleration of the collar P just as it reaches B .

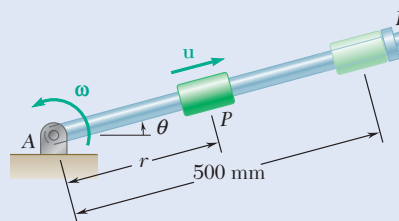


Fig. P15.172

15.173 Pin P slides in a circular slot cut in the plate shown at a constant relative speed $u = 90$ mm/s. Knowing that at the instant shown the plate rotates clockwise about A at the constant rate $\omega = 3$ rad/s, determine the acceleration of the pin if it is located at (a) point A , (b) point B , (c) point C .

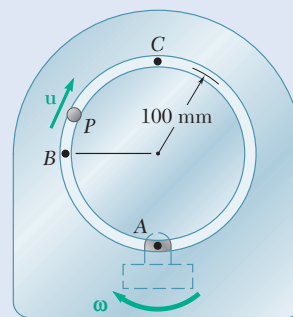


Fig. P15.173

15.174 Rod AD is bent in the shape of an arc of a circle with a radius of $b = 150$ mm. The position of the rod is controlled by pin B that slides in a horizontal slot and also slides along the rod. Knowing that at the instant shown pin B moves to the right at a constant speed of 75 mm/s, determine (a) the angular velocity of the rod, (b) the angular acceleration of the rod.

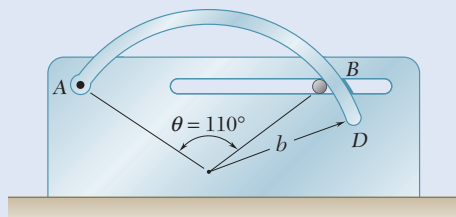


Fig. P15.174

15.175 Solve Prob. 15.174 when $\theta = 90^\circ$.

15.176 Knowing that at the instant shown the rod attached at A has an angular velocity of 5 rad/s counterclockwise and an angular acceleration of 2 rad/s² clockwise, determine the angular velocity and the angular acceleration of the rod attached at B .

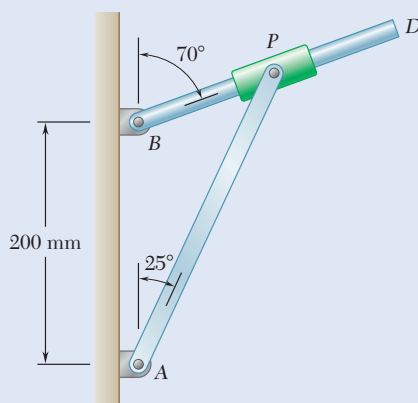


Fig. P15.176

15.177 The Geneva mechanism shown is used to provide an intermittent rotary motion of disk S . Disk D rotates with a constant counterclockwise angular velocity ω_D of 8 rad/s. A pin P is attached to disk D and can slide in one of the six equally spaced slots cut in disk S . It is desirable that the angular velocity of disk S be zero as the pin enters and leaves each of the six slots; this will occur if the distance between the centers of the disks and the radii of the disks are related as shown. Determine the angular velocity and angular acceleration of disk S at the instant when $\phi = 150^\circ$.

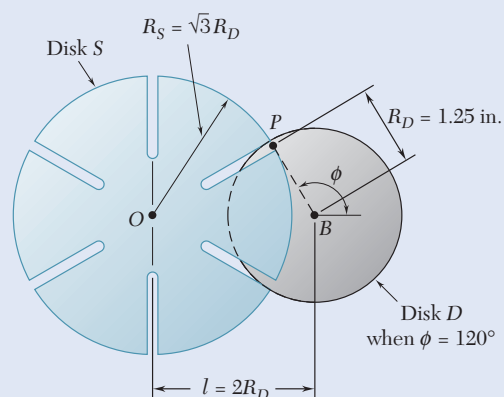


Fig. P15.177

15.178 In Prob. 15.177, determine the angular velocity and angular acceleration of disk S at the instant when $\phi = 135^\circ$.

- 15.179** At the instant shown bar BC has an angular velocity of 3 rad/s and an angular acceleration of 2 rad/s^2 , both counterclockwise; determine the angular acceleration of the plate.

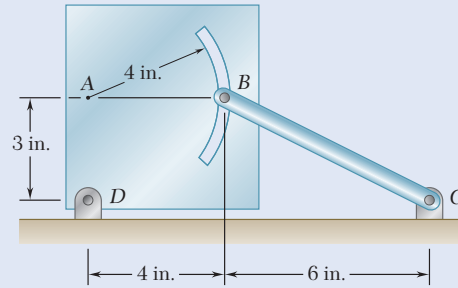


Fig. P15.179 and P15.180

- 15.180** At the instant shown bar BC has an angular velocity of 3 rad/s and an angular acceleration of 2 rad/s^2 , both clockwise; determine the angular acceleration of the plate.

- *15.181** Rod AB passes through a collar that is welded to link DE . Knowing that at the instant shown block A moves to the right at a constant speed of 75 in./s , determine (a) the angular velocity of rod AB , (b) the velocity relative to the collar of the point of the rod in contact with the collar, (c) the acceleration of the point of the rod in contact with the collar. (*Hint: Rod AB and link DE have the same ω and the same α .*)

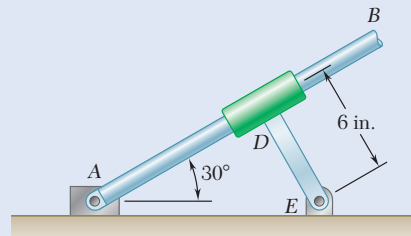


Fig. P15.181

- *15.182** Solve Prob. 15.181 assuming block A moves to the left at a constant speed of 75 in./s .
- *15.183** In Prob. 15.157, determine the acceleration of pin P .

*15.6 MOTION OF A RIGID BODY IN SPACE

Extending the study of motion in two dimensions to analyzing three-dimensional motion uses most of the same concepts as before, but with some added computational complexity. We introduce these ideas in this section and the next, and we will return to them when discussing kinetics of a rigid body in Chapter 18.

15.6A Motion About a Fixed Point

In Sec. 15.1B, we considered the motion of a rigid body constrained to rotate about a fixed axis. Here we examine the more general case of the three-dimensional motion of a rigid body that has a fixed point O . First, we prove:

The most general displacement of a rigid body with a fixed point O is equivalent to a rotation of the body about an axis through O .

This statement is known as Euler's theorem. We analyze the motion of a sphere with a center O ; this analysis can be extended to a rigid body of any shape. Since three points define the position of a solid in space, we let the center O and two points A and B on the surface of the sphere define the position of the sphere and thus the position of the body. Let A_1 and B_1 characterize the position of the sphere at one instant, and let A_2 and B_2 characterize its position at a later instant (Fig. 15.31a). Since the sphere is rigid, the lengths of the arcs of great circles A_1B_1 and A_2B_2 must be equal, but except for this requirement, the positions of A_1 , A_2 , B_1 , and B_2 are arbitrary. We will show that the points A and B can be brought, respectively, from A_1 and B_1 into A_2 and B_2 by a single rotation of the sphere about an axis.

For convenience, and without loss of generality, we select point B so that its initial position coincides with the final position of A ; thus, $B_1 = A_2$ (Fig. 15.31b). We draw the arcs of great circles A_1A_2 , A_2B_2 , and the arcs bisecting, respectively, A_1A_2 and A_2B_2 . Let C be the point of intersection of these last two arcs. We complete the construction by drawing A_1C , A_2C , and B_2C . As pointed out above, because of the rigidity of the sphere, $A_1B_1 = A_2B_2$. Since C is by construction equidistant from A_1 , A_2 , and B_2 , we also have $A_1C = A_2C = B_2C$. As a result, the spherical triangles A_1CA_2 and B_1CB_2 are congruent, and the angles A_1CA_2 and B_1CB_2 are equal. Denoting the common value of these angles by θ , we conclude that the sphere can be brought from its initial position into its final position by a single rotation through θ about the axis OC .

It follows that we can consider the motion during a time interval Δt of a rigid body with a fixed point O as a rotation through $\Delta\theta$ about a certain axis. Drawing a vector with a magnitude of $\Delta\theta/\Delta t$ along that axis and letting Δt approach zero, we obtain in the limit the **instantaneous axis of rotation** and the angular velocity ω of the body at the instant considered (Fig. 15.32). We can then obtain the velocity of a particle P

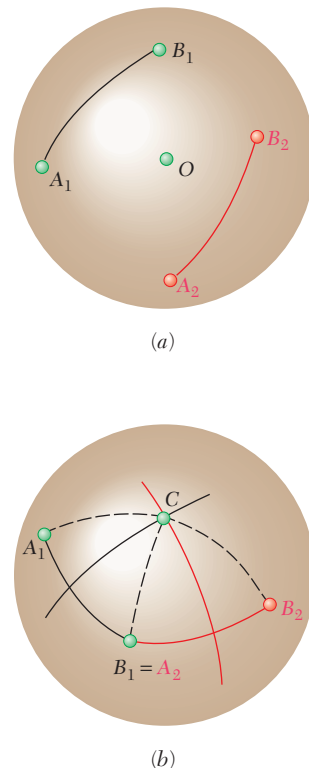


Fig. 15.31 (a) Positions of two points on a rotating sphere; (b) the sphere can be brought into this new position by a single rotation.

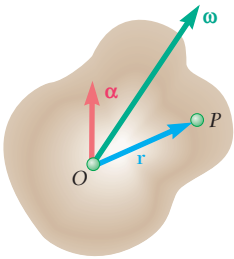


Fig. 15.32 Angular velocity and angular acceleration of a rigid body moving about a fixed point O .

of the body, as in Sec. 15.1B, by forming the vector product of $\boldsymbol{\omega}$ and of the position vector $\mathbf{r}$ of the particle:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \boldsymbol{\omega} \times \mathbf{r} \quad (15.37)$$

We obtain the acceleration of the particle by differentiating Eq. (15.37) with respect to t . As in Sec. 15.1B, we have

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \quad (15.38)$$

Here we have defined the angular acceleration $\boldsymbol{\alpha}$ as the derivative

$$\boldsymbol{\alpha} = \frac{d\boldsymbol{\omega}}{dt} \quad (15.39)$$

of the angular velocity $\boldsymbol{\omega}$.

In the case of the motion of a rigid body with a fixed point, the direction of $\boldsymbol{\omega}$ and of the instantaneous axis of rotation changes from one instant to the next. The angular acceleration $\boldsymbol{\alpha}$ therefore reflects the change in direction of $\boldsymbol{\omega}$ as well as its change in magnitude. Thus, in general, $\boldsymbol{\alpha}$ is *not directed along the instantaneous axis of rotation*. Although the particles of the body located on the instantaneous axis of rotation have zero velocity at the instant considered, they do not have zero acceleration. Also, the accelerations of the various particles of the body *cannot* be determined as if the body were rotating permanently about the instantaneous axis.

Recalling the definition of the velocity of a particle with position vector $\mathbf{r}$, we note that the angular acceleration $\boldsymbol{\alpha}$, as expressed in Eq. (15.39), represents the velocity of the tip of vector $\boldsymbol{\omega}$. This property may be useful in determining the angular acceleration of a rigid body. For example, it follows that vector $\boldsymbol{\alpha}$ is tangent to the curve described in space by the tip of vector $\boldsymbol{\omega}$.

Note that vector $\boldsymbol{\omega}$ moves within the body, as well as in space. It thus generates two cones called, respectively, the **body cone** and the **space cone** (Fig. 15.33).[†] It can be shown that, at any given instant, the two cones are tangent along the instantaneous axis of rotation and that, as the body moves, the body cone appears to *roll* on the space cone.

Before concluding our analysis of the motion of a rigid body with a fixed point, we should prove that angular velocities are actually vectors. Some quantities, such as the *finite rotations* of a rigid body, have magnitude and direction but do not obey the parallelogram law of addition; these quantities cannot be considered to be vectors. In contrast, angular velocities (and also *infinitesimal rotations*), as we demonstrate presently, do obey the parallelogram law and thus are truly vector quantities.

Consider a rigid body with a fixed point O that rotates at a given instant simultaneously about the axes OA and OB with angular velocities $\boldsymbol{\omega}_1$ and $\boldsymbol{\omega}_2$ (Fig. 15.34a). We know that this motion must be equivalent at the instant considered to a single rotation of angular velocity $\boldsymbol{\omega}$. We propose to show that

$$\boldsymbol{\omega} = \boldsymbol{\omega}_1 + \boldsymbol{\omega}_2 \quad (15.40)$$

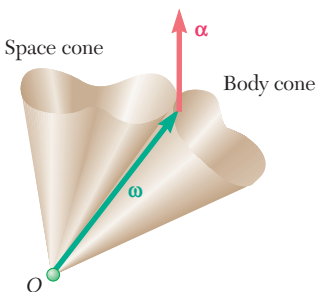


Fig. 15.33 The angular velocity vector generates a body cone and a space cone as it changes direction.

[†]Recall that a cone is, by definition, a surface generated by a straight line passing through a fixed point. In general, the cones considered here are not circular cones.

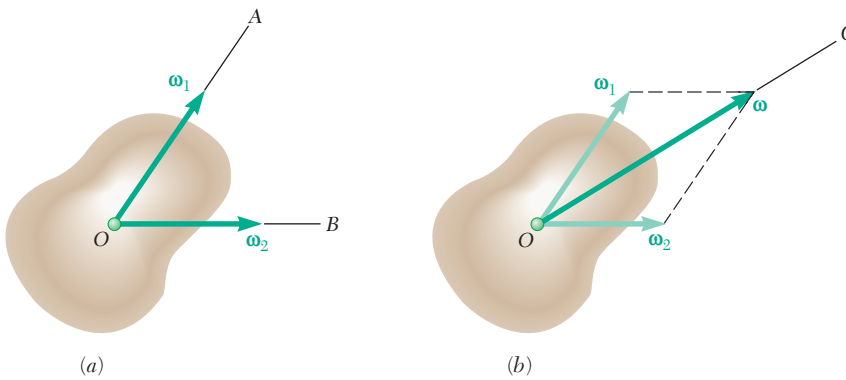


Fig. 15.34 (a) A rigid body rotating about two axes simultaneously; (b) the motion is equivalent to a single rotation with angular velocity equal to the vector sum of the initial angular velocities.

i.e., that we can obtain the resulting angular velocity by adding ω_1 and ω_2 using the parallelogram law (Fig. 15.34b).

Consider a particle P of the body, defined by the position vector $\mathbf{r}$. Denoting the velocity of P when the body rotates about OA only, about OB only, and about both axes simultaneously, by $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}$, respectively, we have

$$\mathbf{v} = \omega \times \mathbf{r} \quad \mathbf{v}_1 = \omega_1 \times \mathbf{r} \quad \mathbf{v}_2 = \omega_2 \times \mathbf{r} \quad (15.41)$$

But the vectorial character of *linear* velocities is well established (since they represent the derivatives of position vectors). We therefore have

$$\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$$

where the plus sign indicates vector addition. Substituting from Eq. (15.41), we obtain

$$\begin{aligned} \omega \times \mathbf{r} &= \omega_1 \times \mathbf{r} + \omega_2 \times \mathbf{r} \\ \omega \times \mathbf{r} &= (\omega_1 + \omega_2) \times \mathbf{r} \end{aligned}$$

where the plus sign still indicates vector addition. Since the relation obtained holds for an arbitrary $\mathbf{r}$, we conclude that Eq. (15.40) must be true.

*15.6B General Motion

We now consider the most general motion of a rigid body in space. Let A and B be two particles of the body. Recall from Sec. 11.4D that we can express the velocity of B with respect to the fixed frame of reference $OXYZ$ as

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \quad (15.42)$$

where $\mathbf{v}_{B/A}$ is the velocity of B relative to a frame $AX'Y'Z'$ attached to A and of fixed orientation (Fig. 15.35). Since A is fixed in this frame, the motion of the body relative to $AX'Y'Z'$ is the motion of a body with a fixed point. Therefore we can obtain the relative velocity $\mathbf{v}_{B/A}$ from Eq. (15.37) after replacing $\mathbf{r}$ by the position vector $\mathbf{r}_{B/A}$ of B relative to A . Substituting for $\mathbf{v}_{B/A}$ into Eq. (15.42), we have

$$\mathbf{v}_B = \mathbf{v}_A + \omega \times \mathbf{r}_{B/A} \quad (15.43)$$

where ω is the angular velocity of the body at the instant considered.



Photo 15.8 You can obtain the angular velocity of a fire truck ladder rotating about its fixed base by adding the angular velocities that correspond to simultaneous rotations about two different axes.

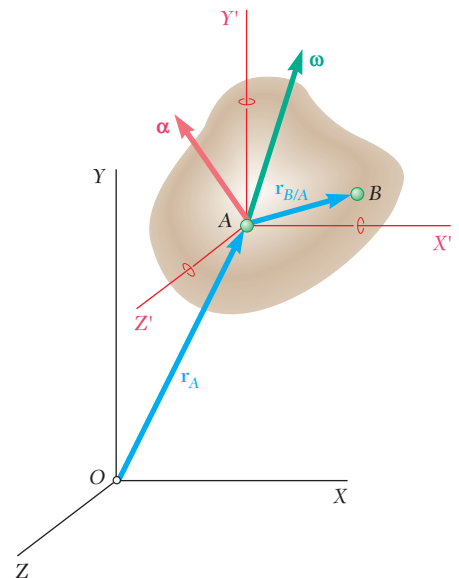


Fig. 15.35 A rigid body moving relative to a fixed reference frame $OXYZ$ and a reference frame attached to the body but with fixed orientation, $OX'Y'Z'$.

We can obtain the acceleration of B by a similar reasoning. We first write

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

and, from Eq. (15.38),

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{B/A}) \quad (15.44)$$

where $\boldsymbol{\alpha}$ is the angular acceleration of the body at the instant considered.

The angular acceleration $\boldsymbol{\alpha}$ represents the rate of change $(\dot{\boldsymbol{\omega}})_{OXYZ}$ of vector $\boldsymbol{\omega}$ with respect to a fixed frame of reference $OXYZ$ and reflects both a change in magnitude and a change in direction of the angular velocity. When computing $\boldsymbol{\alpha}$, you will usually find it convenient to first compute the rate of change $(\dot{\boldsymbol{\omega}})_{Oxyz}$ of $\boldsymbol{\omega}$ with respect to a rotating frame of reference $Oxyz$ of your choice and use Eq. (15.31) to obtain $\boldsymbol{\alpha}$. You have

$$\boldsymbol{\alpha} = (\dot{\boldsymbol{\omega}})_{OXYZ} = (\dot{\boldsymbol{\omega}})_{Oxyz} + \boldsymbol{\Omega} \times \boldsymbol{\omega}$$

where $\boldsymbol{\Omega}$ is the angular velocity of the rotating frame $Oxyz$ [Sample Prob. 15.21].

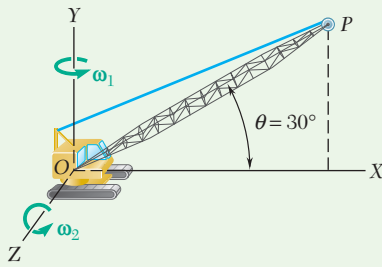
Equations (15.43) and (15.44) show that **the most general motion of a rigid body is equivalent, at any given instant, to the sum of a translation** (in which all of the particles of the body have the same velocity and acceleration as a reference particle A) **and of a motion in which particle A is assumed to be fixed.**[†]

By solving Eqs. (15.43) and (15.44) for $\mathbf{v}_A$ and $\mathbf{a}_A$, it can be shown that the motion of the body with respect to a frame attached to B would be characterized by the same vectors $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ as its motion relative to $AX'Y'Z'$. Thus, the angular velocity and angular acceleration of a rigid body at a given instant are independent of the choice of reference point. If $AX'Y'Z'$ is a non-rotating frame, you should keep in mind that whether the moving frame is attached to A or to B , it should maintain a fixed orientation; that is, it should remain parallel to the fixed reference frame $OXYZ$ throughout the motion of the rigid body.

In many problems, it will be more convenient to use a moving frame that is allowed to rotate as well as to translate. We discuss the use of such moving frames in Sec. 15.7.

[†]Recall from Sec 15.6A that, in general, vectors $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ are not collinear and that the accelerations of the particles of the body in their motion relative to the frame $AX'Y'Z'$ cannot be determined as if the body were rotating permanently about the instantaneous axis through A .

Sample Problem 15.21



The crane shown rotates horizontally with a constant angular velocity ω_1 of 0.30 rad/s. Simultaneously, the boom is being raised with a constant angular velocity ω_2 of 0.50 rad/s relative to the cab. Knowing that the length of the boom OP is $l = 12$ m, determine (a) the angular velocity ω of the boom, (b) the angular acceleration α of the boom, (c) the velocity $\mathbf{v}$ of the tip of the boom, (d) the acceleration $\mathbf{a}$ of the tip of the boom.

STRATEGY: There are multiple rotational axes, so you need to use the general motion velocity and acceleration kinematic equations. Add the given angular velocities vectorially to find the overall angular velocity of the boom, and differentiate that to find the angular acceleration.

MODELING and ANALYSIS:

a. Angular Velocity of Boom. Add the angular velocity ω_1 of the cab and the angular velocity ω_2 of the boom relative to the cab to obtain the angular velocity ω of the boom at the instant considered:

$$\omega = \omega_1 + \omega_2 \quad \omega = (0.30 \text{ rad/s})\mathbf{j} + (0.50 \text{ rad/s})\mathbf{k} \quad \blacktriangleleft$$

b. Angular Acceleration of Boom. Obtain the angular acceleration α of the boom by differentiating ω . Since the vector ω_1 is constant in magnitude and direction, you have

$$\alpha = \dot{\omega} = \dot{\omega}_1 + \dot{\omega}_2 = 0 + \dot{\omega}_2$$

where the rate of change $\dot{\omega}_2$ is to be computed with respect to the fixed frame $OXYZ$. However, it is more convenient to use a frame $Oxyz$ attached to the cab and rotating with it, since the vector ω_2 also rotates with the cab and therefore has zero rate of change with respect to that frame. Using Eq. (15.31) with $\mathbf{Q} = \omega_2$ and $\mathbf{\Omega} = \omega_1$, you have

$$\begin{aligned} (\dot{\mathbf{Q}})_{OXYZ} &= (\dot{\mathbf{Q}})_{Oxyz} + \mathbf{\Omega} \times \mathbf{Q} \\ (\dot{\omega}_2)_{OXYZ} &= (\dot{\omega}_2)_{Oxyz} + \omega_1 \times \omega_2 \\ \alpha &= (\dot{\omega}_2)_{OXYZ} = 0 + (0.30 \text{ rad/s})\mathbf{j} \times (0.50 \text{ rad/s})\mathbf{k} \end{aligned}$$

$$\alpha = (0.15 \text{ rad/s}^2)\mathbf{i} \quad \blacktriangleleft$$

c. Velocity of Tip of Boom. Noting that the position vector of point P is $\mathbf{r} = (10.39 \text{ m})\mathbf{i} + (6 \text{ m})\mathbf{j}$ (Fig. 1) and using the expression found for ω in part (a), you get

$$\mathbf{v} = \omega \times \mathbf{r} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0.30 \text{ rad/s} & 0.50 \text{ rad/s} \\ 10.39 \text{ m} & 6 \text{ m} & 0 \end{vmatrix}$$

$$\mathbf{v} = -(3 \text{ m/s})\mathbf{i} + (5.20 \text{ m/s})\mathbf{j} - (3.12 \text{ m/s})\mathbf{k} \quad \blacktriangleleft$$

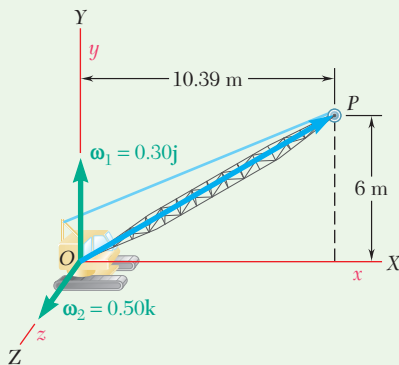


Fig. 1 A rotating frame xyz attached to the cab.

(continued)

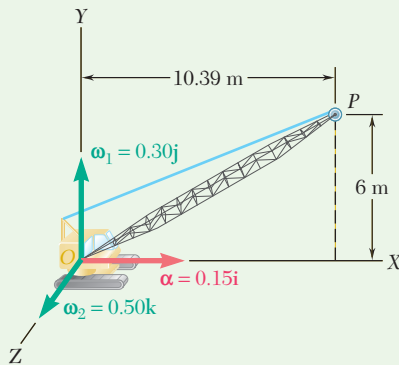


Fig. 2 Angular velocities and accelerations of the boom.

d. Acceleration of Tip of Boom. Recall that $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$. Then, from Fig. 2,

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times \mathbf{v}$$

$$\mathbf{a} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.15 & 0 & 0 \\ 10.39 & 6 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0.30 & 0.50 \\ -3 & 5.20 & -3.12 \end{vmatrix}$$

$$= 0.90\mathbf{k} - 0.94\mathbf{i} - 2.60\mathbf{i} - 1.50\mathbf{j} + 0.90\mathbf{k}$$

$$\mathbf{a} = -(3.54 \text{ m/s}^2)\mathbf{i} - (1.50 \text{ m/s}^2)\mathbf{j} + (1.80 \text{ m/s}^2)\mathbf{k} \quad \blacktriangleleft$$

REFLECT and THINK: The base of the cab acts as the fixed point of the motion. Even though both components of angular velocity are constant, there is an angular acceleration due to the change in direction of the angular velocity $\boldsymbol{\omega}_2$. The angular velocity vector $\boldsymbol{\omega}_2$ changes due to the rotation of the cab, $\boldsymbol{\omega}_1$.

Sample Problem 15.22

The rod AB has a length of 7 in. and is attached to the disk by a ball-and-socket connection and to the collar B by a clevis. The disk rotates in the yz plane at a constant rate of $\omega_1 = 12 \text{ rad/s}$, while the collar is free to slide along the horizontal rod CD . For the position $\theta = 0$, determine (a) the velocity of the collar, (b) the angular velocity of the rod.

STRATEGY: Use the velocity and acceleration kinematic equations to relate the velocities of points A and B .

MODELING and ANALYSIS:

a. Velocity of Collar. Since point A is attached to the disk and since collar B moves in a direction parallel to the x axis, you have (Fig. 1)

$$\mathbf{v}_A = \boldsymbol{\omega}_1 \times \mathbf{r}_A = 12\mathbf{i} \times 2\mathbf{k} = -24\mathbf{j} \quad \mathbf{v}_B = v_B\mathbf{i}$$

Denoting the angular velocity of the rod by $\boldsymbol{\omega}$, you obtain

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

$$v_B\mathbf{i} = -24\mathbf{j} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \omega_x & \omega_y & \omega_z \\ 6 & 3 & -2 \end{vmatrix}$$

$$v_B\mathbf{i} = -24\mathbf{j} + (-2\omega_y - 3\omega_z)\mathbf{i} + (6\omega_z + 2\omega_x)\mathbf{j} + (3\omega_x - 6\omega_y)\mathbf{k}$$

Equating the coefficients of the unit vectors, you get

$$v_B = -2\omega_y - 3\omega_z \quad (1)$$

$$24 = 2\omega_x + 6\omega_z \quad (2)$$

$$0 = 3\omega_x - 6\omega_y \quad (3)$$

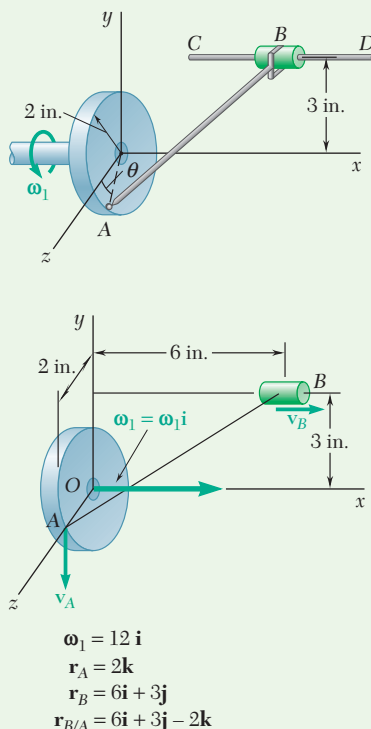


Fig. 1 Angular velocity of the disk and the direction of the velocities of A and B .

You have three equations and four unknowns in these equations. Fortunately, multiplying Eqs. (1), (2), (3), respectively, by 6, 3, -2 and adding gives you

$$6v_B + 72 = 0 \quad v_B = -12 \quad \mathbf{v}_B = -(12 \text{ in./s})\mathbf{i} \quad \blacktriangleleft$$

b. Angular Velocity of Rod AB . Note that you cannot determine the angular velocity from Eqs. (1), (2), and (3) because the determinant formed by the coefficients of ω_x , ω_y , and ω_z is zero. You must therefore obtain an additional equation by considering the constraint imposed by the clevis at B .

The collar–clevis connection at B permits rotation of AB about rod CD and also about an axis perpendicular to the plane containing AB and CD . It prevents rotation of AB about the axis EB , which is perpendicular to CD and lies in the plane containing AB and CD (Fig. 2). Thus, the projection of $\boldsymbol{\omega}$ on $\mathbf{r}_{E/B}$ must be zero, and you have

$$\boldsymbol{\omega} \cdot \mathbf{r}_{E/B} = 0$$

$$(\omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k}) \cdot (-3\mathbf{j} + 2\mathbf{k}) = 0$$

$$-3\omega_y + 2\omega_z = 0 \quad (4)$$

Solving Eqs. (1) through (4) simultaneously, you obtain

$$v_B = -12 \quad \omega_x = 3.69 \quad \omega_y = 1.846 \quad \omega_z = 2.77$$

$$\boldsymbol{\omega} = (3.69 \text{ rad/s})\mathbf{i} + (1.846 \text{ rad/s})\mathbf{j} + (2.77 \text{ rad/s})\mathbf{k} \quad \blacktriangleleft$$

REFLECT and THINK: Note that the direction of EB is that of the vector triple product

$$\mathbf{r}_{B/C} \times (\mathbf{r}_{B/C} \times \mathbf{r}_{B/A})$$

so you could write

$$\boldsymbol{\omega} \cdot [\mathbf{r}_{B/C} \times (\mathbf{r}_{B/C} \times \mathbf{r}_{B/A})] = 0$$

This formulation would be particularly useful if the rod CD were not in a convenient direction.

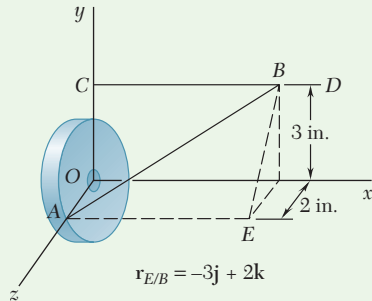


Fig. 2 The collar–clevis prevents rotation about EB .

SOLVING PROBLEMS ON YOUR OWN

In this section, you started the study of the **kinematics of rigid bodies in three dimensions**. You first studied the **motion of a rigid body about a fixed point** and then the **general motion of a rigid body**.

A. Motion of a rigid body about a fixed point. To analyze the motion of a point B of a body rotating about a fixed point O , you may have to take some or all of the following steps.

1. **Determine the position vector $\mathbf{r}$** connecting the fixed point O to point B .
2. **Determine the angular velocity $\boldsymbol{\omega}$ of the body** with respect to a fixed frame of reference. You can often obtain the angular velocity $\boldsymbol{\omega}$ by adding two component angular velocities $\boldsymbol{\omega}_1$ and $\boldsymbol{\omega}_2$ [Sample Prob. 15.21].
3. **Compute the velocity of B** from the equation

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} \quad (15.37)$$

Your computation is usually easier if you express the vector product as a determinant.

4. **Determine the angular acceleration $\boldsymbol{\alpha}$ of the body.** The angular acceleration $\boldsymbol{\alpha}$ represents the rate of change $(\dot{\boldsymbol{\omega}})_{OXYZ}$ of the vector $\boldsymbol{\omega}$ with respect to a fixed frame of reference $OXYZ$ and reflects both a change in magnitude and a change in direction of the angular velocity. However, when computing $\boldsymbol{\alpha}$, you may find it convenient to first compute the rate of change $(\dot{\boldsymbol{\omega}})_{Oxyz}$ of $\boldsymbol{\omega}$ with respect to a rotating frame of reference $Oxyz$ of your choice and use Eq. (15.31). You have

$$\boldsymbol{\alpha} = (\dot{\boldsymbol{\omega}})_{OXYZ} = (\dot{\boldsymbol{\omega}})_{Oxyz} + \boldsymbol{\Omega} \times \boldsymbol{\omega}$$

where $\boldsymbol{\Omega}$ is the angular velocity of the rotating frame $Oxyz$ [Sample Prob. 15.21].

5. **Compute the acceleration of B** by using the equation

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \quad (15.38)$$

Note that the vector product $(\boldsymbol{\omega} \times \mathbf{r})$ represents the velocity of point B and was computed in Step 3. Also, the computation of the first vector product in Eq. (15.38) is often simpler if you express this product in determinant form. Remember that, as was the case with the plane motion of a rigid body, the instantaneous axis of rotation *cannot* be used to determine accelerations.

B. General motion of a rigid body. The general motion of a rigid body may be considered as *the sum of a translation and a rotation*. Keep the following in mind:

a. In the translation part of the motion, all of the points of the body have the *same velocity* $\mathbf{v}_A$ and the *same acceleration* $\mathbf{a}_A$ as point A of the body that has been selected as the reference point.

b. In the rotation part of the motion, the same reference point A is treated as if it were a *fixed point*.

1. To determine the velocity of a point B of the rigid body when you know the velocity $\mathbf{v}_A$ of the reference point A and the angular velocity $\boldsymbol{\omega}$ of the body, you simply add $\mathbf{v}_A$ to the velocity $\mathbf{v}_{B/A} = \boldsymbol{\omega} \times \mathbf{r}_{B/A}$ of B in its rotation about A :

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A} \quad (15.43)$$

As indicated earlier, the computation of the vector product is usually simpler if you express this product in determinant form.

You can also use Eq. (15.43) to determine the magnitude of $\mathbf{v}_B$ when its direction is known, even if $\boldsymbol{\omega}$ is not known. Although the corresponding three scalar equations are linearly dependent and the components of $\boldsymbol{\omega}$ are indeterminate, you can eliminate these components and find $\mathbf{v}_A$ by using an appropriate linear combination of the three equations [Sample Prob. 15.22, part (a)]. Alternatively, you can assign an arbitrary value to one of the components of $\boldsymbol{\omega}$ and solve the equations for $\mathbf{v}_A$. However, you must seek an additional equation in order to determine the true values of the components of $\boldsymbol{\omega}$ [Sample Prob. 15.22, part (b)].

2. To determine the acceleration of a point B of the rigid body when you know the acceleration $\mathbf{a}_A$ of the reference point A and the angular acceleration $\boldsymbol{\alpha}$ of the body, you simply add $\mathbf{a}_A$ to the acceleration of B in its rotation about A , as expressed by Eq. (15.38):

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{B/A}) \quad (15.44)$$

Note that the vector product $(\boldsymbol{\omega} \times \mathbf{r}_{B/A})$ represents the velocity $\mathbf{v}_{B/A}$ of B relative to A and already may have been computed as part of your calculation of $\mathbf{v}_B$.

You can also use the three scalar equations associated with Eq. (15.44) to determine the magnitude of $\mathbf{a}_B$ when its direction is known, even if $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ are not known. Although the components of $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ are indeterminate, you can assign arbitrary values to one of the components of $\boldsymbol{\omega}$ and to one of the components of $\boldsymbol{\alpha}$ and solve the equations for $\mathbf{a}_B$.

Problems

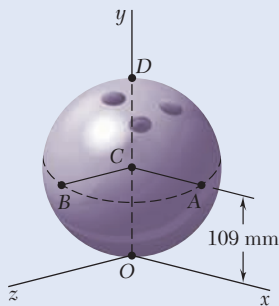


Fig. P15.184 and P15.185

END-OF-SECTION PROBLEMS

15.184 The bowling ball shown rolls without slipping on the horizontal xz plane with an angular velocity $\boldsymbol{\omega} = \omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k}$. Knowing that $\mathbf{v}_A = (4.8 \text{ m/s})\mathbf{i} - (4.8 \text{ m/s})\mathbf{j} + (3.6 \text{ m/s})\mathbf{k}$ and $\mathbf{v}_D = (9.6 \text{ m/s})\mathbf{i} + (7.2 \text{ m/s})\mathbf{k}$, determine (a) the angular velocity of the bowling ball, (b) the velocity of its center C .

15.185 The bowling ball shown rolls without slipping on the horizontal xz plane with an angular velocity $\boldsymbol{\omega} = \omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k}$. Knowing that $\mathbf{v}_B = (3.6 \text{ m/s})\mathbf{i} - (4.8 \text{ m/s})\mathbf{j} + (4.8 \text{ m/s})\mathbf{k}$ and $\mathbf{v}_D = (7.2 \text{ m/s})\mathbf{i} + (9.6 \text{ m/s})\mathbf{k}$, determine (a) the angular velocity of the bowling ball, (b) the velocity of its center C .

15.186 Plate ABD and rod OB are rigidly connected and rotate about the ball-and-socket joint O with an angular velocity $\boldsymbol{\omega} = \omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k}$. Knowing that $\mathbf{v}_A = (80 \text{ mm/s})\mathbf{i} + (360 \text{ mm/s})\mathbf{j} + (v_A)_z \mathbf{k}$ and $\omega_x = 1.5 \text{ rad/s}$, determine (a) the angular velocity of the assembly, (b) the velocity of point D .

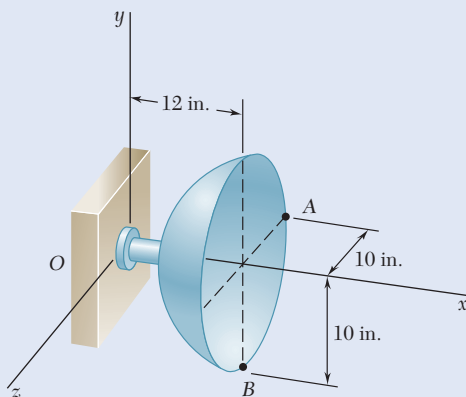


Fig. P15.187

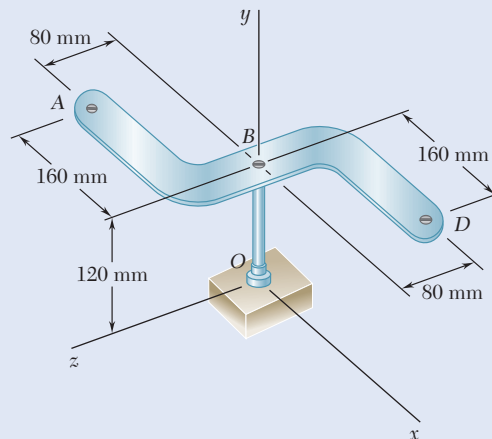


Fig. P15.186

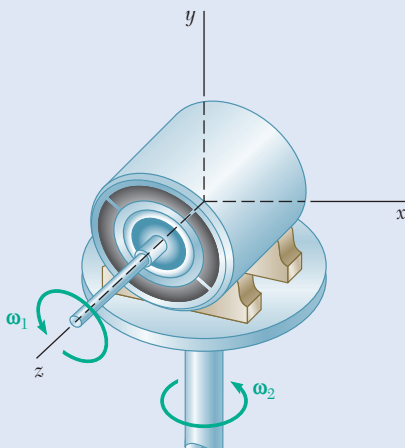


Fig. P15.188

15.187 At the instant considered, the radar antenna shown rotates about the origin of coordinates with an angular velocity $\boldsymbol{\omega} = \omega_x \mathbf{i} + \omega_y \mathbf{j} + \omega_z \mathbf{k}$. Knowing that $(v_A)_y = 15 \text{ in./s}$, $(v_B)_y = 9 \text{ in./s}$, and $(v_B)_z = 18 \text{ in./s}$, determine (a) the angular velocity of the antenna, (b) the velocity of point A .

15.188 The rotor of an electric motor rotates at the constant rate $\omega_1 = 1800 \text{ rpm}$. Determine the angular acceleration of the rotor as the motor is rotated about the y axis with a constant angular velocity ω_2 of 6 rpm counterclockwise when viewed from the positive y axis.

- 15.189** The disk of a portable sander rotates at the constant rate $\omega_1 = 4400$ rpm as shown. Determine the angular acceleration of the disk as a worker rotates the sander about the z axis with an angular velocity of 0.5 rad/s and an angular acceleration of 2.5 rad/s², both clockwise when viewed from the positive z axis.

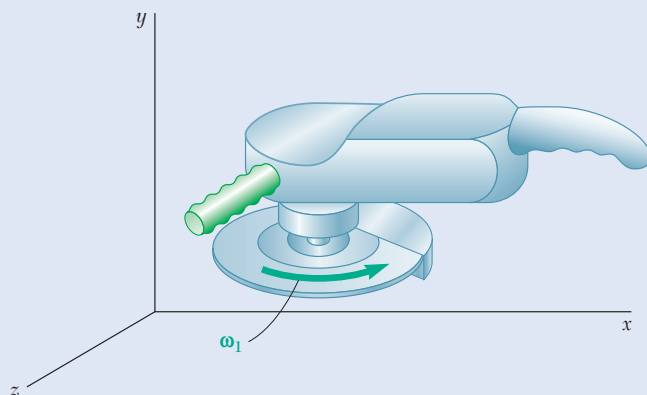


Fig. P15.189

- 15.190** A flight simulator is used to train pilots on how to recognize spatial disorientation. It has four degrees of freedom, and can rotate around a planetary axis as well as in yaw, pitch, and roll. Knowing that the simulator is rotating around the planetary axis with a constant angular velocity of 20 rpm counterclockwise as seen from above, determine the angular acceleration of the cab if (a) the cab has a constant pitch angular velocity of $+3\mathbf{k}$ rad/s, (b) the cab has a constant roll angular velocity of $-4\mathbf{i}$ rad/s.

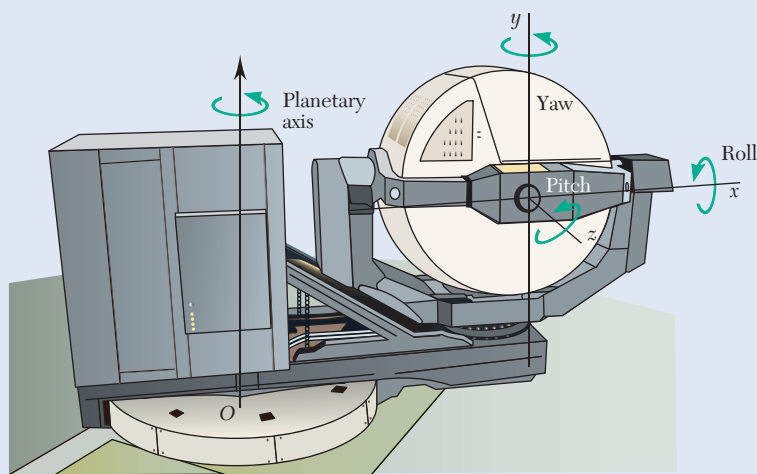


Fig. P15.190

- 15.191** In the system shown, disk A is free to rotate about the horizontal rod OA. Assuming that disk B is stationary ($\omega_2 = 0$), and that shaft OC rotates with a constant angular velocity ω_1 , determine (a) the angular velocity of disk A, (b) the angular acceleration of disk A.
- 15.192** In the system shown, disk A is free to rotate about the horizontal rod OA. Assuming that shaft OC and disk B rotate with constant angular velocities ω_1 and ω_2 , respectively, both counterclockwise, determine (a) the angular velocity of disk A, (b) the angular acceleration of disk A.

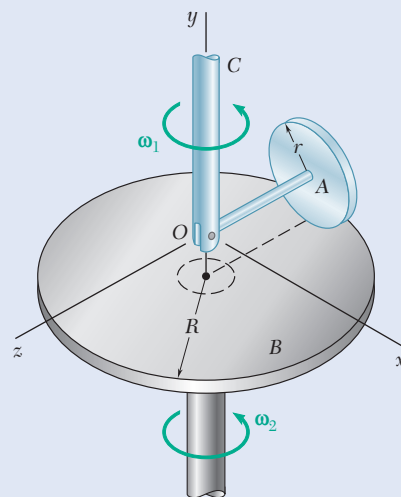


Fig. P15.191 and P15.192

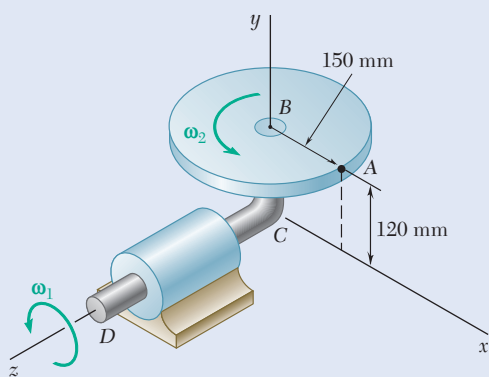


Fig. P15.193

15.193 The L-shaped arm BCD rotates about the z axis with a constant angular velocity $\omega_1 = 5$ rad/s. Knowing that the 150-mm-radius disk rotates about BC with a constant angular velocity $\omega_2 = 4$ rad/s, determine (a) the velocity of point A , (b) the acceleration of point A .

15.194 A gun barrel of length $OP = 4$ m is mounted on a turret as shown. To keep the gun aimed at a moving target, the azimuth angle β is being increased at the rate $d\beta/dt = 30^\circ/\text{s}$ and the elevation angle γ is being increased at the rate $d\gamma/dt = 10^\circ/\text{s}$. For the position $\beta = 90^\circ$ and $\gamma = 30^\circ$, determine (a) the angular velocity of the barrel, (b) the angular acceleration of the barrel, (c) the velocity and acceleration of point P .

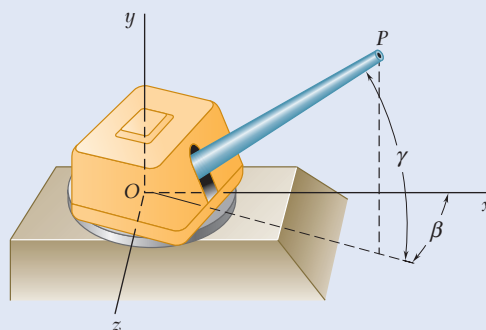


Fig. P15.194

15.195 A 3-in.-radius disk spins at the constant rate $\omega_2 = 4$ rad/s about an axis held by a housing attached to a horizontal rod that rotates at the constant rate $\omega_1 = 5$ rad/s. For the position shown, determine (a) the angular acceleration of the disk, (b) the acceleration of point P on the rim of the disk if $\theta = 0$, (c) the acceleration of point P on the rim of the disk if $\theta = 90^\circ$.

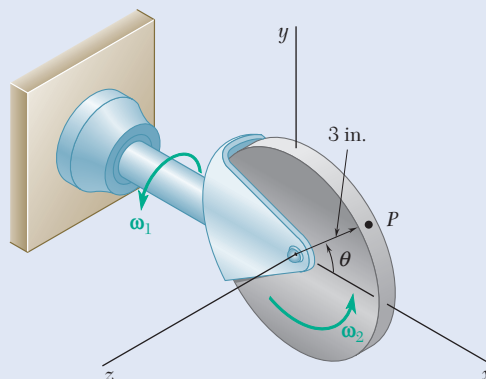


Fig. P15.195 and P15.196

15.196 A 3-in.-radius disk spins at the constant rate $\omega_2 = 4$ rad/s about an axis held by a housing attached to a horizontal rod that rotates at the constant rate $\omega_1 = 5$ rad/s. Knowing that $\theta = 30^\circ$, determine the acceleration of point P on the rim of the disk.

- 15.197** The cone shown rolls on the xz plane with its apex at the origin of coordinates. Denoting by ω_1 the constant angular velocity of the axis OB of the cone about the y axis, determine (a) the rate of spin of the cone about the axis OB , (b) the total angular velocity of the cone, (c) the angular acceleration of the cone.

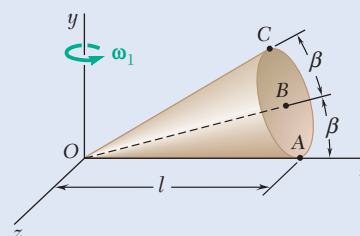


Fig. P15.197

- 15.198** At the instant shown, the robotic arm ABC is being rotated simultaneously at the constant rate $\omega_1 = 0.15$ rad/s about the y axis, and at the constant rate $\omega_2 = 0.25$ rad/s about the z axis. Knowing that the length of arm ABC is 1 m, determine (a) the angular acceleration of the arm, (b) the velocity of point C , (c) the acceleration of point C .

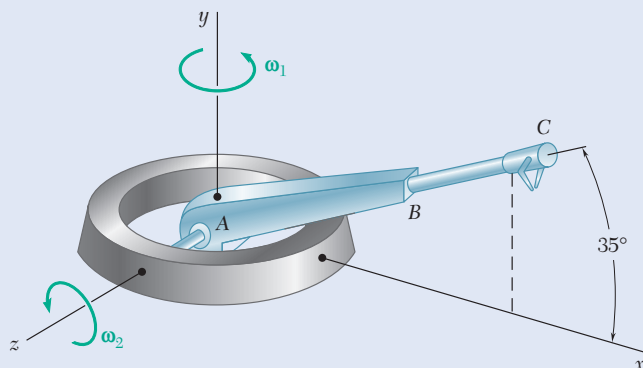


Fig. P15.198

- 15.199** In the planetary gear system shown, gears A and B are rigidly connected to each other and rotate as a unit about the inclined shaft. Gears C and D rotate with constant angular velocities of 30 rad/s and 20 rad/s, respectively (both counterclockwise when viewed from the right). Choosing the x axis to the right, the y axis upward, and the z axis pointing out of the plane of the figure, determine (a) the common angular velocity of gears A and B , (b) the angular velocity of shaft FH , which is rigidly attached to the inclined shaft.

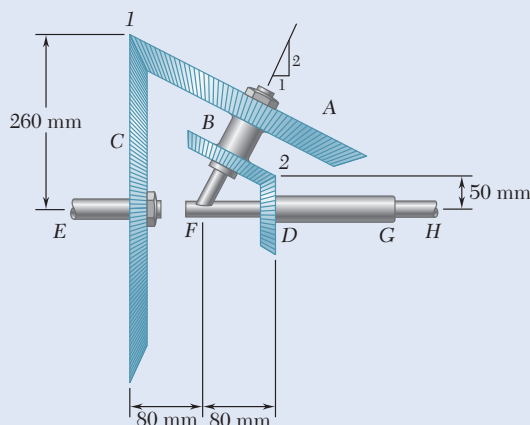


Fig. P15.199

- 15.200** In Prob. 15.199, determine (a) the common angular acceleration of gears A and B , (b) the acceleration of the tooth of gear A that is in contact with gear C at point 1.

- 15.201** Several rods are brazed together to form the robotic guide arm shown that is attached to a ball-and-socket joint at O . Rod OA slides in a straight inclined slot while rod OB slides in a slot parallel to the z axis. Knowing that at the instant shown $\mathbf{v}_B = (9 \text{ in./s})\mathbf{k}$, determine (a) the angular velocity of the guide arm, (b) the velocity of point A , (c) the velocity of point C .

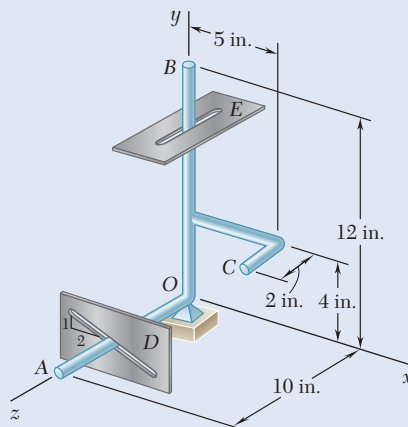


Fig. P15.201

- 15.202** In Prob. 15.201, the speed of point B is known to be constant. For the position shown, determine (a) the angular acceleration of the guide arm, (b) the acceleration of point C .

- 15.203** Rod AB of length 25 in. is connected by ball-and-socket joints to collars A and B , which slide along the two rods shown. Knowing that collar B moves toward point E at a constant speed of 20 in./s, determine the velocity of collar A as collar B passes through point D .

- 15.204** Rod AB has a length of 13 in. and is connected by ball-and-socket joints to collars A and B that slide along the two rods shown. Knowing that collar B moves toward point D at a constant speed of 36 in./s, determine the velocity of collar A when $b = 4$ in.

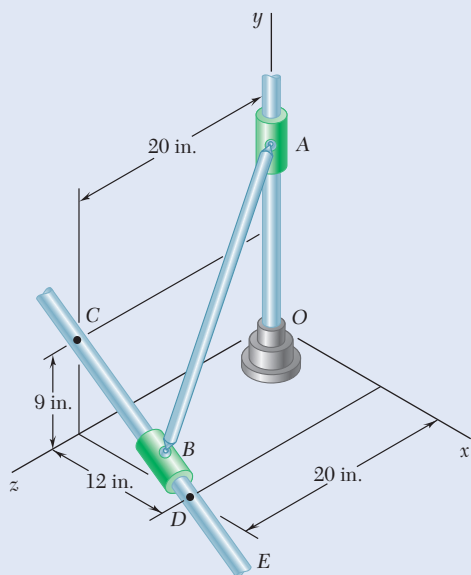


Fig. P15.203

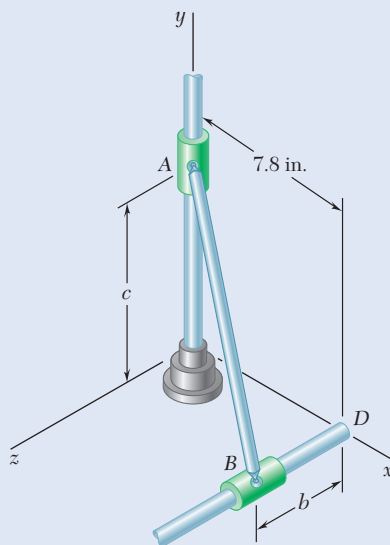


Fig. P15.204

- 15.205** Rod BC and BD are each 840 mm long and are connected by ball-and-socket joints to collars that may slide on the fixed rods shown. Knowing that collar B moves toward A at a constant speed of 390 mm/s, determine the velocity of collar C for the position shown.

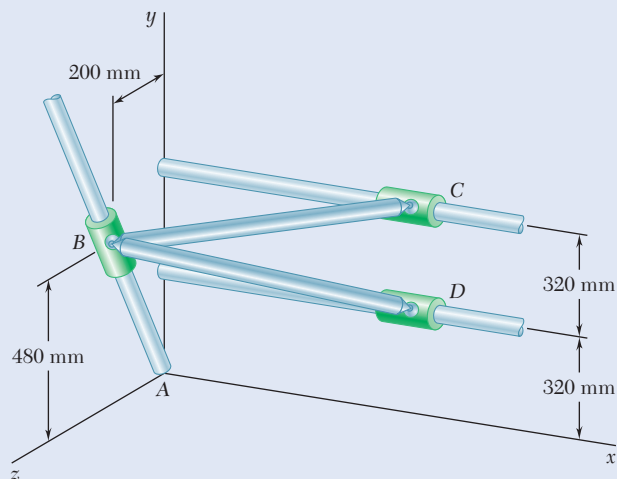


Fig. P15.205

- 15.206** Rod AB is connected by ball-and-socket joints to collar A and to the 16-in.-diameter disk C . Knowing that disk C rotates counterclockwise at the constant rate $\omega_0 = 3$ rad/s in the zx plane, determine the velocity of collar A for the position shown.

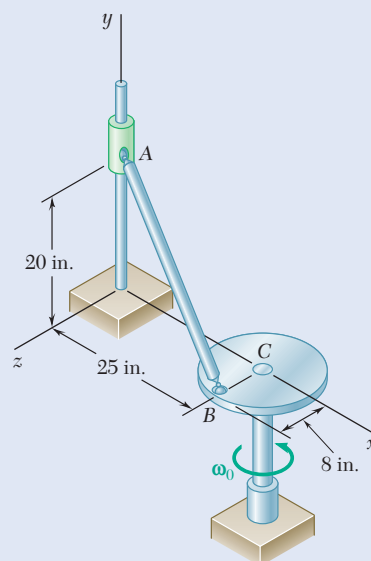


Fig. P15.206

- 15.207** Rod AB of length 29 in. is connected by ball-and-socket joints to the rotating crank BC and to the collar A . Crank BC is of length 8 in. and rotates in the horizontal xz plane at the constant rate $\omega_0 = 10$ rad/s. At the instant shown, when crank BC is parallel to the z axis, determine the velocity of collar A .

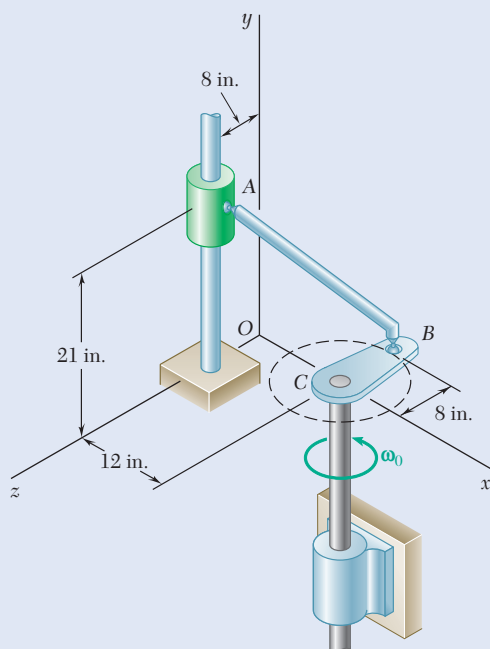


Fig. P15.207

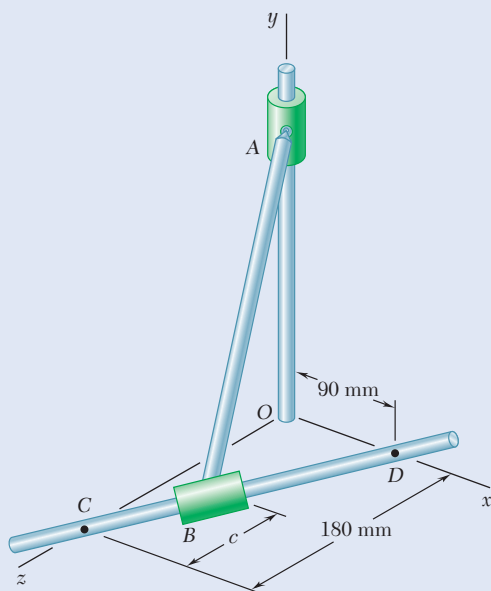


Fig. P15.208 and P15.209

15.208 Rod AB of length 300 mm is connected by ball-and-socket joints to collars A and B , which slide along the two rods shown. Knowing that collar B moves toward point D at a constant speed of 50 mm/s, determine the velocity of collar A when $c = 80$ mm.

15.209 Rod AB of length 300 mm is connected by ball-and-socket joints to collars A and B , which slide along the two rods shown. Knowing that collar B moves toward point D at a constant speed of 50 mm/s, determine the velocity of collar A when $c = 120$ mm.

15.210 Two shafts AC and EG , which lie in the vertical yz plane, are connected by a universal joint at D . Shaft AC rotates with a constant angular velocity ω_1 as shown. At a time when the arm of the crosspiece attached to shaft AC is vertical, determine the angular velocity of shaft EG .

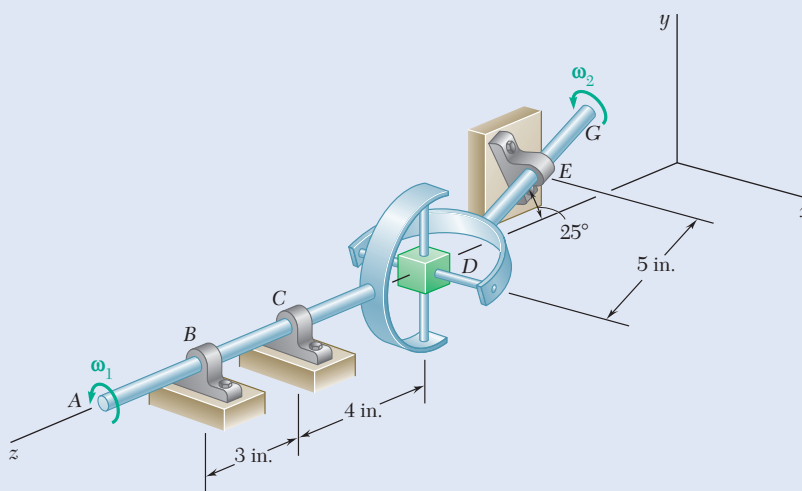


Fig. P15.210

15.211 Solve Prob. 15.210, assuming that the arm of the crosspiece attached to shaft AC is horizontal.

15.212 Rod BC has a length of 42 in. and is connected by a ball-and-socket joint to collar B and by a clevis connection to collar C . Knowing that collar B moves toward A at a constant speed of 19.5 in./s, determine at the instant shown (a) the angular velocity of the rod, (b) the velocity of collar C .

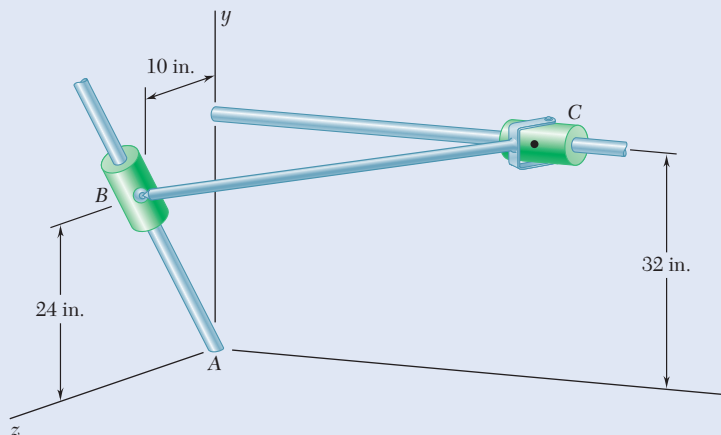


Fig. P15.212

- 15.213** Rod AB has a length of 275 mm and is connected by a ball-and-socket joint to collar A and by a clevis connection to collar B . Knowing that collar B moves down at a constant speed of 1.35 m/s, determine at the instant shown (a) the angular velocity of the rod, (b) the velocity of collar A .

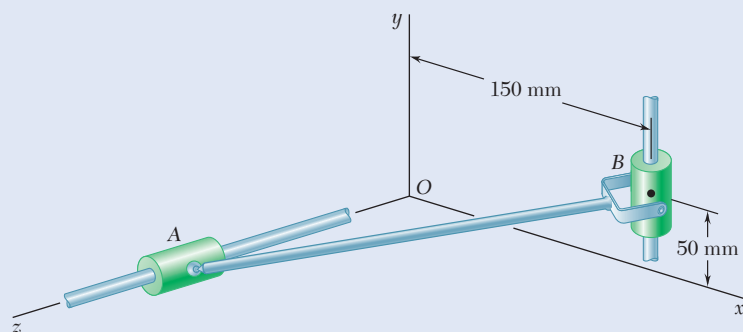


Fig. P15.213

- 15.214** For the mechanism of the Prob. 15.204, determine the acceleration of collar A .
- 15.215** In Prob. 15.205, determine the acceleration of collar C .
- 15.216** In Prob. 15.206, determine the acceleration of collar A .
- 15.217** In Prob. 15.207, determine the acceleration of collar A .
- 15.218** In Prob. 15.208, determine the acceleration of collar A .
- 15.219** In Prob. 15.209, determine the acceleration of collar A .

*15.7 MOTION RELATIVE TO A MOVING REFERENCE FRAME

In this final section of the chapter, we describe motion relative to a moving reference frame—either rotating or in general motion. We will use these results in Chapter 18 when we discuss the kinetics of rigid bodies in three dimensions.

15.7A Three-Dimensional Motion of a Particle Relative to a Rotating Frame

We saw in Sec. 15.5A that given a vector function $\mathbf{Q}(t)$ and two frames of reference centered at O —a fixed frame $OXYZ$ and a rotating frame $Oxyz$ —the rates of change of $\mathbf{Q}$ with respect to the two frames satisfy the relation

$$(\dot{\mathbf{Q}})_{OXYZ} = (\dot{\mathbf{Q}})_{Oxyz} + \boldsymbol{\Omega} \times \mathbf{Q} \quad (15.31)$$

We had assumed at the time that the frame $Oxyz$ was constrained to rotate about a fixed axis OA . However, the derivation given in Sec. 15.5A remains valid when the frame $Oxyz$ is constrained to have only a fixed point O . Under this more general assumption, the axis OA represents the *instantaneous* axis of rotation of the frame $Oxyz$ (Sec. 15.6A) and the vector $\boldsymbol{\Omega}$ represents its angular velocity at the instant considered (Fig. 15.36).

Let us now consider the three-dimensional motion of a particle P relative to a rotating frame $Oxyz$ constrained to have a fixed origin O . Let $\mathbf{r}$ be the position vector of P at a given instant, and let $\boldsymbol{\Omega}$ be the angular velocity of the frame $Oxyz$ with respect to the fixed frame $OXYZ$ at the same instant (Fig. 15.37). The derivations given in Sec. 15.5B for the two-dimensional motion of a particle can be readily extended to the three-dimensional case. Then we can express the absolute velocity $\mathbf{v}_P$ of P (i.e., its velocity with respect to the fixed frame $OXYZ$) as

$$\mathbf{v}_P = \boldsymbol{\Omega} \times \mathbf{r} + (\dot{\mathbf{r}})_{Oxyz} \quad (15.45)$$

where $(\dot{\mathbf{r}})_{Oxyz}$ is the relative velocity of point P with respect to the rotating frame. Sometimes this is also written as $\mathbf{v}_{\text{rel}}$. Denoting the rotating frame $Oxyz$ by $\mathcal{F}$, we can write this relation in the alternative form

$$\mathbf{v}_P = \mathbf{v}_{P'} + \mathbf{v}_{P/\mathcal{F}} \quad (15.46)$$

where $\mathbf{v}_P$ = absolute velocity of particle P

$\mathbf{v}_{P'}$ = velocity of point P' of moving frame $\mathcal{F}$ coinciding with P

$\mathbf{v}_{P/\mathcal{F}}$ = velocity of P relative to moving frame $\mathcal{F}$

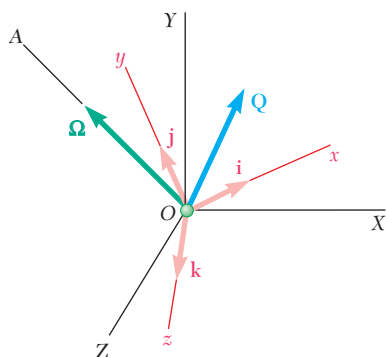


Fig. 15.36 Reference frame $Oxyz$ rotating about an instantaneous axis in fixed frame $OXYZ$ with angular velocity $\boldsymbol{\Omega}$.

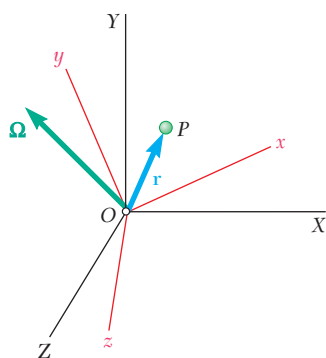


Fig. 15.37 A particle P moving relative to the rotating frame.

The absolute acceleration $\mathbf{a}_P$ of P can be expressed as

$$\mathbf{a}_P = \dot{\boldsymbol{\Omega}} \times \mathbf{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) + 2\boldsymbol{\Omega} \times (\dot{\mathbf{r}})_{Oxyz} + (\ddot{\mathbf{r}})_{Oxyz} \quad (15.47)$$

An alternative form is

$$\mathbf{a}_P = \mathbf{a}_{P'} + \mathbf{a}_{P/\mathcal{F}} + \mathbf{a}_C \quad (15.48)$$

where $\mathbf{a}_P$ = absolute acceleration of particle P

$\mathbf{a}_{P'}$ = acceleration of point P' of moving frame $\mathcal{F}$ coinciding with P

$\mathbf{a}_{P/\mathcal{F}}$ = acceleration of P relative to moving frame $\mathcal{F}$

$\mathbf{a}_C = 2\boldsymbol{\Omega} \times (\dot{\mathbf{r}})_{Oxyz} = 2\boldsymbol{\Omega} \times \mathbf{v}_{P/\mathcal{F}}$ = Coriolis acceleration

Note the difference between this equation and Eq. (15.21) of Sec. 15.4A, and recall the discussion following Eq. (15.36) of Sec. 15.5B.

Also note that the Coriolis acceleration is perpendicular to the vectors $\boldsymbol{\Omega}$ and $\mathbf{v}_{P/\mathcal{F}}$. However, since these vectors are usually not perpendicular to each other, the magnitude of $\mathbf{a}_C$ is in general *not* equal to $2\Omega v_{P/\mathcal{F}}$ —as was the case for the plane motion of a particle. We further note that the Coriolis acceleration reduces to zero when the vectors $\boldsymbol{\Omega}$ and $\mathbf{v}_{P/\mathcal{F}}$ are parallel or when either of them is zero.

Rotating frames of reference are particularly useful in the study of the three-dimensional motion of rigid bodies. If a rigid body has a fixed point O —as was the case for the crane of Sample Prob. 15.21—we can use a frame $Oxyz$ that can rotate. Denoting the angular velocity of the frame $Oxyz$ by $\boldsymbol{\Omega}$, we then resolve the angular velocity $\boldsymbol{\omega}$ of the body into the components $\boldsymbol{\Omega}$ and $\boldsymbol{\omega}_{B/\mathcal{F}}$, where the second component represents the angular velocity of the body relative to the frame $Oxyz$ (see Sample Prob. 15.24). An appropriate choice of a rotating frame often leads to a simpler analysis of the motion of the rigid body than would be possible with axes of fixed orientation. This is especially true in the case of the general three-dimensional motion of a rigid body, i.e., when the rigid body under consideration has no fixed point (see Sample Prob. 15.25).

*15.7B Frame of Reference in General Motion

Consider a fixed frame of reference $OXYZ$ and a frame $Axyz$ that moves in a known, but arbitrary, fashion with respect to $OXYZ$ (Fig. 15.38). Let P be a particle moving in space. The position of P is defined at any instant by the vector $\mathbf{r}_P$ in the fixed frame and by the vector $\mathbf{r}_{P/A}$ in the moving frame. Denoting the position vector of A in the fixed frame by $\mathbf{r}_A$, we have

$$\mathbf{r}_P = \mathbf{r}_A + \mathbf{r}_{P/A} \quad (15.49)$$

We obtain the absolute velocity $\mathbf{v}_P$ of the particle by differentiating, as

$$\mathbf{v}_P = \dot{\mathbf{r}}_P = \dot{\mathbf{r}}_A + \dot{\mathbf{r}}_{P/A} \quad (15.50)$$

where the derivatives are defined with respect to the fixed frame $OXYZ$. The first term in the right-hand side of Eq. (15.50) thus represents the velocity $\mathbf{v}_A$ of the origin A of the moving axes. Since the rate of change

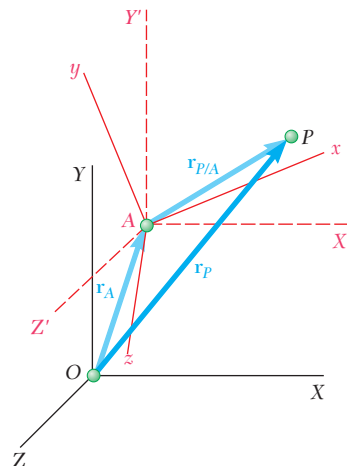


Fig. 15.38 Reference frame $Axyz$ moves arbitrarily relative to fixed frame $OXYZ$.

of a vector is the same with respect to both a fixed frame and a frame in translation (See. 11.4B), we can regard the second term as the velocity $\mathbf{v}_{P/A}$ of P relative to the frame $AX'Y'Z'$ with the same orientation as $OXYZ$ and the same origin as $Axyz$. We therefore have

$$\mathbf{v}_P = \mathbf{v}_A + \mathbf{v}_{P/A} \quad (15.51)$$

However, we can obtain the velocity $\mathbf{v}_{P/A}$ of P relative to $AX'Y'Z'$ from Eq. (15.45) by substituting $\mathbf{r}_{P/A}$ for $\mathbf{r}$ in that equation. We get

$$\mathbf{v}_P = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{P/A} + (\dot{\mathbf{r}}_{P/A})_{Axyz} \quad (15.52)$$

where $\boldsymbol{\Omega}$ is the angular velocity of the frame $Axyz$ at the instant considered.

We obtain the absolute acceleration $\mathbf{a}_P$ of the particle by differentiating Eq. (15.51), as

$$\mathbf{a}_P = \dot{\mathbf{v}}_P = \dot{\mathbf{v}}_A + \dot{\mathbf{v}}_{P/A} \quad (15.53)$$

where the derivatives are defined with respect to either of the frames $OXYZ$ or $AX'Y'Z'$. Thus, the first term in the right-hand side of Eq. (15.53) represents the acceleration $\mathbf{a}_A$ of the origin A of the moving axes, and the second term represents the acceleration $\mathbf{a}_{P/A}$ of P relative to the frame $AX'Y'Z'$. We can obtain this acceleration from Eq. (15.47) by substituting $\mathbf{r}_{P/A}$ for $\mathbf{r}$. We therefore have

$$\begin{aligned} \mathbf{a}_P = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{P/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{P/A}) \\ + 2\boldsymbol{\Omega} \times (\dot{\mathbf{r}}_{P/A})_{Axyz} + (\ddot{\mathbf{r}}_{P/A})_{Axyz} \end{aligned} \quad (15.54)$$



Photo 15.9 The motion of air particles in a hurricane can be considered as motion relative to a frame of reference attached to the Earth and rotating with it.

Formulas (15.52) and (15.54) enable us to determine the velocity and acceleration of a given particle with respect to a fixed frame of reference when we know the motion of the particle with respect to a moving frame. These formulas become more significant, and considerably easier to remember, if we note that the sum of the first two terms in Eq. (15.52) represents the velocity of the point P' of the moving frame that coincides with P at the instant considered and that the sum of the first three terms in Eq. (15.54) represents the acceleration of the same point. Thus, relations in Eqs. (15.46) and (15.48) of the preceding section are still valid in the case of a reference frame in general motion, and we have

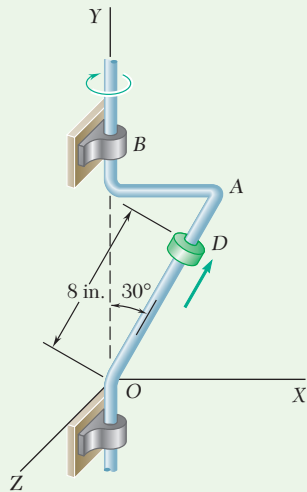
$$\mathbf{v}_P = \mathbf{v}_{P'} + \mathbf{v}_{P'/\mathcal{F}} \quad (15.46)$$

$$\mathbf{a}_P = \mathbf{a}_{P'} + \mathbf{a}_{P'/\mathcal{F}} + \mathbf{a}_C \quad (15.48)$$

where the various vectors involved were defined earlier.

Note that if the moving reference frame $\mathcal{F}$ (or $Axyz$) is in translation, the velocity and acceleration of the point P' of the frame that coincides with P become, respectively, equal to the velocity and acceleration of the origin A of the frame. On the other hand, since the frame maintains a fixed orientation, $\mathbf{a}_C$ is zero, and the relations in Eqs. (15.46) and (15.48) reduce, respectively, to the relations in Eqs. 11.32 and 11.33 derived in Sec. 11.4D.

Sample Problem 15.23



The bent rod OAB rotates about the vertical axis OB . At the instant considered, its angular velocity and angular acceleration are, respectively, 20 rad/s and 200 rad/s^2 , which are both clockwise when viewed from the positive Y axis. The collar D moves along the rod, and at the instant considered, $OD = 8 \text{ in.}$ The velocity and acceleration of the collar relative to the rod are, respectively, 50 in./s and 600 in./s^2 , where both are upward. Determine (a) the velocity of the collar, (b) the acceleration of the collar.

STRATEGY: Use rigid-body kinematics with a rotating coordinate system since collar D is moving relative to the bent rod. Attach the rotating reference frame to the bent rod; then you can calculate its motion relative to the fixed frame and the collar's motion relative to the rotating frame.

MODELING:

Frames of Reference. The angular velocity and angular acceleration of the bent rod (and rotating frame $Oxyz$) relative to the fixed frame $OXYZ$ are $\boldsymbol{\Omega} = (-20 \text{ rad/s})\mathbf{j}$ and $\dot{\boldsymbol{\Omega}} = (-200 \text{ rad/s}^2)\mathbf{j}$, respectively (Fig. 1). The position vector of D is

$$\mathbf{r} = (8 \text{ in.})(\sin 30^\circ \mathbf{i} + \cos 30^\circ \mathbf{j}) = (4 \text{ in.})\mathbf{i} + (6.93 \text{ in.})\mathbf{j}$$

ANALYSIS:

a. Velocity $\mathbf{v}_D$. Denote the point of the rod that coincides with D by D' and the rotating frame $Oxyz$ by $\mathcal{F}$. Then from Eq. (15.46) you have

$$\mathbf{v}_D = \mathbf{v}_{D'} + \mathbf{v}_{D/\mathcal{F}} \quad (1)$$

where

$$\mathbf{v}_{D'} = \boldsymbol{\Omega} \times \mathbf{r} = (-20 \text{ rad/s})\mathbf{j} \times [(4 \text{ in.})\mathbf{i} + (6.93 \text{ in.})\mathbf{j}] = (80 \text{ in./s})\mathbf{k}$$

$$\mathbf{v}_{D/\mathcal{F}} = (50 \text{ in./s})(\sin 30^\circ \mathbf{i} + \cos 30^\circ \mathbf{j}) = (25 \text{ in./s})\mathbf{i} + (43.3 \text{ in./s})\mathbf{j}$$

Substituting the values obtained for $\mathbf{v}_{D'}$ and $\mathbf{v}_{D/\mathcal{F}}$ into Eq. (1) gives

$$\mathbf{v}_D = (25 \text{ in./s})\mathbf{i} + (43.3 \text{ in./s})\mathbf{j} + (80 \text{ in./s})\mathbf{k} \quad \blacktriangleleft$$

b. Acceleration $\mathbf{a}_D$. From Eq. (15.48), you have

$$\mathbf{a}_D = \mathbf{a}_{D'} + \mathbf{a}_{D/\mathcal{F}} + \mathbf{a}_C \quad (2)$$

where

$$\begin{aligned} \mathbf{a}_{D'} &= \dot{\boldsymbol{\Omega}} \times \mathbf{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) \\ &= (-200 \text{ rad/s}^2)\mathbf{j} \times [(4 \text{ in.})\mathbf{i} + (6.93 \text{ in.})\mathbf{j}] - (20 \text{ rad/s})\mathbf{j} \times (80 \text{ in./s})\mathbf{k} \\ &= +(800 \text{ in./s}^2)\mathbf{k} - (1600 \text{ in./s}^2)\mathbf{i} \end{aligned}$$

$$\mathbf{a}_{D/\mathcal{F}} = (600 \text{ in./s}^2)(\sin 30^\circ \mathbf{i} + \cos 30^\circ \mathbf{j}) = (300 \text{ in./s}^2)\mathbf{i} + (520 \text{ in./s}^2)\mathbf{j}$$

$$\begin{aligned} \mathbf{a}_C &= 2\boldsymbol{\Omega} \times \mathbf{v}_{D/\mathcal{F}} \\ &= 2(-20 \text{ rad/s})\mathbf{j} \times [(25 \text{ in./s})\mathbf{i} + (43.3 \text{ in./s})\mathbf{j}] = (1000 \text{ in./s}^2)\mathbf{k} \end{aligned}$$

(continued)

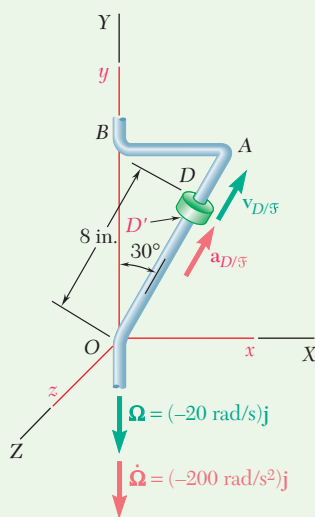


Fig. 1 The rotating coordinate system xyz is attached to rod OAB .

Substituting the values obtained for $\mathbf{a}_{D'}$, $\mathbf{a}_{D/\mathcal{F}}$, and $\mathbf{a}_c$ into Eq. (2), you obtain

$$\mathbf{a}_D = -(1300 \text{ in./s}^2)\mathbf{i} + (520 \text{ in./s}^2)\mathbf{j} + (1800 \text{ in./s}^2)\mathbf{k} \quad \blacktriangleleft$$

REFLECT and THINK: For this problem, the $800 \text{ in./s}^2 \mathbf{k}$ in the $\mathbf{a}_{D'}$ term corresponds to a tangential acceleration due to $\dot{\boldsymbol{\Omega}}$, while the $-1600 \text{ in./s}^2 \mathbf{i}$ corresponds to a normal acceleration toward the axis of rotation. The Coriolis term reflects the fact that the $\mathbf{v}_{D/\mathcal{F}}$ term is changing its direction due to $\boldsymbol{\Omega}$. When solving three-dimensional problems like this, the vector algebra approach is clearly superior to the method discussed in Sample Problem 15.20, since it is very difficult to visualize the direction of the acceleration terms.

Sample Problem 15.24

The crane shown rotates with a constant angular velocity $\boldsymbol{\omega}_1$ of 0.30 rad/s. Simultaneously, the boom is being raised with a constant angular velocity $\boldsymbol{\omega}_2$ of 0.50 rad/s relative to the cab. Knowing that the length of the boom OP is $l = 12 \text{ m}$, determine (a) the velocity of the tip of the boom, (b) the acceleration of the tip of the boom.

STRATEGY: Use rigid body kinematics with a rotating coordinate system because $\boldsymbol{\omega}_2$ is given relative to the cab. Attach a rotating reference frame to the cab; then you can calculate its motion relative to the fixed frame and the motion of the crane tip relative to the rotating frame.

MODELING:

Frames of Reference. The angular velocity of the cab (and rotating frame $Oxyz$) with respect to the fixed frame $OXYZ$ is $\boldsymbol{\Omega} = \boldsymbol{\omega}_1 = (0.30 \text{ rad/s})\mathbf{j}$ (Fig. 1). The angular velocity of the boom relative to the cab and the rotating frame $Oxyz$ (or $\mathcal{F}$ for short) is $\boldsymbol{\omega}_{B/\mathcal{F}} = \boldsymbol{\omega}_2 = (0.50 \text{ rad/s})\mathbf{k}$.

ANALYSIS:

a. Velocity $\mathbf{v}_P$. From Eq. (15.46), you have

$$\mathbf{v}_P = \mathbf{v}_{P'} + \mathbf{v}_{P/\mathcal{F}} \quad (1)$$

where $\mathbf{v}_{P'}$ is the velocity of the point P' of the rotating frame that coincides with P as

$$\mathbf{v}_{P'} = \boldsymbol{\Omega} \times \mathbf{r} = (0.30 \text{ rad/s})\mathbf{j} \times [(10.39 \text{ m})\mathbf{i} + (6 \text{ m})\mathbf{j}] = -(3.12 \text{ m/s})\mathbf{k}$$

and where $\mathbf{v}_{P/\mathcal{F}}$ is the velocity of P relative to the rotating frame $Oxyz$. However, you know that the angular velocity of the boom relative to $Oxyz$ is $\boldsymbol{\omega}_{B/\mathcal{F}} = (0.50 \text{ rad/s})\mathbf{k}$. The velocity of its tip P relative to $Oxyz$ is therefore

$$\begin{aligned} \mathbf{v}_{P/\mathcal{F}} &= \boldsymbol{\omega}_{B/\mathcal{F}} \times \mathbf{r} = (0.50 \text{ rad/s})\mathbf{k} \times [(10.39 \text{ m})\mathbf{i} + (6 \text{ m})\mathbf{j}] \\ &= -(3 \text{ m/s})\mathbf{i} + (5.20 \text{ m/s})\mathbf{j} \end{aligned}$$

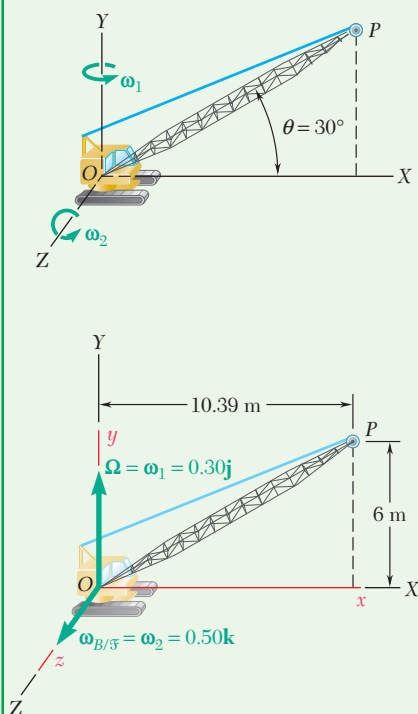


Fig. 1 The rotating coordinate system xyz is attached to the cab.

Substituting the values obtained for $\mathbf{v}_{P'}$ and $\mathbf{v}_{B/\mathcal{F}}$ into Eq. (1), you find

$$\mathbf{v}_P = -(3 \text{ m/s})\mathbf{i} + (5.20 \text{ m/s})\mathbf{j} - (3.12 \text{ m/s})\mathbf{k} \quad \blacktriangleleft$$

b. Acceleration $\mathbf{a}_P$. From Eq. (15.48), you have

$$\mathbf{a}_P = \mathbf{a}_{P'} + \mathbf{a}_{P/\mathcal{F}} + \mathbf{a}_C \quad (2)$$

Since $\boldsymbol{\Omega}$ and $\boldsymbol{\omega}_{B/\mathcal{F}}$ are both constant, you obtain

$$\mathbf{a}_{P'} = \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) = (0.30 \text{ rad/s})\mathbf{j} \times (-3.12 \text{ m/s})\mathbf{k} = -(0.94 \text{ m/s}^2)\mathbf{i}$$

$$\begin{aligned} \mathbf{a}_{P/\mathcal{F}} &= \boldsymbol{\omega}_{B/\mathcal{F}} \times (\boldsymbol{\omega}_{B/\mathcal{F}} \times \mathbf{r}) \\ &= (0.50 \text{ rad/s})\mathbf{k} \times [-(3 \text{ m/s})\mathbf{i} + (5.20 \text{ m/s})\mathbf{j}] \\ &= -(1.50 \text{ m/s}^2)\mathbf{j} - (2.60 \text{ m/s}^2)\mathbf{i} \end{aligned}$$

$$\begin{aligned} \mathbf{a}_C &= 2\boldsymbol{\Omega} \times \mathbf{v}_{P/\mathcal{F}} \\ &= 2(0.30 \text{ rad/s})\mathbf{j} \times [-(3 \text{ m/s})\mathbf{i} + (5.20 \text{ m/s})\mathbf{j}] = (1.80 \text{ m/s}^2)\mathbf{k} \end{aligned}$$

Substituting for $\mathbf{a}_{P'}$, $\mathbf{a}_{P/\mathcal{F}}$, and $\mathbf{a}_C$ into Eq. (2), you find

$$\mathbf{a}_P = -(3.54 \text{ m/s}^2)\mathbf{i} - (1.50 \text{ m/s}^2)\mathbf{j} + (1.80 \text{ m/s}^2)\mathbf{k} \quad \blacktriangleleft$$

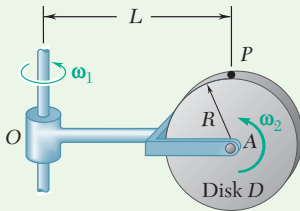
REFLECT and THINK: You also could have attached your reference frame to rotate with the boom:

$$\boldsymbol{\Omega}_B = \boldsymbol{\omega}_{\mathcal{F}} + \boldsymbol{\omega}_{B/\mathcal{F}} = (0.30 \text{ rad/s})\mathbf{j} + (0.50 \text{ rad/s})\mathbf{k}$$

and used Eq. 15.52 for

$$\begin{aligned} \mathbf{v}_P &= \boldsymbol{\Omega}_B \times \mathbf{r} = [(0.30 \text{ rad/s})\mathbf{j} + (0.5 \text{ rad/s})\mathbf{k}] \times [(10.39 \text{ m})\mathbf{i} + (6 \text{ m})\mathbf{j}] \\ &= -(3.0 \text{ m/s})\mathbf{i} + (5.20 \text{ m/s})\mathbf{j} - (3.12 \text{ m/s})\mathbf{k} \end{aligned}$$

which is the same answer you found previously. Similarly, you could use Eq. 15.54 to solve for the acceleration. If the crane were moving forward, you would just add its translational velocity and acceleration to that due to the rotations.



Sample Problem 15.25

Disk D has a radius R and is pinned to end A of the arm OA . OA has a length L and is located in the plane of the disk. The arm rotates about a vertical axis through O at the constant rate ω_1 , and the disk rotates about A at the constant rate ω_2 . Determine (a) the velocity of point P located directly above A , (b) the acceleration of P , (c) the angular velocity and angular acceleration of the disk.

STRATEGY: Use rigid-body kinematics with a rotating coordinate system, since disk D is moving relative to the arm OA .

MODELING:

Frames of Reference. Attach a moving frame $Axyz$ to arm OA . Its angular velocity with respect to the fixed frame $OXYZ$ is therefore $\boldsymbol{\Omega} = \omega_1\mathbf{j}$

(continued)

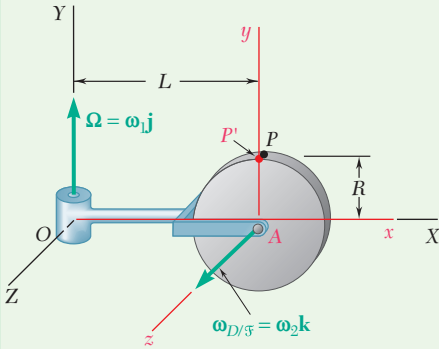


Fig. 1 The rotating coordinate system xyz is attached to arm OA at point A .

(Fig. 1). The angular velocity of disk D relative to the moving frame $Axyz$ (or $\mathcal{F}$ for short) is $\omega_{D/\mathcal{F}} = \omega_2 \mathbf{k}$. The position vector of P relative to O is $\mathbf{r} = L\mathbf{i} + R\mathbf{j}$, and its position vector relative to A is $\mathbf{r}_{P/A} = R\mathbf{j}$.

ANALYSIS:

a. Velocity $\mathbf{v}_P$. Denote by P' the point of the moving frame that coincides with P . Then from Eq. (15.46), you have

$$\mathbf{v}_P = \mathbf{v}_{P'} + \mathbf{v}_{P/\mathcal{F}} \quad (1)$$

where $\mathbf{v}_{P'} = \boldsymbol{\Omega} \times \mathbf{r} = \omega_1 \mathbf{j} \times (L\mathbf{i} + R\mathbf{j}) = -\omega_1 L \mathbf{k}$

$$\mathbf{v}_{P/\mathcal{F}} = \omega_{D/\mathcal{F}} \times \mathbf{r}_{P/A} = \omega_2 \mathbf{k} \times R\mathbf{j} = -\omega_2 R \mathbf{i}$$

Substituting the values obtained for $\mathbf{v}_{P'}$ and $\mathbf{v}_{D/\mathcal{F}}$ into Eq. (1), you obtain

$$\mathbf{v}_P = -\omega_2 R \mathbf{i} - \omega_1 L \mathbf{k} \quad \blacktriangleleft$$

b. Acceleration $\mathbf{a}_P$. From Eq. (15.48), you have

$$\mathbf{a}_P = \mathbf{a}_{P'} + \mathbf{a}_{P/\mathcal{F}} + \mathbf{a}_C \quad (2)$$

Since $\boldsymbol{\Omega}$ and $\omega_{D/\mathcal{F}}$ are both constant, you obtain

$$\mathbf{a}_{P'} = \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) = \omega_1 \mathbf{j} \times (-\omega_1 L \mathbf{k}) = -\omega_1^2 L \mathbf{i}$$

$$\mathbf{a}_{P/\mathcal{F}} = \omega_{D/\mathcal{F}} \times (\omega_{D/\mathcal{F}} \times \mathbf{r}_{P/A}) = \omega_2 \mathbf{k} \times (-\omega_2 R \mathbf{i}) = -\omega_2^2 R \mathbf{j}$$

$$\mathbf{a}_C = 2\boldsymbol{\Omega} \times \mathbf{v}_{P/\mathcal{F}} = 2\omega_1 \mathbf{j} \times (-\omega_2 R \mathbf{i}) = 2\omega_1 \omega_2 R \mathbf{k}$$

Substituting these values into Eq. (2), you find

$$\mathbf{a}_P = -\omega_1^2 L \mathbf{i} - \omega_2^2 R \mathbf{j} + 2\omega_1 \omega_2 R \mathbf{k} \quad \blacktriangleleft$$

c. Angular Velocity and Angular Acceleration of Disk.

$$\boldsymbol{\omega} = \boldsymbol{\Omega} + \omega_{D/\mathcal{F}} \quad \boldsymbol{\omega} = \omega_1 \mathbf{j} + \omega_2 \mathbf{k} \quad \blacktriangleleft$$

Using Eq. (15.31) with $\boldsymbol{\Omega} = \boldsymbol{\omega}$, you obtain

$$\begin{aligned} \boldsymbol{\alpha} &= (\dot{\boldsymbol{\omega}})_{OXYZ} = (\dot{\boldsymbol{\omega}})_{Axyz} + \boldsymbol{\Omega} \times \boldsymbol{\omega} \\ &= 0 + \omega_1 \mathbf{j} \times (\omega_1 \mathbf{j} + \omega_2 \mathbf{k}) \end{aligned}$$

$$\boldsymbol{\alpha} = \omega_1 \omega_2 \mathbf{i} \quad \blacktriangleleft$$

REFLECT and THINK: Knowing the absolute angular velocity of the disk is equal to $\omega_1 \mathbf{j} + \omega_2 \mathbf{k}$, you could have determined the velocity of P by attaching the rotating axes to the disk and using Eq. 15.52,

$$\begin{aligned} \mathbf{v}_P &= \mathbf{v}_A + \boldsymbol{\Omega}_D \times \mathbf{r}_{P/A} + \mathbf{v}_{P/A} = \omega_1 \mathbf{j} \times L\mathbf{i} + (\omega_1 \mathbf{j} + \omega_2 \mathbf{k}) \times R\mathbf{j} + 0 \\ &= -\omega_1 L \mathbf{k} - \omega_2 R \mathbf{i} \end{aligned}$$

which is the same answer we found earlier. Similarly,

$$\begin{aligned} \mathbf{a}_P &= \mathbf{a}_A + \dot{\boldsymbol{\Omega}}_D \times \mathbf{r}_{P/A} + \boldsymbol{\Omega}_D \times (\boldsymbol{\Omega}_D \times \mathbf{r}_{P/A}) + 2\boldsymbol{\Omega}_D \times \dot{\mathbf{r}}_{P/A} + \ddot{\mathbf{r}}_{P/A} \\ &= -\omega_1^2 L \mathbf{i} + \omega_1 \omega_2 \mathbf{i} \times R\mathbf{j} + (\omega_1 \mathbf{j} + \omega_2 \mathbf{k}) \times [(\omega_1 \mathbf{j} + \omega_2 \mathbf{k}) \times R\mathbf{j}] + 0 + 0 \\ &= -\omega_1^2 L \mathbf{i} + \omega_1 \omega_2 R \mathbf{k} + (\omega_1 \mathbf{j} + \omega_2 \mathbf{k}) \times (-\omega_2 R \mathbf{i}) \\ &= -\omega_1^2 L \mathbf{i} - \omega_2^2 R \mathbf{j} + 2\omega_1 \omega_2 R \mathbf{k} \end{aligned}$$

which, again, is the same answer shown previously.

SOLVING PROBLEMS ON YOUR OWN

In this section, we concluded our presentation of the kinematics of rigid bodies by showing you how to use an auxiliary frame of reference $\mathcal{F}$ to analyze the three-dimensional motion of a rigid body. This auxiliary frame may be a *rotating frame* with a fixed origin O or it may be a *frame in general motion*.

A. Using a rotating frame of reference. As you approach a problem involving the use of a rotating frame $\mathcal{F}$, you should take the following steps.

1. Select the rotating frame $\mathcal{F}$ that you wish to use and draw the corresponding coordinate axes x , y , and z from the fixed point O .

2. Determine the angular velocity Ω of the frame $\mathcal{F}$ with respect to a fixed frame $OXYZ$. In most cases, you will have selected a frame that is attached to some rotating element of the system; Ω is then the angular velocity of that element.

3. Designate as P' the point of the rotating frame $\mathcal{F}$ that coincides with the point P of interest at the instant you are considering. Determine the velocity $\mathbf{v}_{P'}$ and the acceleration $\mathbf{a}_{P'}$ of point P' . Since P' is part of $\mathcal{F}$ and has the same position vector $\mathbf{r}$ as P , you will find

$$\mathbf{v}_{P'} = \Omega \times \mathbf{r} \quad \text{and} \quad \mathbf{a}_{P'} = \alpha \times \mathbf{r} + \Omega \times (\Omega \times \mathbf{r})$$

where α is the angular acceleration of $\mathcal{F}$.

4. Determine the velocity and acceleration of point P with respect to the frame $\mathcal{F}$. As you are trying to determine $\mathbf{v}_{P/\mathcal{F}}$ and $\mathbf{a}_{P/\mathcal{F}}$, you will find it useful to visualize the motion of P on frame $\mathcal{F}$ when the frame is not rotating. If P is a point of a rigid body $\mathcal{B}$ that has an angular velocity $\omega_{\mathcal{B}}$ and an angular acceleration $\alpha_{\mathcal{B}}$ relative to $\mathcal{F}$ [Sample Prob. 15.24], you will find that

$$\mathbf{v}_{P/\mathcal{F}} = \omega_{\mathcal{B}} \times \mathbf{r} \quad \text{and} \quad \mathbf{a}_{P/\mathcal{F}} = \alpha_{\mathcal{B}} \times \mathbf{r} + \omega_{\mathcal{B}} \times (\omega_{\mathcal{B}} \times \mathbf{r})$$

5. Determine the Coriolis acceleration. Considering the angular velocity Ω of frame $\mathcal{F}$ and the velocity $\mathbf{v}_{P/\mathcal{F}}$ of point P relative to that frame, which was computed in Step 4, you have

$$\mathbf{a}_C = 2\Omega \times \mathbf{v}_{P/\mathcal{F}}$$

6. The velocity and the acceleration of P with respect to the fixed frame $OXYZ$ can now be obtained by adding the expressions you have determined:

$$\mathbf{v}_P = \mathbf{v}_{P'} + \mathbf{v}_{P/\mathcal{F}} \quad (15.46)$$

$$\mathbf{a}_P = \mathbf{a}_{P'} + \mathbf{a}_{P/\mathcal{F}} + \mathbf{a}_C \quad (15.48)$$

(continued)

B. Using a frame of reference in general motion. The steps that you will take differ only slightly from those listed under part A. They consist of the following:

1. Select the frame $\mathcal{F}$ that you wish to use and a reference point A in that frame from which you will draw the coordinate axes, x , y , and z , defining that frame. Consider the motion of the frame as the sum of a **translation with A and a rotation about A** .

2. Determine the velocity $\mathbf{v}_A$ of point A and the angular velocity $\mathbf{\Omega}$ of the frame. In most cases, you will have selected a frame that is attached to some element of the system; $\mathbf{\Omega}$ is then the angular velocity of that element.

3. Designate as P' the point of frame $\mathcal{F}$ that coincides with the point P of interest at the instant you are considering, and determine the velocity $\mathbf{v}_{P'}$ and the acceleration $\mathbf{a}_{P'}$ of that point. In some cases, you can do this by visualizing the motion of P if that point were prevented from moving with respect to $\mathcal{F}$ [Sample Prob. 15.25]. A more general approach is to recall that the motion of P' is the sum of a translation with the reference point A and a rotation about A . You can obtain the velocity $\mathbf{v}_{P'}$ and the acceleration $\mathbf{a}_{P'}$ of P' , therefore, by adding $\mathbf{v}_A$ and $\mathbf{a}_A$, respectively, to the expressions found in part A, Step 3, and replacing the position vector $\mathbf{r}$ by the vector $\mathbf{r}_{P/A}$ drawn from A to P :

$$\mathbf{v}_{P'} = \mathbf{v}_A + \mathbf{\Omega} \times \mathbf{r}_{P/A} \quad \mathbf{a}_{P'} = \mathbf{a}_A + \mathbf{\alpha} \times \mathbf{r}_{P/A} + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}_{P/A})$$

4, 5, and 6 are the same as in part A of this summary, except that the vector $\mathbf{r}$ should again be replaced by $\mathbf{r}_{P/A}$. Thus, Eqs. (15.46) and (15.48) can still be used to obtain the velocity and the acceleration of P with respect to the fixed frame of reference $OXYZ$.

C. Alternative approach using a frame of reference in general motion. As shown in the sample problems, you can also use Eqs. (15.52) and (15.54) to determine the velocity and acceleration of point P , respectively.

$$\mathbf{v}_P = \mathbf{v}_A + \mathbf{\Omega} \times \mathbf{r}_{P/A} + (\dot{\mathbf{r}}_{P/A})_{Axyz} \quad (15.52)$$

$$\begin{aligned} \mathbf{a}_P = \mathbf{a}_A + \dot{\mathbf{\Omega}} \times \mathbf{r}_{P/A} + \mathbf{\Omega} \times (\mathbf{\Omega} \times \mathbf{r}_{P/A}) \\ + 2\mathbf{\Omega} \times (\dot{\mathbf{r}}_{P/A})_{Axyz} + (\ddot{\mathbf{r}}_{P/A})_{Axyz} \end{aligned} \quad (15.54)$$

You first need to determine a reference point A and attach your rotating frame of reference at that point; generally this is attached to a specific part of the object under consideration (e.g., the cab or boom of a crane). Define the angular velocity of the frame as $\mathbf{\Omega}$ and the angular acceleration of the frame as $\dot{\mathbf{\Omega}}$. The terms $(\dot{\mathbf{r}}_{P/A})_{Axyz}$ and $(\ddot{\mathbf{r}}_{P/A})_{Axyz}$ represent the velocity and acceleration of point P relative to the rotating frame of reference A_{xyz} .

Problems

END-OF-SECTION PROBLEMS

- 15.220** A flight simulator is used to train pilots on how to recognize spatial disorientation. It has four degrees of freedom and can rotate around a planetary axis as well as in yaw, pitch, and roll. The pilot is seated so that her head B is located at $\mathbf{r} = 2\mathbf{i} + 1\mathbf{j}$ ft with respect to the center of the cab A . Knowing that the cab is rotating about the planetary axis with a constant angular velocity of 20 rpm counterclockwise as seen from above, and pitches with a constant angular velocity of $+3\mathbf{k}$ rad/s, determine (a) the velocity of the pilot's head, (b) the angular acceleration of the cab, (c) the acceleration of the pilot's head.

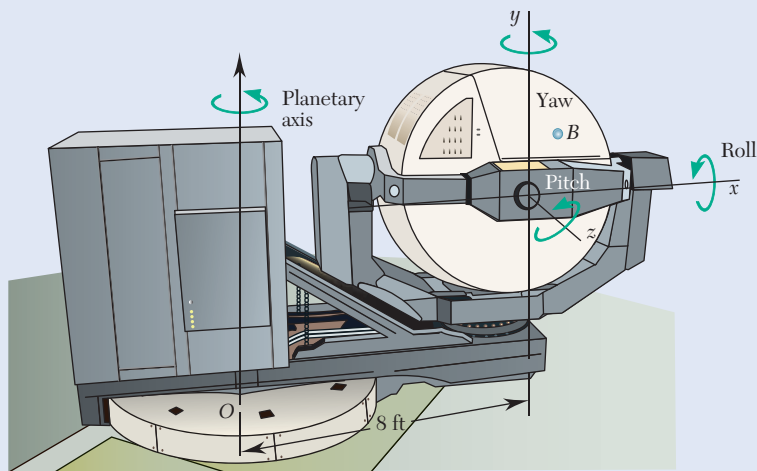


Fig. P15.220 and P15.221

- 15.221** A flight simulator is used to train pilots on how to recognize spatial disorientation. It has four degrees of freedom and can rotate around a planetary axis as well as in yaw, pitch, and roll. The pilot is seated so that her head B is located at $\mathbf{r} = 2\mathbf{i} + 1\mathbf{j}$ ft with respect to the center of the cab A . The cab is rotating about the planetary axis with an angular velocity of 20 rpm counterclockwise as seen from above and this is increasing by 1 rad/s^2 . Knowing that the cab rolls with a constant angular velocity of $-4\mathbf{i}$ rad/s, determine (a) the velocity of the pilot's head, (b) the angular acceleration of the cab, (c) the acceleration of the pilot's head.

- 15.222 and 15.223** The rectangular plate shown rotates at the constant rate $\omega_2 = 12 \text{ rad/s}$ with respect to arm AE , which itself rotates at the constant rate $\omega_1 = 9 \text{ rad/s}$ about the Z axis. For the position shown, determine the velocity and acceleration of the point of the plate indicated.

15.222 Corner B

15.223 Corner C

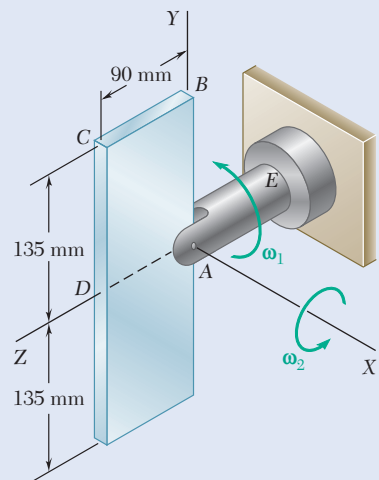


Fig. P15.222 and P15.223

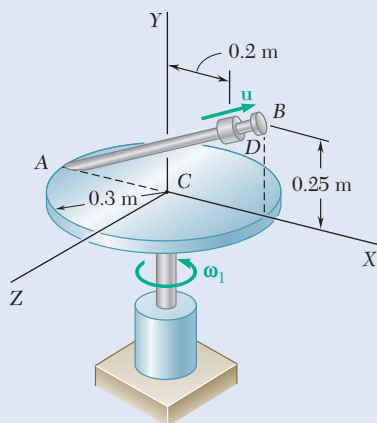


Fig. P15.224

15.224 Rod AB is welded to the 0.3-m-radius plate that rotates at the constant rate $\omega_1 = 6$ rad/s. Knowing that collar D moves toward end B of the rod at a constant speed $u = 1.3$ m/s, determine, for the position shown, (a) the velocity of D , (b) the acceleration of D .

15.225 The bent rod shown rotates at the constant rate of $\omega_1 = 5$ rad/s and collar C moves toward point B at a constant relative speed of $u = 39$ in./s. Knowing that collar C is halfway between points B and D at the instant shown, determine its velocity and acceleration.

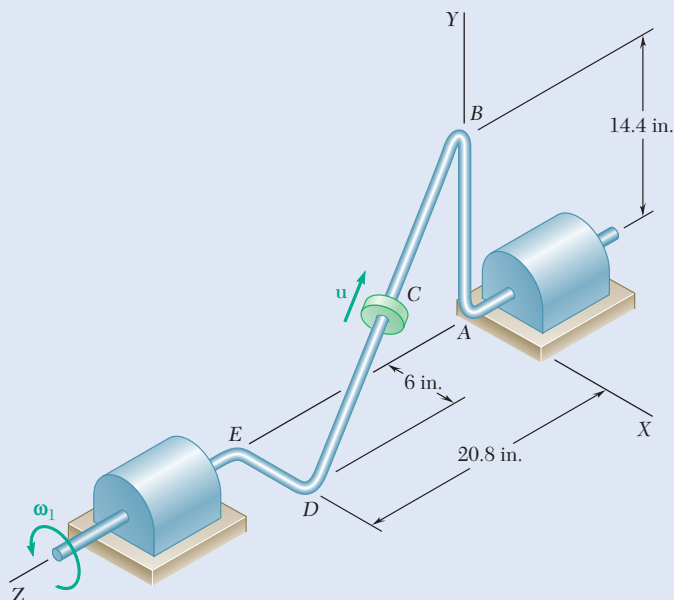


Fig. P15.225

15.226 The bent pipe shown rotates at the constant rate $\omega_1 = 10$ rad/s. Knowing that a ball bearing D moves in portion BC of the pipe toward end C at a constant relative speed $u = 2$ ft/s, determine at the instant shown (a) the velocity of D , (b) the acceleration of D .

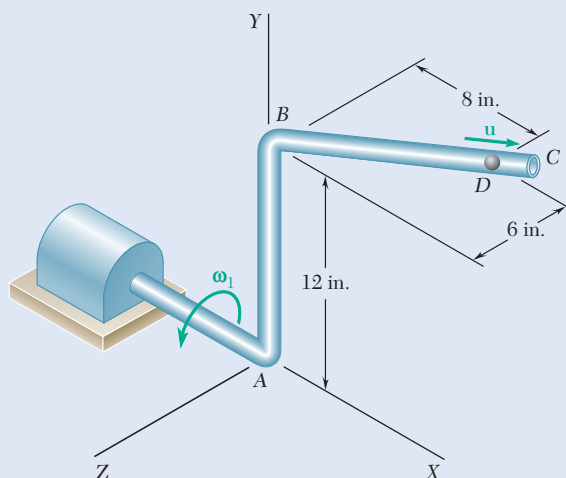


Fig. P15.226

15.227 The circular plate shown rotates about its vertical diameter at the constant rate $\omega_1 = 10 \text{ rad/s}$. Knowing that in the position shown the disk lies in the XY plane and point D of strap CD moves upward at a constant relative speed $u = 1.5 \text{ m/s}$, determine (a) the velocity of D , (b) the acceleration of D .

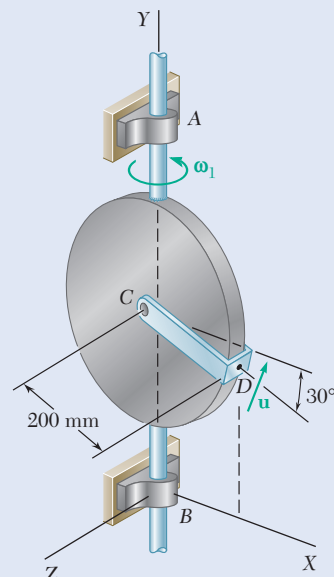


Fig. P15.227

15.228 Manufactured items are spray-painted as they pass through the automated work station shown. Knowing that the bent pipe ACE rotates at the constant rate $\omega_1 = 0.4 \text{ rad/s}$ and that at point D the paint moves through the pipe at a constant relative speed $u = 150 \text{ mm/s}$, determine, for the position shown, (a) the velocity of the paint at D , (b) the acceleration of the paint at D .

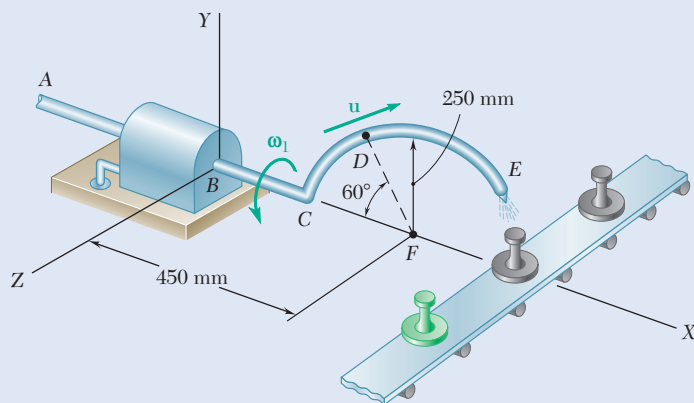


Fig. P15.228

15.229 Solve Prob. 15.227, assuming that at the instant shown the angular velocity ω_1 of the plate is 10 rad/s and is decreasing at the rate of 25 rad/s^2 , while the relative speed u of point D of strap CD is 1.5 m/s and is decreasing at the rate of 3 m/s^2 .

15.230 Solve Prob. 15.225, assuming that at the instant shown the angular velocity ω_1 of the rod is 5 rad/s and is increasing at the rate of 10 rad/s^2 while the relative speed u of the collar C is 39 in./s and is decreasing at the rate of 260 in./s^2 .

15.231 Using the method of Sec. 15.7A, solve Prob. 15.192.

15.232 Using the method of Sec. 15.7A, solve Prob. 15.196.

15.233 Using the method of Sec. 15.7A, solve Prob. 15.198.

15.234 The 400-mm bar AB is made to rotate at the constant rate of $\omega_2 = d\theta/dt = 8 \text{ rad/s}$ with respect to the frame CD that rotates at the constant rate of $\omega_1 = 12 \text{ rad/s}$ about the Y axis. Knowing that $\theta = 60^\circ$ at the instant shown, determine the velocity and acceleration of point A .

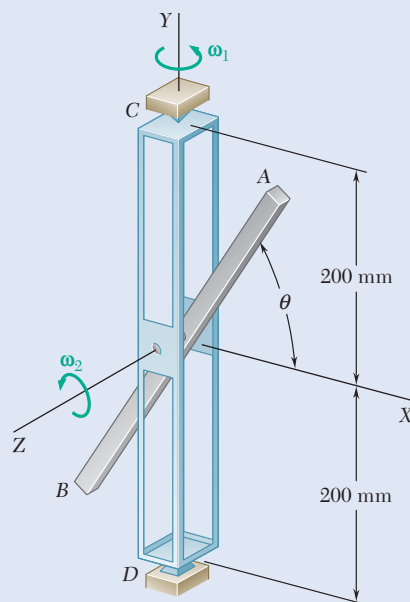


Fig. P15.234 and P15.235

15.235 The 400-mm bar AB is made to rotate at the rate $\omega_2 = d\theta/dt$ with respect to the frame CD that rotates at the rate ω_1 about the Y axis. At the instant shown, $\omega_1 = 12 \text{ rad/s}$, $d\omega_1/dt = -16 \text{ rad/s}^2$, $\omega_2 = 8 \text{ rad/s}$, $d\omega_2/dt = 10 \text{ rad/s}^2$, and $\theta = 60^\circ$. Determine the velocity and acceleration of point A at this instant.

15.240 The vertical plate shown is welded to arm EFG , and the entire unit rotates at the constant rate $\omega_1 = 1.6 \text{ rad/s}$ about the Y axis. At the same time, a continuous link belt moves around the perimeter of the plate at a constant speed $u = 4.5 \text{ in./s}$. For the position shown, determine the acceleration of the link of the belt located (a) at point A , (b) at point B .

15.241 The vertical plate shown is welded to arm EFG , and the entire unit rotates at the constant rate $\omega_1 = 1.6 \text{ rad/s}$ about the Y axis. At the same time, a continuous link belt moves around the perimeter of the plate at a constant speed $u = 4.5 \text{ in./s}$. For the position shown, determine the acceleration of the link of the belt located (a) at point C , (b) at point D .

15.242 A disk of 180-mm radius rotates at the constant rate $\omega_2 = 12 \text{ rad/s}$ with respect to arm CD , which itself rotates at the constant rate $\omega_1 = 8 \text{ rad/s}$ about the Y axis. Determine at the instant shown the velocity and acceleration of point A on the rim of the disk.

15.243 A disk of 180-mm radius rotates at the constant rate $\omega_2 = 12 \text{ rad/s}$ with respect to arm CD , which itself rotates at the constant rate $\omega_1 = 8 \text{ rad/s}$ about the Y axis. Determine at the instant shown the velocity and acceleration of point B on the rim of the disk.

15.244 A square plate of side $2r$ is welded to a vertical shaft that rotates with a constant angular velocity ω_1 . At the same time, rod AB of length r rotates about the center of the plate with a constant angular velocity ω_2 with respect to the plate. For the position of the plate shown, determine the acceleration of end B of the rod if (a) $\theta = 0$, (b) $\theta = 90^\circ$, (c) $\theta = 180^\circ$.

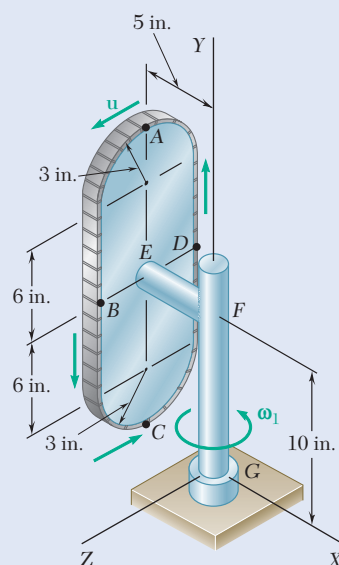


Fig. P15.240 and P15.241

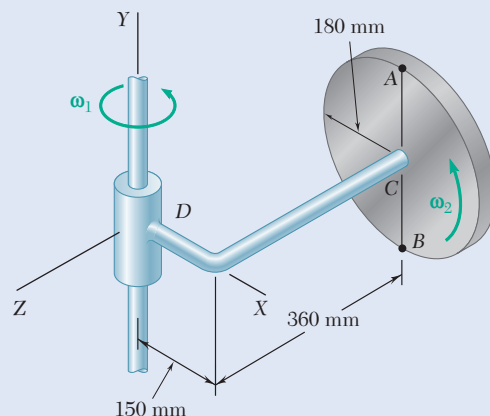


Fig. P15.242 and P15.243

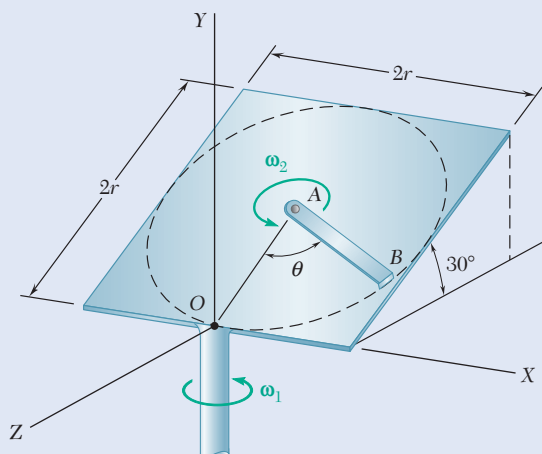


Fig. P15.244

- 15.245** Two disks, each of 130-mm radius, are welded to the 500-mm rod CD . The rod-and-disks unit rotates at the constant rate $\omega_2 = 3$ rad/s with respect to arm AB . Knowing that at the instant shown $\omega_1 = 4$ rad/s, determine the velocity and acceleration of (a) point E , (b) point F .

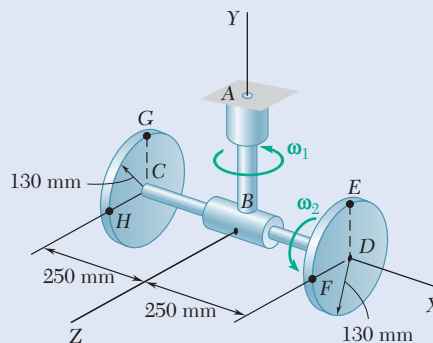


Fig. P15.245

- 15.246** In Prob. 15.245, determine the velocity and acceleration of (a) point G , (b) point H .

- 15.247** The position of the stylus tip A is controlled by the robot shown. In the position shown, the stylus moves at a constant speed $u = 180$ mm/s relative to the solenoid BC . At the same time, arm CD rotates at the constant rate $\omega_2 = 1.6$ rad/s with respect to component DEG . Knowing that the entire robot rotates about the X axis at the constant rate $\omega_1 = 1.2$ rad/s, determine (a) the velocity of A , (b) the acceleration of A .

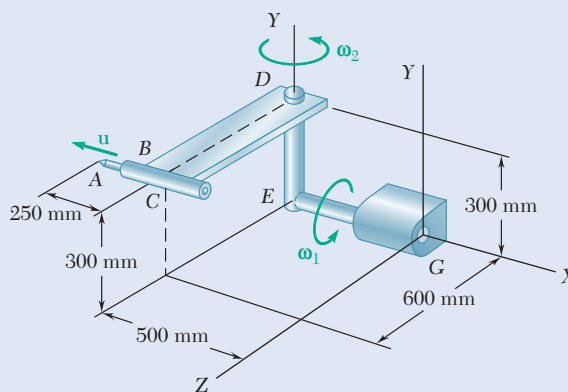


Fig. P15.247

Review and Summary

This chapter was devoted to the study of the kinematics of rigid bodies.

Rigid Body in Translation

We first considered the **translation** of a rigid body [Sec. 15.1A] and observed that in such a motion **all points of the body have the same velocity and the same acceleration at any given instant.**

Rigid Body in Rotation About a Fixed Axis

We next considered the **rotation** of a rigid body about a fixed axis [Sec. 15.1B]. The position of the body is defined by the angle θ that the line BP , drawn from the axis of rotation to a point P of the body, forms with a fixed plane (Fig. 15.39). We found that the magnitude of the velocity of P is

$$v = \frac{ds}{dt} = r\dot{\theta} \sin \phi \quad (15.4)$$

where $\dot{\theta}$ is the time derivative of θ . We then expressed the velocity of P as

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \boldsymbol{\omega} \times \mathbf{r} \quad (15.5)$$

where the vector

$$\boldsymbol{\omega} = \omega \mathbf{k} = \dot{\theta} \mathbf{k} \quad (15.6)$$

is directed along the fixed axis of rotation and represents the angular velocity of the body.

Denoting the derivative $d\boldsymbol{\omega}/dt$ of the angular velocity by $\boldsymbol{\alpha}$, we expressed the acceleration of P as

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \quad (15.8)$$

Differentiating Eq. (15.6) and recalling that $\mathbf{k}$ is constant in magnitude and direction, we found that

$$\boldsymbol{\alpha} = \alpha \mathbf{k} = \dot{\omega} \mathbf{k} = \ddot{\theta} \mathbf{k} \quad (15.9)$$

The vector $\boldsymbol{\alpha}$ represents the angular acceleration of the body and is directed along the fixed axis of rotation [Sample Prob. 15.2].

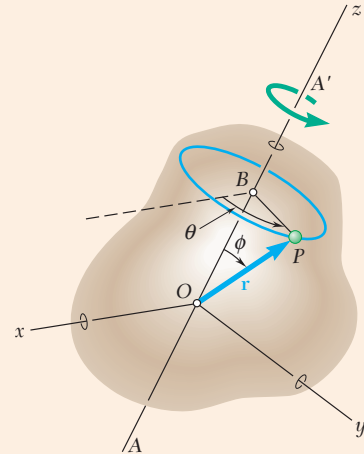


Fig. 15.39

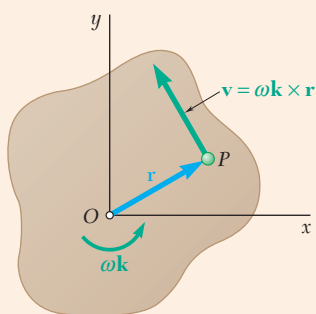


Fig. 15.40

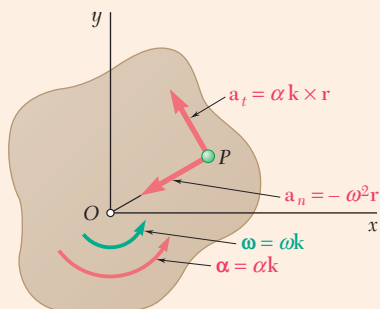


Fig. 15.41

Rotation of a Representative Slab:

Tangential and Normal Components

Next we considered the motion of a representative slab located in a plane perpendicular to the axis of rotation of the body (Fig. 15.40). Since the angular velocity is perpendicular to the slab, we expressed the velocity of a point P of the slab as

$$\mathbf{v} = \boldsymbol{\omega} \mathbf{k} \times \mathbf{r} \quad (15.10)$$

where $\mathbf{v}$ is contained in the plane of the slab. Substituting $\boldsymbol{\omega} = \omega \mathbf{k}$ and $\boldsymbol{\alpha} = \alpha \mathbf{k}$ into Eq. (15.8), we found that we could resolve the acceleration of P into tangential and normal components (Fig. 15.41) respectively equal to

$$\begin{aligned} \mathbf{a}_t &= \boldsymbol{\alpha} \mathbf{k} \times \mathbf{r} & a_t &= r\alpha \\ \mathbf{a}_n &= -\omega^2 \mathbf{r} & a_n &= r\omega^2 \end{aligned} \quad (15.11')$$

Angular Velocity and Angular Acceleration of a Rotating Rigid Body

Recalling Eqs. (15.6) and (15.9), we obtained the following expressions for the *angular velocity* and the *angular acceleration* of the rigid body [Sec. 15.1C]:

$$\boldsymbol{\omega} = \frac{d\theta}{dt} \mathbf{k} \quad (15.12)$$

$$\boldsymbol{\alpha} = \frac{d\boldsymbol{\omega}}{dt} = \frac{d^2\theta}{dt^2} \mathbf{k} \quad (15.13)$$

or

$$\alpha = \omega \frac{d\omega}{d\theta} \quad (15.14)$$

We noted that these expressions are similar to those obtained in Chap. 11 for the rectilinear motion of a particle.

Two particular cases of rotation are frequently encountered: *uniform rotation* and *uniformly accelerated rotation*. You can solve problems involving either of these motions by using equations similar to those used in Sec. 11.2 for the uniform rectilinear motion and the uniformly accelerated rectilinear motion of a particle, but where x , v , and a are replaced by θ , ω , and α , respectively [Sample Prob. 15.1].

Velocities in Plane Motion

We can consider the **most general plane motion** of a rigid body as the **sum of a translation and a rotation** [Sec. 15.2A]. For example, the body shown in Fig. 15.42 can be assumed to translate with point A , while simultaneously rotating about A . It follows [Sec. 15.2B] that the velocity of any point B of the rigid body can be expressed as

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \quad (15.17)$$

where $\mathbf{v}_A$ is the velocity of A and $\mathbf{v}_{B/A}$ is the relative velocity of B with respect to A or, more precisely, with respect to axes $x'y'$ translating with A . Denoting the position vector of B relative to A by $\mathbf{r}_{B/A}$, we found that

$$\mathbf{v}_{B/A} = \boldsymbol{\omega} \mathbf{k} \times \mathbf{r}_{B/A} \quad v_{B/A} = r\omega \quad (15.18)$$

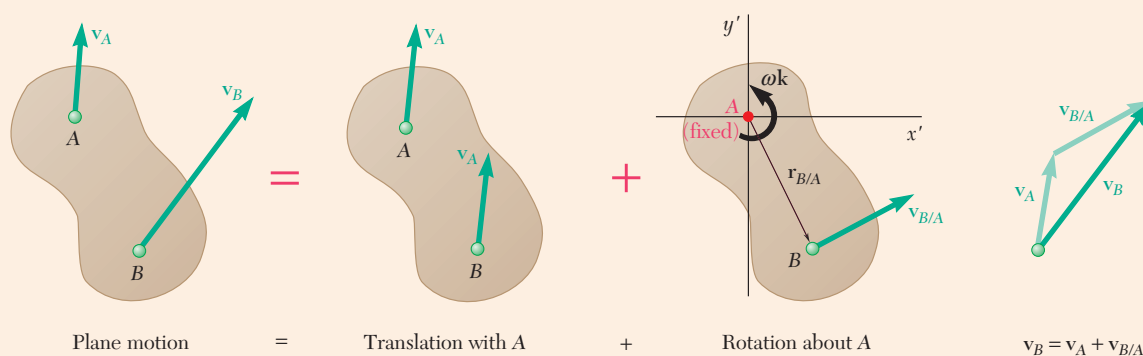


Fig. 15.42

The fundamental equation (15.17) relating the absolute velocities of points A and B and the relative velocity of B with respect to A was expressed in the form of a vector diagram, which can be used to solve problems involving the motion of various types of mechanisms [Sample Probs. 15.6 and 15.7].

Instantaneous Center of Rotation

We presented another approach to the solution of problems involving the velocities of the points of a rigid body in plane motion in Sec. 15.3 and used it in Sample Probs. 15.9 through 15.11. It is based on the determination of the **instantaneous center of rotation** C of the rigid body (Fig. 15.43).

Accelerations in Plane Motion

In Sec. 15.4A, we used the fact that any plane motion of a rigid body can be considered the sum of a translation of the body with a reference point A and a rotation about A . Knowing this, we can find the absolute acceleration of A by adding the relative acceleration of B with respect to A to the absolute acceleration of B .

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A} \quad (15.21)$$

where $\mathbf{a}_{B/A}$ consisted of a *normal component* $(\mathbf{a}_{B/A})_n$ with a magnitude $r\omega^2$ directed toward A and a *tangential component* $(\mathbf{a}_{B/A})_t$ with a magnitude $r\alpha$

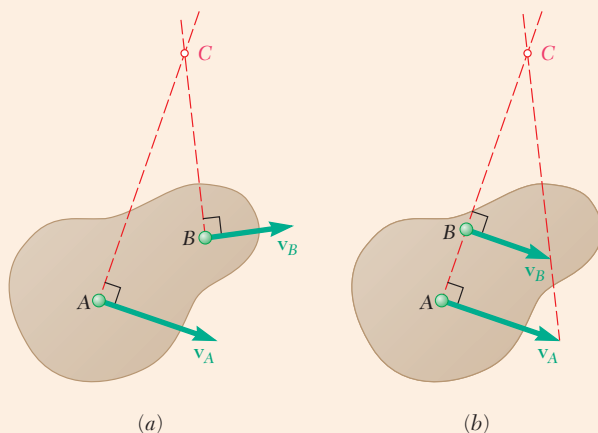


Fig. 15.43

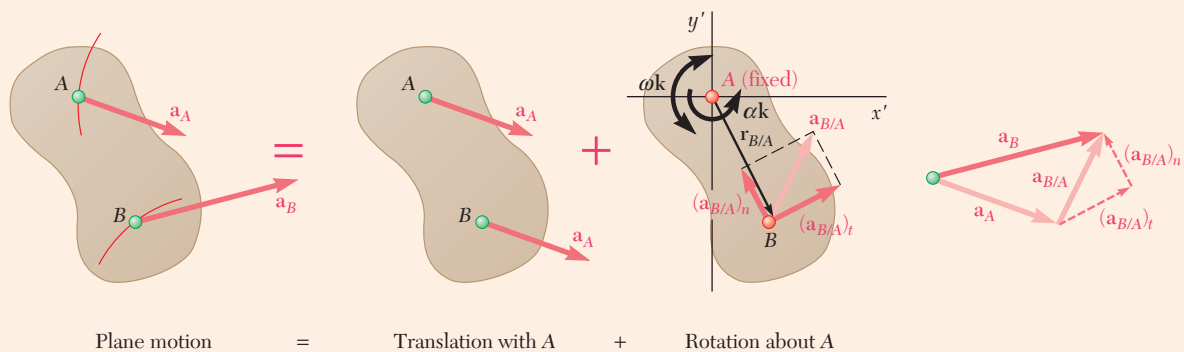


Fig. 15.44

perpendicular to the line AB (Fig. 15.44). We expressed the fundamental relation in Eq. (15.21) in terms of vector diagrams or vector equations and used them to determine the accelerations of given points of various mechanisms [Sample Probs. 15.12 through 15.16]. We noted that we cannot use the instantaneous center of rotation C considered in Sec. 15.3 for the determination of accelerations, since point C , in general, does *not* have zero acceleration.

Coordinates Expressed in Terms of a Parameter

In the case of certain mechanisms, it is possible to express the coordinates x and y of all significant points of the mechanism by means of simple analytic expressions containing a *single parameter*. We can obtain the components of the absolute velocity and acceleration of a given point by differentiating twice with respect to the time t the coordinates x and y of that point [Sec. 15.4B].

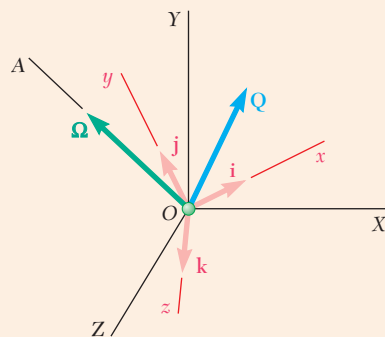


Fig. 15.45

Rate of Change of a Vector with Respect to a Rotating Frame

The rate of change of a vector is the same with respect to a fixed frame of reference and with respect to a frame in translation, but the rate of change of a vector with respect to a rotating frame is different. Therefore, in order to study the motion of a particle relative to a rotating frame, we first had to compare the rates of change of a general vector $\mathbf{Q}$ with respect to a fixed frame $OXYZ$ and with respect to a frame $Oxyz$ rotating with an angular velocity Ω [Sec. 15.5A] (Fig. 15.45). We obtained the fundamental relation

$$(\dot{\mathbf{Q}})_{OXYZ} = (\dot{\mathbf{Q}})_{Oxyz} + \Omega \times \mathbf{Q} \quad (15.31)$$

and we concluded that the rate of change of the vector $\mathbf{Q}$ with respect to the fixed frame $OXYZ$ consists of two parts: The first part represents the rate of change of $\mathbf{Q}$ with respect to the rotating frame $Oxyz$; the second part, $\Omega \times \mathbf{Q}$, is induced by the rotation of the frame $Oxyz$.

Plane Motion of a Particle Relative to a Rotating Frame

The next section [Sec. 15.5B] was devoted to the two-dimensional kinematic analysis of a particle P moving with respect to a frame $\mathcal{F}$ rotating with an angular velocity Ω about a fixed axis (Fig. 15.46). We found that the absolute velocity of P could be expressed as

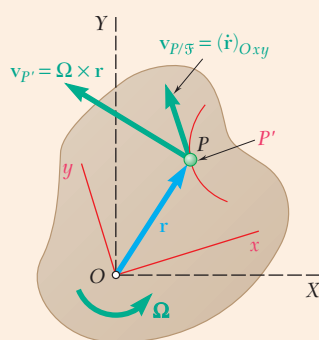


Fig. 15.46

$$\mathbf{v}_P = \mathbf{v}_{P'} + \mathbf{v}_{P/\mathcal{F}} \quad (15.33)$$

or

$$\mathbf{v}_P = \mathbf{v}_O + \boldsymbol{\Omega} \times \mathbf{r} + \mathbf{v}_{\text{rel}} \quad (15.32')$$

where $\mathbf{v}_P$ = absolute velocity of particle P

$\mathbf{v}_{P'} = \mathbf{v}_O + \boldsymbol{\Omega} \times \mathbf{r}$ = velocity of point P' of moving frame $\mathcal{F}$
coinciding with P

$\mathbf{v}_{P/\mathcal{F}} = \mathbf{v}_{\text{rel}}$ = velocity of P relative to moving frame $\mathcal{F}$

We noted that we obtain the same expression for $\mathbf{v}_P$ if the frame is in translation rather than in rotation. However, when the frame is in rotation, the expression for the acceleration of P contains an additional term $\mathbf{a}_C$ called the **Coriolis acceleration**. We have

$$\mathbf{a}_P = \mathbf{a}_{P'} + \mathbf{a}_{P/\mathcal{F}} + \mathbf{a}_C \quad (15.36)$$

or

$$\mathbf{a}_P = \mathbf{a}_O + \dot{\boldsymbol{\Omega}} \times \mathbf{r} - \Omega^2 \mathbf{r} + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}}$$

where $\mathbf{a}_P$ = absolute acceleration of particle P

$\mathbf{a}_{P'} = \mathbf{a}_O + \dot{\boldsymbol{\Omega}} \times \mathbf{r} - \Omega^2 \mathbf{r}$ = acceleration of point P' of moving
frame $\mathcal{F}$ coinciding with P

$\mathbf{a}_{P/\mathcal{F}} = \mathbf{a}_{\text{rel}}$ = acceleration of P relative to moving frame $\mathcal{F}$

$\mathbf{a}_C = 2\boldsymbol{\Omega} \times (\dot{\mathbf{r}})_{Oxyz} = 2\boldsymbol{\Omega} \times \mathbf{v}_{P/\mathcal{F}} = 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}}$ = Coriolis acceleration

Since $\boldsymbol{\Omega}$ and $\mathbf{v}_{P/\mathcal{F}}$ are perpendicular to each other in the case of plane motion, the Coriolis acceleration has a magnitude $a_C = 2\Omega v_{P/\mathcal{F}}$ and points in the direction obtained by rotating the vector $\mathbf{v}_{P/\mathcal{F}}$ through 90° in the sense of rotation of the moving frame. We can use formulas (15.33) and (15.36) to analyze the motion of mechanisms that contain parts sliding on each other [Sample Probs. 15.17 through 15.20].

Motion of a Rigid Body with a Fixed Point

In the last part of this chapter, we studied the kinematics of rigid bodies in three dimensions. We first considered the motion of a rigid body with a fixed point [Sec. 15.6A]. After proving that the most general displacement of a rigid body with a fixed point O is equivalent to a rotation of the body about an axis through O , we were able to define the angular velocity $\boldsymbol{\omega}$ and the **instantaneous axis of rotation** of the body at a given instant. The velocity of a point P of the body (Fig. 15.47) again could be expressed as

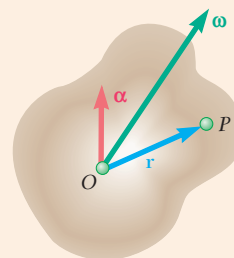


Fig. 15.47

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \boldsymbol{\omega} \times \mathbf{r} \quad (15.37)$$

Differentiating this expression gave

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \quad (15.38)$$

However, since the direction of $\boldsymbol{\omega}$ changes from one instant to the next, the angular acceleration $\boldsymbol{\alpha}$ is, in general, not directed along the instantaneous axis of rotation [Sample Prob. 15.21].

General Motion in Space

We showed in Sec. 15.6B that **the most general motion of a rigid body in space is equivalent, at any given instant, to the sum of a translation and a rotation**. Considering two particles A and B of the body, we found that

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \quad (15.42)$$

where $\mathbf{v}_{B/A}$ is the velocity of B relative to a frame $AX'Y'Z'$ attached to A and of fixed orientation (Fig. 15.48). Denoting by $\mathbf{r}_{B/A}$ the position vector of B relative to A , we have

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A} \quad (15.43)$$

where $\boldsymbol{\omega}$ is the angular velocity of the body at the instant considered [Sample Prob. 15.22]. We obtained the acceleration of B using a similar reasoning. We first wrote

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

and, recalling Eq. (15.38),

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{B/A}) \quad (15.44)$$

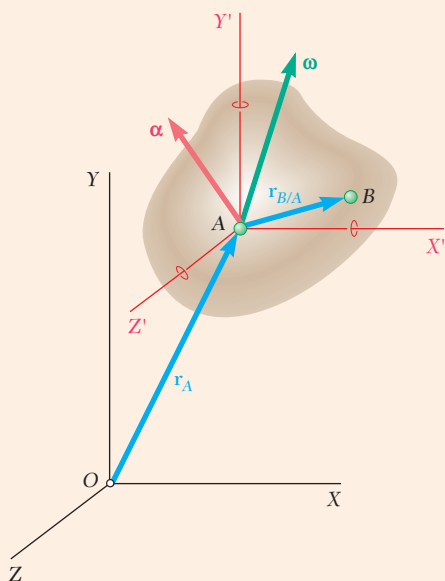


Fig. 15.48

Three-Dimensional Motion of a Particle Relative to a Rotating Frame

In the final section of this chapter, we considered the three-dimensional motion of a particle P relative to a frame $Oxyz$ rotating with an angular velocity $\boldsymbol{\Omega}$ with respect to a fixed frame $OXYZ$ (Fig. 15.49). In Sec. 15.7A, we expressed the absolute velocity $\mathbf{v}_P$ of P as

$$\mathbf{v}_P = \boldsymbol{\Omega} \times \mathbf{r} + (\dot{\mathbf{r}})_{Oxyz} \quad (15.45)$$

or alternatively as

$$\mathbf{v}_P = \mathbf{v}_{P'} + \mathbf{v}_{P/\mathcal{F}} \quad (15.46)$$

where $\mathbf{v}_P$ = absolute velocity of particle P

$\mathbf{v}_{P'}$ = velocity of point P' of moving frame $\mathcal{F}$ coinciding with P

$\mathbf{v}_{P/\mathcal{F}}$ = velocity of P relative to moving frame $\mathcal{F}$

The absolute acceleration $\mathbf{a}_P$ of P can be expressed as

$$\mathbf{a}_P = \dot{\boldsymbol{\Omega}} \times \mathbf{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) + 2\boldsymbol{\Omega} \times (\dot{\mathbf{r}})_{Oxyz} + (\ddot{\mathbf{r}})_{Oxyz} \quad (15.47)$$

or alternatively

$$\mathbf{a}_P = \mathbf{a}_{P'} + \mathbf{a}_{P/\mathcal{F}} + \mathbf{a}_C \quad (15.48)$$

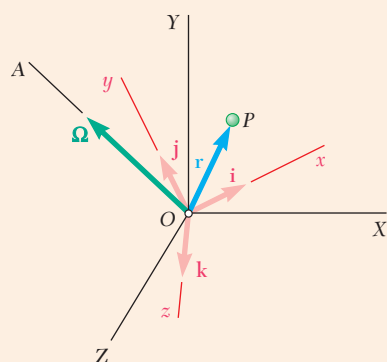


Fig. 15.49

where $\mathbf{a}_P$ = absolute acceleration of particle P

$\mathbf{a}_{P'}$ = acceleration of point P' of moving frame $\mathcal{F}$ coinciding with P

$\mathbf{a}_{P/\mathcal{F}}$ = acceleration of P relative to moving frame $\mathcal{F}$

$\mathbf{a}_C = 2\boldsymbol{\Omega} \times (\dot{\mathbf{r}})_{Oxyz} = 2\boldsymbol{\Omega} \times \mathbf{v}_{P/\mathcal{F}} = \text{Coriolis acceleration}$

We noted that the magnitude a_c of the Coriolis acceleration is not equal to $2\Omega v_{P/\mathcal{F}}$ [Sample Prob. 15.23] except in the special case when $\boldsymbol{\Omega}$ and $\mathbf{v}_{P/\mathcal{F}}$ are perpendicular to each other. Additionally, we usually will have to use Eq. 15.31 to determine the angular acceleration $\dot{\boldsymbol{\Omega}}$ of the rotating frame.

Frame of Reference in General Motion

We also observed [Sec. 15.7B] that Eqs. (15.46) and (15.48) remain valid when the frame $Axyz$ moves in a known—but arbitrary—fashion with respect to the fixed frame $OXYZ$ (Fig. 15.50), provided that the motion of A is included in the terms $\mathbf{v}_{P'}$ and $\mathbf{a}_{P'}$ representing the absolute velocity and acceleration of the coinciding point P' . We obtained

$$\mathbf{v}_P = \mathbf{v}_A + \boldsymbol{\Omega} \times \mathbf{r}_{P/A} + (\dot{\mathbf{r}}_{P/A})_{Axyz} \quad (15.52)$$

and

$$\begin{aligned} \mathbf{a}_P = \mathbf{a}_A + \dot{\boldsymbol{\Omega}} \times \mathbf{r}_{P/A} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}_{P/A}) \\ + 2\boldsymbol{\Omega} \times (\dot{\mathbf{r}}_{P/A})_{Axyz} + (\ddot{\mathbf{r}}_{P/A})_{Axyz} \end{aligned} \quad (15.54)$$

Rotating frames of reference are particularly useful in the study of the three-dimensional motion of rigid bodies. Indeed, in many cases, an appropriate choice of the rotating frame leads to a simpler analysis of the motion of the rigid body than would be possible with axes of fixed orientation [Sample Probs. 15.24 and 15.25].

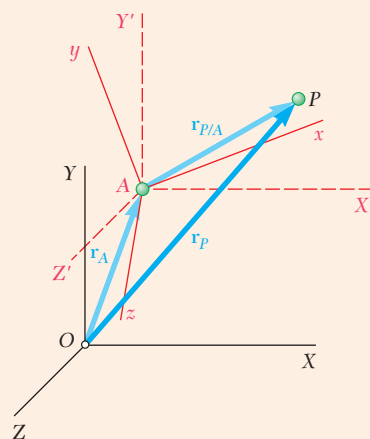


Fig. 15.50

Review Problems

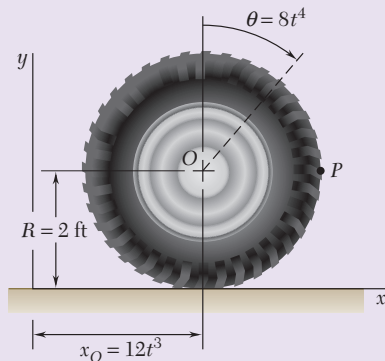


Fig. P15.248

15.248 A wheel moves in the xy plane in such a way that the location of its center is given by the equations $x_O = 12t^3$ and $y_O = R = 2$, where x_O and y_O are measured in feet and t is measured in seconds. The angular displacement of a radial line measured from a vertical reference line is $\theta = 8t^4$, where θ is measured in radians. Determine the velocity of point P located on the horizontal diameter of the wheel at $t = 1$ s.

15.249 Two blocks and a pulley are connected by inextensible cords as shown. The relative velocity of block A with respect to block B is 2.5 ft/s to the left at time $t = 0$ and 1.25 ft/s to the left when $t = 0.25$ s. Knowing that the angular acceleration of the pulley is constant, find (a) the relative acceleration of block A with respect to block B , (b) the distance block A moves relative to block B during the interval $0 \leq t \leq 0.25$ s.

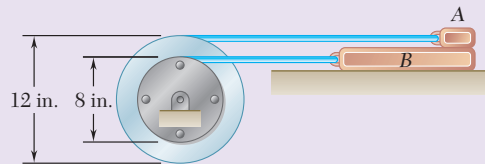


Fig. P15.249

15.250 A baseball pitching machine is designed to deliver a baseball with a ball speed of 70 mph and a ball rotation of 300 rpm clockwise. Knowing that there is no slipping between the wheels and the baseball during the ball launch, determine the angular velocities of wheels A and B .

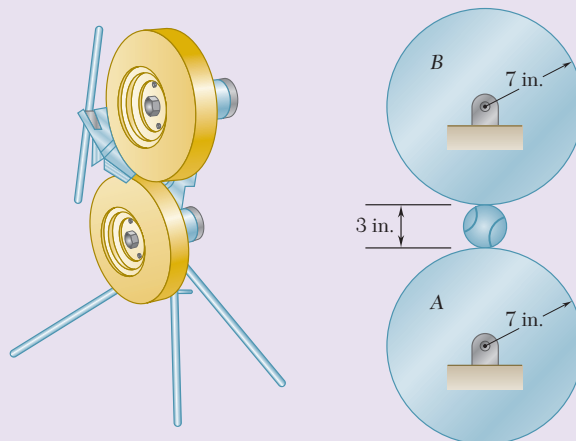


Fig. P15.250

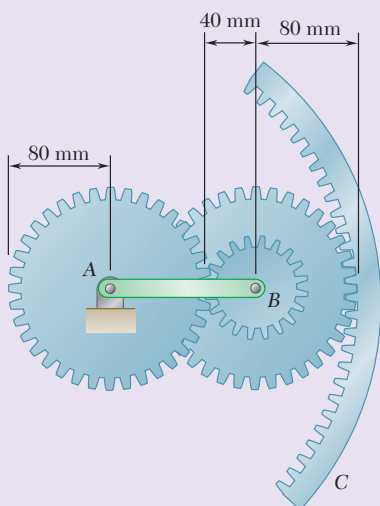


Fig. P15.251

15.251 Knowing that inner gear A is stationary and outer gear C starts from rest and has a constant angular acceleration of 4 rad/s^2 clockwise, determine at $t = 5$ s (a) the angular velocity of arm AB , (b) the angular velocity of gear B , (c) the acceleration of the point on gear B that is in contact with gear A .

- 15.252** Knowing that at the instant shown bar AB has an angular velocity of 10 rad/s clockwise and it is slowing down at a rate of 2 rad/s^2 , determine the angular accelerations of bar BD and bar DE .

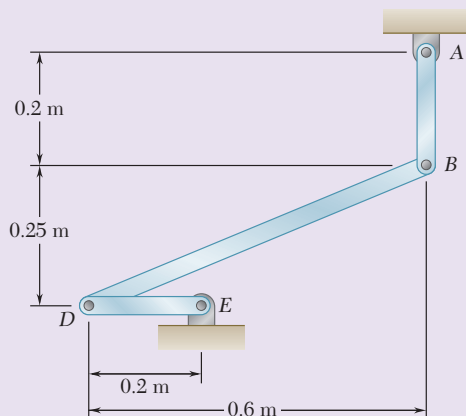


Fig. P15.252

- 15.253** Knowing that at the instant shown rod AB has zero angular acceleration and an angular velocity of 15 rad/s counterclockwise, determine (a) the angular acceleration of arm DE , (b) the acceleration of point D .

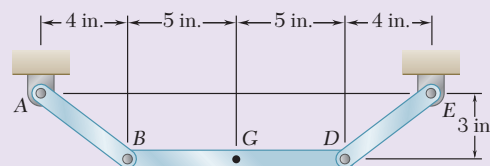


Fig. P15.253

- 15.254** Rod AB is attached to a collar at A and is fitted with a wheel at B that has a radius $r = 15 \text{ mm}$. Knowing that when $\theta = 60^\circ$ the collar has a velocity of 250 mm/s upward and it is slowing down at a rate of 150 mm/s^2 , determine (a) the angular acceleration of rod AB , (b) the angular acceleration of the wheel.

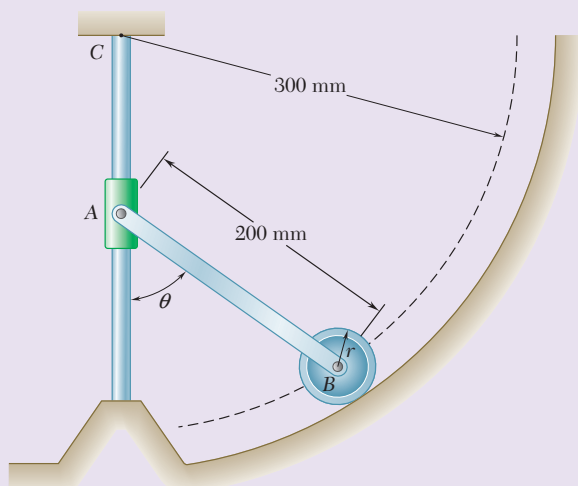


Fig. P15.254

- 15.255** Water flows through a curved pipe AB that rotates with a constant clockwise angular velocity of 90 rpm . If the velocity of the water relative to the pipe is 8 m/s , determine the total acceleration of a particle of water at point P .

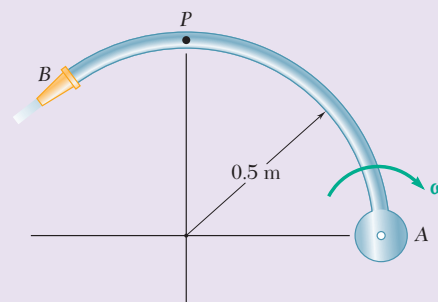


Fig. P15.255

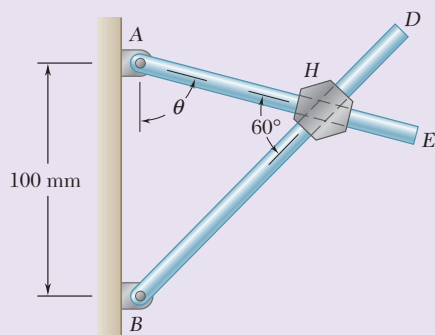


Fig. P15.257

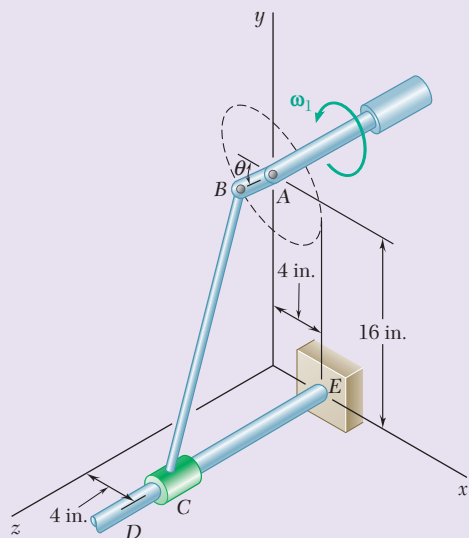


Fig. P15.258

15.256 A disk of 0.15-m radius rotates at the constant rate ω_2 with respect to plate BC , which itself rotates at the constant rate ω_1 about the y axis. Knowing that $\omega_1 = \omega_2 = 3$ rad/s, determine, for the position shown, the velocity and acceleration (a) of point D , (b) of point F .

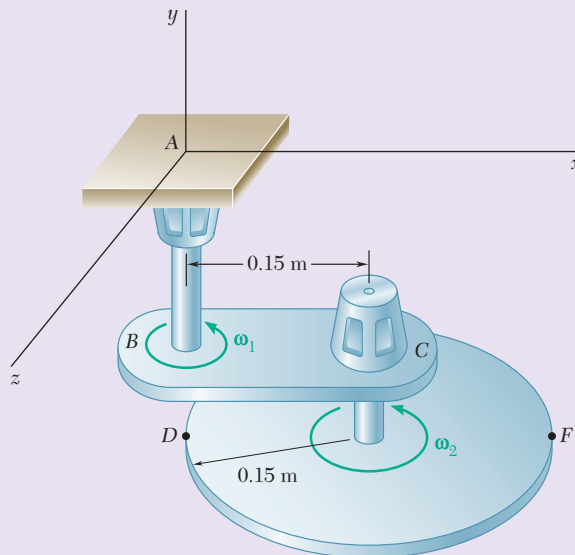


Fig. P15.256

15.257 Two rods AE and BD pass through holes drilled into a hexagonal block. (The holes are drilled in different planes so that the rods will not touch each other.) Knowing that rod AE has an angular velocity of 20 rad/s clockwise and an angular acceleration of 4 rad/s² counterclockwise when $\theta = 90^\circ$, determine (a) the relative velocity of the block with respect to each rod, (b) the relative acceleration of the block with respect to each rod.

15.258 Rod BC of length 24 in. is connected by ball-and-socket joints to a rotating arm AB and to a collar C that slides on the fixed rod DE . Knowing that the length of arm AB is 4 in. and that it rotates at the constant rate $\omega_1 = 10$ rad/s, determine the velocity of collar C when $\theta = 0$.

15.259 In the position shown the thin rod moves at a constant speed $u = 3$ in./s out of the tube BC . At the same time, tube BC rotates at the constant rate $\omega_2 = 1.5$ rad/s with respect to arm CD . Knowing that the entire assembly rotates about the X axis at the constant rate $\omega_1 = 1.2$ rad/s, determine the velocity and acceleration of end A of the rod.

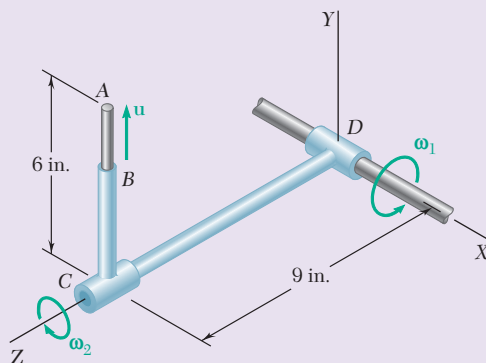


Fig. P15.259



16

Plane Motion of Rigid Bodies: Forces and Accelerations

The blades of the wind turbines shown in this picture are subjected to large forces and moments during motion. In this chapter, you will learn to analyze the motion of a rigid body by considering the motion of its mass center, the motion relative to its mass center, and the external forces acting on it.

Introduction

16.1 KINETICS OF A RIGID BODY

16.1A Equations of Motion for a Rigid Body

16.1B Angular Momentum of a Rigid Body in Plane Motion

16.1C Plane Motion of a Rigid Body

*16.1D A Remark on the Axioms of the Mechanics of Rigid Bodies

16.1E Solution of Problems Involving the Motion of a Rigid Body

16.1F Systems of Rigid Bodies

16.2 CONSTRAINED PLANE MOTION

Objectives

- **Discuss** how the mass and mass moment of inertia affect the linear and angular accelerations of a rigid body.
- **Model** physical systems involving rigid bodies by drawing correct free-body diagrams and kinetic diagrams.
- Using rigid-body kinetics principles, **determine** whether a body slips or tips and if a wheel rolls with or without slip.
- **Apply** appropriate kinetic equations and kinematics relationships to solve kinetics problems for a rigid body undergoing translation, centroidal rotation, or general plane motion.
- **Analyze** systems of connected rigid bodies using appropriate kinetic and kinematic equations.
- **Analyze** constrained motion of rigid bodies, including fixed-axis rotation and rolling disks and wheels.

Introduction

In this chapter and in Chaps. 17 and 18, you will study the **kinetics of rigid bodies**; i.e., the relations between the forces acting on a rigid body, the shape and mass of the body, and the motion produced. You studied similar relations in Chaps. 12 and 13, assuming then that you could consider the body as a particle, with its mass concentrated in one point and all forces acting at that point. Now you have to take into account the shape of the body, as well as the exact location of the points of application of the forces. You also will be concerned not only with the motion of the body as a whole but with the motion of the body about its mass center.

Our approach will be to consider rigid bodies as made up of large numbers of particles and to use the results obtained in Chap. 14 for the motion of systems of particles. Specifically, we use two equations from Chap. 14: Eq. (14.16), $\Sigma \mathbf{F} = m\mathbf{\bar{a}}$, which relates the resultant of the external forces and the acceleration of the mass center G of the system of particles, and Eq. (14.23), $\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G$, which relates the resultant moment of the external forces and the angular momentum of the system of particles about G .

Except for Sec. 16.1A, which applies to the most general case of the motion of a rigid body, the results derived in this chapter are limited in two ways: (1) They are restricted to the *plane motion* of rigid bodies, i.e., where all motion occurs in a single two-dimensional reference plane. (2) The rigid bodies considered consist only of plane rigid bodies and of bodies that are symmetrical with respect to a reference plane (or more generally, bodies that have a principal centroidal axis of inertia perpendicular to a reference plane). The study of the plane motion of nonsymmetrical three-dimensional bodies and, more generally, the motion of rigid bodies in three-dimensional space will be postponed until Chap. 18.

In Sec. 16.1B, we define the angular momentum of a rigid body in plane motion and show that the rate of change of the angular momentum $\mathbf{H}_G$ about the mass center is equal to the product $\bar{I}\alpha$ of the centroidal mass moment of inertia $\bar{I}$ and the angular acceleration α of the body. We then prove that the external forces acting on a rigid body are equivalent to a vector $m\bar{\mathbf{a}}$ attached at the mass center and a couple of moment $\bar{I}\alpha$.

We also derive the principle of transmissibility using only the parallelogram law and Newton's laws of motion, allowing us to remove this principle from the list of axioms (*Statics*, Sec. 1.2) required for the study of the statics and dynamics of rigid bodies. We then discuss the use of the free-body diagram and kinetic diagram in the solution of all problems involving the plane motion of rigid bodies.

We consider the plane motion of connected rigid bodies in Sec. 16.1F, which will prepare you to solve a variety of problems involving the translation, centroidal rotation, and unconstrained motion of rigid bodies. In the remaining part of this chapter, we present the solutions of problems involving noncentroidal rotation, rolling motion, and other partially constrained plane motions of rigid bodies.

16.1 KINETICS OF A RIGID BODY

As we saw in Chapter 15, we can generally consider the motion of a rigid body to be a combination of translation of the body and rotation about its mass center. We use this same idea to analyze the relationship between forces and moments acting on a rigid body and the body's linear and angular acceleration.

16.1A Equations of Motion for a Rigid Body

Consider a rigid body acted upon by several external forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots$ (Fig. 16.1). We can assume that the body is made of a large number n of particles of mass Δm_i ($i = 1, 2, \dots, n$) and apply the results obtained in Chap. 14 for a system of particles (Fig. 16.2). Consider first the motion of the mass center G of the body with respect to the newtonian frame of reference $Oxyz$. From Eq. (14.16), we have

**Translational equation
of motion**

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}} \quad (16.1)$$

where m is the mass of the body and $\bar{\mathbf{a}}$ is the acceleration of the mass center G . Turning now to the motion of the body relative to the centroidal frame of reference $Gx'y'z'$, from Eq. (14.23), we have

**Rotational equation
of motion**

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G \quad (16.2)$$

where $\dot{\mathbf{H}}_G$ represents the rate of change of $\mathbf{H}_G$, which is the angular momentum about G of the system of particles forming the rigid body.

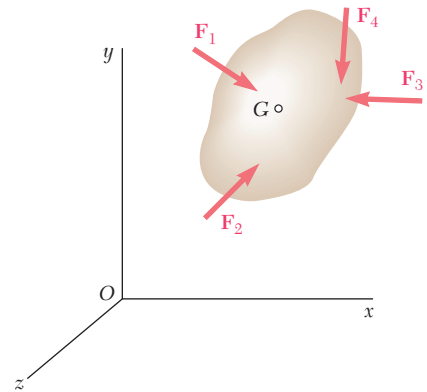


Fig. 16.1 A rigid body acted on by several external forces.

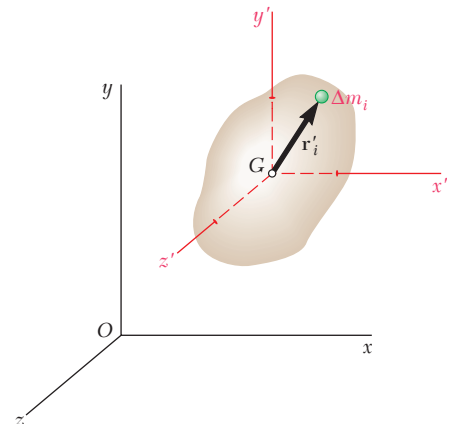


Fig. 16.2 A particle of a rigid body in relation to the mass center G .

In the following discussion, we refer to $\mathbf{H}_G$ simply as the **angular momentum of the rigid body about its mass center G** . Together, Eqs. (16.1) and (16.2) express that

The system of the external forces and moments is equipollent to the system consisting of the vector $m\bar{\mathbf{a}}$ attached at G and the couple of moment $\dot{\mathbf{H}}_G$ (Fig. 16.3).

As you will see in Chap. 18, Eqs. (16.1) and (16.2) apply to general three-dimensional motion of a rigid body. In the rest of this chapter, however, we limit our analysis to the **plane motion** of rigid bodies, i.e., to a motion in which each particle remains within a fixed reference plane. We also assume that the rigid bodies considered consist only of plane rigid bodies and of bodies that are symmetrical with respect to the plane of motion. Further study of the plane motion of nonsymmetrical three-dimensional bodies and of the motion of rigid bodies in three-dimensional space will be postponed until Chap. 18.

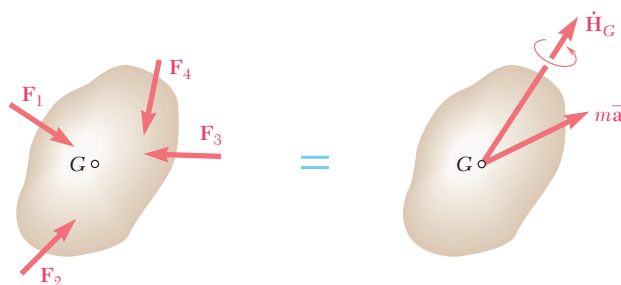


Fig. 16.3 A system of external forces is equipollent to an inertial vector $m\bar{\mathbf{a}}$ and a couple of moment $\dot{\mathbf{H}}_G$ acting at the mass center.

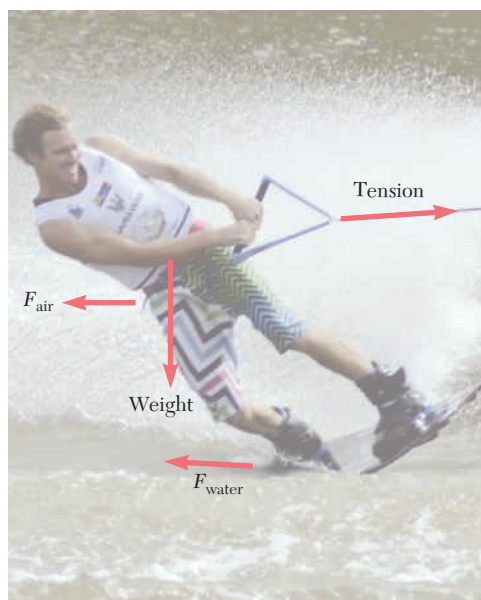


Photo 16.1 The system of external forces acting on the man and wakeboard includes the weights, the tension in the tow rope, and the forces exerted by the water and the air.

16.1B Angular Momentum of a Rigid Body in Plane Motion

Consider a rigid body in plane motion. Assume that the body is made of a large number n of particles P_i with a mass Δm_i . Then from Eq. (14.24) of Sec. 14.1D, we can compute the angular momentum $\mathbf{H}_G$ of the rigid body about its mass center G by taking the moments about G of the momenta of the particles of the body with respect to either of the frames Oxy or $Gx'y'$ (Fig. 16.4). Choosing the second option gives

$$\mathbf{H}_G = \sum_{i=1}^n (\mathbf{r}'_i \times \mathbf{v}'_i \Delta m_i) \quad (16.3)$$

where $\mathbf{r}'_i$ and $\mathbf{v}'_i \Delta m_i$ denote, respectively, the position vector and the linear momentum of the particle P_i relative to the centroidal frame of reference $Gx'y'$. However, since the particle is part of the rigid body, we have $\mathbf{v}'_i = \boldsymbol{\omega} \times \mathbf{r}'_i$, where $\boldsymbol{\omega}$ is the angular velocity of the body at the instant considered. We have

$$\mathbf{H}_G = \sum_{i=1}^n [\mathbf{r}'_i \times (\boldsymbol{\omega} \times \mathbf{r}'_i) \Delta m_i]$$

Referring to Fig. 16.4, we easily verify that this expression represents a vector of the same direction as $\boldsymbol{\omega}$ (i.e., perpendicular to the body) and with a magnitude of $\omega \sum r_i'^2 \Delta m_i$. Recalling that the sum $\sum r_i'^2 \Delta m_i$ represents the moment of inertia $\bar{I}$ of the rigid body about a centroidal axis perpendicular to the body, we conclude that the angular momentum $\mathbf{H}_G$ of the rigid body about its mass center is

Angular momentum of a rigid body about G

$$\mathbf{H}_G = \bar{I} \boldsymbol{\omega} \quad (16.4)$$

Differentiating both sides of Eq. (16.4), we obtain

Rate of change of angular momentum about G

$$\dot{\mathbf{H}}_G = \bar{I} \dot{\boldsymbol{\omega}} = \bar{I} \boldsymbol{\alpha} \quad (16.5)$$

Thus, the rate of change of the angular momentum of the rigid body is represented by a vector in the same direction as $\boldsymbol{\alpha}$ (i.e., perpendicular to the body) with a magnitude $\bar{I} \alpha$.

Keep in mind that the results obtained in this section have been derived for a rigid body in plane motion. As you will see in Chap. 18, they remain valid in the case of the plane motion of rigid bodies that are symmetrical with respect to a reference plane (or, more generally, bodies that have a principal centroidal axis of inertia perpendicular to a reference plane). However, they do not apply in the case of nonsymmetrical bodies or in the case of three-dimensional motion.

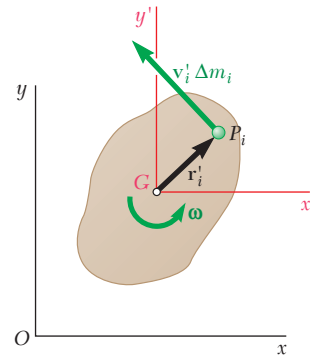


Fig. 16.4 The angular momentum about G of a particle of a rigid body is $\mathbf{r}'_i \times \mathbf{v}'_i \Delta m_i$.



Photo 16.2 The hard disk and pick-up arm of a computer hard drive undergo fixed-axis rotation.

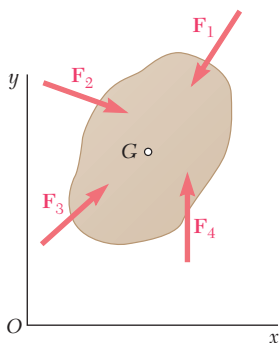


Fig. 16.5 A rigid body acted upon by several external forces in the plane of the body.

16.1C Plane Motion of a Rigid Body

Consider a rigid body with a mass m moving under the action of several external forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots$ contained in the plane of the body (Fig. 16.5). Substituting $\bar{\mathbf{H}}_G$ from Eq. (16.5) into Eq. (16.2) and writing the fundamental equations of motion from Eqs. (16.1) and (16.2) in scalar form, we have

$$\Sigma F_x = m\bar{a}_x \quad \Sigma F_y = m\bar{a}_y \quad \Sigma M_G = \bar{I}\alpha \quad (16.6)$$

Equations (16.6) show that we can obtain the acceleration of the mass center G of the rigid body and its angular acceleration α once we have determined the resultant of the external forces acting on the body and their moment resultant about G . Given appropriate initial conditions, we can then obtain the coordinates $\bar{x}$ and $\bar{y}$ of the mass center and the angular coordinate θ of the body by integration at any instant t . Thus,

The motion of the rigid body is completely defined by the resultant force and resultant moment about G acting on the body.

Since the motion of a rigid body depends only upon the resultant and resultant moment of the external forces acting on it, it follows that **two systems of forces that are equipollent** (i.e., that have the same resultant and the same moment resultant) **are also equivalent**. That is, they have exactly the same effect on a given rigid body.

Consider in particular the system of external forces acting on a rigid body (Fig. 16.6a) and the system of inertial terms associated with the particles forming the rigid body (Fig. 16.6b). We showed in Sec. 14.1A that the two systems thus defined are equipollent. But since the particles we are considering now form a rigid body, it follows from the discussion above that the two systems are also equivalent. We can thus state that

The external forces acting on a rigid body are equivalent to the inertial terms of the various particles forming the body.

The fact that the system of external forces is equivalent to the system of inertial terms has been emphasized by the use of red equal signs in Fig. 16.6 and also in Fig. 16.7. Here, using results obtained earlier in this section, we replaced the inertial terms by a vector $m\bar{\mathbf{a}}$ attached at the mass center G of the rigid body and the rotational inertial term $\bar{I}\alpha$.

Let's look at three examples of rigid-body plane motion.

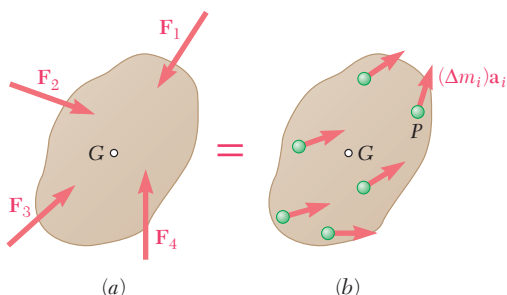


Fig. 16.6 The external forces acting on the rigid body are equivalent to the inertial terms of the particles of the body.

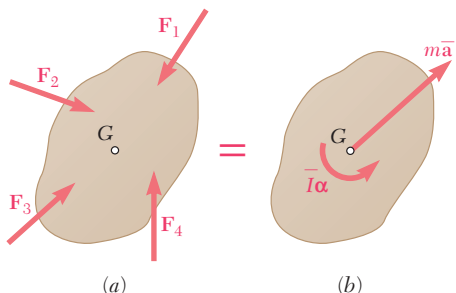


Fig. 16.7 The external forces acting on the rigid body are also equivalent to a vector $m\bar{\mathbf{a}}$ attached to the mass center G and a rotational inertia $\bar{I}\alpha$.

Translation. In the case of a body in translation, the angular acceleration of the body is equal to zero and its inertial terms reduce to the vector $m\bar{\mathbf{a}}$ attached at G (Fig. 16.8). Thus, the resultant of the external forces acting on a rigid body in translation passes through the mass center of the body and is equal to $m\bar{\mathbf{a}}$.

Centroidal Rotation. When a rigid body, or more generally, a body symmetrical with respect to a reference plane, rotates about a fixed axis perpendicular to the reference plane and passing through its mass center G , we say that the body is in *centroidal rotation*. Since the acceleration $\bar{\mathbf{a}}$ is identically equal to zero, the inertial terms of the body reduce to the couple $\bar{I}\alpha$ (Fig. 16.9). Thus, the external forces acting on a body in centroidal rotation are equivalent to the rotational inertia $\bar{I}\alpha$.

General Plane Motion. Comparing Fig. 16.7 with Figs. 16.8 and 16.9, we observe that, from the point of view of *kinetics*, the most general plane motion of a rigid body symmetrical with respect to a reference plane can be replaced by the sum of a translation and a centroidal rotation. Note that this statement is more restrictive than the similar statement made earlier from the point of view of *kinematics* (Sec. 15.2A), since we now require that the mass center of the body be selected as the reference point.

Referring to Eqs. (16.6), we observe that the first two equations are identical with the equations of motion of a particle of mass m acted upon by the given forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots$. We thus check that

The mass center G of a rigid body in plane motion moves as if the entire mass of the body were concentrated at that point, and as if all the external forces act on it.

Recall that we already obtained this result in Sec. 14.1C in the general case of a system of particles with the particles being not necessarily rigidly connected. We also note, as we did earlier, that the system of the external forces does not, in general, reduce to a single vector $m\bar{\mathbf{a}}$ attached at G . Therefore, in the general case of the plane motion of a rigid body, **the resultant of the external forces acting on the body does not pass through the mass center of the body.**

Finally, note that the last of Eqs. (16.6) would still be valid if the rigid body, while subjected to the same applied forces, were constrained to rotate about a fixed axis through G . Thus, **a rigid body in plane motion rotates about its mass center as if this point were fixed.**

*16.1D A Remark on the Axioms of the Mechanics of Rigid Bodies

The fact that two equipollent systems of external forces acting on a rigid body are also equivalent—i.e., have the same effect on that rigid body—has already been established in *Statics*, Sec. 3.4B. However, there we derived it from the *principle of transmissibility*, which is one of the axioms used in our study of the statics of rigid bodies. We have not used this axiom in the present chapter because Newton's second and third laws of motion make its use unnecessary in the study of the dynamics of rigid bodies.

In fact, we can now derive the principle of transmissibility from the other axioms used in the study of mechanics. This principle stated, without proof (Sec. 3.1B), that the conditions of equilibrium or motion of a rigid body remain unchanged if a force $\mathbf{F}$ acting at a given point of the rigid body is replaced by a force $\mathbf{F}'$ of the same magnitude and same direction—but acting at a different point—provided that the two forces have the same line of action. But since $\mathbf{F}$ and $\mathbf{F}'$ have the same moment about any given point, it is clear that they form two equipollent systems of external forces. Thus, we may now *prove*, as a result of what we established in the preceding section, that $\mathbf{F}$ and $\mathbf{F}'$ have the same effect on the rigid body (see Fig. 3.3 repeated here).

We can therefore remove the principle of transmissibility from the list of axioms required for the study of the mechanics of rigid bodies. These axioms are reduced to the parallelogram law of addition of vectors and to Newton's laws of motion.

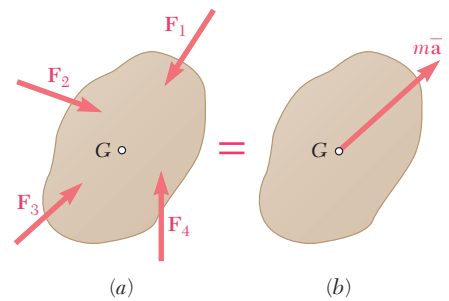


Fig. 16.8 A rigid body in translation has a vector $m\bar{\mathbf{a}}$ attached to the mass center G but no rotational inertia $\bar{I}\alpha$.

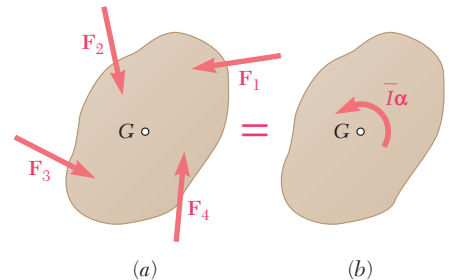


Fig. 16.9 A rigid body in centroidal rotation has a rotational inertia $\bar{I}\alpha$ but no $m\bar{\mathbf{a}}$.

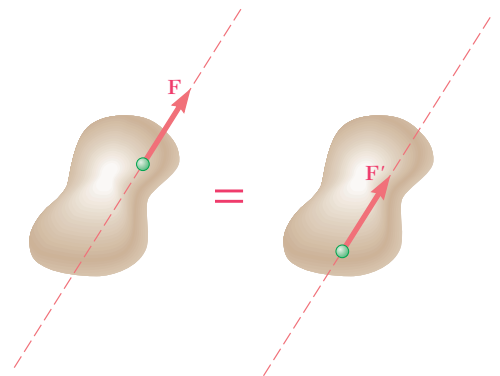


Fig. 3.3 (repeated)

16.1E Solution of Problems Involving the Motion of a Rigid Body

We saw in Sec. 16.1C that when a rigid body is in plane motion a fundamental relation exists between the forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots$, acting on the body, the acceleration $\bar{\mathbf{a}}$ of its mass center, and the angular acceleration α of the body. This relation is represented in Fig. 16.7 in the form of a **free-body diagram** and a **kinetic diagram**. We can use these diagrams to determine the acceleration $\bar{\mathbf{a}}$ and the angular acceleration α produced by a given system of forces acting on a rigid body or, conversely, to determine the forces that produce a given motion of the rigid body.

We can use the three algebraic equations of Eq. (16.6) to solve problems of plane motion.[†] However, our experience in statics suggests that the solution of many problems involving rigid bodies can be simplified by an appropriate choice of the point about which we compute the moments of the forces. It is therefore preferable to remember the relation between the forces and the accelerations in the pictorial form shown in Fig. 16.7 and to derive from this fundamental relation the component or moment equations that best fit the solution of the problem under consideration.

Drawing a free-body diagram for rigid bodies follows the same basic steps as we discussed in Chapter 12. For rigid bodies, however, it is important to draw your forces at their location of action, since you will be summing moments about specific points. Labeling different dimensions on your free-body diagram is particularly helpful when summing these moments.

The kinetic diagram is also slightly modified from Chap. 12. The translational inertial term $m\bar{\mathbf{a}}$ is always located at the center of mass of the body. We are now concerned with the rotational inertia of the body, so we include an additional term on our kinetic diagram, $\bar{I}\alpha$. This is also located at the center of mass of the body.

We can apply the steps from Chap. 12 to the pendulum shown in Fig. 16.10, where a moment M is applied to the bar. These steps include:

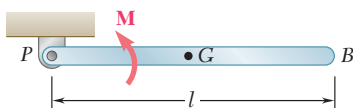


Fig. 16.10 A pendulum with mass m , length l , and an applied moment M .

1. Isolating the **body**
2. Defining the **axes**
3. Replacing constraints with **support forces**
4. Adding **applied forces and moments**, as well as **body forces** to the diagram
5. Labeling the free-body diagram with **dimensions**

For the kinetic diagram, we typically draw the translational inertial term in component form (e.g., $m\bar{a}_x$ and $m\bar{a}_y$) at the center of mass of the body and add the rotational inertial term $\bar{I}\alpha$. Using these steps gives you the free-body diagram and kinetic diagram shown in Fig. 16.11.

We use the pendulum shown in Fig. 16.10 to illustrate an alternative form of the moment equation. It is straightforward to apply Eq. 16.6 to this problem, where the sum of moments about the center of mass results in

$$+\circlearrowleft \Sigma M_G = \bar{I}\alpha : \quad M - P_y \left(\frac{L}{2} \right) = \bar{I}\alpha$$

[†]Recall that the last of Eq. (16.6) is valid only in the case of the plane motion of a rigid body symmetrical with respect to a reference plane. In all other cases, you need to use the methods of Chap. 18.

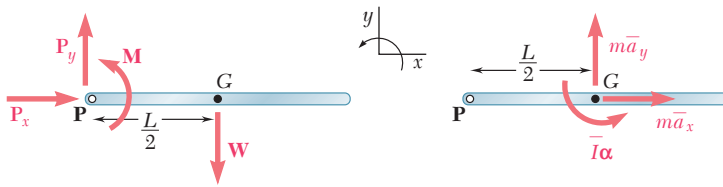


Fig. 16.11 Free-body diagram and kinetic diagram for a pendulum with an external moment applied.

Alternatively, we could choose an arbitrary point P about which to sum moments. If we choose P to be at the left end of the rod, then we also have to sum the moments about P due to the inertial terms. In this case, we obtain

$$+\circlearrowleft \Sigma M_P = \bar{I}\alpha + m\bar{a}d_{\perp} : \quad M - W\left(\frac{L}{2}\right) = \bar{I}\alpha + m\bar{a}_y\left(\frac{L}{2}\right) + m\bar{a}_x(0)$$

where $d_{\perp}$ is the perpendicular distance from point P to the line of action of the resultant acceleration vector $\bar{\mathbf{a}}$. As in statics, you can also determine the moment about a point P by using vector products, as

$$m\bar{a}d_{\perp} = \mathbf{r}_{G/P} \times m\bar{\mathbf{a}}$$

where $\mathbf{r}_{G/P}$ is the vector from point P to the center of mass of the body. Therefore, we can also write Eq (16.6) as

$$\begin{aligned} \Sigma F_x &= m\bar{a}_x & \Sigma F_y &= m\bar{a}_y \\ \Sigma M_G &= \bar{I}\alpha \text{ or } \Sigma M_P = \bar{I}\alpha + m\bar{a}d_{\perp} & \text{or } \Sigma M_P &= \bar{I}\alpha + \mathbf{r}_{G/P} \times m\bar{\mathbf{a}} \end{aligned} \quad (16.6')$$

The use of a free-body diagram and a kinetic diagram, showing vectorially the relationship between the forces applied on the rigid body and the resulting linear and angular accelerations, presents considerable advantages over the blind application of formulas (16.6). We can summarize these advantages as follows.

1. The use of a pictorial representation provides a much clearer understanding of the effect of the forces on the motion of the body.
2. This approach makes it possible to divide the solution of a dynamics problem into two parts: In the first part, the analysis of the kinematic and kinetic characteristics of the problem leads to the free-body diagram and the kinetic diagram of Fig. 16.7; in the second part, you can use the diagrams to analyze the various forces and vectors involved.
3. A unified approach is provided for the analysis of the plane motion of a rigid body, regardless of the particular type of motion involved. Although the kinematics of the various motions considered may vary from one case to the other, the approach to the kinetics of the motion is consistently the same. In every case, you draw a diagram showing the external forces, the vector $m\bar{\mathbf{a}}$ associated with the motion of G , and the couple $\bar{I}\alpha$ associated with the rotation of the body about G .
4. The resolution of the plane motion of a rigid body into a translation and a centroidal rotation, which we use here, is a basic concept that can be applied effectively throughout the study of mechanics. We will use it again in Chap. 17 with both the method of work and energy and the method of impulse and momentum.

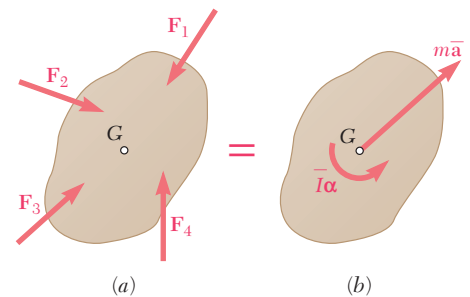


Fig. 16.7 (repeated)

5. As you will see in Chap. 18, we can extend this approach to the study of the general three-dimensional motion of a rigid body. The motion of the body is again resolved into a translation and a rotation about the mass center, and we use free-body diagrams and kinetic diagrams to indicate the relationship between the external forces and the rates of change of the linear and angular momenta of the body.

16.1F Systems of Rigid Bodies

The method just described also can be used in problems involving the plane motion of several connected rigid bodies. For each part of the system, you draw a diagram similar to Fig. 16.7. You can obtain the equations of motion from these diagrams and solve them simultaneously.

In some cases, as in Sample Prob. 16.4, you can draw a single diagram for the entire system. This diagram should include all of the external forces as well as the vectors $m\bar{\mathbf{a}}$ and the couples $\bar{I}\boldsymbol{\alpha}$ associated with the various parts of the system. However, you can omit internal forces, such as the forces exerted by connecting cables, because they occur in pairs of equal and opposite forces and are thus equipollent to zero. The equations obtained by expressing that the system of external forces is equipollent to the system of inertial terms can be solved for the remaining unknowns (note that we cannot speak of equivalent systems because we are not dealing with a single rigid body). For systems involving multiple rigid bodies, the general equation of motion is written as

$$\Sigma \mathbf{F} = \Sigma m_i \bar{\mathbf{a}}_i \quad \text{and} \quad \Sigma \mathbf{M}_P = \dot{\mathbf{H}}_P$$

where

$$\dot{\mathbf{H}}_P = \Sigma \bar{I}_i \boldsymbol{\alpha}_i + \Sigma m_i \bar{\mathbf{a}}_i (d_{\perp})_i = \Sigma \bar{I}_i \boldsymbol{\alpha}_i + \Sigma [(\mathbf{r}_{G/P})_i \times m_i \bar{\mathbf{a}}_i]$$

Historically, sometimes these equations have been written as

$$\Sigma \mathbf{F} = \Sigma \mathbf{F}_{\text{eff}} \quad \text{and} \quad \Sigma \mathbf{M}_P = \Sigma (\mathbf{M}_P)_{\text{eff}}$$

where the left-hand sides of these equations come from the free-body diagram and the right-hand sides come from the kinetic diagram. We have chosen not to use this notation because the terms on the right-hand side are due to the inertial terms and not due to external forces and moments.

It is not possible to include more than one rigid body in your system in problems involving more than three unknowns, since only three equations of motion are available when a single diagram is used. We will not elaborate upon this point, since the discussion involved would be completely similar to that given in Sec. 6.3B in the case of the equilibrium of a system of rigid bodies.



Photo 16.3 The forklift and moving load can be analyzed as a system of two connected rigid bodies in plane motion.

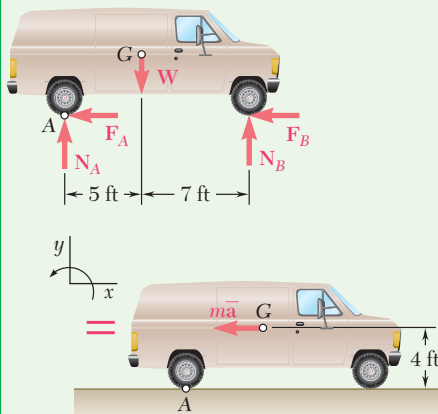
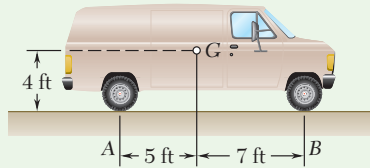


Fig. 1 Free-body diagram and kinetic diagram for the van.

Sample Problem 16.1

When the forward speed of the van shown is 30 ft/s, the brakes are suddenly applied, causing all four wheels to stop rotating. The van skids to rest in 20 ft. Determine the magnitude of the normal reaction and of the friction force at each wheel as the van skids to rest.

STRATEGY: You are given enough information to determine the acceleration and you want to find forces, so use Newton's second law. The motion described is pure translation, so the angular acceleration is zero.

MODELING: Choose the van to be your system and model it as a rigid body. A free-body diagram and a kinetic diagram for this system are shown in Fig. 1. The external forces consist of the weight $\mathbf{W}$ of the truck and of the normal reactions and friction forces at the wheels. The vectors $\mathbf{N}_A$ and $\mathbf{F}_A$ represent the sum of the reactions at the rear wheels, while $\mathbf{N}_B$ and $\mathbf{F}_B$ represent the sum of the reactions at the front wheels. Since the truck is in translation, $\alpha = 0$ and the inertial terms reduce to the vector $m\bar{\mathbf{a}}$ attached at G .

ANALYSIS:

Kinematics of Motion. Choose the positive sense to the right and use the equations of uniformly accelerated motion. You have

$$\begin{aligned} \bar{v}_0 &= +30 \text{ ft/s} & \bar{v}^2 &= \bar{v}_0^2 + 2\bar{a}\bar{x} & 0 &= (30)^2 + 2\bar{a}(20) \\ \bar{a} &= -22.5 \text{ ft/s}^2 & \bar{\mathbf{a}} &= 22.5 \text{ ft/s}^2 \leftarrow \end{aligned}$$

Equations of Motion. You can obtain three equations of motion by expressing that the system of the external forces from your free-body diagram is equivalent to the inertial terms from your kinetic diagram. Applying Newton's second law in the x and y directions gives

$$+\uparrow \Sigma F_y = m\bar{a}_y: \quad N_A + N_B - W = 0 \quad (1)$$

$$+\rightarrow \Sigma F_x = m\bar{a}_x: \quad -(F_A + F_B) = -m\bar{a} \quad (2)$$

Taking moments about any point gives you a third equation. For moments about point A , you find

$$+\curvearrowright \Sigma M_A = \bar{I}\alpha + m\bar{a}d_{\perp}: \quad -W(5 \text{ ft}) + N_B(12 \text{ ft}) = m\bar{a}(4 \text{ ft}) \quad (3)$$

In these three equations you have five unknowns, N_A , N_B , F_A , F_B , and $\bar{a}$. Since $F_A = \mu_k N_A$ and $F_B = \mu_k N_B$, where μ_k is the coefficient of kinetic friction, you have from Eq. (1)

$$F_A + F_B = \mu_k(N_A + N_B) = \mu_k W$$

Substituting into Eq. (2) and using $m = W/g$ gives

$$-\mu_k W = -\frac{W}{32.2 \text{ ft/s}^2} \bar{a} = -\frac{W}{32.2 \text{ ft/s}^2} (22.5 \text{ ft/s}^2)$$

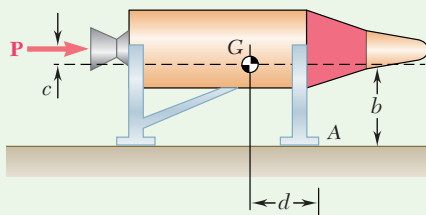
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or $\mu_k = 0.699$. Solving Eq. (3) for N_B gives you $N_B = 0.640W$. Substituting this into Eq. (1), you find $N_A = 0.350W$. The friction forces are easily determined once you know the normal forces $F_A = \mu_k N_A = (0.699)(0.350W) = 0.245W$ and $F_B = \mu_k N_B = (0.699)(0.650W) = 0.454W$.

Reactions at Each Wheel. Recall that the values computed here represent the sum of the reactions at the two front wheels or the two rear wheels. You obtain the magnitude of the reactions at each wheel by writing

$$\begin{aligned} N_{\text{front}} &= \frac{1}{2}N_B = 0.325W & N_{\text{rear}} &= \frac{1}{2}N_A = 0.175W \quad \blacktriangleleft \\ F_{\text{front}} &= \frac{1}{2}F_B = 0.227W & F_{\text{rear}} &= \frac{1}{2}F_A = 0.122W \quad \blacktriangleleft \end{aligned}$$

REFLECT and THINK: Note that even though the angular acceleration of the van is zero, the sum of the moments about point A is not equal to zero, since from the kinetic diagram, $m\bar{a}$ produces a moment about A. Rather than taking moments about point A, you also could have chosen to take moments about the center of mass, G. In this case, the sum of the moments would have been equal to zero. You only get three independent equations for a rigid body in plane motion: ΣF_x , ΣF_y , and one moment equation.



Sample Problem 16.2

A sled is jet-propelled along a straight track by a force P that increases linearly with time according to $P = kt$, where k is a constant. The coefficient of sliding friction between the sled runners and the track is μ_k , the coefficient of static friction is μ_s , and the mass of the sled is m . Determine (a) the time at which the tip of the rocket begins to rotate downward, (b) the acceleration of the sled at this instant. Neglect loss of mass due to fuel consumption and assume that the sled will slide before it tips.

STRATEGY: Since you are given a force, use Newton's second law to find the acceleration required for the rocket to begin rotating forward. You can then find the time using $P = kt$.

MODELING: Choose the sled as your system and model it as a rigid body. The rocket force must overcome the static friction force before it begins moving. Define this time to be t_0 . Figure 1 shows a free-body diagram when the motion is impending. In this case, both of the friction forces are set equal to the maximum allowable friction force $\mu_s N$. Free-body and kinetic diagrams for when the sled is about to tip are shown in Fig. 2. Just as the sled starts to tip, the normal force on the rear of the sled goes to zero.

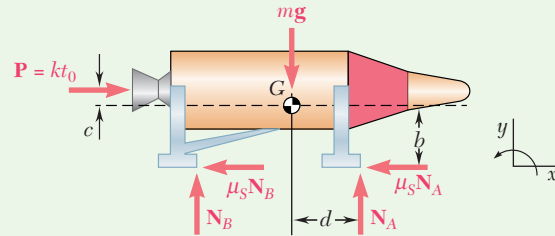


Fig. 1 Free-body diagram when motion is impending.

ANALYSIS: Using Fig. 1 and applying Newton's second law in the y - and x -directions gives

$$+\uparrow \Sigma F_y = m\bar{a}_y: \quad N_A + N_B - mg = 0 \quad \text{or} \quad N_A + N_B = mg$$

$$+\rightarrow \Sigma F_x = m\bar{a}_x: \quad kt_0 - (\mu_s N_A + \mu_s N_B) = 0$$

or

$$kt_0 = \mu_s(N_A + N_B) = \mu_s mg \quad (1)$$

Now that you know when the sled begins to slide, you can determine the time it will start to tip using Fig. 2.

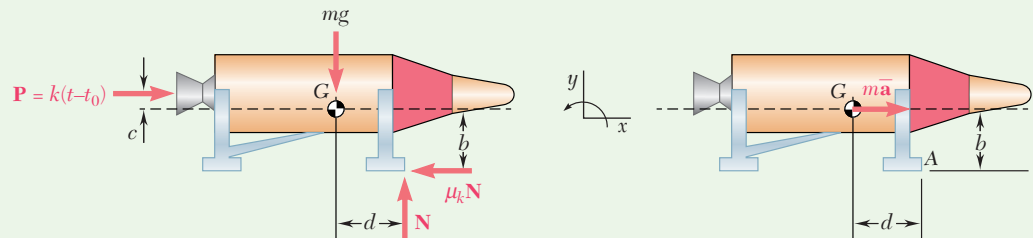


Fig. 2 Free-body diagram and kinetic diagram for the sled after it begins to move.

From this diagram, you can apply Newton's second law in the x and y directions and sum moments about any point. If you choose to take moments about G , you find

$$+\rightarrow \Sigma F_x = m\bar{a}_x: \quad k(t - t_0) - \mu_k N = m\bar{a} \quad (2)$$

$$+\uparrow \Sigma F_y = m\bar{a}_y: \quad N - mg = 0 \quad (3)$$

$$+\curvearrowright \Sigma M_G = I\bar{\alpha}: \quad Nd - \mu_k Nb - k(t - t_0)c = 0 \quad (4)$$

Solving Eqs. (1), (2), (3), and (4) for t_0 , t , N , $\bar{a}$, you find $N = mg$, $t_0 = \mu_s mg/k$ and

$$t = \frac{mg(d + c\mu_s - b\mu_k)}{kc} \quad \blacktriangleleft$$

$$\bar{a} = \frac{g(d - c\mu_k - b\mu_k)}{c} \quad \blacktriangleleft$$

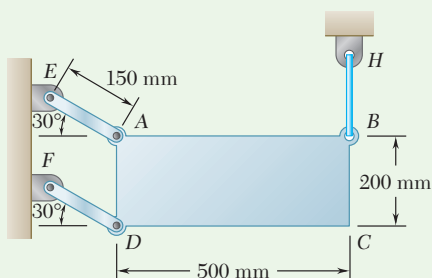
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REFLECT and THINK: Rather than taking moments about G , you could have chosen any other point. For example, for moments about A , you have

$$+\circlearrowleft \Sigma M_A = \bar{I}\alpha + m\bar{a}d: \quad mgd - k(t - t_0)(b + c) = -m\bar{a}b$$

Using this equation rather than Eq. (4) will give you the same answer. To check the assumption that the sled slides before it tips, you would need to use Fig. 1 and show that both N_A and N_B are positive for the given value of $P = kt_0$.

Sample Problem 16.3



The thin plate $ABCD$ has a mass of 8 kg and is held in the position shown by the wire BH and two links AE and DF . Neglecting the mass of the links, determine immediately after wire BH has been cut (a) the acceleration of the plate, (b) the force in each link.

STRATEGY: Since you are asked to determine the acceleration and forces, use Newton's second law. After wire BH has been cut, corners A and D move along parallel circles, each with a radius of 150 mm centered, respectively, at E and F . The motion of the plate is thus a curvilinear translation (Fig. 1); the particles forming the plate move along parallel circles each with a radius of 150 mm.

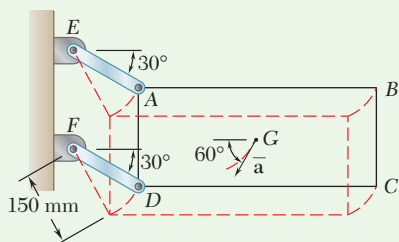


Fig. 1 Curvilinear translation of the plate.

MODELING: Choose the plate to be your system and model it as a rigid body. To draw the kinetic diagram, you need to consider the kinematics of the motion. At the instant wire BH is cut, the velocity of the plate is zero. Thus, the acceleration of the mass center G of the plate is tangent to the circular path described by G (Fig. 1). The free-body diagram and kinetic diagram for this system are shown in Fig. 2. The external forces consist of the weight $\mathbf{W}$ and the forces $\mathbf{F}_{AE}$ and $\mathbf{F}_{DF}$ exerted by the links. Since the plate is in translation, the kinetic diagram is the vector $m\bar{\mathbf{a}}$ attached at G and directed along the t axis.

ANALYSIS:

a. Acceleration of the Plate.

$$+\circlearrowleft \Sigma F_t = m\bar{a}_t;$$

$$W \cos 30^\circ = m\bar{a}$$

$$mg \cos 30^\circ = m\bar{a}$$

$$\bar{a} = g \cos 30^\circ = (9.81 \text{ m/s}^2) \cos 30^\circ \quad (1)$$

$$\bar{a} = 8.50 \text{ m/s}^2 \nearrow 60^\circ \quad \blacktriangleleft$$

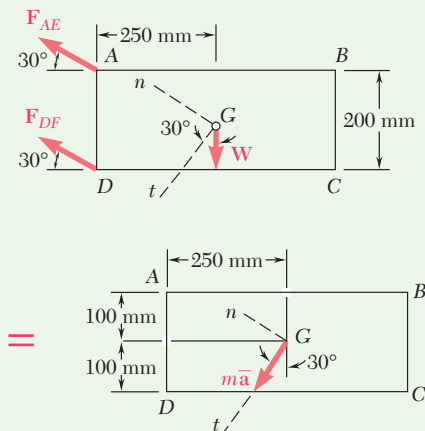


Fig. 2 Free-body diagram and kinetic diagram for the plate.

b. Forces in Links *AE* and *DF*.

$$+\nearrow \Sigma F_n = m\bar{a}_n: \quad F_{AE} + F_{DF} - W \sin 30^\circ = 0 \quad (2)$$

$$+\downarrow \Sigma M_G = \bar{I}\alpha:$$

$$(F_{AE} \sin 30^\circ)(250 \text{ mm}) - (F_{AE} \cos 30^\circ)(100 \text{ mm}) \\ + (F_{DF} \sin 30^\circ)(250 \text{ mm}) + (F_{DF} \cos 30^\circ)(100 \text{ mm}) = 0 \\ 38.4F_{AE} + 211.6F_{DF} = 0$$

$$F_{DF} = -0.1815F_{AE} \quad (3)$$

Substituting F_{DF} from Eq. (3) into Eq. (2), you have

$$F_{AE} - 0.1815F_{AE} - W \sin 30^\circ = 0$$

$$F_{AE} = 0.6109W$$

$$F_{DF} = -0.1815(0.6109W) = -0.1109W$$

Noting that $W = mg = (8 \text{ kg})(9.81 \text{ m/s}^2) = 78.48 \text{ N}$, you have

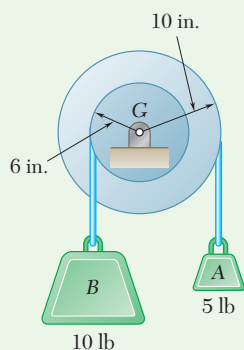
$$F_{AE} = 0.6109(78.48 \text{ N}) \quad F_{AE} = 47.9 \text{ N } T \quad \blacktriangleleft$$

$$F_{DF} = -0.1109(78.48 \text{ N}) \quad F_{DF} = 8.70 \text{ N } C \quad \blacktriangleleft$$

where bar *AE* is in tension and bar *DF* is in compression.

REFLECT and THINK: If *AE* and *DF* had been cables rather than links, the answers you just determined indicate that *DF* would have gone slack (i.e., you can't push on a rope), since the analysis showed that it would be in compression. Therefore, the plate would not be undergoing curvilinear translation, but it would have been undergoing general plane motion. It is important to note that there is always more than one way to solve problems like this, since you can choose to take moments about any point you wish. In this case, you took them about *G*, but you could have also chosen to take them about *A* or *D*.

Sample Problem 16.4



A pulley weighing 12 lb and having a radius of gyration of 8 in. is connected to two blocks as shown. Assuming no axle friction, determine the angular acceleration of the pulley and the acceleration of each block.

STRATEGY: Since you want to determine accelerations and are given the weights, use Newton's second law.

MODELING: Choose the pulley and the two blocks as a single system. The pulley moves in pure rotation and each block moves in pure translation.

(continued)

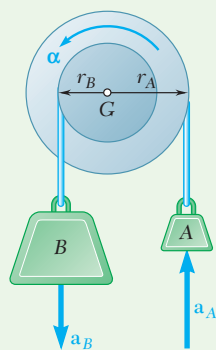


Fig. 1 Acceleration directions assuming a CCW angular acceleration.

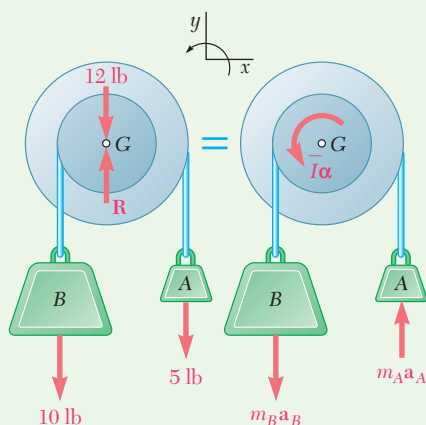


Fig. 2 Free-body diagram and kinetic diagram for the system.

Sense of Motion. Although you can assume an arbitrary sense of motion as shown in Fig. 1 (since no friction forces are involved) and later check it by the sign of the answer, you may prefer to determine the actual sense of rotation of the pulley. First determine the weight of block B , W'_B , required to maintain the equilibrium of the pulley when it is acted upon by the 5-lb block A .

$$+\uparrow \Sigma M_G = 0: \quad W'_B(6 \text{ in.}) - (5 \text{ lb})(10 \text{ in.}) = 0 \quad W'_B = 8.33 \text{ lb}$$

Since block B actually weighs 10 lb, the pulley rotates counterclockwise. The free-body and kinetic diagrams for this system are shown in Fig. 2. The forces external to the system consist of the weights of the pulley and the two blocks and of the reaction at G (Fig. 2). The forces exerted by the cables on the pulley and on the blocks are internal to the system and cancel out. Since the motion of the pulley is a centroidal rotation and the motion of each block is a translation, the inertial terms reduce to the couple $\bar{I}\alpha$ and the two vectors $m\mathbf{a}_A$ and $m\mathbf{a}_B$.

ANALYSIS:

Kinematics of Motion. Assuming α is counterclockwise and noting that $a_A = r_A\alpha$ and $a_B = r_B\alpha$, you obtain

$$\mathbf{a}_A = \left(\frac{10}{12} \text{ ft}\right)\alpha \uparrow \quad \mathbf{a}_B = \left(\frac{6}{12} \text{ ft}\right)\alpha \downarrow$$

Equations of Motion. The centroidal moment of inertia of the pulley is

$$\bar{I} = m\bar{k}^2 = \frac{W}{g}\bar{k}^2 = \frac{12 \text{ lb}}{32.2 \text{ ft/s}^2} \left(\frac{8}{12} \text{ ft}\right)^2 = 0.1656 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$$

Since the system of external forces is equivalent to the system of inertial terms, you have

$$\begin{aligned} +\uparrow \Sigma M_G &= \dot{H}_G: \\ (10 \text{ lb})\left(\frac{6}{12} \text{ ft}\right) - (5 \text{ lb})\left(\frac{10}{12} \text{ ft}\right) &= +\bar{I}\alpha + m_B a_B \left(\frac{6}{12} \text{ ft}\right) + m_A a_A \left(\frac{10}{12} \text{ ft}\right) \\ (10)\left(\frac{6}{12}\right) - (5)\left(\frac{10}{12}\right) &= 0.1656\alpha + \frac{10}{32.2}\left(\frac{6}{12}\alpha\right)\left(\frac{6}{12}\right) + \frac{5}{32.2}\left(\frac{10}{12}\alpha\right)\left(\frac{10}{12}\right) \\ \alpha &= +2.374 \text{ rad/s}^2 & \alpha &= 2.37 \text{ rad/s}^2 \uparrow \quad \blacktriangleleft \\ a_A &= r_A\alpha = \left(\frac{10}{12} \text{ ft}\right)(2.374 \text{ rad/s}^2) & \mathbf{a}_A &= 1.978 \text{ ft/s}^2 \uparrow \quad \blacktriangleleft \\ a_B &= r_B\alpha = \left(\frac{6}{12} \text{ ft}\right)(2.374 \text{ rad/s}^2) & \mathbf{a}_B &= 1.187 \text{ ft/s}^2 \downarrow \quad \blacktriangleleft \end{aligned}$$

REFLECT and THINK: You could also solve this problem by considering the pulley and each block as separate systems, but you would have more resulting equations. You would have to use this approach if you wanted to know the forces in the cables.

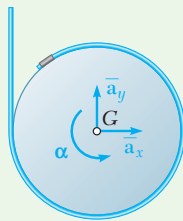
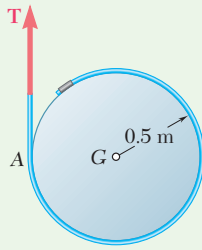


Fig. 1 Assumed directions for the angular acceleration and the acceleration of the center of mass.

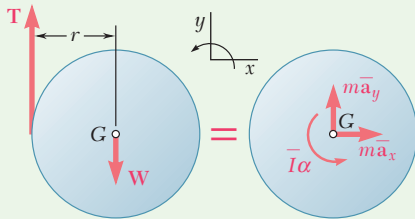


Fig. 2 Free-body diagram and kinetic diagram for the disk.

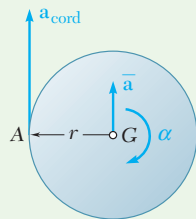


Fig. 3 Acceleration of points A and G on the disk.

Sample Problem 16.5

A cord is wrapped around a homogeneous disk with a radius of $r = 0.5$ m and a mass of $m = 15$ kg. If the cord is pulled upward with a force $\mathbf{T}$ of magnitude 180 N, determine (a) the acceleration of the center of the disk, (b) the angular acceleration of the disk, (c) the acceleration of the cord.

STRATEGY: Since you have forces and are interested in determining accelerations, use Newton's second law.

MODELING: Choose the disk and the cord as your system. Assume that the components $\bar{a}_x$ and $\bar{a}_y$ of the acceleration of the center are directed, respectively, to the right and upward and that the angular acceleration of the disk is counterclockwise (Fig. 1). A free-body diagram and kinetic diagram for this system are shown in Fig. 2. The external forces acting on the disk consist of its weight $\mathbf{W}$ and the force $\mathbf{T}$ exerted by the cord.

ANALYSIS:

Equations of Motion. Applying Newton's second law in the x and y directions gives

$$\begin{aligned} \rightarrow \Sigma F_x &= m\bar{a}_x: & 0 &= m\bar{a}_x & \bar{a}_x &= 0 \quad \blacktriangleleft \\ +\uparrow \Sigma F_y &= m\bar{a}_y: & T - W &= m\bar{a}_y \\ & & \bar{a}_y &= \frac{T - W}{m} \end{aligned}$$

Since $T = 180$ N, $m = 15$ kg, and $W = (15 \text{ kg})(9.81 \text{ m/s}^2) = 147.1$ N, you have

$$\bar{a}_y = \frac{180 \text{ N} - 147.1 \text{ N}}{15 \text{ kg}} = +2.19 \text{ m/s}^2 \quad \bar{a}_y = 2.19 \text{ m/s}^2 \uparrow \quad \blacktriangleleft$$

Now taking moments about the center of gravity, you get

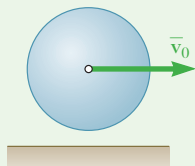
$$\begin{aligned} +\curvearrowright \Sigma M_G &= \bar{I}\alpha: & -Tr &= \bar{I}\alpha \\ & & -Tr &= (\tfrac{1}{2}mr^2)\alpha \\ \alpha &= -\frac{2T}{mr} = -\frac{2(180 \text{ N})}{(15 \text{ kg})(0.5 \text{ m})} = -48.0 \text{ rad/s}^2 \\ & & \alpha &= 48.0 \text{ rad/s}^2 \downarrow \quad \blacktriangleleft \end{aligned}$$

Acceleration of Cord. The acceleration of the cord is equal to the tangential component of the acceleration of point A on the disk, so you have (Fig. 3)

$$\begin{aligned} \mathbf{a}_{\text{cord}} &= (\mathbf{a}_A)_t = \bar{\mathbf{a}} + (\mathbf{a}_{A/G})_t \\ &= [2.19 \text{ m/s}^2 \uparrow] + [(0.5 \text{ m})(48 \text{ rad/s}^2) \uparrow] \\ & \mathbf{a}_{\text{cord}} = 26.2 \text{ m/s}^2 \uparrow \quad \blacktriangleleft \end{aligned}$$

REFLECT and THINK: The angular acceleration is clockwise, as we would expect. A similar analysis would apply in many practical situations, such as pulling wire off a spool or paper off a roll. In such cases, you would need to be sure that the tension pulling on the disk is not larger than the tensile strength of the material.

Sample Problem 16.6



A uniform sphere with mass m and radius r is projected along a rough horizontal surface with a linear velocity $\bar{v}_0$ and no angular velocity. Denoting the coefficient of kinetic friction between the sphere and the floor by μ_k , determine (a) the time t_1 at which the sphere starts rolling without sliding, (b) the linear velocity and angular velocity of the sphere at time t_1 .

STRATEGY: Since you have forces acting on the sphere, use Newton's second law. To relate the acceleration to the velocity, you need to use the basic kinematic relationships. The sphere starts out rotating and sliding; it stops sliding when the instantaneous point of contact with the ground has a velocity of zero.

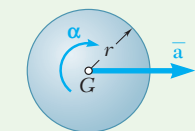


Fig. 1 Assumed directions for the angular and linear acceleration of the sphere.

MODELING: Choose the sphere as your system and model it as a rigid body. The assumed positive directions for the acceleration of the mass center and the angular acceleration are shown in Fig. 1. Free-body and kinetic diagrams for this system are shown in Fig. 2. Since the point of the sphere in contact with the surface is sliding to the right, the friction force $\mathbf{F}$ is directed to the left. While the sphere is sliding, the magnitude of the friction force is $F = \mu_k N$.

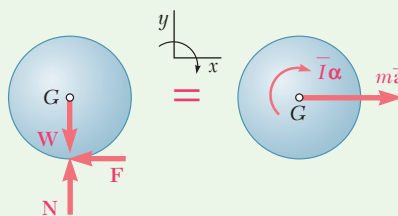


Fig. 2 Free-body diagram and kinetic diagram for the sphere.

ANALYSIS:

Equations of Motion. Applying Newton's second law in the x and y directions gives

$$\begin{aligned}
 +\uparrow \Sigma F_y &= m\bar{a}_y: & N - W &= 0 \\
 & & N = W = mg & & F = \mu_k N = \mu_k mg \\
 +\rightarrow \Sigma F_x &= m\bar{a}_x: & -F &= m\bar{a} & -\mu_k mg = m\bar{a} & \bar{a} = -\mu_k g
 \end{aligned}$$

Now taking moments about the center of gravity, you get

$$+\downarrow \Sigma M_G = \bar{I}\alpha: \quad Fr = \bar{I}\alpha$$

Noting that $\bar{I} = \frac{2}{5}mr^2$ and substituting the given value for F , you have

$$(\mu_k mg)r = \frac{2}{5}mr^2\alpha \quad \alpha = \frac{5}{2} \frac{\mu_k g}{r}$$

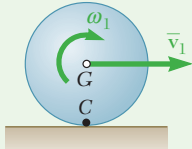


Fig. 3 The point of contact has zero velocity when the sphere starts rolling.

Kinematics of Motion. As long as the sphere both rotates and slides, its linear and angular accelerations are constant. Therefore, you can use the constant-acceleration equations to relate these accelerations to the linear velocity and angular velocity.

$$t = 0, \bar{v} = \bar{v}_0 \quad \bar{v} = \bar{v}_0 + \bar{a}t = \bar{v}_0 - \mu_k g t \quad (1)$$

$$t = 0, \omega_0 = 0 \quad \omega = \omega_0 + \alpha t = 0 + \left(\frac{5 \mu_k g}{2 r} \right) t \quad (2)$$

The sphere starts rolling without sliding when the velocity $\mathbf{v}_C$ of the point of contact C is zero (Fig. 3). At that time, $t = t_1$, point C becomes the instantaneous center of rotation, and you have

$$\bar{v}_1 = r \omega_1 \quad (3)$$

Substituting in Eq. (3) the values obtained for $\bar{v}_1$ and ω_1 by making $t = t_1$ in Eqs. (1) and (2), respectively, you obtain

$$\bar{v}_0 - \mu_k g t_1 = r \left(\frac{5 \mu_k g}{2 r} t_1 \right) \quad t_1 = \frac{2}{7} \frac{\bar{v}_0}{\mu_k g} \quad \blacktriangleleft$$

Substituting for t_1 into Eq. (2), you have

$$\begin{aligned} \omega_1 &= \frac{5 \mu_k g}{2 r} t_1 = \frac{5 \mu_k g}{2 r} \left(\frac{2}{7} \frac{\bar{v}_0}{\mu_k g} \right) & \omega_1 &= \frac{5}{7} \frac{\bar{v}_0}{r} & \omega_1 &= \frac{5}{7} \frac{\bar{v}_0}{r} \downarrow \quad \blacktriangleleft \\ \bar{v}_1 &= r \omega_1 = r \left(\frac{5}{7} \frac{\bar{v}_0}{r} \right) & \bar{v}_1 &= \frac{5}{7} \bar{v}_0 & \mathbf{v}_1 &= \frac{5}{7} \bar{v}_0 \rightarrow \quad \blacktriangleleft \end{aligned}$$

REFLECT and THINK: Notice we chose a different coordinate system than we usually do, with the positive rotation going clockwise. This means that you will not be able to use vector algebra solutions since it is not a right-handed coordinate system.

You could use this type of analysis to determine how long it takes a bowling ball to begin to roll without slip or to see how the coefficient of friction affects this motion. Instead of taking moments about the center of gravity, you could have chosen to take moments about point C , in which case your third equation would have been $\Sigma M_C = \dot{H}_C \longrightarrow 0 = m \bar{a} r + \bar{I} \alpha$.

SOLVING PROBLEMS ON YOUR OWN

This chapter deals with the **plane motion of rigid bodies**, and in this first section, we considered rigid bodies that are free to move under the action of applied forces.

1. Free-body diagram and kinetic diagram. After choosing a system, your first step in the solution of a problem is to draw a *free-body diagram* and a *kinetic diagram*.

a. A free-body diagram shows ***the forces exerted on the body***, including the applied forces and moments, the reactions at the supports, and the weight of the body.

b. A kinetic diagram shows the **inertial terms**: vector $m\bar{\mathbf{a}}$ and the couple $\bar{I}\alpha$.

2. Using your free-body diagram and kinetic diagram, generate the equations of motion for the system. Drawing good free-body and kinetic diagrams will allow you to *sum components in any direction and to sum moments about any point*. For a single body, you can obtain a maximum of three independent equations (two translational and one moment) that can be used to help analyze the system. Noting that the external forces and moments are equivalent to the inertial terms, we wrote

$$\begin{aligned}\Sigma F_x &= m\bar{a}_x & \Sigma F_y &= m\bar{a}_y \\ \Sigma M_G &= \bar{I}\alpha \text{ or } \Sigma M_P = \bar{I}\alpha + m\bar{a}d_\perp & \Sigma M_P &= \bar{I}\alpha + \mathbf{r}_{G/P} \times m\bar{\mathbf{a}}\end{aligned}\quad (16.6')$$

where G is the center of mass of the body, P is any arbitrary point, and $d_\perp$ is the perpendicular distance between point P and the line of action of the acceleration of the center of mass.

3. Apply kinematic relationships. Often, you will have more than three unknowns and will need to generate additional equations. You can usually do this by applying kinematic relationships, such as $a_n = r\omega^2$ and $a_t = r\alpha$ or for a rigid body undergoing fixed-axis rotation or the more general expression relating the acceleration of two points on a rigid body, as

$$\mathbf{a}_B = \mathbf{a}_A + \alpha \mathbf{k} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A} \quad (15.21')$$

4. Plane motion of a rigid body. The problems that you will be asked to solve will fall into one of the following categories.

a. Rigid body in translation. For a body in translation, the angular acceleration is zero. The kinetic diagram, therefore, is simply the vector $m\bar{\mathbf{a}}$ applied at the mass center [Sample Probs. 16.1 through 16.3].

b. Rigid body in centroidal rotation. For a body in centroidal rotation, the linear acceleration of the mass center is zero. Therefore, the kinetic diagram is simply the couple $\bar{I}\alpha$ [Sample Prob. 16.4].

c. Rigid body in general plane motion. You can consider the general plane motion of a rigid body to be the sum of a translation and a centroidal rotation. The kinetic diagram contains the vector $m\bar{\mathbf{a}}$ and the couple $\bar{I}\alpha$ [Sample Probs. 16.5 and 16.6].

5. Plane motion of a system of rigid bodies. You first should draw a free-body diagram and a kinetic diagram that includes all of the rigid bodies of the system. A vector $m\bar{\mathbf{a}}$ and a couple $\bar{I}\alpha$ are attached to each body. However, the forces exerted on each other by the various bodies of the system can be omitted, since they occur in pairs of equal and opposite forces.

a. If no more than three unknowns are involved, you can use the free-body and kinetic diagrams to sum components in any direction and sum moments about any point, obtaining equations that can be solved for the desired unknowns [Sample Prob. 16.4].

b. If more than three unknowns are involved, you must choose a new system, use kinematics, or use additional information in the problem statement to find additional equations.

Problems

CONCEPT QUESTIONS

- 16.CQ1** Two pendulums, *A* and *B*, with the masses and lengths shown are released from rest. Which system has a larger mass moment of inertia about its pivot point?
- A*
 - B*
 - They are the same.

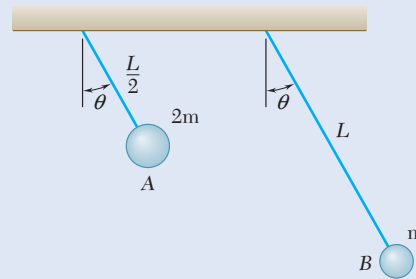


Fig. P16.CQ1 and P16.CQ2

- 16.CQ2** Two pendulums, *A* and *B*, with the masses and lengths shown are released from rest. Which system has a larger angular acceleration immediately after release?
- A*
 - B*
 - They are the same.
- 16.CQ3** Two solid cylinders, *A* and *B*, have the same mass m and the radii $2r$ and r , respectively. Each is accelerated from rest with a force applied as shown. In order to impart identical angular accelerations to both cylinders, what is the relationship between F_1 and F_2 ?
- $F_1 = 0.5F_2$
 - $F_1 = F_2$
 - $F_1 = 2F_2$
 - $F_1 = 4F_2$
 - $F_1 = 8F_2$

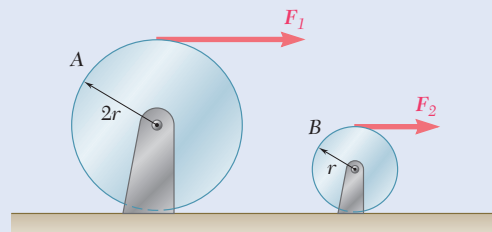


Fig. P16.CQ3

FREE-BODY PRACTICE PROBLEMS

16.F1 A 6-ft board is placed in a truck with one end resting against a block secured to the floor and the other leaning against a vertical partition. Draw the FBD and KD necessary to determine the maximum allowable acceleration of the truck if the board is to remain in the position shown.

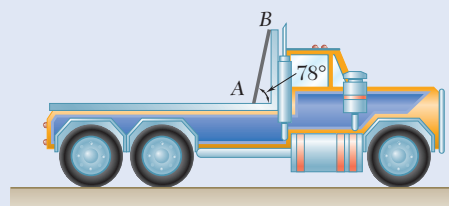


Fig. P16.F1

16.F2 A uniform circular plate of mass 3 kg is attached to two links AC and BD of the same length. Knowing that the plate is released from rest in the position shown, in which lines joining G to A and B are, respectively, horizontal and vertical, draw the FBD and KD for the plate.

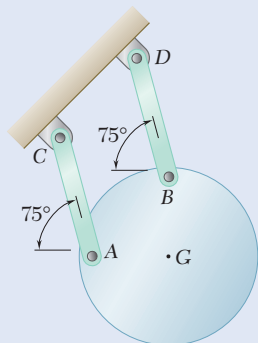


Fig. P16.F2

16.F3 Two uniform disks and two cylinders are assembled as indicated. Disk A weighs 20 lb and disk B weighs 12 lb. Knowing that the system is released from rest, draw the FBD and KD for the whole system.

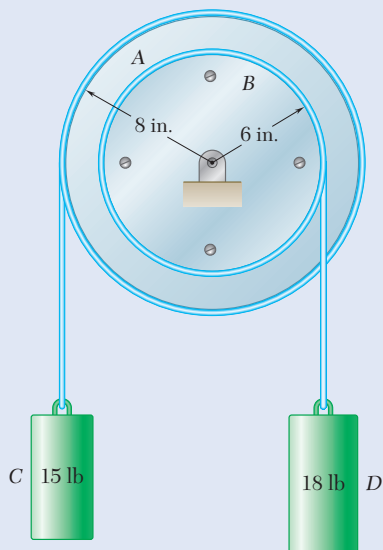


Fig. P16.F3

16.F4 The 400-lb crate shown is lowered by means of two overhead cranes. Knowing the tension in each cable, draw the FBD and KD that can be used to determine the angular acceleration of the crate and the acceleration of the center of gravity.

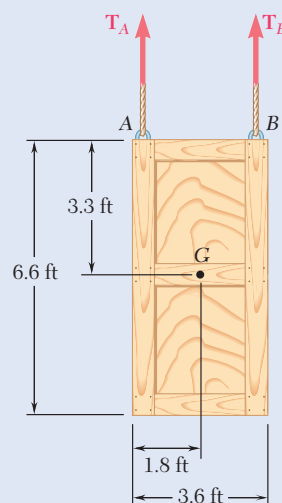


Fig. P16.F4

END-OF-SECTION PROBLEMS

- 16.1** A 60-lb uniform thin panel is placed in a truck with end A resting on a rough horizontal surface and end B supported by a smooth vertical surface. Knowing that the deceleration of the truck is 12 ft/s^2 , determine (a) the reactions at ends A and B , (b) the minimum required coefficient of static friction at end A .

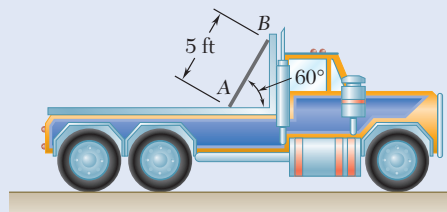


Fig. P16.1 and P16.2

- 16.2** A 60-lb uniform thin panel is placed in a truck with end A resting on a rough horizontal surface and end B supported by a smooth vertical surface. Knowing that the panel remains in the position shown, determine (a) the maximum allowable acceleration of the truck, (b) the corresponding minimum required coefficient of static friction at end A .
- 16.3** Knowing that the coefficient of static friction between the tires and the road is 0.80 for the automobile shown, determine the maximum possible acceleration on a level road, assuming (a) four-wheel drive, (b) rear-wheel drive, (c) front-wheel drive.

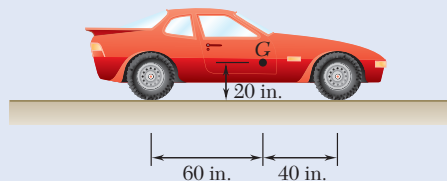


Fig. P16.3

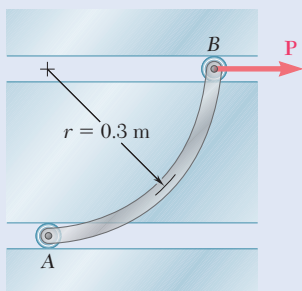


Fig. P16.4

- 16.4** The motion of the 2.5-kg rod AB is guided by two small wheels that roll freely in horizontal slots. If a force $\mathbf{P}$ of magnitude 8 N is applied at B , determine (a) the acceleration of the rod, (b) the reactions at A and B .
- 16.5** A uniform rod BC of mass 4 kg is connected to a collar A by a 250-mm cord AB . Neglecting the mass of the collar and cord, determine (a) the smallest constant acceleration $\mathbf{a}_A$ for which the cord and the rod will lie in a straight line, (b) the corresponding tension in the cord.

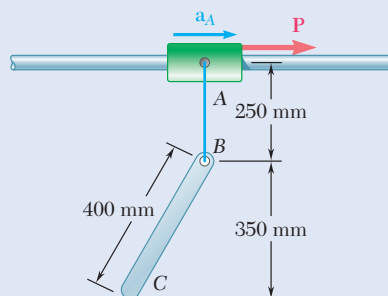


Fig. P16.5

- 16.6** A 2000-kg truck is being used to lift a 400-kg boulder B that is on a 50-kg pallet A . Knowing the acceleration of the rear-wheel-drive truck is 1 m/s^2 , determine (a) the reaction at each of the front wheels, (b) the force between the boulder and the pallet.

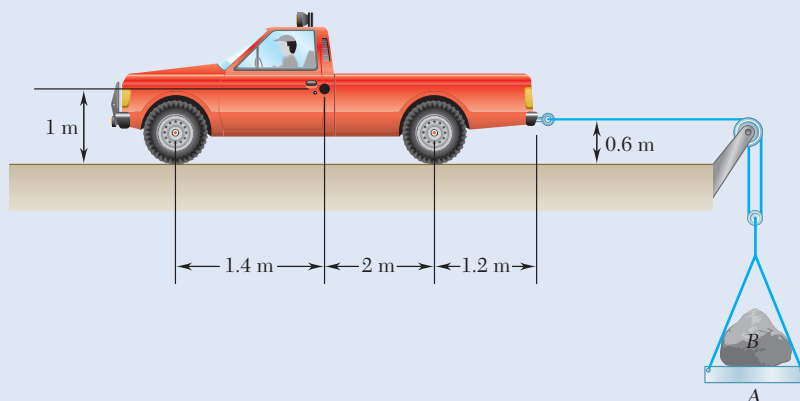


Fig. P16.6

- 16.7** The support bracket shown is used to transport a cylindrical can from one elevation to another. Knowing that $\mu_s = 0.25$ between the can and the bracket, determine (a) the magnitude of the upward acceleration $\mathbf{a}$ for which the can will slide on the bracket, (b) the smallest ratio h/d for which the can will tip before it slides.

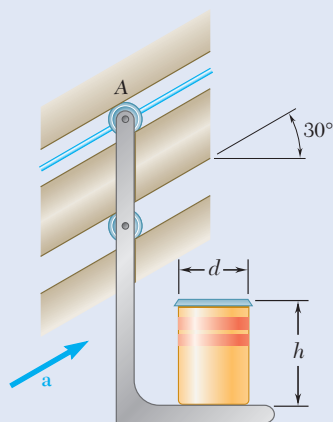


Fig. P16.7

- 16.8** Solve Prob. 16.7, assuming that the acceleration $\mathbf{a}$ of the bracket is directed downward.
- 16.9** A 20-kg cabinet is mounted on casters that allow it to move freely ($\mu = 0$) on the floor. If a 100-N force is applied as shown, determine (a) the acceleration of the cabinet, (b) the range of values of h for which the cabinet will not tip.
- 16.10** Solve Prob. 16.9, assuming that the casters are locked and slide on the rough floor ($\mu_k = 0.25$).

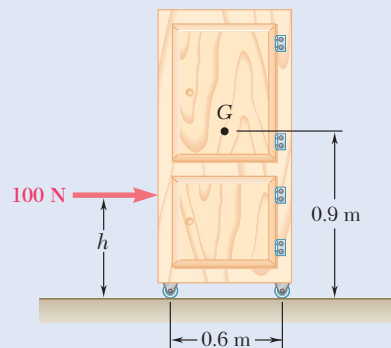


Fig. P16.9

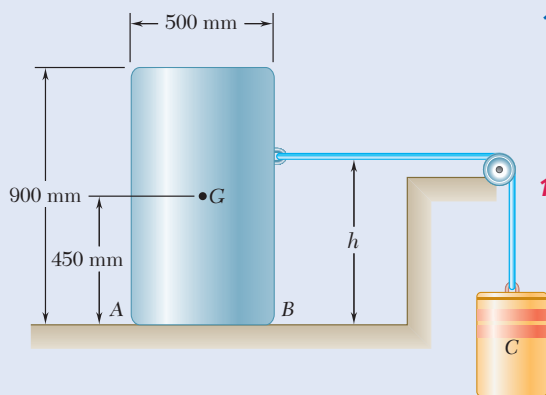


Fig. P16.11

16.11 A completely filled barrel and its contents have a combined mass of 90 kg. A cylinder C is connected to the barrel at a height $h = 550$ mm as shown. Knowing $\mu_s = 0.40$ and $\mu_k = 0.35$, determine the maximum mass of C so the barrel will not tip.

16.12 A 40-kg vase has a 200-mm-diameter base and is being moved using a 100-kg utility cart as shown. The cart moves freely ($\mu = 0$) on the ground. Knowing the coefficient of static friction between the vase and the cart is $\mu_s = 0.4$, determine the maximum force F that can be applied if the vase is not to slide or tip.

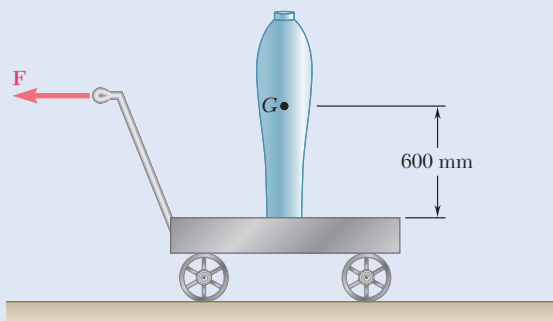


Fig. P16.12

16.13 The retractable shelf shown is supported by two identical linkage-and-spring systems; only one of the systems is shown. A 20-kg machine is placed on the shelf so that half of its weight is supported by the system shown. If the springs are removed and the system is released from rest, determine (a) the acceleration of the machine, (b) the tension in link AB . Neglect the weight of the shelf and links.

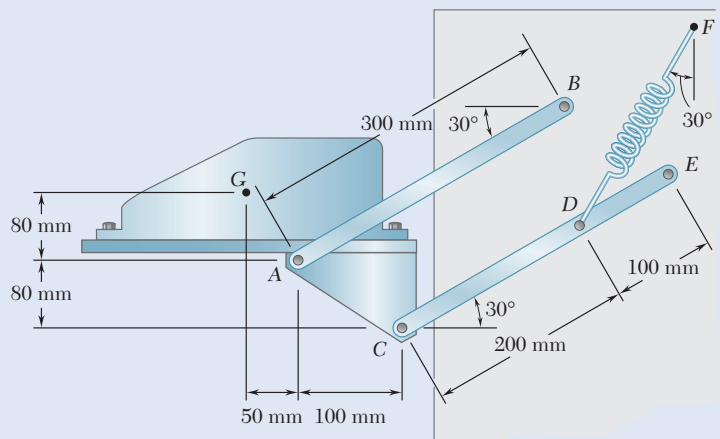


Fig. P16.13

- 16.14** Bars AB and BE , each with a mass of 4 kg, are welded together and are pin-connected to two links AC and BD . Knowing that the assembly is released from rest in the position shown and neglecting the masses of the links, determine (a) the acceleration of the assembly, (b) the forces in the links.

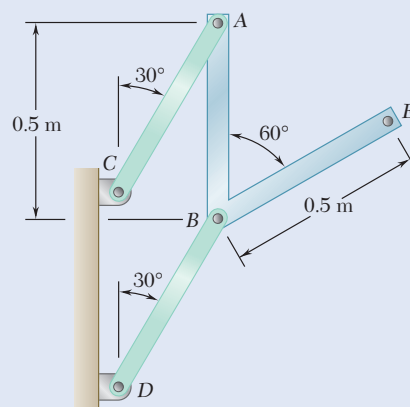


Fig. P16.14

- 16.15** At the instant shown, the tensions in the vertical ropes AB and DE are 300 N and 200 N, respectively. Knowing that the mass of the uniform bar BE is 5 kg, determine, at this instant, (a) the force $\mathbf{P}$, (b) the magnitude of the angular velocity of each rope, (c) the angular acceleration of each rope.

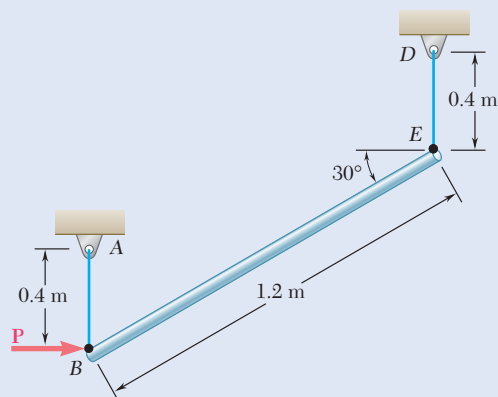


Fig. P16.15

- 16.16** Three bars, each of mass 3 kg, are welded together and pin-connected to two links BE and CF . Neglecting the weight of the links, determine the force in each link immediately after the system is released from rest.

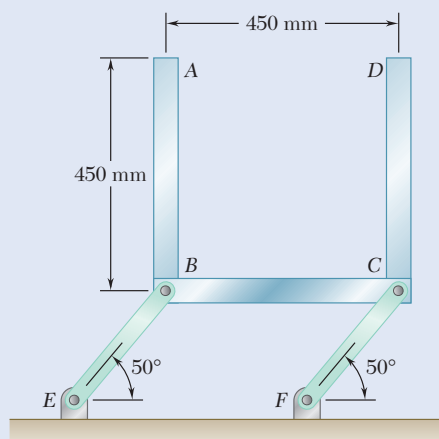


Fig. P16.16

- 16.17** Members ACE and DCB are each 600 mm long and are connected by a pin at C . The mass center of the 10-kg member AB is located at G . Determine (a) the acceleration of AB immediately after the system has been released from rest in the position shown, (b) the corresponding force exerted by roller A on member AB . Neglect the weight of members ACE and DCB .

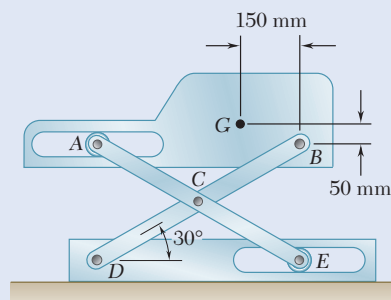


Fig. P16.17

- 16.18** A prototype rotating bicycle rack is designed to save space at a train station. The combined weight of platform BD and the bicycle is 40 lbs and is centered 1 ft above the midpoint of the platform. The motor at A causes the support beam AB to have an angular velocity of 10 rpm and zero angular acceleration at $\theta = 30^\circ$. At this instant, determine the vertical components of the forces exerted on platform BD by the pins at B and D .

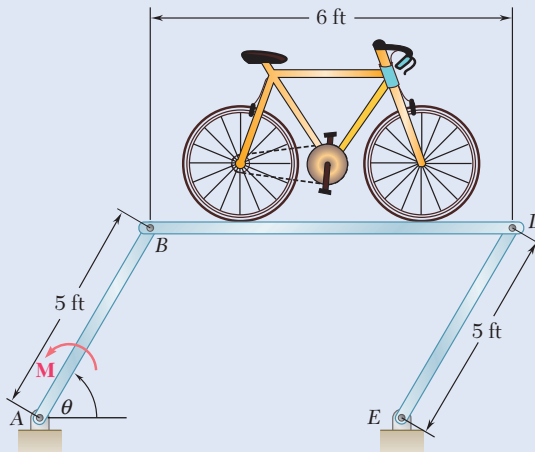


Fig. P16.18

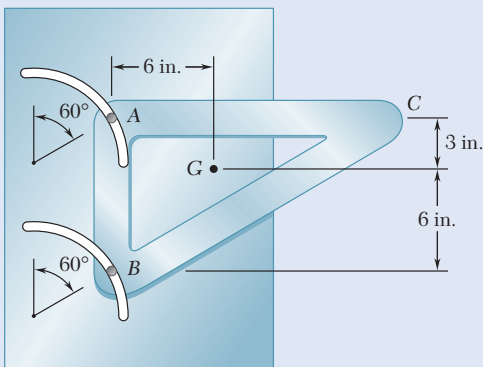


Fig. P16.19

- 16.19** The triangular weldment ABC is guided by two pins that slide freely in parallel curved slots of radius 6 in. cut in a vertical plate. The weldment weighs 16 lb and its mass center is located at point G . Knowing that at the instant shown the velocity of each pin is 30 in./s downward along the slots, determine (a) the acceleration of the weldment, (b) the reactions at A and B .

- 16.20** The coefficients of friction between the 30-lb block and the 5-lb platform BD are $\mu_s = 0.50$ and $\mu_k = 0.40$. Determine the accelerations of the block and of the platform immediately after wire AB has been cut.

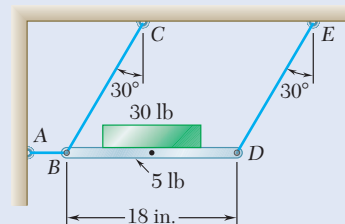


Fig. P16.20

- 16.21** Draw the shear and bending-moment diagrams for the vertical rod AB of Prob. 16.16.

- *16.22** Draw the shear and bending-moment diagrams for each of the bars AB and BE of Prob. 16.14.

16.23 For a rigid body in translation, show that the system of the inertial terms consists of vectors $(\Delta m_i)\bar{\mathbf{a}}$ attached to the various particles of the body, where $\bar{\mathbf{a}}$ is the acceleration of the mass center G of the body. Further show, by computing their sum and the sum of their moments about G , that the inertial terms reduce to a single vector $m\bar{\mathbf{a}}$ attached at G .

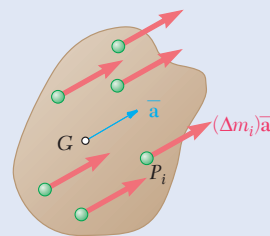


Fig. P16.23

16.24 For a rigid body in centroidal rotation, show that the system of the inertial terms consists of vectors $-(\Delta m_i)\omega^2 \mathbf{r}'_i$ and $(\Delta m_i)(\boldsymbol{\alpha} \times \mathbf{r}'_i)$ attached to the various particles P_i of the body, where $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ are the angular velocity and angular acceleration of the body, and where $\mathbf{r}'_i$ denotes the position vector of the particle P_i relative to the mass center G of the body. Further show, by computing their sum and the sum of their moments about G , that the inertial terms reduce to a couple $\bar{I}\boldsymbol{\alpha}$.

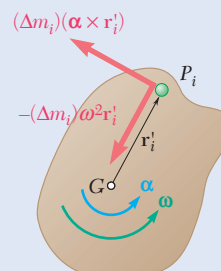


Fig. P16.24

16.25 It takes 10 min for a 2.4-Mg flywheel to coast to rest from an angular velocity of 300 rpm. Knowing that the radius of gyration of the flywheel is 1 m, determine the average magnitude of the couple due to kinetic friction in the bearing.

16.26 The rotor of an electric motor has an angular velocity of 3600 rpm when the load and power are cut off. The 120-lb rotor, which has a centroidal radius of gyration of 9 in., then coasts to rest. Knowing that kinetic friction results in a couple of magnitude 2.5 lb·ft exerted on the rotor, determine the number of revolutions that the rotor executes before coming to rest.

16.27 The 8-in.-radius brake drum is attached to a larger flywheel that is not shown. The total mass moment of inertia of the drum and the flywheel is 14 lb·ft·s² and the coefficient of kinetic friction between the drum and the brake shoe is 0.35. Knowing that the angular velocity of the flywheel is 360 rpm counterclockwise when a force $\mathbf{P}$ of magnitude 75 lb is applied to the pedal C , determine the number of revolutions executed by the flywheel before it comes to rest.

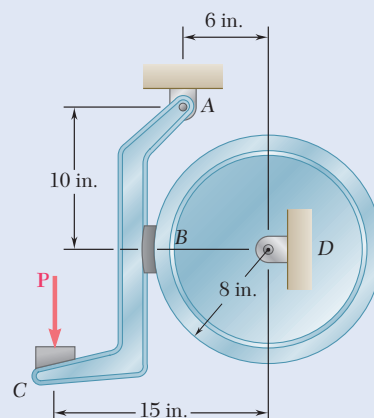


Fig. P16.27

16.28 Solve Prob. 16.27, assuming that the initial angular velocity of the flywheel is 360 rpm clockwise.

16.29 The 100-mm-radius brake drum is attached to a flywheel that is not shown. The drum and flywheel together have a mass of 300 kg and a radius of gyration of 600 mm. The coefficient of kinetic friction between the brake band and the drum is 0.30. Knowing that a force $\mathbf{P}$ of magnitude 50 N is applied at A when the angular velocity is 180 rpm counterclockwise, determine the time required to stop the flywheel when $a = 200$ mm and $b = 160$ mm.

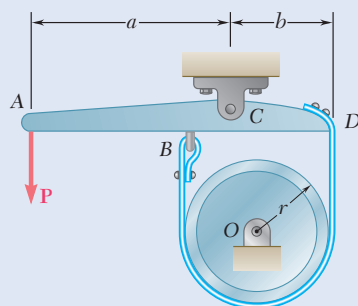


Fig. P16.29

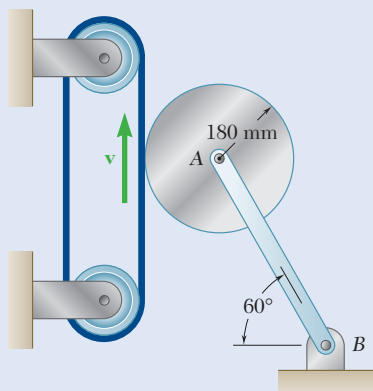


Fig. P16.30

16.30 The 180-mm-radius disk is at rest when it is placed in contact with a belt moving at a constant speed. Neglecting the weight of the link AB and knowing that the coefficient of kinetic friction between the disk and the belt is 0.40, determine the angular acceleration of the disk while slipping occurs.

16.31 Solve Prob. 16.30, assuming that the direction of motion of the belt is reversed.

16.32 In order to determine the mass moment of inertia of a flywheel of radius 600 mm, a 12-kg block is attached to a wire that is wrapped around the flywheel. The block is released and is observed to fall 3 m in 4.6 s. To eliminate bearing friction from the computation, a second block of mass 24 kg is used and is observed to fall 3 m in 3.1 s. Assuming that the moment of the couple due to friction remains constant, determine the mass moment of inertia of the flywheel.

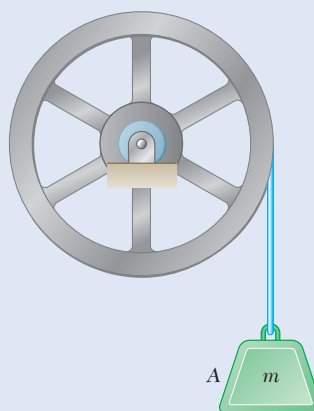


Fig. P16.32 and P16.33

16.33 The flywheel shown has a radius of 20 in., a weight of 250 lb, and a radius of gyration of 15 in. A 30-lb block A is attached to a wire that is wrapped around the flywheel, and the system is released from rest. Neglecting the effect of friction, determine (a) the acceleration of block A , (b) the speed of block A after it has moved 5 ft.

16.34 Each of the double pulleys shown has a mass moment of inertia of $15 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$ and is initially at rest. The outside radius is 18 in., and the inner radius is 9 in. Determine (a) the angular acceleration of each pulley, (b) the angular velocity of each pulley after point A on the cord has moved 10 ft.

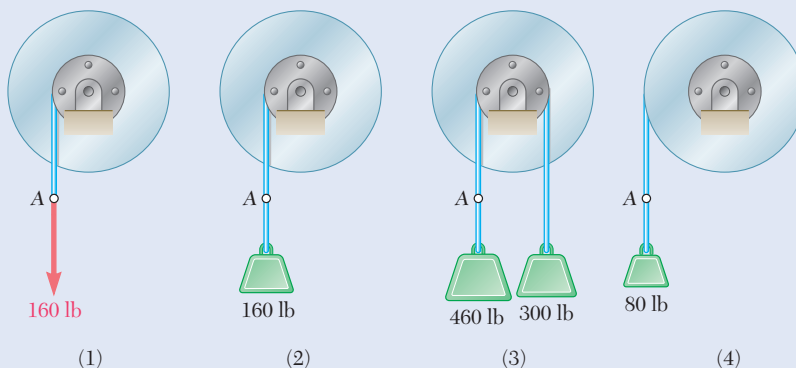


Fig. P16.34

- 16.35** Each of the gears *A* and *B* has a mass of 9 kg and has a radius of gyration of 200 mm; gear *C* has a mass of 3 kg and has a radius of gyration of 75 mm. If a couple **M** of constant magnitude 5 N·m is applied to gear *C*, determine (a) the angular acceleration of gear *A*, (b) the tangential force that gear *C* exerts on gear *A*.

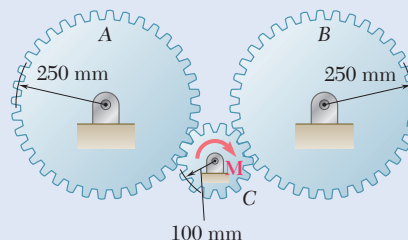


Fig. P16.35

- 16.36** Solve Prob. 16.35, assuming that the couple **M** is applied to disk *A*.

- 16.37** Gear *A* weighs 1 lb and has a radius of gyration of 1.3 in.; gear *B* weighs 6 lb and has a radius of gyration of 3 in.; gear *C* weighs 9 lb and has a radius of gyration of 4.3 in. Knowing a couple **M** of constant magnitude of 40 lb·in. is applied to gear *A*, determine (a) the angular acceleration of gear *C*, (b) the tangential force that gear *B* exerts on gear *C*.

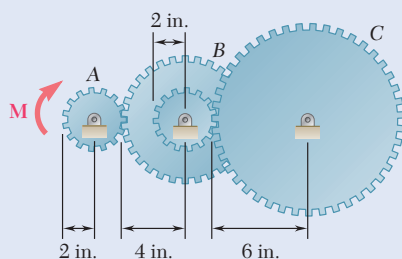


Fig. P16.37

- 16.38** The 25-lb double pulley shown is at rest and in equilibrium when a constant 3.5 lb·ft couple **M** is applied. Neglecting the effect of friction and knowing that the radius of gyration of the double pulley is 6 in., determine (a) the angular acceleration of the double pulley, (b) the tension in each rope.

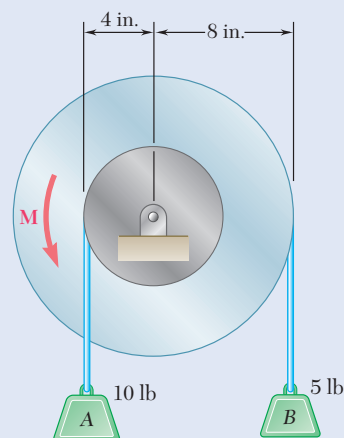


Fig. P16.38

- 16.39** A belt of negligible mass passes between cylinders *A* and *B* and is pulled to the right with a force **P**. Cylinders *A* and *B* weigh, respectively, 5 and 20 lb. The shaft of cylinder *A* is free to slide in a vertical slot and the coefficients of friction between the belt and each of the cylinders are $\mu_s = 0.50$ and $\mu_k = 0.40$. For $P = 3.6$ lb, determine (a) whether slipping occurs between the belt and either cylinder, (b) the angular acceleration of each cylinder.

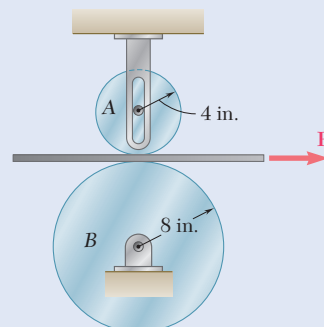


Fig. P16.39

- 16.40** Solve Prob. 16.39 for $P = 2.00$ lb.

- 16.41** Disk *A* has a mass of 6 kg and an initial angular velocity of 360 rpm clockwise; disk *B* has a mass of 3 kg and is initially at rest. The disks are brought together by applying a horizontal force of magnitude 20 N to the axle of disk *A*. Knowing that $\mu_k = 0.15$ between the disks and neglecting bearing friction, determine (a) the angular acceleration of each disk, (b) the final angular velocity of each disk.

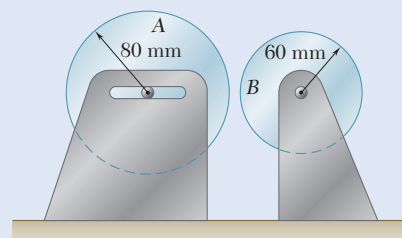


Fig. P16.41

- 16.42** Solve Prob. 16.41, assuming that initially disk *A* is at rest and disk *B* has an angular velocity of 360 rpm clockwise.

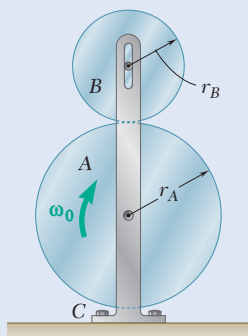


Fig. P16.43 and P16.44

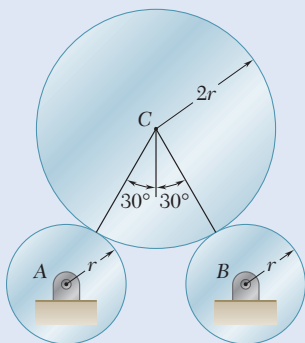


Fig. P16.45

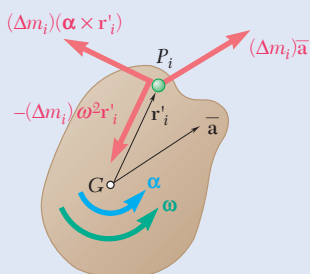


Fig. P16.47

16.43 Disk A has a mass $m_A = 4$ kg, a radius $r_A = 300$ mm, and an initial angular velocity $\omega_0 = 300$ rpm clockwise. Disk B has a mass $m_B = 1.6$ kg, a radius $r_B = 180$ mm, and is at rest when it is brought into contact with disk A. Knowing that $\mu_k = 0.35$ between the disks and neglecting bearing friction, determine (a) the angular acceleration of each disk, (b) the reaction at the support C.

16.44 Disk B is at rest when it is brought into contact with disk A, which has an initial angular velocity ω_0 . (a) Show that the final angular velocities of the disks are independent of the coefficient of friction μ_k between the disks as long as $\mu_k \neq 0$. (b) Express the final angular velocity of disk A in terms of ω_0 and the ratio of the masses of the two disks m_A/m_B .

16.45 Cylinder A has an initial angular velocity of 720 rpm clockwise, and cylinders B and C are initially at rest. Disks A and B each weigh 5 lb and have radius $r = 4$ in. Disk C weighs 20 lb and has a radius of 8 in. The disks are brought together when C is placed gently onto A and B. Knowing that $\mu_k = 0.25$ between A and C and no slipping occurs between B and C, determine (a) the angular acceleration of each disk, (b) the final angular velocity of each disk.

16.46 Show that the system of the inertial terms for a rigid body in plane motion reduces to a single vector, and express the distance from the mass center G of the body to the line of action of this vector in terms of the centroidal radius of gyration $\bar{k}$ of the body, the magnitude $\bar{a}$ of the acceleration of G , and the angular acceleration α .

16.47 For a rigid body in plane motion, show that the system of the inertial terms consists of vectors $(\Delta m_i)\bar{a}$, $-(\Delta m_i)\omega^2 \mathbf{r}'_i$, and $(\Delta m_i)(\alpha \times \mathbf{r}'_i)$ attached to the various particles P_i of the body, where $\bar{a}$ is the acceleration of the mass center G of the body, ω is the angular velocity of the body, α is its angular acceleration, and $\mathbf{r}'_i$ denotes the position vector of the particle P_i , relative to G . Further show, by computing their sum and the sum of their moments about G , that the inertial terms reduce to a vector $m\bar{a}$ attached at G and a couple $\bar{I}\alpha$.

16.48 A uniform slender rod AB rests on a frictionless horizontal surface, and a force $\mathbf{P}$ of magnitude 0.25 lb is applied at A in a direction perpendicular to the rod. Knowing that the rod weighs 1.75 lb, determine (a) the acceleration of point A, (b) the acceleration of point B, (c) the location of the point on the bar that has zero acceleration.

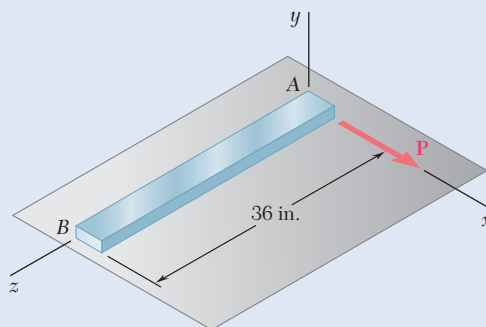


Fig. P16.48

- 16.49** (a) In Prob. 16.48, determine the point of the rod AB at which the force $\mathbf{P}$ should be applied if the acceleration of point B is to be zero. (b) Knowing that $P = 0.25$ lb, determine the corresponding acceleration of point A .

- 16.50 and 16.51** A force $\mathbf{P}$ with a magnitude of 3 N is applied to a tape wrapped around the body indicated. Knowing that the body rests on a frictionless horizontal surface, determine the acceleration of (a) point A , (b) point B .

16.50 A thin hoop of mass 2.4 kg.

16.51 A uniform disk of mass 2.4 kg.

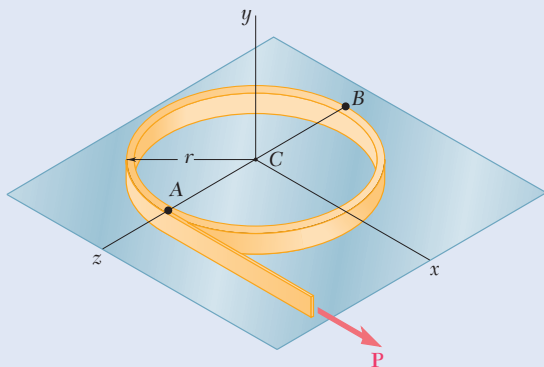


Fig. P16.50

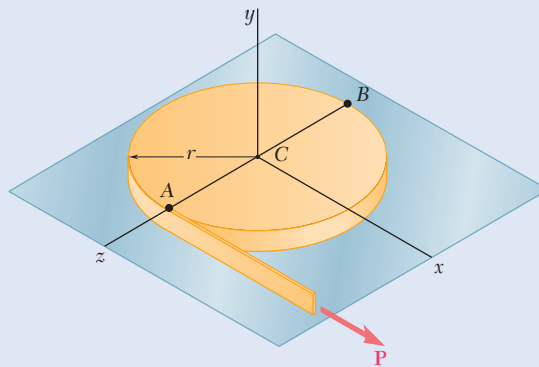


Fig. P16.51

- 16.52** A 250-lb satellite has a radius of gyration of 24 in. with respect to the y axis and is symmetrical with respect to the zx plane. Its orientation is changed by firing four small rockets A , B , C , and D , each of which produces a 4-lb thrust $\mathbf{T}$ directed as shown. Determine the angular acceleration of the satellite and the acceleration of its mass center G (a) when all four rockets are fired, (b) when all rockets except D are fired.

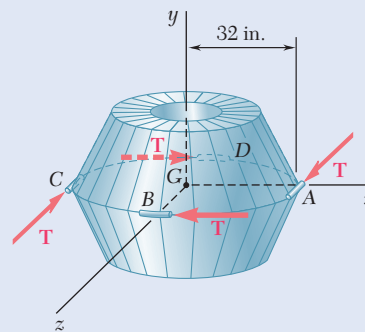


Fig. P16.52

- 16.53** A rectangular plate of mass 5 kg is suspended from four vertical wires, and a force $\mathbf{P}$ of magnitude 6 N is applied to corner C as shown. Immediately after $\mathbf{P}$ is applied, determine the acceleration of (a) the midpoint of edge BC , (b) corner B .

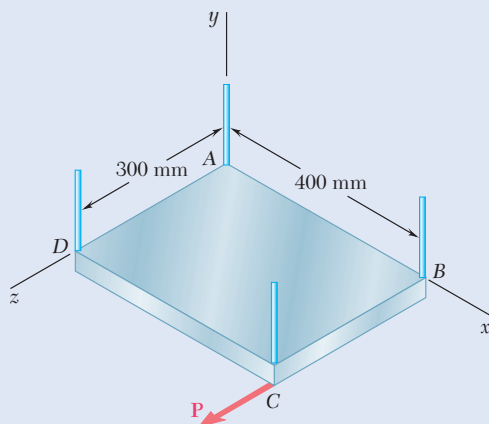


Fig. P16.53

- 16.54** A uniform semicircular plate with a mass of 6 kg is suspended from three vertical wires at points A , B , and C , and a force $\mathbf{P}$ with a magnitude of 5 N is applied to point B . Immediately after $\mathbf{P}$ is applied, determine the acceleration of (a) the mass center of the plate, (b) point C .

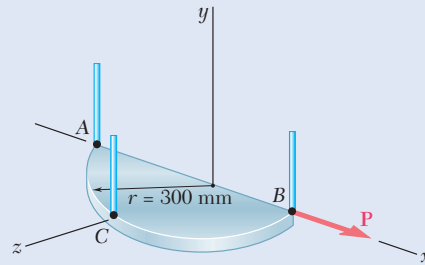


Fig. P16.54

- 16.55** A drum with a 200-mm radius is attached to a disk with a radius of $r_A = 150$ mm. The disk and drum have a combined mass of 5 kg and a combined radius of gyration of 120 mm and are suspended by two cords. Knowing that $T_A = 35$ N and $T_B = 25$ N, determine the accelerations of points A and B on the cords.

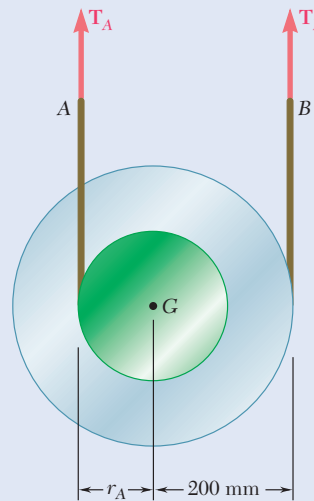


Fig. P16.55 and P16.56

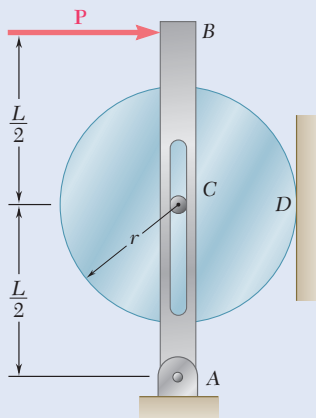


Fig. P16.57

- 16.56** A drum with a 200-mm radius is attached to a disk with a radius of $r_A = 140$ mm. The disk and drum have a combined mass of 5 kg and are suspended by two cords. Knowing that the acceleration of point B on the cord is zero, $T_A = 40$ N, and $T_B = 20$ N, determine the combined radius of gyration of the disk and drum.
- 16.57** The 12-lb uniform disk shown has a radius of $r = 3.2$ in. and rotates counterclockwise. Its center C is constrained to move in a slot cut in the vertical member AB , and a 11-lb horizontal force $\mathbf{P}$ is applied at B to maintain contact at D between the disk and the vertical wall. The disk moves downward under the influence of gravity and the friction at D . Knowing that the coefficient of kinetic friction between the disk and the wall is 0.12 and neglecting friction in the vertical slot, determine (a) the angular acceleration of the disk, (b) the acceleration of the center C of the disk.

16.58 The steel roll shown has a mass of 1200 kg, a centroidal radius of gyration of 150 mm, and is lifted by two cables looped around its shaft. Knowing that for each cable $T_A = 3100$ N and $T_B = 3300$ N, determine (a) the angular acceleration of the roll, (b) the acceleration of its mass center.

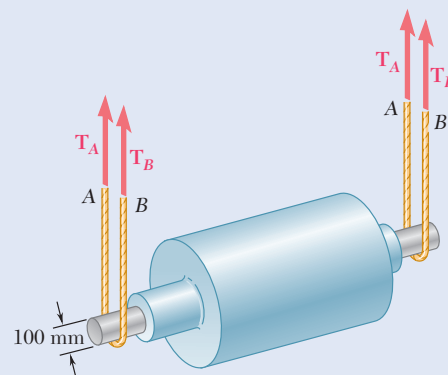


Fig. P16.58 and P16.59

16.59 The steel roll shown has a mass of 1200 kg, has a centroidal radius of gyration of 150 mm, and is lifted by two cables looped around its shaft. Knowing that at the instant shown the acceleration of the roll is 150 mm/s^2 downward and that for each cable $T_A = 3000$ N, determine (a) the corresponding tension T_B , (b) the angular acceleration of the roll.

16.60 and 16.61 A 15-ft beam weighing 500 lb is lowered by means of two cables unwinding from overhead cranes. As the beam approaches the ground, the crane operators apply brakes to slow the unwinding motion. Knowing that the deceleration of cable A is 20 ft/s^2 and the deceleration of cable B is 2 ft/s^2 , determine the tension in each cable.

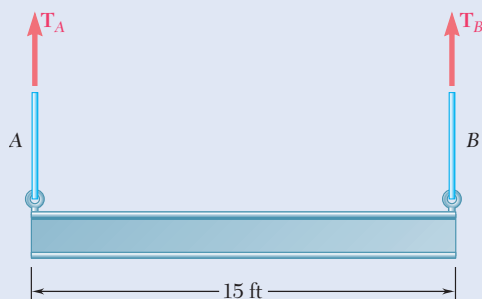


Fig. P16.60

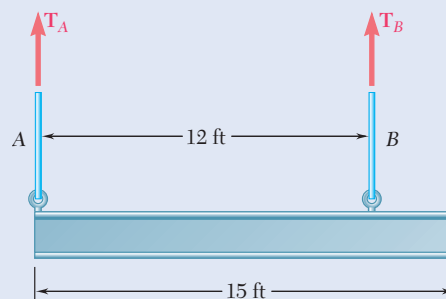


Fig. P16.61

16.62 Two uniform cylinders, each of weight $W = 14 \text{ lb}$ and radius $r = 5 \text{ in.}$, are connected by a belt as shown. If the system is released from rest, determine (a) the angular acceleration of each cylinder, (b) the tension in the portion of belt connecting the two cylinders, (c) the velocity of the center of the cylinder A after it has moved through 3 ft.

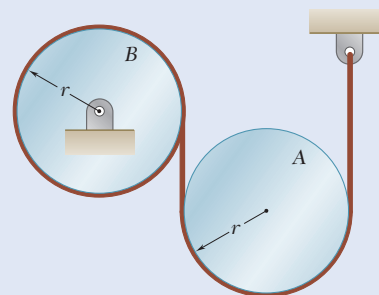


Fig. P16.62

16.63 and 16.64 A beam AB with a mass m and of uniform cross section is suspended from two springs as shown. If spring 2 breaks, determine at that instant (a) the angular acceleration of the beam, (b) the acceleration of point A, (c) the acceleration of point B.

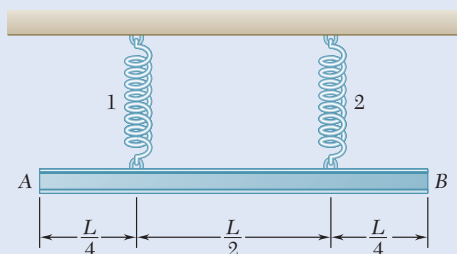


Fig. P16.63

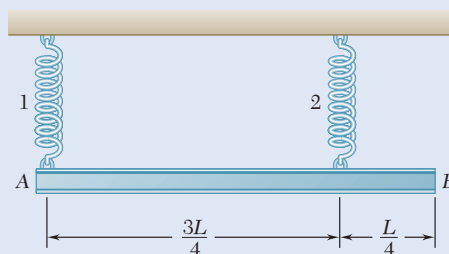


Fig. P16.64

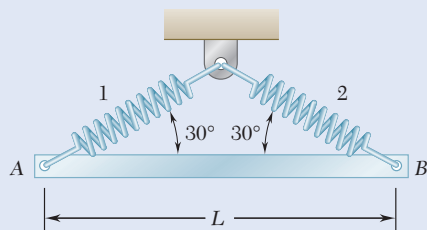


Fig. P16.65

16.65 A uniform slender bar AB with a mass m is suspended from two springs as shown. If spring 2 breaks, determine at that instant (a) the angular acceleration of the bar, (b) the acceleration of point A , (c) the acceleration of point B .

16.66 through 16.68 A thin plate of the shape indicated and of mass m is suspended from two springs as shown. If spring 2 breaks, determine the acceleration at that instant (a) of point A , (b) of point B .

16.66 A square plate of side b

16.67 A thin hoop of diameter b

16.68 A rectangular plate of height b and width a

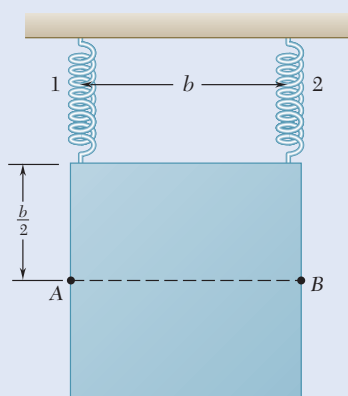


Fig. P16.66

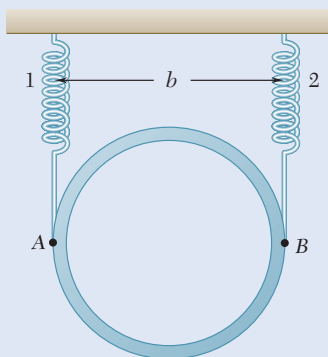


Fig. P16.67

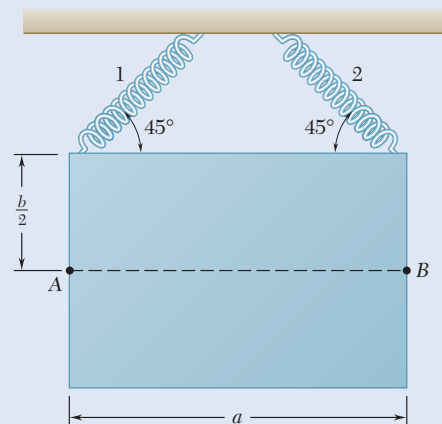


Fig. P16.68

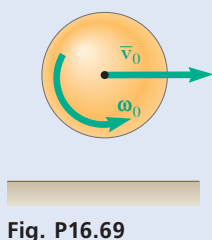


Fig. P16.69

16.69 A sphere of radius r and mass m is projected along a rough horizontal surface with the initial velocities indicated. If the final velocity of the sphere is to be zero, express, in terms of v_0 , r , and μ_k , (a) the required magnitude of ω_0 , (b) the time t_1 required for the sphere to come to rest, (c) the distance the sphere will move before coming to rest.

16.70 Solve Prob. 16.69, assuming that the sphere is replaced by a uniform thin hoop of radius r and mass m .

16.71 A bowler projects an 8-in.-diameter ball weighing 12 lb along an alley with a forward velocity $\mathbf{v}_0$ of 15 ft/s and a backspin ω_0 of 9 rad/s. Knowing that the coefficient of kinetic friction between the ball and the alley is 0.10, determine (a) the time t_1 at which the ball will start rolling without sliding, (b) the speed of the ball at time t_1 , (c) the distance the ball will have traveled at time t_1 .

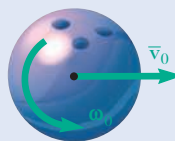


Fig. P16.71

16.72 Solve Prob. 16.71, assuming that the bowler projects the ball with the same forward velocity but with a backspin of 18 rad/s.

- 16.73** A uniform sphere of radius r and mass m is placed with no initial velocity on a belt that moves to the right with a constant velocity v_1 . Denoting by μ_k the coefficient of kinetic friction between the sphere and the belt, determine (a) the time t_1 at which the sphere will start rolling without sliding, (b) the linear and angular velocities of the sphere at time t_1 .

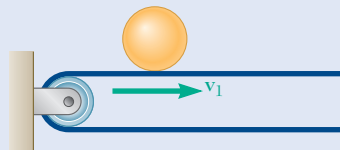


Fig. P16.73

- 16.74** A sphere of radius r and mass m has a linear velocity v_0 directed to the left and no angular velocity as it is placed on a belt moving to the right with a constant velocity v_1 . If after first sliding on the belt the sphere is to have no linear velocity relative to the ground as it starts rolling on the belt without sliding, determine in terms of v_1 and the coefficient of kinetic friction μ_k between the sphere and the belt (a) the required value of v_0 , (b) the time t_1 at which the sphere will start rolling on the belt, (c) the distance the sphere will have moved relative to the ground at time t_1 .

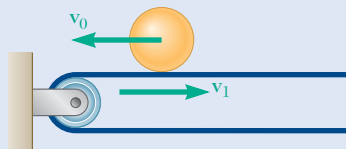


Fig. P16.74

16.2 CONSTRAINED PLANE MOTION

Most engineering applications deal with rigid bodies that are moving under given constraints. For example, cranks must rotate about a fixed axis, wheels must roll without sliding, and connecting rods must describe certain prescribed motions. In all such cases, definite relations exist between the components of the acceleration $\bar{\mathbf{a}}$ of the mass center G of the body considered and its angular acceleration α . The corresponding motion is said to be a **constrained motion**.

As discussed in the previous section, we draw our free-body and kinetic diagrams (Fig. 16.13) and then write the equations of motion. The solution of a problem involving a constrained plane motion also calls for a *kinematic analysis* of the problem. Consider, for example, a slender rod AB with a length l and a mass m , where the extremities are connected to blocks of negligible mass that slide along horizontal and vertical frictionless tracks. The rod is pulled by a force $\mathbf{P}$ applied at A (Fig. 16.12). We know from Sec. 15.4A that we can determine the acceleration $\bar{\mathbf{a}}$ of the mass center G of the rod at any given instant from the position of the rod, its angular velocity, and its angular acceleration at that instant. Suppose, for example, that we know the values of θ , ω , and α at a given instant, and we wish to determine the corresponding value of the force $\mathbf{P}$ as well as the reactions at A and B . We should first *determine the components* $\bar{a}_x$ and $\bar{a}_y$ *of the acceleration of the mass center* G using the method in Sec. 15.4A. We next solve our equations of motion using the expressions obtained for $\bar{a}_x$ and $\bar{a}_y$. We can then find the unknown forces $\mathbf{P}$, $\mathbf{N}_A$, and $\mathbf{N}_B$ by solving the appropriate equations.

Suppose now that we know the applied force $\mathbf{P}$, the angle θ , and the angular velocity ω of the rod at a given instant and that we wish to find the angular acceleration α of the rod and the components $\bar{a}_x$ and $\bar{a}_y$ of the acceleration of its mass center at that instant, as well as the reactions at A and B . The preliminary kinematic study of the problem will aim *to express the components* $\bar{a}_x$ and $\bar{a}_y$ *of the acceleration of* G *in terms of the angular acceleration* α *of the rod*. This is done by first expressing the acceleration of a suitable reference point such as A in terms of the angular acceleration α . We can then determine the components $\bar{a}_x$ and $\bar{a}_y$ of the acceleration of G in terms of α and carry these expressions into Fig. 16.13. We can then derive three equations in terms of α , N_A , and N_B and solve for the three unknowns (see Sample Prob. 16.12).

When a mechanism consists of several moving parts, we can use the approach just described with each part of the mechanism. The procedure required to determine the various unknowns is then similar to the procedure followed in the case of the equilibrium of a system of connected rigid bodies (Sec. 6.3B).

Earlier, we analyzed two particular cases of constrained plane motion: translation of a rigid body, in which the angular acceleration of the body is constrained to be zero, and centroidal rotation, in which the acceleration $\bar{\mathbf{a}}$ of the mass center of the body is constrained to be zero. Two other particular cases of constrained plane motion are of special interest: *noncentroidal rotation* of a rigid body and *rolling motion* of a disk or wheel. We can analyze these two cases using one of the general

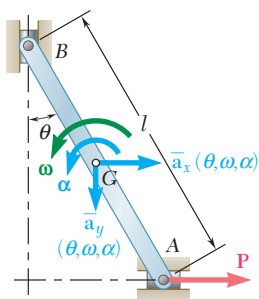


Fig. 16.12 Kinematic variables for a constrained rod pulled to the right.

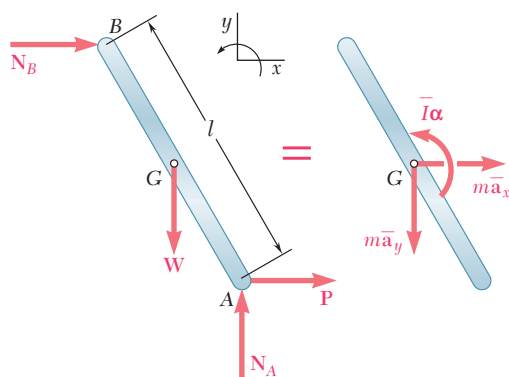


Fig. 16.13 Free-body diagram and kinetic diagram for the rod in Fig. 16.12.

methods described previously. However, in view of the range of their applications, they deserve a few special comments.

Noncentroidal Rotation. The motion of a rigid body constrained to rotate about a fixed axis that does not pass through its mass center is called **noncentroidal rotation**. The mass center G of the body moves along a circle with a radius $\bar{r}$ centered at point O , where the axis of rotation intersects the plane of reference (Fig. 16.14). Denoting the angular velocity and the angular acceleration of the line OG by ω and α , respectively, we obtain the following expressions for the tangential and normal components of the acceleration of G :

$$\bar{a}_t = \bar{r}\alpha \quad \bar{a}_n = \bar{r}\omega^2 \quad (16.7)$$

Since line OG belongs to the body, its angular velocity ω and its angular acceleration α also represent the angular velocity and the angular acceleration of the body. Equations (16.7) define, therefore, the kinematic relation between the motion of the mass center G and the motion of the body about G .

We obtain an interesting relation by equating the moments about the fixed point O of the forces and vectors shown, respectively, in Fig. 16.15a and b. We have

$$+\sum M_O = \bar{I}\alpha + (m\bar{r}\alpha)\bar{r} = (\bar{I} + m\bar{r}^2)\alpha$$

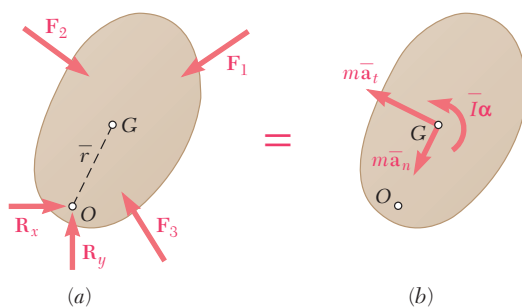


Fig. 16.15 Free-body diagram and kinetic diagram for the rigid body in Fig. 16.14.

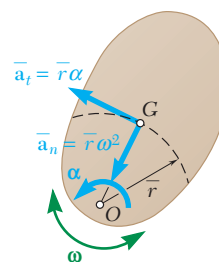


Fig. 16.14 For noncentroidal fixed-axis rotation, the center of mass has a tangential and a normal component of acceleration.

But according to the parallel-axis theorem, we have $\bar{I} + m\bar{r}^2 = I_O$, where I_O denotes the moment of inertia of the rigid body about the fixed axis. We therefore obtain

Moments about a fixed axis

$$\Sigma M_O = I_O \alpha \quad (16.8)$$

Although formula (16.8) expresses an important relation between the sum of the moments of the external forces about the fixed point O and the product $I_O \alpha$, we will still need to apply Eq. (16.1) to find the forces at O .

A particular case of noncentroidal rotation is of special interest—the case of *uniform rotation*, in which the angular velocity ω is constant. Since α is zero, the inertia couple in Fig. 16.15 vanishes, and the inertia vector reduces to its normal component. This component (also called *centrifugal force* in layman's terms) represents the tendency of the rigid body to break away from the axis of rotation.

Rolling Motion. Another important case of plane motion is the motion of a disk or wheel rolling on a plane surface. If the disk is constrained to roll without sliding, the acceleration $\bar{a}$ of its mass center G and its angular acceleration α are not independent. Assuming that the disk is balanced so that its mass center and its geometric center coincide, the distance $\bar{x}$ traveled by G during a rotation θ of the disk is $\bar{x} = r\theta$, where r is the radius of the disk. Differentiating this relation twice, we have

$$\bar{a} = r\alpha \quad (16.9)$$

Recall that the system of the inertial terms in plane motion reduces to a vector $m\bar{a}$ and a couple $\bar{I}\alpha$. We find that, in the particular case of the rolling motion of a balanced disk, these terms reduce to a vector of magnitude $m r \alpha$ attached at G and to a couple with a magnitude of $\bar{I}\alpha$. We may thus say that the external forces are equivalent to the vector and couple shown in Fig. 16.16.

When a disk **rolls without sliding**, there is no relative motion between the point of the disk in contact with the ground and the ground itself. Thus, as far as the computation of the friction force $\mathbf{F}$ is concerned, a rolling disk can be compared with a block at rest on a surface. The magnitude F of the friction force can have any value, as long as this value does not exceed the maximum value $F_m = \mu_s N$, where μ_s is the coefficient of static friction and N is the magnitude of the normal force. In the case of a rolling disk, the magnitude F of the friction force therefore should be determined independently of N by solving the equation obtained from Fig. 16.16.

When *sliding is impending*, the friction force reaches its maximum value $F_m = \mu_s N$ and can be obtained after solving for N .



Photo 16.4 As the ball hits the bowling alley, it first spins and slides, then rolls without sliding.

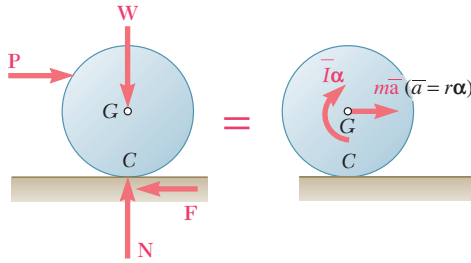


Fig. 16.16 Free-body diagram and kinetic diagram for a disk rolling without slipping on a fixed surface.

When the disk *rotates and slides* at the same time, a relative motion exists between the point of the disk in contact with the ground and the ground itself. The force of friction has the magnitude $F_k = \mu_k N$, where μ_k is the coefficient of kinetic friction. In this case, however, the motion of the mass center G of the disk and the rotation of the disk about G are independent, and $\bar{a}$ is not equal to $r\alpha$.

We can summarize these three different cases as

Rolling, no sliding	$F \leq \mu_s N$	$\bar{a} = r\alpha$
Rolling, sliding impending	$F = \mu_s N$	$\bar{a} = r\alpha$
Rotating and sliding	$F = \mu_k N$	$\bar{a}$ and α independent

If you do not know whether or not a disk slides, you should first assume that the disk rolls without sliding. You will then be able to solve your system of equations by assuming that $\bar{a} = r\alpha$. If F is found to be smaller than or equal to $\mu_s N$, the assumption is proved correct. If F is found to be larger than $\mu_s N$, the assumption is incorrect, and you should start the problem again, assuming rotating, sliding, and that $F = \mu_k N$.

When a disk is *unbalanced*, i.e., when its mass center G does not coincide with its geometric center O , the relation in Eq. (16.9) does not hold between $\bar{a}$ and α . However, a similar relation holds between the magnitude a_O of the acceleration of the geometric center and the angular acceleration α of an unbalanced disk that rolls without sliding. We have

$$a_O = r\alpha \quad (16.10)$$

To determine $\bar{a}$ in terms of the angular acceleration α and the angular velocity ω of the disk, we can use the relative-acceleration formula, as

$$\begin{aligned} \bar{\mathbf{a}} &= \bar{\mathbf{a}}_G = \mathbf{a}_O + \mathbf{a}_{G/O} \\ &= \mathbf{a}_O + (\mathbf{a}_{G/O})_t + (\mathbf{a}_{G/O})_n \end{aligned} \quad (16.11)$$

where the three component accelerations have the directions indicated in Fig. 16.17 and the magnitudes $a_O = r\alpha$, $(a_{G/O})_t = (OG)\alpha$, and $(a_{G/O})_n = (OG)\omega^2$. These terms also can be solved using the relationship between two points on a rigid body undergoing plane motion:

$$\bar{\mathbf{a}} = \mathbf{a}_O + \boldsymbol{\alpha} \times \mathbf{r}_{G/O} - \omega^2 \mathbf{r}_{G/O} \quad (16.12)$$

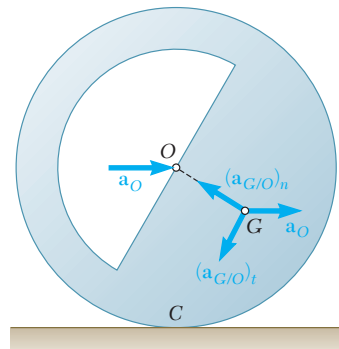


Fig. 16.17 Accelerations of the geometric center O and center of mass G for a rolling unbalanced disk.

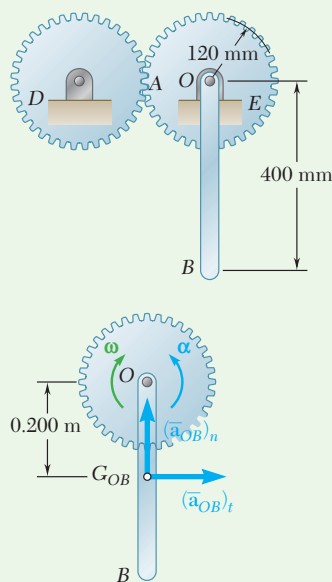


Fig. 1 Acceleration of the center of gravity of the bar.

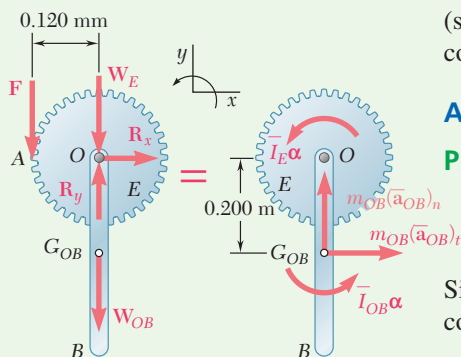


Fig. 2 Free-body diagram and kinetic diagram for the system.

Sample Problem 16.7

The portion AOB of a mechanism consists of a 400-mm steel rod OB welded to a gear E with a radius of 120 mm that can rotate about a horizontal shaft O . It is actuated by a gear D and, at the instant shown, has a clockwise angular velocity of 8 rad/s and a counterclockwise angular acceleration of 40 rad/s². Knowing that rod OB has a mass of 3 kg and gear E has a mass of 4 kg and a radius of gyration of 85 mm, determine (a) the tangential force exerted by gear D on gear E , (b) the components of the reaction at point O on the shaft.

STRATEGY: Since you are asked to determine forces, use Newton's second law.

MODELING: For your system, choose the single object that consists of the steel rod OB and the gear E . Since these two objects are welded together, they have the same angular velocity and angular acceleration. Rather than finding the center of mass for this object, use the center of mass for gear E and for rod OB separately in your kinetic diagram. Therefore, first determine the components of the acceleration of the mass center G_{OB} of the rod (Fig. 1) as

$$\begin{aligned}(\bar{a}_{OB})_t &= \bar{r}\alpha = (0.200 \text{ m})(40 \text{ rad/s}^2) = 8 \text{ m/s}^2 \\(\bar{a}_{OB})_n &= \bar{r}\omega^2 = (0.200 \text{ m})(8 \text{ rad/s})^2 = 12.8 \text{ m/s}^2\end{aligned}$$

A free-body diagram and kinetic diagram for the system are shown in Fig. 2. The inertial terms on your kinetic diagram include a couple $\bar{I}_E\alpha$ (since gear E is in centroidal rotation), a couple $\bar{I}_{OB}\alpha$, and two vector components $m_{OB}(\bar{a}_{OB})_n$ and $m_{OB}(\bar{a}_{OB})_t$ at the mass center of OB .

ANALYSIS:

Preliminary Calculations: The magnitudes of the weights are

$$\begin{aligned}W_E &= m_E g = (4 \text{ kg})(9.81 \text{ m/s}^2) = 39.2 \text{ N} \\W_{OB} &= m_{OB} g = (3 \text{ kg})(9.81 \text{ m/s}^2) = 29.4 \text{ N}\end{aligned}$$

Since you know the accelerations, you can compute the magnitudes of the components and couples on your kinetic diagram, as

$$\begin{aligned}\bar{I}_E\alpha &= m_E \bar{k}_E^2 \alpha = (4 \text{ kg})(0.085 \text{ m})^2 (40 \text{ rad/s}^2) = 1.156 \text{ N}\cdot\text{m} \\m_{OB}(\bar{a}_{OB})_t &= (3 \text{ kg})(8 \text{ m/s}^2) = 24.0 \text{ N} \\m_{OB}(\bar{a}_{OB})_n &= (3 \text{ kg})(12.8 \text{ m/s}^2) = 38.4 \text{ N} \\\bar{I}_{OB}\alpha &= (\tfrac{1}{12} m_{OB} L^2) \alpha = \tfrac{1}{12} (3 \text{ kg})(0.400 \text{ m})^2 (40 \text{ rad/s}^2) = 1.600 \text{ N}\cdot\text{m}\end{aligned}$$

Equations of Motion. Setting the system of the external forces shown in your free-body diagram equal to the inertia terms in your kinetic diagram, you obtain the following equations, which you can solve as

$$+\circlearrowleft \Sigma M_O = \dot{H}_O:$$

$$\begin{aligned}F(0.120 \text{ m}) &= \bar{I}_E\alpha + m_{OB}(\bar{a}_{OB})_t(0.200 \text{ m}) + \bar{I}_{OB}\alpha \\F(0.120 \text{ m}) &= 1.156 \text{ N}\cdot\text{m} + (24.0 \text{ N})(0.200 \text{ m}) + 1.600 \text{ N}\cdot\text{m}\end{aligned}$$

$$F = 63.0 \text{ N}$$

$$\mathbf{F} = 63.0 \text{ N} \downarrow \quad \blacktriangleleft$$

$$\begin{aligned}
 \rightarrow \Sigma F_x &= \Sigma m\bar{a}_x: & R_x &= m_{OB}(\bar{a}_{OB})_t \\
 & & R_x &= 24.0 \text{ N} & \mathbf{R}_x &= 24.0 \text{ N} \rightarrow \blacktriangleleft \\
 +\uparrow \Sigma F_y &= \Sigma m\bar{a}_y: & R_y - F - W_E - W_{OB} &= m_{OB}(\bar{a}_{OB})_n \\
 & & R_y - 63.0 \text{ N} - 39.2 \text{ N} - 29.4 \text{ N} &= 38.4 \text{ N} \\
 & & R_y &= 170.0 \text{ N} & \mathbf{R}_y &= 170.0 \text{ N} \uparrow \blacktriangleleft
 \end{aligned}$$

REFLECT and THINK: When you drew your kinetic diagram, you put your inertia terms at the center of mass for the gear and for the rod. Alternatively, you could have found the center of mass for the system and put the vectors $\bar{I}_{AOB}\alpha$, $m_{AOB}\bar{a}_x$ and $m_{AOB}\bar{a}_y$ on the diagram. Finally, you could have found an overall I_O for the combined gear and rod and used Eq. 16.8 to solve for force F .

Sample Problem 16.8

A 6×8 in. rectangular plate weighing 60 lb is suspended from two pins A and B . If pin B is suddenly removed, determine (a) the angular acceleration of the plate, (b) the components of the reaction at pin A immediately after pin B has been removed.

STRATEGY: You are asked to determine forces and the angular acceleration of the plate, so use Newton's second law.

MODELING: Choose the plate to be your system and model it as a rigid body. Observe that as the plate rotates about point A , its mass center G describes a circle with a radius $\bar{r}$ and its center at A (Fig. 1). The free-body diagram and kinetic diagram for this system are shown in Fig. 2. The plate is released from rest ($\omega = 0$), so the normal component of the acceleration of G is zero. The magnitude of the acceleration $\bar{a}$ of the mass center G is thus $\bar{a} = \bar{r}\alpha$.

ANALYSIS:

a. Angular Acceleration. Using your free-body diagram and kinetic diagram, you can take moments about A to find

$$+\downarrow \Sigma M_A = \bar{I}\alpha + m\bar{a}d_{\perp}: \quad W\bar{x} = \bar{I}\alpha + (m\bar{a})\bar{r}$$

Since $\bar{a} = \bar{r}\alpha$, you have

$$W\bar{x} = \bar{I}\alpha + (m\bar{r}\alpha)\bar{r} \quad \alpha = \frac{W\bar{x}}{\frac{W}{g}\bar{r}^2 + \bar{I}} \quad (1)$$

(continued)

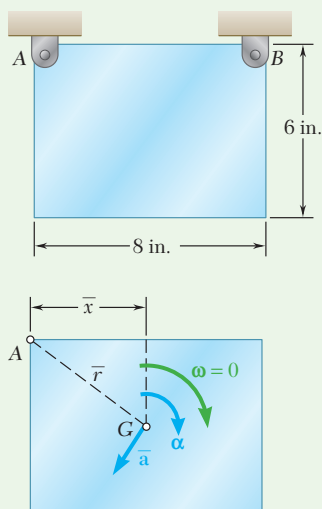


Fig. 1 The plate travels in a circle about A .

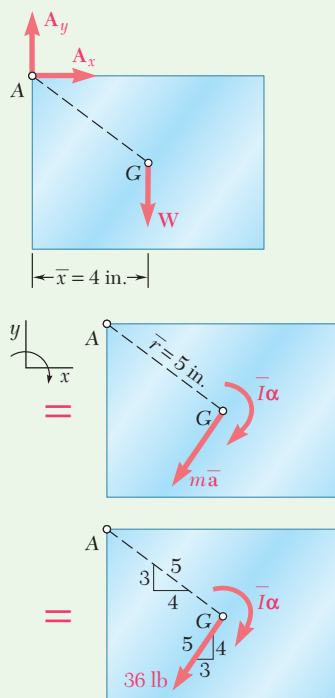


Fig. 2 Free-body diagram and kinetic diagram for the plate.

The centroidal moment of inertia of the plate is

$$\bar{I} = \frac{m}{12}(a^2 + b^2) = \frac{60 \text{ lb}}{12(32.2 \text{ ft/s}^2)}[(\frac{8}{12} \text{ ft})^2 + (\frac{6}{12} \text{ ft})^2] \\ = 0.1078 \text{ lb} \cdot \text{ft} \cdot \text{s}^2$$

Substituting this value of $\bar{I}$ together with $W = 60 \text{ lb}$, $\bar{r} = \frac{5}{12} \text{ ft}$, and $\bar{x} = \frac{4}{12} \text{ ft}$ into Eq. (1), you obtain

$$\alpha = +46.4 \text{ rad/s}^2 \quad \alpha = 46.4 \text{ rad/s}^2 \downarrow \quad \blacktriangleleft$$

b. Reaction at A. Using the computed value of α , determine the magnitude of the vector $m\bar{a}$ attached at G as

$$m\bar{a} = m\bar{r}\alpha = \frac{60 \text{ lb}}{32.2 \text{ ft/s}^2}(\frac{5}{12} \text{ ft})(46.4 \text{ rad/s}^2) = 36.0 \text{ lb}$$

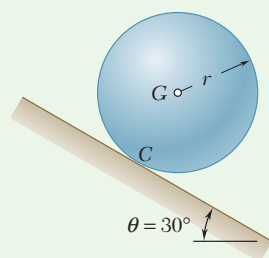
Applying Newton's second law in the x and y directions gives

$$+\rightarrow \Sigma F_x = m\bar{a}_x: \quad A_x = -\frac{3}{5}(36 \text{ lb}) \\ = -21.6 \text{ lb} \quad A_x = 21.6 \text{ lb} \leftarrow \quad \blacktriangleleft$$

$$+\uparrow \Sigma F_y = m\bar{a}_y: \quad A_y - 60 \text{ lb} = -\frac{4}{5}(36 \text{ lb}) \\ A_y = +31.2 \text{ lb} \quad A_y = 31.2 \text{ lb} \uparrow \quad \blacktriangleleft$$

REFLECT and THINK: If you had chosen to take moments about the center of gravity rather than point A , the two reaction forces A_x and A_y would have been in the resulting equation; that is, you would have had one equation and three unknowns, and you could not solve for α directly. Therefore, you would also need to use the equations from the x and y directions to solve for the three unknowns. Note that for convenience, we used a non-right handed coordinate system.

Sample Problem 16.9



A sphere with a radius r and a weight W is released with no initial velocity on an incline and rolls without slipping. Determine (a) the minimum value of the coefficient of static friction compatible with the rolling motion, (b) the velocity of the center G of the sphere after the sphere has rolled 10 ft, (c) the velocity of G if the sphere were to move 10 ft down a frictionless 30° incline.

STRATEGY: Use Newton's second law to determine the acceleration of the center of gravity. Then determine the velocity from kinematics.

MODELING: Choose the sphere to be your system and model it as a rigid body. Recall that for rolling motion, the instantaneous point of contact has a velocity of zero, which leads to $\bar{a} = r\alpha$ (Fig. 1). A free-body diagram and kinetic diagram for this system are shown in Fig. 2. The external forces $\mathbf{W}$, $\mathbf{N}$, and $\mathbf{F}$ form a system equivalent to the inertial terms represented by the vector $m\bar{a}$ and the couple $\bar{I}\alpha$.

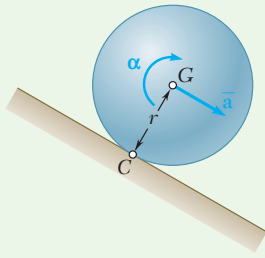


Fig. 1 The acceleration of G down the incline.

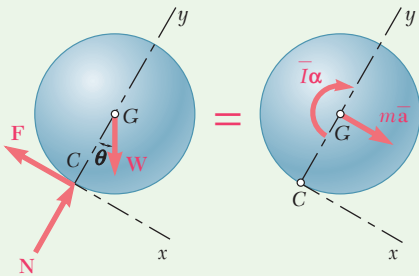


Fig. 2 Free-body diagram and kinetic diagram for the sphere.

ANALYSIS:

a. Minimum μ_s for Rolling Motion. Since the sphere rolls without sliding, you have $\bar{a} = r\alpha$ and can sum moments about C :

$$+\circlearrowleft \Sigma M_C = \bar{I}\alpha + m\bar{a}d_{\perp}: \quad (W \sin \theta)r = \bar{I}\alpha + (m\bar{a})r$$

$$(W \sin \theta)r = \bar{I}\alpha + (mr\alpha)r$$

Noting that $m = W/g$ and $\bar{I} = \frac{2}{5}mr^2$, you have

$$(W \sin \theta)r = \frac{2}{5} \frac{W}{g} r^2 \alpha + \left(\frac{W}{g} r \alpha \right) r \quad \alpha = + \frac{5g \sin \theta}{7r}$$

$$\bar{a} = r\alpha = \frac{5g \sin \theta}{7} = \frac{5(32.2 \text{ ft/s}^2) \sin 30^\circ}{7} = 11.50 \text{ ft/s}^2$$

Applying Newton's second law in the x and y directions gives

$$+\searrow \Sigma F_x = m\bar{a}_x: \quad W \sin \theta - F = m\bar{a}$$

$$W \sin \theta - F = \frac{W}{g} \frac{5g \sin \theta}{7}$$

$$F = +\frac{2}{7}W \sin \theta = \frac{2}{7}W \sin 30^\circ \quad \mathbf{F} = 0.143W \nearrow 30^\circ$$

$$+\nearrow \Sigma F_y = m\bar{a}_y: \quad N - W \cos \theta = 0$$

$$N = W \cos \theta = 0.866W \quad \mathbf{N} = 0.866W \nearrow 60^\circ$$

$$\mu_s = \frac{F}{N} = \frac{0.143W}{0.866W} \quad \mu_s = 0.165 \quad \blacktriangleleft$$

b. Velocity of Rolling Sphere. This is a case of uniformly accelerated motion, so

$$\bar{v}_0 = 0 \quad \bar{a} = 11.50 \text{ ft/s}^2 \quad \bar{x} = 10 \text{ ft} \quad \bar{x}_0 = 0$$

$$\bar{v}^2 = \bar{v}_0^2 + 2\bar{a}(\bar{x} - \bar{x}_0) \quad \bar{v}^2 = 0 + 2(11.50 \text{ ft/s}^2)(10 \text{ ft})$$

$$\bar{v} = 15.17 \text{ ft/s} \quad \mathbf{\bar{v} = 15.17 \text{ ft/s} \nearrow 30^\circ} \quad \blacktriangleleft$$

c. Velocity of Sliding Sphere. Now assuming no friction, you have $F = 0$ and obtain

$$+\circlearrowleft \Sigma M_G = \bar{I}\alpha: \quad 0 = \bar{I}\alpha \quad \alpha = 0$$

$$+\searrow \Sigma F_x = m\bar{a}_x: \quad W \sin 30^\circ = m\bar{a} \quad 0.50W = \frac{W}{g} \bar{a}$$

$$\bar{a} = +16.1 \text{ ft/s}^2 \quad \mathbf{\bar{a} = 16.1 \text{ ft/s}^2 \nearrow 30^\circ}$$

Substituting $\bar{a} = 16.1 \text{ ft/s}^2$ into the equations for uniformly accelerated motion, you obtain

$$\bar{v}^2 = \bar{v}_0^2 + 2\bar{a}(\bar{x} - \bar{x}_0) \quad \bar{v}^2 = 0 + 2(16.1 \text{ ft/s}^2)(10 \text{ ft})$$

$$\bar{v} = 17.94 \text{ ft/s} \quad \mathbf{\bar{v} = 17.94 \text{ ft/s} \nearrow 30^\circ} \quad \blacktriangleleft$$

REFLECT and THINK: Note that the sphere moving down a frictionless surface has a higher velocity than the rolling sphere, as you would expect. It is also interesting to note that the expression you obtained for the acceleration of the center of mass, that is, $\bar{a} = 5g \sin \theta / 7$, is independent of the radius of the sphere and the mass of the sphere. This means that any two solid spheres, as long as they are rolling without sliding, have the same linear acceleration.

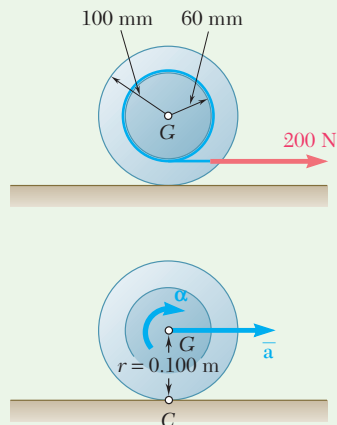


Fig. 1 Linear and angular acceleration of the wheel.

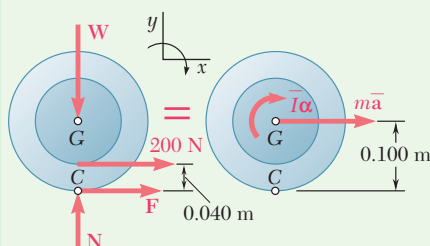


Fig. 2 Free-body diagram and kinetic diagram for the wheel assuming the friction force is to the right.

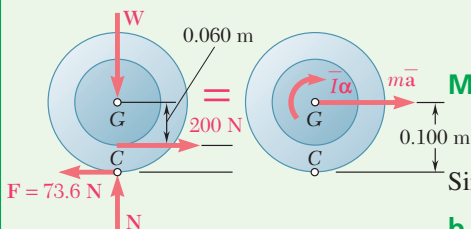


Fig. 3 Free-body diagram and kinetic diagram for the wheel when it is sliding and rotating.

Sample Problem 16.10

A cord is wrapped around the inner drum of a wheel and pulled horizontally with a force of 200 N. The wheel has a mass of 50 kg and a radius of gyration of 70 mm. Knowing that the coefficients of friction are $\mu_s = 0.20$ and $\mu_k = 0.15$, determine the acceleration of G and the angular acceleration of the wheel.

STRATEGY: Since you have forces acting on the wheel and are interested in accelerations, use Newton's second law. Assume the wheel rolls without sliding and compare the friction force needed with the maximum possible friction force. If the force needed exceeds the force available, redo the problem assuming rotation and sliding.

MODELING: Choose the wheel as your system and model it as a rigid body. The acceleration of G is to the right and the angular acceleration is clockwise (Fig. 1). The free-body and kinetic diagrams for this system are shown in (Fig. 2).

ANALYSIS:

a. Assume Rolling without Sliding. In this case, you have

$$\bar{a} = r\alpha = (0.100 \text{ m})\alpha$$

The moment of inertia of the wheel is

$$\bar{I} = m\bar{k}^2 = (50 \text{ kg})(0.070 \text{ m})^2 = 0.245 \text{ kg}\cdot\text{m}^2$$

Equations of Motion. Setting the system of external forces in your free-body diagram equal to the system of inertial terms in your kinetic diagram, you obtain

$$+\downarrow \Sigma M_C = \bar{I}\alpha + m\bar{a}d_{\perp}: \quad (200 \text{ N})(0.040 \text{ m}) = \bar{I}\alpha + (m\bar{a})(0.100 \text{ m})$$

$$8.00 \text{ N}\cdot\text{m} = (0.245 \text{ kg}\cdot\text{m}^2)\alpha + (50 \text{ kg})(0.100 \text{ m})\alpha(0.100 \text{ m})$$

$$\alpha = +10.74 \text{ rad/s}^2$$

$$\bar{a} = r\alpha = (0.100 \text{ m})(10.74 \text{ rad/s}^2) = 1.074 \text{ m/s}^2$$

$$+\rightarrow \Sigma F_x = m\bar{a}_x: \quad F + 200 \text{ N} = m\bar{a}$$

$$F + 200 \text{ N} = (50 \text{ kg})(1.074 \text{ m/s}^2)$$

$$F = -146.3 \text{ N}$$

$$\mathbf{F} = 146.3 \text{ N} \leftarrow$$

$$+\uparrow \Sigma F_y = m\bar{a}_y:$$

$$N - W = 0 \quad N - W = mg = (50 \text{ kg})(9.81 \text{ m/s}^2) = 490.5 \text{ N}$$

$$\mathbf{N} = 490.5 \text{ N} \uparrow$$

Maximum Possible Friction Force.

$$F_{\max} = \mu_s N = 0.20(490.5 \text{ N}) = 98.1 \text{ N}$$

Since $F > F_{\max}$, the assumed motion is impossible.

b. Rotating and Sliding. Since the wheel must rotate and slide at the same time, we draw new free-body and kinetic diagrams (Fig. 3), where $\bar{a}$ and α are independent and

$$F = F_k = \mu_k N = 0.15(490.5 \text{ N}) = 73.6 \text{ N}$$

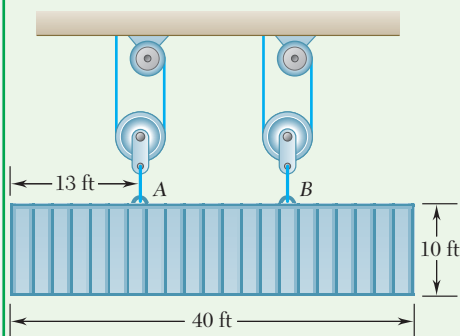
From the computation of part (a), you found that $\mathbf{F}$ is directed to the left. You can obtain and solve the following equations of motion as

$$\begin{aligned} \rightarrow \Sigma F_x = m\bar{a}_x: \quad 200 \text{ N} - 73.6 \text{ N} &= (50 \text{ kg})\bar{a} \\ \bar{a} &= +2.53 \text{ m/s}^2 \end{aligned} \quad \bar{\mathbf{a}} = 2.53 \text{ m/s}^2 \rightarrow \blacktriangleleft$$

$$\begin{aligned} +\curvearrowright \Sigma M_G = \bar{I}\alpha: \\ (73.6 \text{ N})(0.100 \text{ m}) - (200 \text{ N})(0.060 \text{ m}) &= (0.245 \text{ kg}\cdot\text{m}^2)\alpha \\ \alpha &= -18.94 \text{ rad/s}^2 \end{aligned} \quad \alpha = 18.94 \text{ rad/s}^2 \curvearrowright \blacktriangleleft$$

REFLECT and THINK: The wheel has larger linear and angular accelerations under conditions of rotating while sliding than when rolling without sliding.

Sample Problem 16.11



Overhead cranes are often used to move large containers in shipyards. A simplified model of a 60,000-lb container and crane is shown. The uniform container is at rest when the connection at B fails. Determine the tension in the cable connecting the pulley to the container at A .

STRATEGY: Since you are asked to find a tension, use Newton's second law.

MODELING: Start by choosing the container to be your system. After the connection at B fails, the only external forces acting on the container are the tension in the cable at A and the weight. A free-body diagram and kinetic diagram for this system immediately after the connection at B fails are shown in Fig. 1. Since the container is undergoing general plane motion, in the kinetic diagram you can represent the acceleration of the center of mass as having a vertical and a horizontal component.

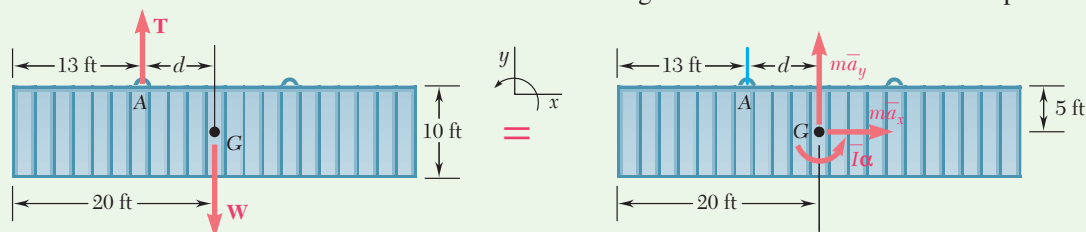


Fig. 1 Free-body diagram and kinetic diagram for the container.

ANALYSIS: Using Fig. 1 and applying Newton's second law in the x -direction and y -direction and summing moments about point G gives you

$$\rightarrow \Sigma F_x = m\bar{a}_x: \quad 0 = m\bar{a}_x \quad (1)$$

$$+\uparrow \Sigma F_y = m\bar{a}_y: \quad T - W = m\bar{a}_y \quad (2)$$

$$+\curvearrowright \Sigma M_G = \bar{I}\alpha: \quad -Td = \bar{I}\alpha \quad (3)$$

(continued)

where

$d = 7$ ft and

$$m = \frac{W}{g} = \frac{60,000 \text{ lb}}{32.2 \text{ ft/s}^2} = 1863 \text{ lb} \cdot \text{s}^2/\text{ft}$$

$$\bar{I} = \frac{1}{12}m(b^2 + c^2) = \frac{1}{12}(1863 \text{ lb} \cdot \text{s}^2/\text{ft})[(40 \text{ ft})^2 + (10 \text{ ft})^2] = 26,400 \text{ lb} \cdot \text{ft} \cdot \text{s}^2$$

In Eqs. (1) through (3), you have four unknowns: T , $\bar{a}_x$, $\bar{a}_y$, and α . You can use kinematics to obtain additional equations. You want to relate the acceleration of the center of mass to that of another point on the container. At the instant the cable breaks, the angular velocity of the cable is zero, so point A has no normal acceleration, but it has an acceleration perpendicular to the cable. The accelerations of A and G are related by

$$\mathbf{a}_G = \mathbf{a}_A + \mathbf{a}_{G/A} = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{G/A} - \omega^2 \mathbf{r}_{G/A}$$

Substituting in known values and letting $\omega = 0$ and $\boldsymbol{\alpha} = \alpha \mathbf{k}$ gives you

$$\bar{a}_x \mathbf{i} + \bar{a}_y \mathbf{j} = a_A \mathbf{i} + \alpha \mathbf{k} \times [d \mathbf{i} - 5 \mathbf{j}] - 0 = a_A \mathbf{i} + (d\alpha) \mathbf{j} + (5\alpha) \mathbf{i}$$

Equating components gives

$$\mathbf{i}: \bar{a}_x = a_A + 5\alpha \quad (4)$$

$$\mathbf{j}: \bar{a}_y = d\alpha \quad (5)$$

Solving Eqs. (1–5) for T , $\bar{a}_x$, $\bar{a}_y$, a_A , and α gives you $T = 44,580$ lb, $\bar{a}_y = -8.275 \text{ ft/s}^2$, $\bar{a}_x = 0$, $a_A = 5.911 \text{ ft/s}^2$, and $\alpha = -1.182 \text{ rad/s}^2$.

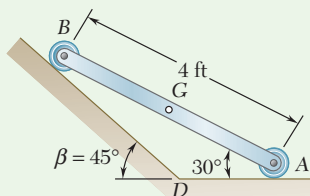
$$\mathbf{T} = 44,600 \text{ lb} \uparrow \quad \blacktriangleleft$$

REFLECT and THINK: You don't need all five equations to solve for the required unknowns; that is, you could have chosen to just use Eqs. (2), (3), and (5). The acceleration of the center of gravity is only in the vertical direction at the instant the cable breaks. When the container was at rest, the force in the cable at A was 30,000 lb. The tension increased when the connection at B failed. What would have happened if A had been at the upper left edge of the container? Your analysis would be identical except that d would be equal to 20 ft rather than 7 ft. Substituting this into your equations and solving gives you $T = 15,690$ lb, which is less than 30,000 lb.

Sample Problem 16.12

The ends of a 4-ft rod weighing 50 lb can move freely and with no friction along two straight tracks as shown. If the rod is released from rest at the position shown, determine (a) the angular acceleration of the rod, (b) the reactions at A and B .

STRATEGY: Since you are asked to determine forces and accelerations, use Newton's second law. The motion is constrained, so the acceleration of G must be related to the angular acceleration α . To obtain this relation, first determine the magnitude of the acceleration $\mathbf{a}_A$ of point A in terms of α .



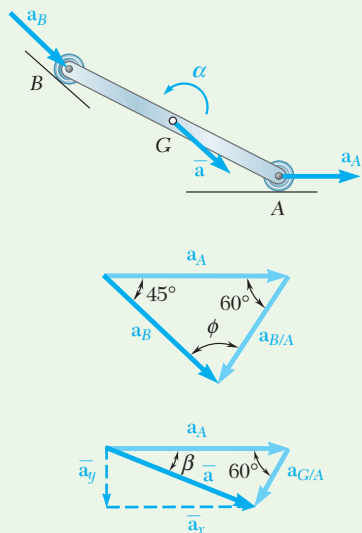


Fig. 1 Vector diagrams for accelerations of points on the rod.

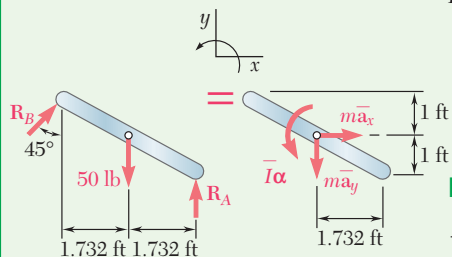


Fig. 2 Free-body diagram and kinetic diagram for the rod assuming a downward acceleration.

MODELING and ANALYSIS: Choose the rod to be your system and model it as a rigid body. Before drawing the kinetic diagram, you need to relate the acceleration of G to the angular acceleration of the rod. You can do this using kinematics.

Kinematics of Motion. Assume that α is directed counterclockwise. Noting that $a_{B/A} = 4\alpha$, you have (Fig. 1)

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

$$[a_B \searrow 45^\circ] = [a_A \rightarrow] + [4\alpha \nearrow 60^\circ]$$

Noting that $\phi = 75^\circ$ and using the law of sines, you obtain

$$a_A = 5.46\alpha \quad a_B = 4.90\alpha$$

Now you can find the acceleration of G from

$$\bar{\mathbf{a}} = \mathbf{a}_G = \mathbf{a}_A + \mathbf{a}_{G/A}$$

$$\bar{\mathbf{a}} = [5.46\alpha \rightarrow] + [2\alpha \nearrow 60^\circ]$$

Resolving $\bar{\mathbf{a}}$ into x and y components, you obtain

$$\bar{a}_x = 5.46\alpha - 2\alpha \cos 60^\circ = 4.46\alpha \quad \bar{a}_x = 4.46\alpha \rightarrow$$

$$\bar{a}_y = -2\alpha \sin 60^\circ = -1.732\alpha \quad \bar{a}_y = 1.732\alpha \downarrow$$

Kinetics of Motion. Draw a free-body diagram and kinetic diagram for your system (Fig. 2). Compute the following magnitudes.

$$\bar{I} = \frac{1}{12}ml^2 = \frac{1}{12} \frac{50 \text{ lb}}{32.2 \text{ ft/s}^2} (4 \text{ ft})^2 = 2.07 \text{ lb} \cdot \text{ft} \cdot \text{s}^2 \quad \bar{I}\alpha = 2.07\alpha$$

$$m\bar{a}_x = \frac{50}{32.2} (4.46\alpha) = 6.93\alpha \quad m\bar{a}_y = -\frac{50}{32.2} (1.732\alpha) = -2.69\alpha$$

Equations of Motion.

$$+\circlearrowleft \Sigma M_B = \bar{I}\alpha + m\bar{a}_\perp:$$

$$R_A(4 \cos 30^\circ \text{ ft}) - W(2 \cos 30^\circ \text{ ft}) = \bar{I}\alpha + (m\bar{a}_x)(2 \sin 30^\circ \text{ ft}) - (m\bar{a}_y)(2 \cos 30^\circ \text{ ft})$$

$$R_A(3.464) - (50 \text{ lb})(1.732) = 2.07\alpha + (6.93\alpha)(1.000) - (2.69\alpha)(1.732) \quad (1)$$

$$+\rightarrow \Sigma F_x = m\bar{a}_x: \quad R_B \sin 45^\circ = 6.93\alpha \quad (2)$$

$$+\uparrow \Sigma F_y = m\bar{a}_y: \quad R_A + R_B \cos 45^\circ - 50 = -2.69\alpha \quad (3)$$

Solving these equations gives

$$\alpha = 2.30 \text{ rad/s}^2 \nearrow$$

$$\mathbf{R}_B = 22.5 \text{ lb} \nearrow 45^\circ$$

$$\mathbf{R}_A = 27.9 \text{ lb} \uparrow$$

REFLECT and THINK: For the kinematics, you could have used the vector algebra approach rather than the method demonstrated in this example. Using the vector algebra approach, you can write

$$\mathbf{a}_B = \mathbf{a}_A + \alpha \mathbf{k} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A}$$

(continued)

Substituting the directions assumed in Fig. 1, you find

$$\begin{aligned}\frac{a_B}{\sqrt{2}}\mathbf{i} - \frac{a_B}{\sqrt{2}}\mathbf{j} &= a_A\mathbf{i} + \alpha\mathbf{k} \times (-3.464\mathbf{i} + 2\mathbf{j}) + 0 \\ &= a_A\mathbf{i} + (-3.464\alpha\mathbf{j} - 2\alpha\mathbf{i})\end{aligned}$$

Equating components gives

$$\begin{aligned}\mathbf{i}: \quad \frac{a_B}{\sqrt{2}} &= a_A - 2\alpha \\ \mathbf{j}: \quad \frac{a_B}{\sqrt{2}} &= -3.46\alpha\end{aligned}$$

Solving these, you find $a_B = 4.90\alpha$ and $a_A = 5.46\alpha$, which are similar to the approach shown previously. You can determine the acceleration of the center of gravity in terms of the angular acceleration using $\mathbf{a}_G = \mathbf{a}_A + \alpha\mathbf{k} \times \mathbf{r}_{G/A} - \omega^2\mathbf{r}_{G/A}$. Substituting the directions assumed in Fig. 1, you find

$$\bar{a}_x\mathbf{i} + \bar{a}_y\mathbf{j} = a_A\mathbf{i} + \alpha\mathbf{k} \times (-1.732\mathbf{i} + 1\mathbf{j}) + 0 = a_A\mathbf{i} + (-1.732\alpha\mathbf{j} - 1\alpha\mathbf{i})$$

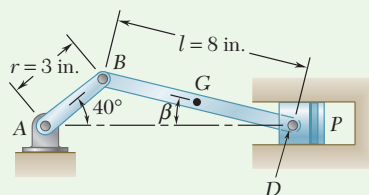
Equating components gives

$$\begin{aligned}\mathbf{i}: \quad \bar{a}_x &= a_A - 1\alpha = 4.46\alpha \\ \mathbf{j}: \quad \bar{a}_y &= -1.732\alpha\end{aligned}$$

These are identical to the answers determined previously.

Sample Problem 16.13

In the engine system from Sample Prob. 15.15, the crank AB has a constant clockwise angular velocity of 2000 rpm. Knowing that the connecting rod BD weighs 4 lb and the piston P weighs 5 lb, determine the forces on the connecting rod at B and D . Assume the center of mass of BD is at its geometric center and it can be treated as a uniform, slender rod.



STRATEGY: Since you are asked to find forces at the instant shown, use Newton's second law.

MODELING: Since you want to determine the forces at B and D , start by choosing the connecting rod BD as your system. The pin forces at B and D are represented by horizontal and vertical components, and since

the rod is undergoing general plane motion, you can represent the acceleration of the center of mass in the kinetic diagram as having a vertical and a horizontal component. The free-body and kinetic diagrams for this system are shown in Fig. 1, where $\ell = 8 \text{ in.} = 0.6667 \text{ ft}$ and $\beta = 13.95^\circ$.

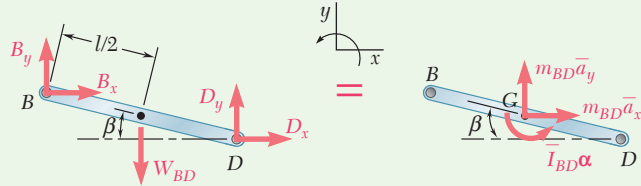


Fig. 1

ANALYSIS: Using Fig. 1, applying Newton's second law in the x -direction and y -direction, and summing moments about point G gives

$$\rightarrow \Sigma F_x = m\bar{a}_x: \quad B_x + D_x = m_{BD}\bar{a}_x \quad (1)$$

$$+\uparrow \Sigma F_y = m\bar{a}_y: \quad B_y + D_y - W_{BD} = m_{BD}\bar{a}_y \quad (2)$$

$$+\curvearrowright \Sigma M_G = \bar{I}\alpha: \quad -B_y(\ell/2)\cos\beta - B_x(\ell/2)\sin\beta + D_y(\ell/2)\cos\beta + D_x(\ell/2)\sin\beta = \bar{I}_{BD}\alpha_{BD} \quad (3)$$

where

$$m_{BD} = \frac{W_{BD}}{g} = \frac{4 \text{ lb}}{32.2 \text{ ft/s}^2} = 0.1242 \text{ lb}\cdot\text{s}^2/\text{ft}$$

$$\bar{I}_{BD} = \frac{1}{12}m_{BD}\ell^2 = \frac{1}{12}(0.1242 \text{ lb}\cdot\text{s}^2/\text{ft})(0.6667 \text{ ft})^2 = 0.004601 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$$

In Eqs. (1) through (3), you have seven unknowns: B_x , B_y , D_x , D_y , $\bar{a}_x$, $\bar{a}_y$, and α_{BD} . Therefore, you need more equations. You can get them from kinematics or by choosing another system. Choose the piston to be your system, model it as a particle, and draw its free-body and kinetic diagrams (Fig. 2).

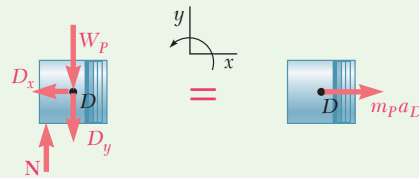


Fig. 2 Free-body diagram and kinetic diagram for the piston.

Note that you must draw D_x and D_y in the opposite directions to what you drew for the connecting rod. Using Fig. 2 and applying Newton's second law in the x -direction and y -direction gives

$$\rightarrow \Sigma F_x = m\bar{a}_x: \quad -D_x = m_P a_D \quad (4)$$

$$+\uparrow \Sigma F_y = m\bar{a}_y: \quad -D_y + N - W_P = 0 \quad (5)$$

where

$$m_P = \frac{W_P}{g} = \frac{5 \text{ lb}}{32.2 \text{ ft/s}^2} = 0.1553 \text{ lb}\cdot\text{s}^2/\text{ft}$$

You now have five equations and nine unknowns: N , B_x , B_y , D_x , D_y , $\bar{a}_x$, $\bar{a}_y$, α_{BD} , and a_D . You could choose crank AB as another system, but since this will introduce three additional unknowns (the reactions at A and the driving torque) and you are not provided its mass, you should turn to kinematics for additional equations. From Sample Prob. 15.15, you obtained $\omega_{BD} = 62.0 \text{ rad/s} \uparrow$, $\mathbf{a}_D = 9290 \text{ ft/s}^2 \leftarrow$, and $\alpha_{BD} = 9940 \text{ rad/s}^2 \uparrow$. These reduce the number of unknowns by two, so you have five equations and seven unknowns: N , B_x , B_y , D_x , D_y , $\bar{a}_x$, and $\bar{a}_y$. You can find two more equations by relating the acceleration of the center of mass of the connecting rod to the acceleration of D ,

$$\mathbf{a}_G = \mathbf{a}_D + \mathbf{a}_{G/D} = \mathbf{a}_D + \boldsymbol{\alpha} \times \mathbf{r}_{G/D} - \omega_{BD}^2 \mathbf{r}_{G/D}$$

Substituting in known and assumed values (Fig. 1) $\mathbf{a}_D = a_D \mathbf{i}$, where $a_D = -9290 \text{ ft/s}^2$, and $\boldsymbol{\alpha}_{BD} = \alpha_{BD} \mathbf{k}$, where $\alpha_{BD} = 9940 \text{ rad/s}^2$, gives

$$\begin{aligned} \bar{a}_x \mathbf{i} + \bar{a}_y \mathbf{j} &= a_D \mathbf{i} + \alpha_{BD} \mathbf{k} \times \left[-\frac{\ell}{2} \cos \beta \mathbf{i} + \frac{\ell}{2} \sin \beta \mathbf{j} \right] - \omega_{BD}^2 \left[-\frac{\ell}{2} \cos \beta \mathbf{i} + \frac{\ell}{2} \sin \beta \mathbf{j} \right] \\ &= a_D \mathbf{i} - \alpha_{BD} \frac{\ell}{2} \cos \beta \mathbf{j} - \alpha_{BD} \frac{\ell}{2} \sin \beta \mathbf{i} + \omega_{BD}^2 \frac{\ell}{2} \cos \beta \mathbf{i} - \omega_{BD}^2 \frac{\ell}{2} \sin \beta \mathbf{j} \end{aligned}$$

Equating components, you have

$$\mathbf{i}: \bar{a}_x = a_D - \alpha_{BD} \frac{\ell}{2} \sin \beta + \omega_{BD}^2 \frac{\ell}{2} \cos \beta \quad (6)$$

$$\mathbf{j}: \bar{a}_y = -\alpha_{BD} \frac{\ell}{2} \cos \beta - \omega_{BD}^2 \frac{\ell}{2} \sin \beta \quad (7)$$

You now have seven equations and seven unknowns. Substituting in numerical values and solving these equations using your calculator or software such as MathCad, Maple, Matlab, or Mathematica gives you $B_x = -2541 \text{ lb}$, $B_y = 207.2 \text{ lb}$, $D_x = 1442 \text{ lb}$, $D_y = -641 \text{ lb}$, $N = -636 \text{ lb}$, $\bar{a}_x = -8845 \text{ ft/s}^2$, and $\bar{a}_y = -3524 \text{ ft/s}^2$.

$$B_x = 2541 \text{ lb} \leftarrow \quad B_y = 207 \text{ lb} \uparrow \quad \blacktriangleleft$$

$$D_x = 1442 \text{ lb} \rightarrow \quad D_y = 641 \text{ lb} \downarrow \quad \blacktriangleleft$$

REFLECT and THINK: The calculated forces are much larger than the weight of the piston and the connecting rod. This problem required multiple systems and rigid body kinematics to solve, most of which was done in Sample Prob. 15.15. In problems like this, it is a good practice to focus on the problem formulation and to keep track of equations and unknowns. Once you have enough equations to solve for all the unknowns, using a computer or calculator to solve the resulting equations is often the easiest approach.

SOLVING PROBLEMS ON YOUR OWN

In this section, we considered the **plane motion of rigid bodies under constraints**. We found that the types of constraints involved in engineering problems vary widely. For example, a rigid body may be constrained to rotate about a fixed axis or to roll on a given surface, or it may be pin-connected to collars or to other bodies.

1. Your solution of a problem involving the constrained motion of a rigid body consists, in general, of three steps. First, you should model your system by drawing the free-body diagram and the kinetic diagram. Second, use these diagrams to write out your equations of motion. Finally, you will generally need to consider the *kinematics of the motion* to have enough equations to solve the problem. Sometimes it is helpful to examine the kinematics first to help you draw the kinetic diagram and choose an appropriate coordinate system.

2. Free-body diagram and kinetic diagram. Your first step in the solution of a problem is to draw a *free-body diagram* and a *kinetic diagram*.

a. A free-body diagram shows *the forces exerted on the body*, including the applied forces, the reactions at the supports, and the weight of the body.

b. A kinetic diagram shows the **inertial terms**: vector $m\bar{a}$ and couple $\bar{I}\alpha$.

3. Using your free-body diagram and kinetic diagram, generate the equations of motion for the system. Drawing good free-body and kinetic diagrams will allow you to *sum components in any direction* and to *sum moments about any point*. For a single body, you can obtain a maximum of three independent equations (two translational and one moment) that can be used to help analyze the system.

$$\Sigma F_x = m\bar{a}_x \quad \Sigma F_y = m\bar{a}_y$$

$$\Sigma M_G = \bar{I}\alpha \quad \text{or} \quad \Sigma M_O = I_O\alpha \quad \text{or} \quad \Sigma M_P = \bar{I}\alpha + m\bar{a}d_{\perp} \quad \text{or} \quad \Sigma M_P = \bar{I}\alpha + r_{G/P} \times m\bar{a}$$

where G is the center of mass of the body, O is a fixed axis of rotation, P is any arbitrary point, and $d_{\perp}$ is the perpendicular distance between point P and the line of action of the acceleration of the center of mass.

4. The kinematic analysis of the motion uses the methods you learned in Chap. 15. Due to the constraints, linear and angular accelerations are related. You should establish relationships among the accelerations (angular as well as linear), and your goal should be to express all accelerations in terms of a single unknown acceleration.

a. For a body in noncentroidal rotation about a fixed axis, the components of the acceleration of the mass center are $\bar{a}_t = \bar{r}\alpha$ and $\bar{a}_n = \bar{r}\omega^2$, where ω is generally known [Sample Probs. 16.7 and 16.8].

(continued)

b. For a rolling disk or wheel, the acceleration of the geometric center is $\bar{a} = r\alpha$ [Sample Prob. 16.9].

c. For a body in general plane motion, your best course of action if neither $\bar{a}$ nor α is known or readily obtainable is to express $\bar{a}$ in terms of α [Sample Probs. 16.10 through 16.13]. This can be done by relating the acceleration of the center of mass to some reference point:

$$\bar{\mathbf{a}} = \mathbf{a}_A + \alpha \mathbf{k} \times \mathbf{r}_{G/A} - \omega^2 \mathbf{r}_{G/A}$$

5. When solving problems involving rolling disks or wheels, keep in mind the following situations.

a. If sliding is impending, the friction force exerted on the rolling body has reached its maximum value, so $F_m = \mu_s N$, where N is the normal force exerted on the body and μ_s is the coefficient of static friction between the surfaces of contact.

b. If sliding is not impending, the friction force F can have any value smaller than F_m and therefore should be considered an independent unknown. After you have determined F , be sure to check that it is smaller than F_m ; if it is not, the body does not roll but rotates and slides as described in the next paragraph.

c. If the body rotates and slides at the same time, then the body is not rolling, and the acceleration $\bar{a}$ of the mass center is *independent* of the angular acceleration α of the body: $\bar{a} \neq r\alpha$. On the other hand, the friction force has a well-defined value, $F = \mu_k N$, where μ_k is the coefficient of kinetic friction between the surfaces of contact.

d. For an unbalanced rolling disk or wheel, the relation $\bar{a} = r\alpha$ between the acceleration $\bar{a}$ of the mass center G and the angular acceleration α of the disk or wheel does not hold any more. However, a similar relation holds between the acceleration a_O of the geometric center O and the angular acceleration α of the disk or wheel: $a_O = r\alpha$. This relation can be used to express $\bar{a}$ in terms of α and ω (Fig. 16.17).

6. For a system of connected rigid bodies, the goal of your kinematic analysis should be to determine all the accelerations from the given data or to express them all in terms of a single unknown. For systems with several degrees of freedom, you will need to use as many unknowns as there are degrees of freedom.

Your kinetic analysis will sometimes be carried out by drawing a free-body diagram and a kinetic diagram for the entire system. If you only have three unknowns, this is usually the best approach. In most cases, however, it will be necessary to analyze each rigid body separately in order to obtain enough equations to solve for all the unknown quantities in the problem.

Problems

CONCEPT QUESTIONS

16.CQ4 A cord is attached to a spool when a force $\mathbf{P}$ is applied to the cord as shown. Assuming the spool rolls without slipping, what direction does the spool move for each case?

Case 1: **a.** left **b.** right **c.** It would not move.

Case 2: **a.** left **b.** right **c.** It would not move.

Case 3: **a.** left **b.** right **c.** It would not move.

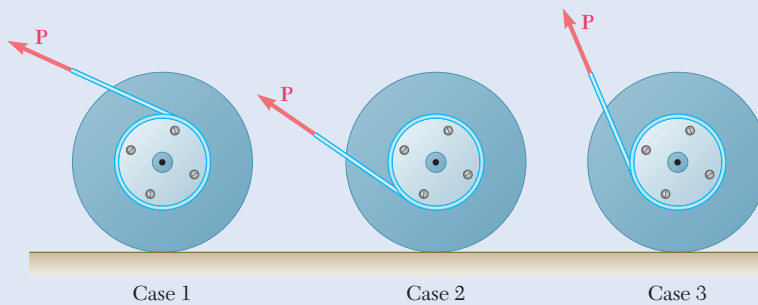


Fig. P16.CQ4 and P16.CQ5

16.CQ5 A cord is attached to a spool when a force $\mathbf{P}$ is applied to the cord as shown. Assuming the spool rolls without slipping, in what direction does the friction force act for each case?

Case 2: **a.** left **b.** right **c.** The friction force would be zero.

Case 3: **a.** left **b.** right **c.** The friction force would be zero.

16.CQ6 A front-wheel-drive car starts from rest and accelerates to the right. Knowing that the tires do not slip on the road, what is the direction of the friction force the road applies to the front tires?

a. left

b. right

c. The friction force is zero.

16.CQ7 A front-wheel-drive car starts from rest and accelerates to the right. Knowing that the tires do not slip on the road, what is the direction of the friction force the road applies to the rear tires?

a. left

b. right

c. The friction force is zero.

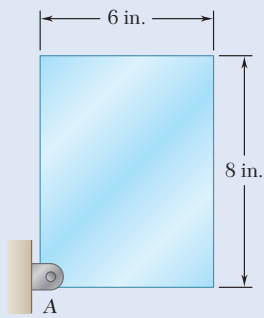


Fig. P16.F5

FREE-BODY PRACTICE PROBLEMS

16.F5 A uniform 6×8 -in. rectangular plate of mass m is pinned at A . Knowing the angular velocity of the plate at the instant shown is ω , draw the FBD and KD.

16.F6 Two identical 4-lb slender rods AB and BC are connected by a pin at B and by the cord AC . The assembly rotates in a vertical plane under the combined effect of gravity and a couple M applied to rod AB . Knowing that in the position shown the angular velocity of the assembly is ω , draw the FBD and KD that can be used to determine the angular acceleration of the assembly.

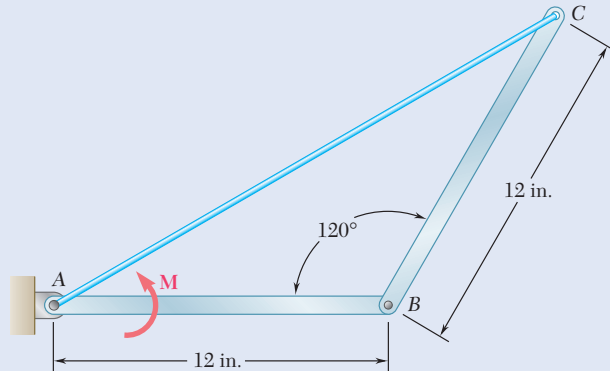


Fig. P16.F6

16.F7 The 4-lb uniform rod AB is attached to collars of negligible mass that slide without friction along the fixed rods shown. Rod AB is at rest in the position $\theta = 25^\circ$ when a horizontal force P is applied to collar A causing it to start moving to the left. Draw the FBD and KD for the rod.

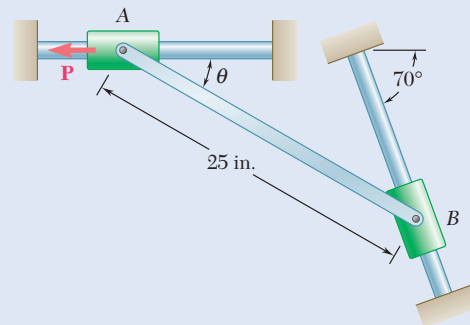


Fig. P16.F7

16.F8 A uniform disk of mass $m = 4$ kg and radius $r = 150$ mm is supported by a belt $ABCD$ that is bolted to the disk at B and C . If the belt suddenly breaks at a point located between A and B , draw the FBD and KD for the disk immediately after the break.

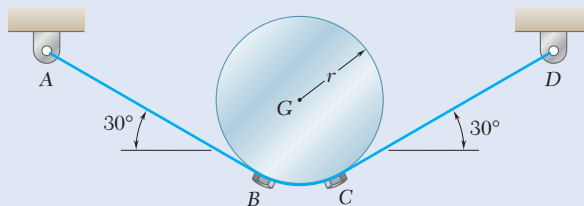


Fig. P16.F8

END-OF-SECTION PROBLEMS

- 16.75** Show that the couple $\bar{I}\alpha$ of Fig. 16.15 can be eliminated by attaching the vectors $m\bar{\mathbf{a}}_t$ and $m\bar{\mathbf{a}}_n$ at a point P called the *center of percussion*, located on line OG at a distance $GP = \bar{k}^2/\bar{r}$ from the mass center of the body.

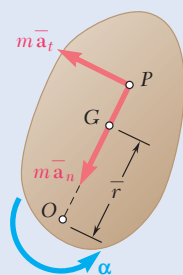


Fig. P16.75

- 16.76** A uniform slender rod of length $L = 900$ mm and mass $m = 4$ kg is suspended from a hinge at C . A horizontal force $\mathbf{P}$ of magnitude 75 N is applied at end B . Knowing that $\bar{r} = 225$ mm, determine (a) the angular acceleration of the rod, (b) the components of the reaction at C .
- 16.77** In Prob. 16.76, determine (a) the distance $\bar{r}$ for which the horizontal component of the reaction at C is zero, (b) the corresponding angular acceleration of the rod.
- 16.78** A uniform slender rod of length $L = 36$ in. and weight $W = 4$ lb hangs freely from a hinge at A . If a force $\mathbf{P}$ of magnitude 1.5 lb is applied at B horizontally to the left ($h = L$), determine (a) the angular acceleration of the rod, (b) the components of the reaction at A .
- 16.79** In Prob. 16.78, determine (a) the distance h for which the horizontal component of the reaction at A is zero, (b) the corresponding angular acceleration of the rod.
- 16.80** An athlete performs a leg extension on a machine using a 20-kg mass at A located 400 mm away from the knee joint at center O . Biomechanical studies show that the patella tendon inserts at B , which is 100 mm below point O and 20 mm from the center line of the tibia (see figure). The mass of the lower leg and foot is 5 kg, the center of gravity of this segment is 300 mm from the knee, and the radius of gyration about the knee is 350 mm. Knowing that the leg is moving at a constant angular velocity of 30 degrees per second when $\theta = 60^\circ$, determine (a) the force $\mathbf{F}$ in the patella tendon, (b) the magnitude of the joint force at the knee joint center O .

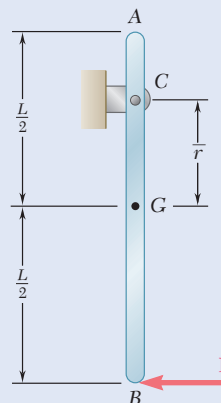


Fig. P16.76

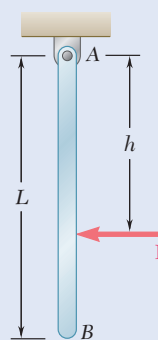


Fig. P16.78

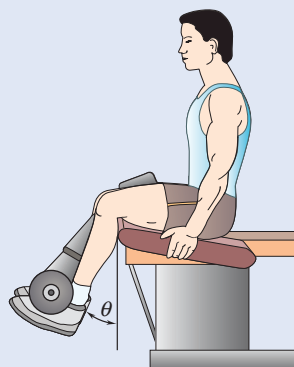
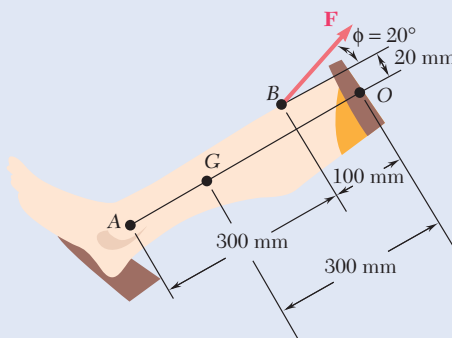


Fig. P16.80



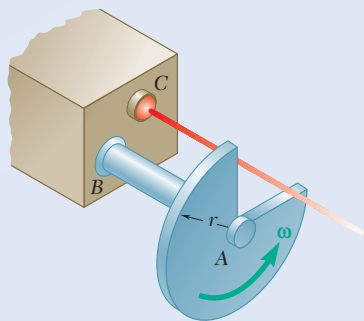


Fig. P16.81

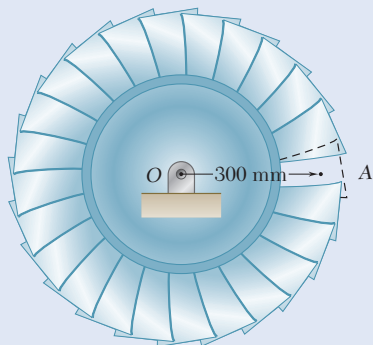


Fig. P16.83

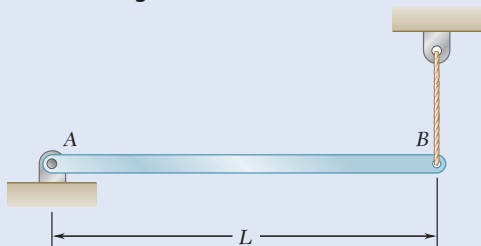


Fig. P16.84

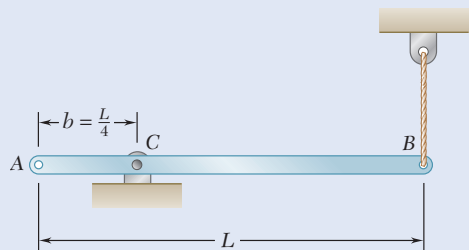


Fig. P16.85

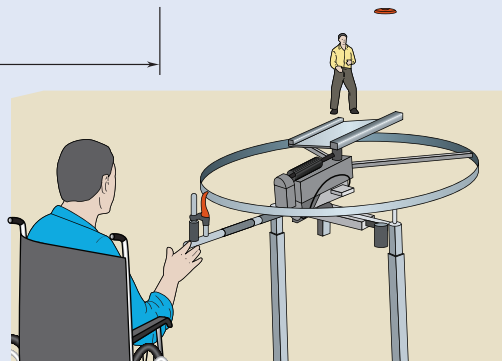


Fig. P16.86

16.81 The shutter shown was formed by removing one quarter of a disk of 0.75-in. radius and is used to interrupt a beam of light emanating from a lens at C. Knowing that the shutter weighs 0.125 lb and rotates at the constant rate of 24 cycles per second, determine the magnitude of the force exerted by the shaft on the shaft at A.

16.82 A 6-in.-diameter hole is cut as shown in a thin disk of 15-in. diameter. The disk rotates in a horizontal plane about its geometric center A at the constant rate of 480 rpm. Knowing that the disk has a mass of 60 lb after the hole has been cut, determine the horizontal component of the force exerted by the shaft on the disk at A.

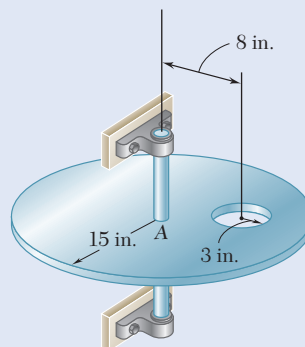
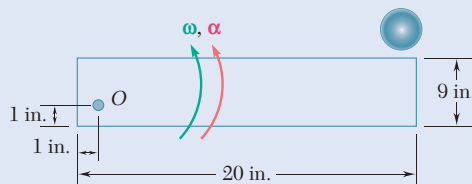


Fig. P16.82

16.83 A turbine disk of mass 26 kg rotates at a constant rate of 9600 rpm. Knowing that the mass center of the disk coincides with the center of rotation O, determine the reaction at O immediately after a single blade at A, of mass 45 g, becomes loose and is thrown off.

16.84 and 16.85 A uniform rod of length L and mass m is supported as shown. If the cable attached at end B suddenly breaks, determine (a) the acceleration of end B, (b) the reaction at the pin support.

16.86 An adapted launcher uses a torsional spring about point O to help people with mobility impairments throw a Frisbee®. Just after the Frisbee leaves the arm, the angular velocity of the throwing arm is 200 rad/s and its acceleration is 10 rad/s²; both are counterclockwise. The rotation point O is located 1 in. from the two sides. Assume that you can model the 2-lb throwing arm as a uniform rectangle. Just after the Frisbee leaves the arm, determine (a) the moment about O caused by the spring, (b) the forces on the pin at O.



- 16.87** A 1.5-kg slender rod is welded to a 5-kg uniform disk as shown. The assembly swings freely about C in a vertical plane. Knowing that in the position shown the assembly has an angular velocity of 10 rad/s clockwise, determine (a) the angular acceleration of the assembly, (b) the components of the reaction at C .

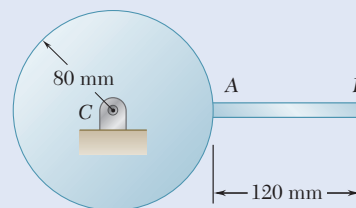


Fig. P16.87

- 16.88** Two identical 4-lb slender rods AB and BC are connected by a pin at B and by the cord AC . The assembly rotates in a vertical plane under the combined effect of gravity and a 6 lb·ft couple $\mathbf{M}$ applied to rod AB . Knowing that in the position shown the angular velocity of the assembly is zero, determine (a) the angular acceleration of the assembly, (b) the tension in cord AC .

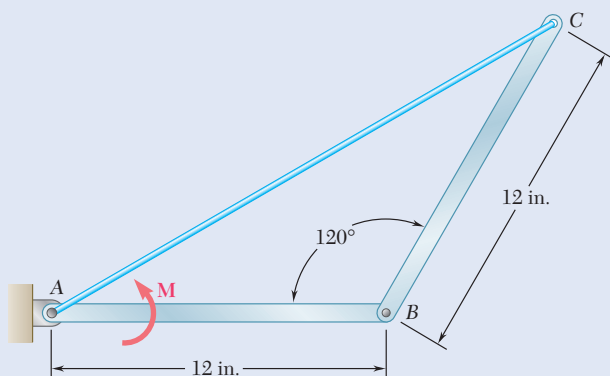


Fig. P16.88

- 16.89** The object ABC consists of two slender rods welded together at point B . Rod AB has a weight of 2 lb and bar BC has a weight of 4 lb. Knowing the magnitude of the angular velocity of ABC is 10 rad/s when $\theta = 0^\circ$, determine the components of the reaction at point C at this location.

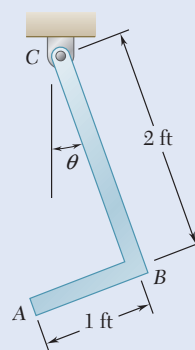


Fig. P16.89

- 16.90** A 3.5-kg slender rod AB and a 2-kg slender rod BC are connected by a pin at B and by the cord AC . The assembly can rotate in a vertical plane under the combined effect of gravity and a couple $\mathbf{M}$ applied to rod BC . Knowing that in the position shown the angular velocity of the assembly is zero and the tension in cord AC is equal to 25 N, determine (a) the angular acceleration of the assembly, (b) the magnitude of the couple $\mathbf{M}$.

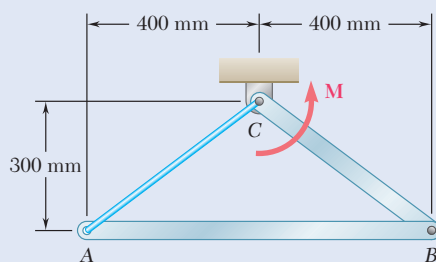


Fig. P16.90

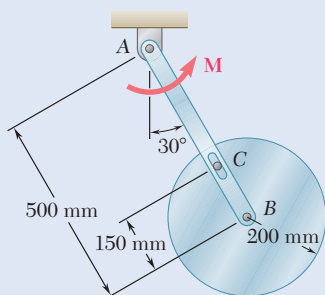


Fig. P16.91

16.91 A 9-kg uniform disk is attached to the 5-kg slender rod AB by means of frictionless pins at B and C . The assembly rotates in a vertical plane under the combined effect of gravity and of a couple $\mathbf{M}$ that is applied to rod AB . Knowing that at the instant shown the assembly has an angular velocity of 6 rad/s and an angular acceleration of 25 rad/s^2 , both counterclockwise, determine (a) the couple $\mathbf{M}$, (b) the force exerted by pin C on member AB .

16.92 Derive the equation $\Sigma M_C = I_C \alpha$ for the rolling disk of Fig. 16.16, where ΣM_C represents the sum of the moments of the external forces about the instantaneous center C , and I_C is the moment of inertia of the disk about C .

16.93 Show that in the case of an unbalanced disk, the equation derived in Prob. 16.92 is valid only when the mass center G , the geometric center O , and the instantaneous center C happen to lie in a straight line.

16.94 A wheel of radius r and centroidal radius of gyration $\bar{k}$ is released from rest on the incline and rolls without sliding. Derive an expression for the acceleration of the center of the wheel in terms of r , $\bar{k}$, β , and g .

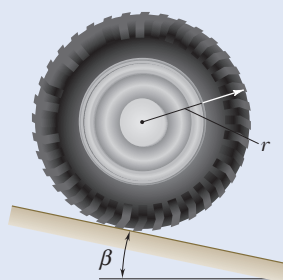


Fig. P16.94

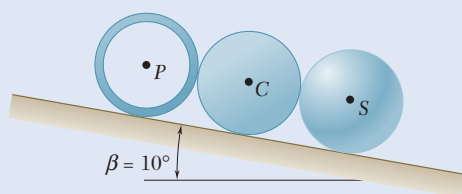


Fig. P16.95

16.95 A homogeneous sphere S , a uniform cylinder C , and a thin pipe P are in contact when they are released from rest on the incline shown. Knowing that all three objects roll without slipping, determine, after 4 s of motion, the distance between (a) the pipe and the cylinder, (b) the cylinder and the sphere.

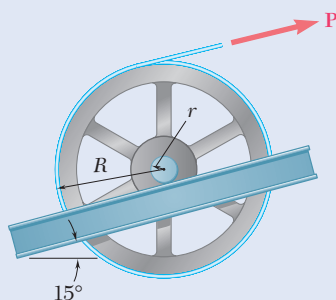


Fig. P16.96 and P16.97

16.96 A 40-kg flywheel of radius $R = 0.5 \text{ m}$ is rigidly attached to a shaft of radius $r = 0.05 \text{ m}$ that can roll along parallel rails. A cord is attached as shown and pulled with a force $\mathbf{P}$ of magnitude 150 N . Knowing the centroidal radius of gyration is $\bar{k} = 0.4 \text{ m}$, determine (a) the angular acceleration of the flywheel, (b) the velocity of the center of gravity after 5 s.

16.97 A 40-kg flywheel of radius $R = 0.5 \text{ m}$ is rigidly attached to a shaft of radius $r = 0.05 \text{ m}$ that can roll along parallel rails. A cord is attached as shown and pulled with a force $\mathbf{P}$. Knowing the centroidal radius of gyration is $\bar{k} = 0.4 \text{ m}$ and the coefficient of static friction is $\mu_s = 0.4$, determine the largest magnitude of force $\mathbf{P}$ for which no slipping will occur.

16.98 through 16.101 A drum of 60-mm radius is attached to a disk of 120-mm radius. The disk and drum have a total mass of 6 kg and a combined radius of gyration of 90 mm. A cord is attached as shown and pulled with a force $\mathbf{P}$ of magnitude 20 N. Knowing that the disk rolls without sliding, determine (a) the angular acceleration of the disk and the acceleration of G , (b) the minimum value of the coefficient of static friction compatible with this motion.

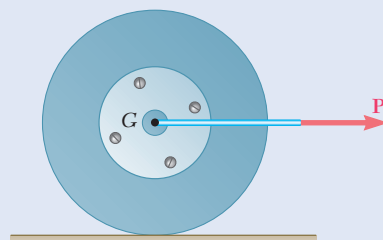


Fig. P16.98 and P16.102

16.102 through 16.105 A drum of 4-in. radius is attached to a disk of 8-in. radius. The disk and drum have a combined weight of 10 lb and a combined radius of gyration of 6 in. A cord is attached as shown and pulled with a force $\mathbf{P}$ of magnitude 5 lb. Knowing that the coefficients of static and kinetic friction are $\mu_s = 0.25$ and $\mu_k = 0.20$, respectively, determine (a) whether or not the disk slides, (b) the angular acceleration of the disk and the acceleration of G .

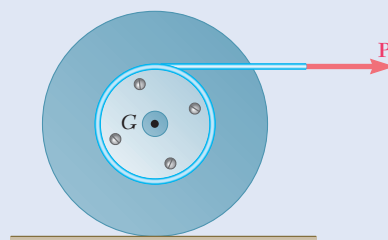


Fig. P16.99 and P16.103

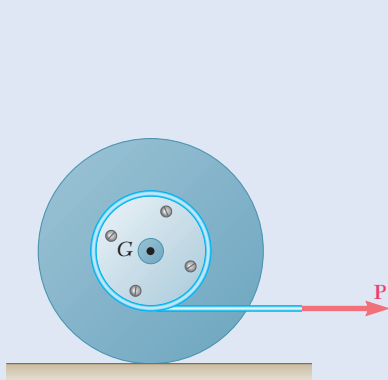


Fig. P16.100 and P16.104

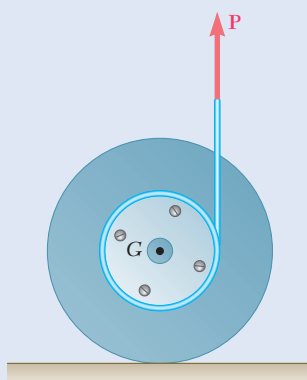


Fig. P16.101 and P16.105

16.106 and 16.107 A 12-in.-radius cylinder of weight 16 lb rests on a 6-lb carriage. The system is at rest when a force $\mathbf{P}$ of magnitude 4 lb is applied. Knowing that the cylinder rolls without sliding on the carriage and neglecting the mass of the wheels of the carriage, determine (a) the acceleration of the carriage, (b) the acceleration of point A, (c) the distance the cylinder has rolled with respect to the carriage after 0.5 s.

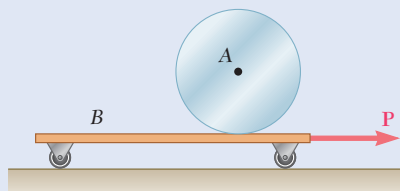


Fig. P16.106

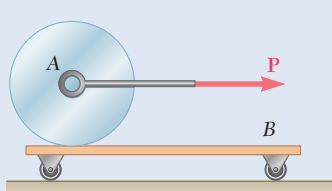


Fig. P16.107

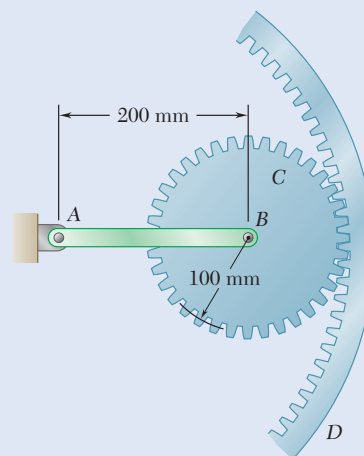


Fig. P16.108

16.108 Gear C has a mass of 5 kg and a centroidal radius of gyration of 75 mm. The uniform bar AB has a mass of 3 kg and gear D is stationary. If the system is released from rest in the position shown, determine (a) the angular acceleration of gear C, (b) the acceleration of point B.

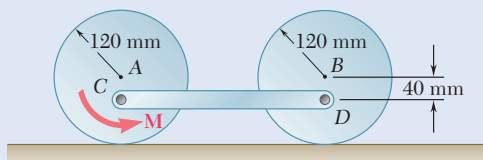


Fig. P16.109

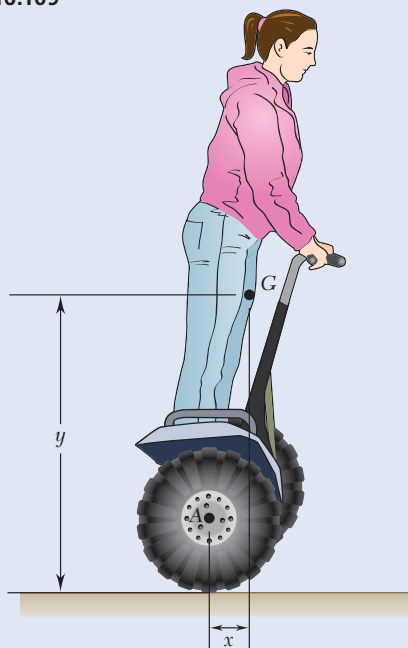


Fig. P16.110

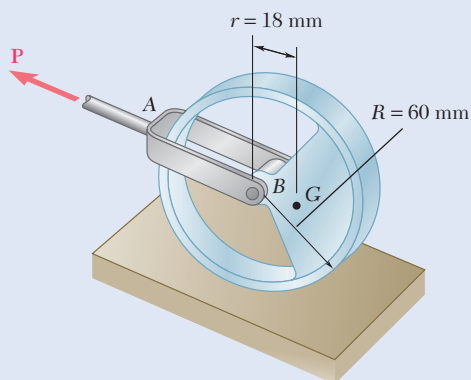


Fig. P16.113

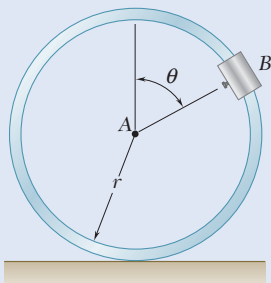


Fig. P16.114 and P16.115

16.109 Two uniform disks A and B , each with a mass of 2 kg, are connected by a 2.5-kg rod CD as shown. A counterclockwise couple M of moment 2.25 N·m is applied to disk A . Knowing that the disks roll without sliding, determine (a) the acceleration of the center of each disk, (b) the horizontal component of the force exerted on disk B by pin D .

16.110 A single-axis personal transport device starts from rest with the rider leaning slightly forward. Together, the two wheels weigh 25 lbs, and each has a radius of 10 in. The mass moment of inertia of the wheels about the axle is 0.15 slug·ft². The combined weight of the rest of the device and the rider (excluding the wheels) is 200 lbs, and the center of gravity G of this weight is located at $x = 4$ in. in front of axle A and $y = 36$ in. above the ground. An initial clockwise torque M is applied by the motor to the wheels. Knowing that the coefficients of static and kinetic friction are 0.7 and 0.6, respectively, determine (a) the torque M that will keep the rider in the same angular position, (b) the corresponding linear acceleration of the rider.

16.111 A hemisphere of weight W and radius r is released from rest in the position shown. Determine (a) the minimum value of μ_s for which the hemisphere starts to roll without sliding, (b) the corresponding acceleration of point B . [Hint: Note that $OG = \frac{3}{8}r$ and that, by the parallel-axis theorem, $\bar{I} = \frac{2}{5}mr^2 - m(OG)^2$.]

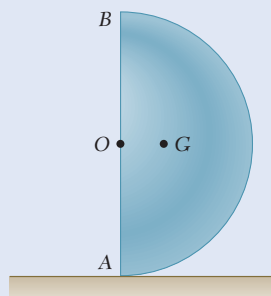


Fig. P16.111

16.112 Solve Prob. 16.111, considering a half cylinder instead of a hemisphere. [Hint: Note that $OG = \frac{4r}{3\pi}$ and that, by the parallel-axis theorem, $\bar{I} = \frac{1}{2}mr^2 - m(OG)^2$.]

16.113 The center of gravity G of a 1.5-kg unbalanced tracking wheel is located at a distance $r = 18$ mm from its geometric center B . The radius of the wheel is $R = 60$ mm and its centroidal radius of gyration is 44 mm. At the instant shown, the center B of the wheel has a velocity of 0.35 m/s and an acceleration of 1.2 m/s², both directed to the left. Knowing that the wheel rolls without sliding and neglecting the mass of the driving yoke AB , determine the horizontal force P applied to the yoke.

16.114 A small clamp of mass m_B is attached at B to a hoop of mass m_h . The system is released from rest when $\theta = 90^\circ$ and rolls without sliding. Knowing that $m_h = 3m_B$, determine (a) the angular acceleration of the hoop, (b) the horizontal and vertical components of the acceleration of B .

16.115 A small clamp of mass m_B is attached at B to a hoop of mass m_h . Knowing that the system is released from rest and rolls without sliding, derive an expression for the angular acceleration of the hoop in terms of m_B , m_h , r , and θ .

- 16.116** A 4-lb bar is attached to a 10-lb uniform cylinder by a square pin, P , as shown. Knowing that $r = 16$ in., $h = 8$ in., $\theta = 20^\circ$, $L = 20$ in., and $\omega = 2$ rad/s at the instant shown, determine the reactions at P at this instant assuming that the cylinder rolls without sliding down the incline.

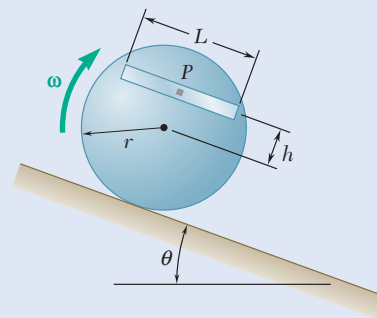


Fig. P16.116

- 16.117** The uniform rod AB with a mass m and a length of $2L$ is attached to collars of negligible mass that slide without friction along fixed rods. If the rod is released from rest in the position shown, derive an expression for (a) the angular acceleration of the rod, (b) the reaction at A .

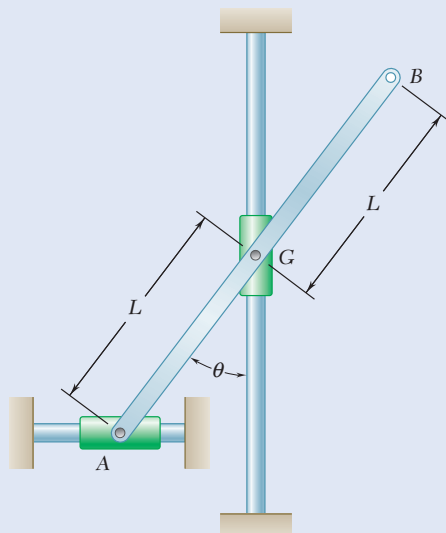


Fig. P16.117 and P16.118

- 16.118** The 10-lb-uniform rod AB has a total length of $2L = 2$ ft and is attached to collars of negligible mass that slide without friction along fixed rods. If rod AB is released from rest when $\theta = 30^\circ$, determine immediately after release (a) the angular acceleration of the rod, (b) the reaction at A .

- 16.119** A 40-lb ladder rests against a wall when the bottom begins to slide out. The ladder is 30 ft long and the coefficient of kinetic friction between the ladder and all surfaces is 0.2. For $\theta = 40^\circ$, determine (a) the angular acceleration of the ladder, (b) the forces at A and B .

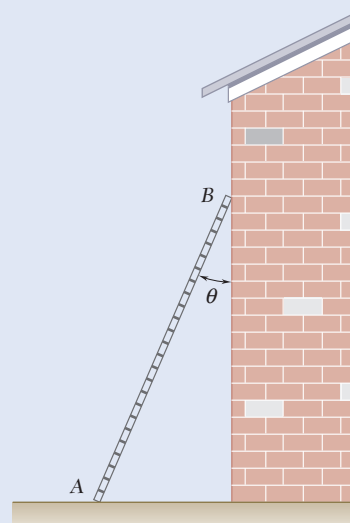


Fig. P16.119

- 16.120** A beam AB of length L and mass m is supported by two cables as shown. If cable BD breaks, determine at that instant the tension in the remaining cable as a function of its initial angular orientation θ .

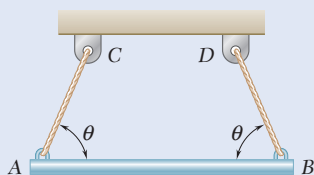


Fig. P16.120

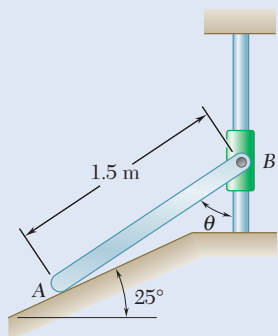


Fig. P16.121 and P16.122

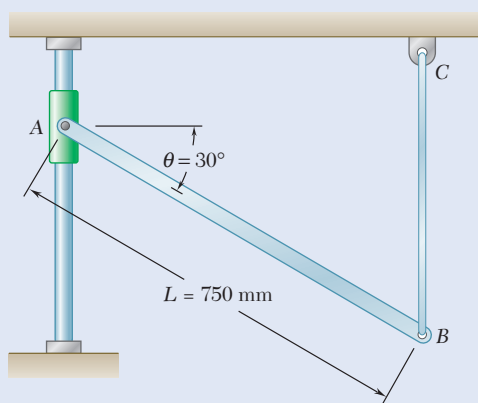


Fig. P16.123

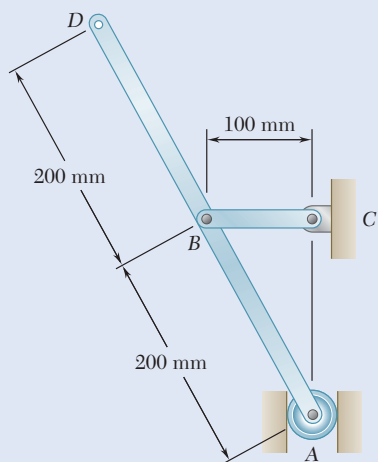


Fig. P16.124

16.121 End A of the 6-kg uniform rod AB rests on the inclined surface, while end B is attached to a collar of negligible mass that can slide along the vertical rod shown. Knowing that the rod is released from rest when $\theta = 35^\circ$ and neglecting the effect of friction, determine immediately after release (a) the angular acceleration of the rod, (b) the reaction at B.

16.122 End A of the 6-kg uniform rod AB rests on the inclined surface, while end B is attached to a collar of negligible mass that can slide along the vertical rod shown. When the rod is at rest, a vertical force $\mathbf{P}$ is applied at B, causing end B of the rod to start moving upward with an acceleration of 4 m/s^2 . Knowing that $\theta = 35^\circ$, determine the force $\mathbf{P}$.

16.123 End A of the 8-kg uniform rod AB is attached to a collar that can slide without friction on a vertical rod. End B of the rod is attached to a vertical cable BC . If the rod is released from rest in the position shown, determine immediately after release (a) the angular acceleration of the rod, (b) the reaction at A.

16.124 The 4-kg uniform rod ABD is attached to the crank BC and is fitted with a small wheel that can roll without friction along a vertical slot. Knowing that at the instant shown crank BC rotates with an angular velocity of 6 rad/s clockwise and an angular acceleration of 15 rad/s^2 counterclockwise, determine the reaction at A.

16.125 The 3-lb uniform rod BD is connected to crank AB and to a collar of negligible weight. A couple (not shown) is applied to crank AB , causing it to rotate with an angular velocity of 12 rad/s counterclockwise and an angular acceleration of 80 rad/s^2 clockwise at the instant shown. Neglecting the effect of friction, determine the reaction at D.

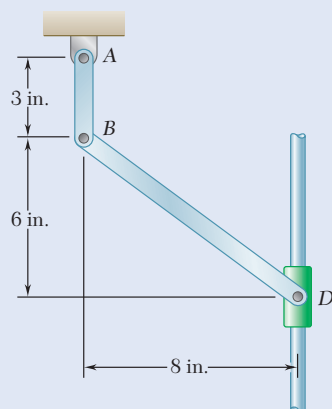


Fig. P16.125 and P16.126

16.126 The 3-lb uniform rod BD is connected to crank AB and to a collar of negligible weight. A couple (not shown) is applied to crank AB causing it to rotate. At the instant shown, crank AB has an angular velocity of 12 rad/s and an angular acceleration of 80 rad/s^2 ; both are counterclockwise. Neglecting the effect of friction, determine the reaction at D.

- 16.127** The test rig shown was developed to perform fatigue testing on fitness trampolines. A motor drives the 200-mm radius flywheel AB , which is pinned at its center point A , in a counterclockwise direction with a constant angular velocity of 120 rpm. The flywheel is attached to slider CD by the 400-mm connecting rod BC . The mass of the connecting rod BC is 5 kg, and the mass of the link CD and foot is 2 kg. At the instant when $\theta = 0^\circ$ and the foot is just above the trampoline, determine the force exerted by pin C on rod BC .

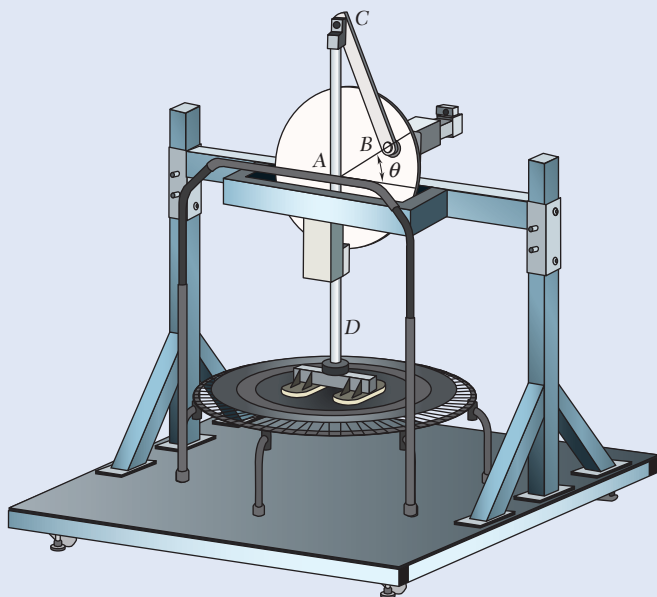


Fig. P16.127

- 16.128** Solve Prob. 16.127 for $\theta = 90^\circ$.

- 16.129** The 4-kg uniform slender bar BD is attached to bar AB and a wheel of negligible mass that rolls on a circular surface. Knowing that at the instant shown bar AB has an angular velocity of 6 rad/s and no angular acceleration, determine the reaction at point D .

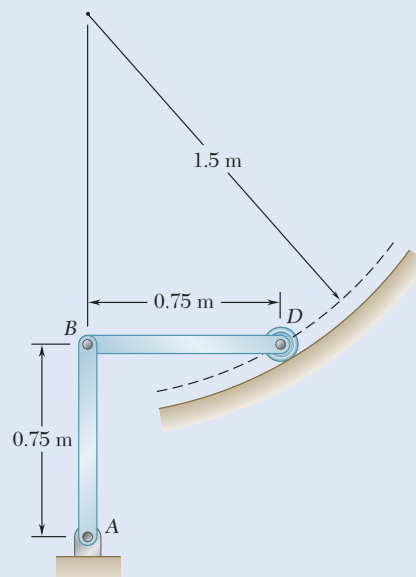


Fig. P16.129

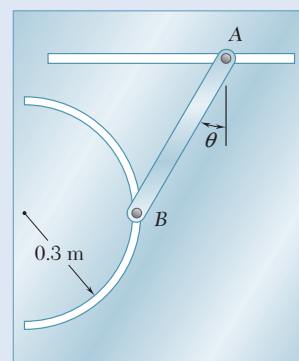


Fig. P16.130

- 16.130** The motion of the uniform slender rod of length $L = 0.5$ m and mass $m = 3$ kg is guided by pins at A and B that slide freely in frictionless slots, circular and horizontal, cut into a vertical plate as shown. Knowing that at the instant shown the rod has an angular velocity of 3 rad/s counterclockwise and $\theta = 30^\circ$, determine the reactions at points A and B .

- 16.131** At the instant shown, the 20-ft-long, uniform 100-lb pole ABC has an angular velocity of 1 rad/s counterclockwise and point C is sliding to the right. A 120-lb horizontal force $\mathbf{P}$ acts at B . Knowing the coefficient of kinetic friction between the pole and the ground is 0.3, determine at this instant (a) the acceleration of the center of gravity, (b) the normal force between the pole and the ground.

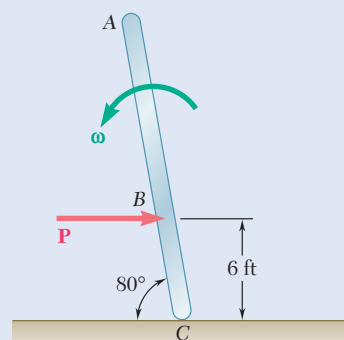


Fig. P16.131

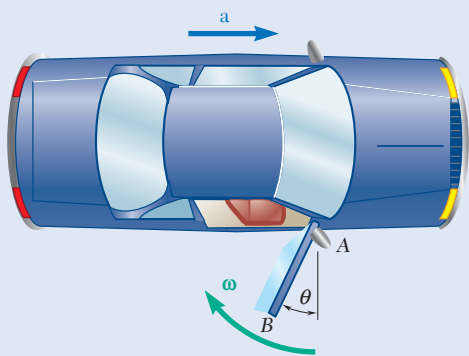
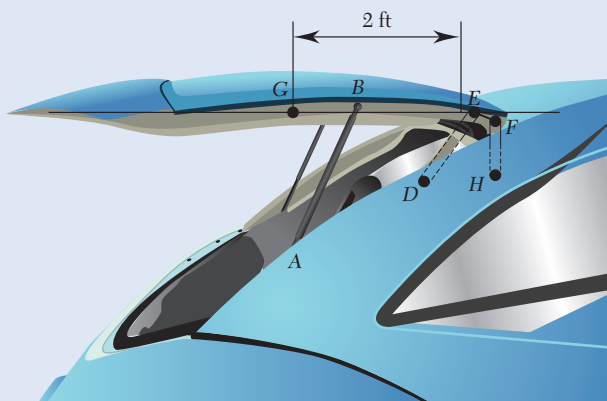


Fig. P16.132

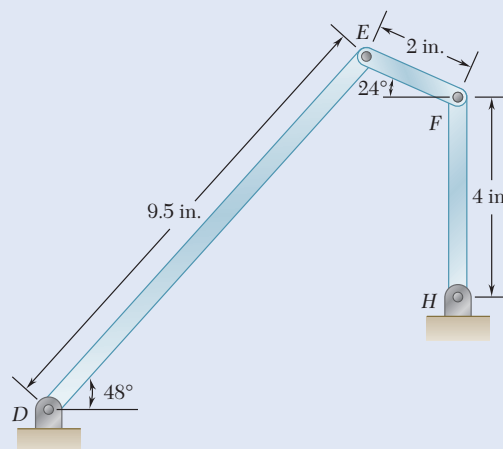
16.132 A driver starts his car with the door on the passenger's side wide open ($\theta = 0$). The 80-lb door has a centroidal radius of gyration $k = 12.5$ in., and its mass center is located at a distance $r = 22$ in. from its vertical axis of rotation. Knowing that the driver maintains a constant acceleration of 6 ft/s^2 , determine the angular velocity of the door as it slams shut ($\theta = 90^\circ$).

16.133 For the car of Prob. 16.132, determine the smallest constant acceleration that the driver can maintain if the door is to close and latch, knowing that as the door hits the frame its angular velocity must be at least 2 rad/s for the latching mechanism to operate.

16.134 The hatchback of a car is positioned as shown to help determine the appropriate size for a damping mechanism AB . The weight of the door is 40 lbs , and its mass moment of inertia about the center of gravity G is $15 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$. The linkage $DEFH$ controls the motion of the hatch and is shown in more detail in part (b) of the figure. Assume that the mass of the links DE , EF , and FH are negligible, compared to the mass of the door. With AB removed, determine (a) the initial angular acceleration of the 40-lb door as it is released from rest, (b) the force on link FH .



(a)



(b)

Fig. P16.134

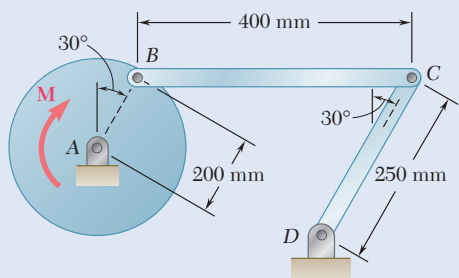


Fig. P16.135 and P16.136

***16.135** The 6-kg rod BC connects a 10-kg disk centered at A to a 5-kg rod CD . The motion of the system is controlled by the couple $\mathbf{M}$ applied to disk A . Knowing that at the instant shown disk A has an angular velocity of 36 rad/s clockwise and no angular acceleration, determine (a) the couple $\mathbf{M}$, (b) the components of the force exerted at C on rod BC .

***16.136** The 6-kg rod BC connects a 10-kg disk centered at A to a 5-kg rod CD . The motion of the system is controlled by the couple $\mathbf{M}$ applied to disk A . Knowing that at the instant shown disk A has an angular velocity of 36 rad/s clockwise and an angular acceleration of 150 rad/s^2 counterclockwise, determine (a) the couple $\mathbf{M}$, (b) the components of the force exerted at C on rod BC .

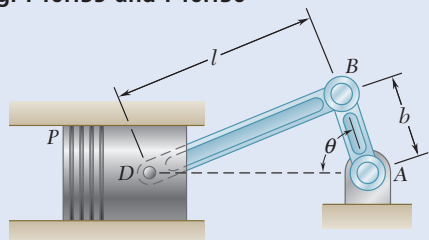


Fig. P16.137

16.137 In the engine system shown $l = 250 \text{ mm}$ and $b = 100 \text{ mm}$. The connecting rod BD is assumed to be a 1.2-kg uniform slender rod and is attached to the 1.8-kg piston P . During a test of the system, crank AB is made to rotate with a constant angular velocity of 600 rpm clockwise with no force applied to the face of the piston. Determine the forces exerted on the connecting rod at B and D when $\theta = 180^\circ$. (Neglect the effect of the weight of the rod.)

16.138 Solve Prob. 16.137 when $\theta = 90^\circ$.

16.139 The 4-lb uniform slender rod AB , the 8-lb uniform slender rod BF , and the 4-lb uniform thin sleeve CE are connected as shown and move without friction in a vertical plane. The motion of the linkage is controlled by the couple $\mathbf{M}$ applied to rod AB . Knowing that at the instant shown the angular velocity of rod AB is 15 rad/s and the magnitude of the couple $\mathbf{M}$ is 5 ft·lb, determine (a) the angular acceleration of rod AB , (b) the reaction at point D .

16.140 The 4-lb uniform slender rod AB , the 8-lb uniform slender rod BF , and the 4-lb uniform thin sleeve CE are connected as shown and move without friction in a vertical plane. The motion of the linkage is controlled by the couple $\mathbf{M}$ applied to rod AB . Knowing that at the instant shown the angular velocity of rod AB is 30 rad/s and the angular acceleration of rod AB is 96 rad/s² clockwise, determine (a) the magnitude of the couple $\mathbf{M}$, (b) the reaction at point D .

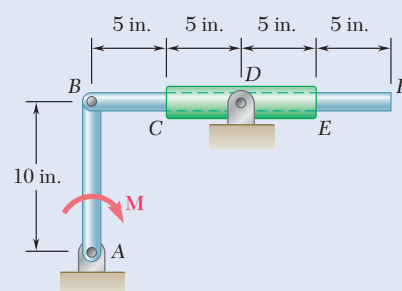


Fig. P16.139 and P16.140

16.141 Two rotating rods in the vertical plane are connected by a slider block P of negligible mass. The rod attached at A has a weight of 1.6 lb and a length of 8 in. Rod BP weighs 2 lb and is 10 in. long and the friction between block P and AE is negligible. The motion of the system is controlled by a couple $\mathbf{M}$ applied to rod BP . Knowing that rod BP has a constant angular velocity of 20 rad/s clockwise, determine (a) the couple $\mathbf{M}$, (b) the components of the force exerted on AE by block P .

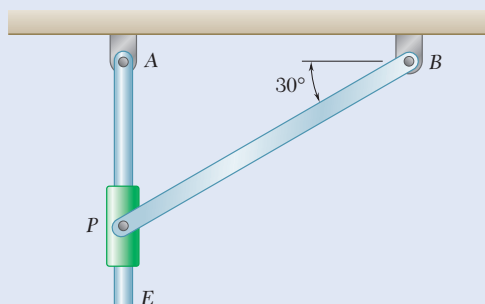


Fig. P16.141 and P16.142

16.142 Two rotating rods in the vertical plane are connected by a slider block P of negligible mass. The rod attached at A has a mass of 0.8 kg and a length of 160 mm. Rod BP has a mass of 1 kg and is 200 mm long and the friction between block P and AE is negligible. The motion of the system is controlled by a couple $\mathbf{M}$ applied to bar BP . Knowing that at the instant shown rod BP has an angular velocity of 20 rad/s clockwise and an angular acceleration of 80 rad/s² clockwise, determine (a) the couple $\mathbf{M}$, (b) the components of the force exerted on AE by block P .

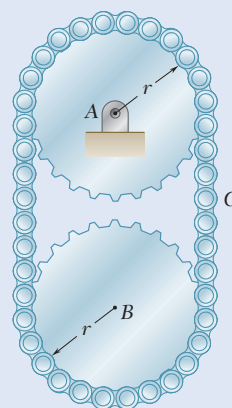


Fig. P16.143

*** 16.143** Two disks, each with a mass m and a radius r , are connected as shown by a continuous chain belt of negligible mass. If a pin at point C of the chain belt is suddenly removed, determine (a) the angular acceleration of each disk, (b) the tension in the left-hand portion of the belt, (c) the acceleration of the center of disk B .

*** 16.144** A uniform slender bar AB of mass m is suspended as shown from a uniform disk of the same mass m . Neglecting the effect of friction, determine the accelerations of points A and B immediately after a horizontal force $\mathbf{P}$ has been applied at B .

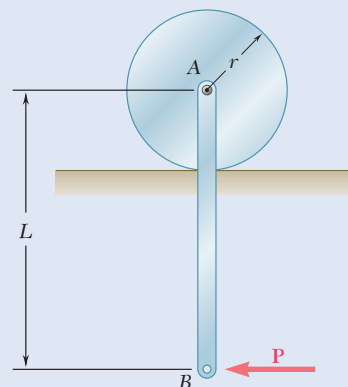


Fig. P16.144

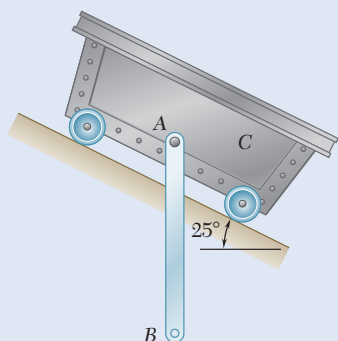


Fig. P16.145

16.145 A uniform rod AB , of mass 15 kg and length 1 m, is attached to the 20-kg cart C . Neglecting friction, determine immediately after the system has been released from rest, (a) the acceleration of the cart, (b) the angular acceleration of the rod.

*** 16.146** The uniform slender 2-kg bar BD is attached to the uniform 6-kg disk by a pin at B and released from rest in the position shown. Assuming that the disk rolls without slipping, determine (a) the initial reaction at the contact point A , (b) the corresponding smallest allowable value of the coefficient of static friction.

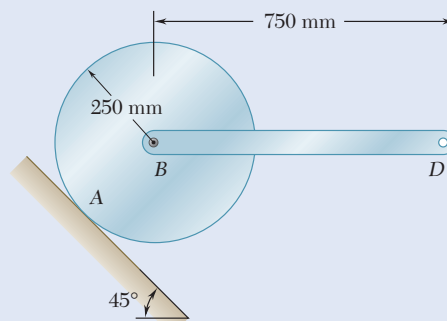


Fig. P16.146

*** 16.147 and * 16.148** The 6-lb cylinder B and the 4-lb wedge A are held at rest in the position shown by cord C . Assuming that the cylinder rolls without slipping on the wedge and neglecting friction between the wedge and the ground, determine, immediately after cord C has been cut, (a) the acceleration of the wedge, (b) the angular acceleration of the cylinder.

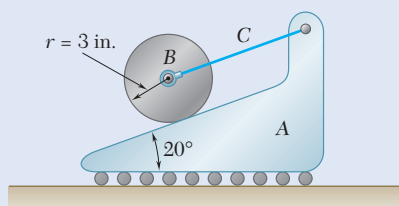


Fig. P16.147

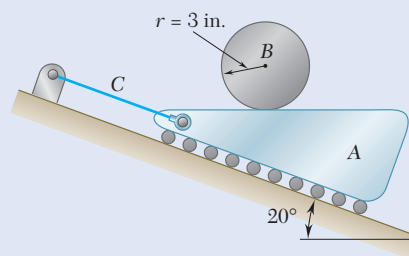


Fig. P16.148

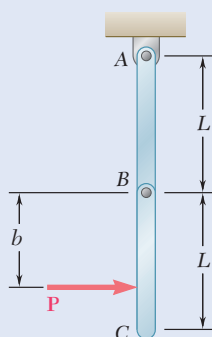


Fig. P16.149 and P16.150

*** 16.149** Each of the 3-kg bars AB and BC is of length $L = 500$ mm. A horizontal force $\mathbf{P}$ of magnitude 20 N is applied to bar BC as shown. Knowing that $b = L$ ($\mathbf{P}$ is applied at C), determine the angular acceleration of each bar.

*** 16.150** Each of the 3-kg bars AB and BC is of length $L = 500$ mm. A horizontal force $\mathbf{P}$ of magnitude 20 N is applied to bar BC . For the position shown, determine (a) the distance b for which the bars move as if they formed a single rigid body, (b) the corresponding angular acceleration of the bars.

*** 16.151** (a) Determine the magnitude and the location of the maximum bending moment in the rod of Prob. 16.78. (b) Show that the answer to part a is independent of the weight of the rod.

*** 16.152** Draw the shear and bending-moment diagrams for the rod of Prob. 16.84 immediately after the cable at B breaks.

Review and Summary

In this chapter, we studied the **kinetics of rigid bodies**, i.e., the relations between the forces acting on a rigid body, the shape and mass of the body, and the motion produced. Except for the first two sections, which apply to the most general case of the motion of a rigid body, our analysis was restricted to the **plane motion of rigid bodies** and rigid bodies symmetrical with respect to the plane of motion. We will study the plane motion of nonsymmetrical rigid bodies and the motion of rigid bodies in three-dimensional space in Chap. 18.

Fundamental Equations of Motion for a Rigid Body

We first recalled [Sec. 16.1A] the two fundamental equations derived in Chap. 14 for the motion of a system of particles and observed that they apply in the most general case of the motion of a rigid body. The first equation defines the motion of the mass center G of the body; we have

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}} \quad (16.1)$$

where m is the mass of the body and $\bar{\mathbf{a}}$ is the acceleration of G . The second equation is related to the motion of the body relative to a centroidal frame of reference; we have

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G \quad (16.2)$$

where $\dot{\mathbf{H}}_G$ is the rate of change of the angular momentum $\mathbf{H}_G$ of the body about its mass center G . Together, Eqs. (16.1) and (16.2) state that **the system of the external forces is equipollent to the system consisting of the vector $m\bar{\mathbf{a}}$ attached at G and the couple of moment $\dot{\mathbf{H}}_G$** (Fig. 16.18).

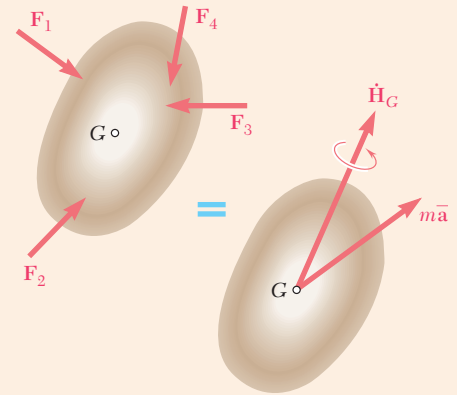


Fig. 16.18

Angular Momentum in Plane Motion

Restricting our analysis at this point and for the rest of the chapter to the plane motion of rigid bodies and rigid bodies symmetrical with respect to the plane of motion, we showed [Sec. 16.1B] that the angular momentum of the body could be expressed as

$$\mathbf{H}_G = \bar{I}\boldsymbol{\omega} \quad (16.4)$$

where $\bar{I}$ is the moment of inertia of the body about a centroidal axis perpendicular to the reference plane and $\boldsymbol{\omega}$ is the angular velocity of the body. Differentiating both sides of Eq. (16.4), we obtained

$$\dot{\mathbf{H}}_G = \bar{I}\dot{\boldsymbol{\omega}} = \bar{I}\boldsymbol{\alpha} \quad (16.5)$$

which shows that, in the restricted case considered here, we can represent the rate of change of the angular momentum of the rigid body by a vector of the same direction as $\boldsymbol{\alpha}$ (i.e., perpendicular to the plane of reference) and of magnitude $\bar{I}\alpha$.

Equations for the Plane Motion of a Rigid Body

It follows from [Sec. 16.1E] that the plane motion of a rigid body or of a rigid body symmetrical with respect to the reference plane is defined by the

three scalar equations. You will have one equation for the x -direction, one for the y -direction, and one moment equation, as

$$\begin{aligned}\Sigma F_x &= m\bar{a}_x & \Sigma F_y &= m\bar{a}_y \\ \Sigma M_G &= \bar{I}\alpha & \text{or } \Sigma M_O &= I_O\alpha & \text{or } \Sigma M_P &= \bar{I}\alpha + m\bar{a}d_\perp & \text{or} \\ \Sigma M_P &= \bar{I}\alpha + \mathbf{r}_{G/P} \times m\bar{\mathbf{a}}\end{aligned}$$

where G is the center of mass of the body, O is a fixed axis of rotation, P is any arbitrary point, and $d_\perp$ is the perpendicular distance between point P and the line of action of the acceleration of the center of mass.

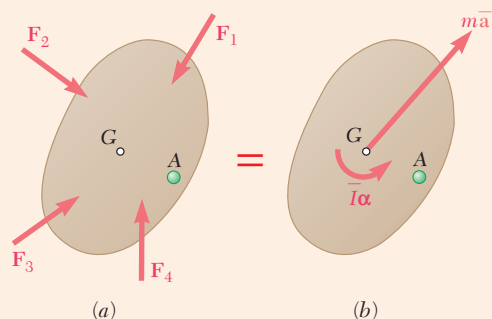


Fig. 16.19

Plane Motion of a Rigid Body

It further follows that *the external forces acting on the rigid body are actually equivalent to the inertial terms of the various particles forming the body*. This statement can be represented by a free-body diagram and kinetic diagram as shown in Fig. 16.19, where the inertial terms have been represented by a vector $m\bar{\mathbf{a}}$ attached at G and a couple $\bar{I}\alpha$. In the particular case of a rigid body in *translation*, the inertial terms shown in part b of this figure reduce to the single vector $m\bar{\mathbf{a}}$, whereas in the particular case of a rigid body in *centroidal rotation*, they reduce to the single couple $\bar{I}\alpha$. In any other case of plane motion, both the vector $m\bar{\mathbf{a}}$ and the couple $\bar{I}\alpha$ should be included.

Free-Body Diagram and Kinetic Diagram

Any kinetics problem involving the plane motion of a rigid body may be solved by drawing a **free-body diagram** and **kinetic diagram** similar to that of Fig. 16.19 [Sec. 16.1E]. You can then obtain three equations of motion (see previous equations) by equating the x components, y components, and moments about a chosen point (such as G or some arbitrary point P) of the forces and vectors involved [Sample Probs. 16.1 through 16.5].

Connected Rigid Bodies

We can also use the method described previously to solve problems involving the plane motion of several connected rigid bodies [Sec. 16.1F]. You draw a free-body diagram and kinetic diagram for each system and solve the equations of motion simultaneously. In some cases, however, you can include multiple objects in your system and draw a single diagram for the entire system, including all of the external forces as well as the vectors $m\bar{\mathbf{a}}$ and the couples $\bar{I}\alpha$ associated with the various parts of the system [Sample Prob. 16.4].

Constrained Plane Motion

In the second section of this chapter, we were concerned with rigid bodies *moving under given constraints* [Sec. 16.2]. Although the kinetic analysis of the constrained plane motion of a rigid body is the same as before, it must be supplemented by a *kinematic analysis* that aims to express the components $\bar{a}_x$ and $\bar{a}_y$ of the acceleration of the mass center G of the body in terms of its angular acceleration α . This often involves using analyses that we examined in Ch. 15, including the relationship between two points on a body undergoing general plane motion:

$$\bar{\mathbf{a}} = \mathbf{a}_A + \alpha \mathbf{k} \times \mathbf{r}_{G/A} - \omega^2 \mathbf{r}_{G/A}$$

Problems solved in this way included the noncentroidal rotation of rods and plates [Sample Probs. 16.7 and 16.8], the rolling motion of spheres and wheels [Sample Probs. 16.9 and 16.10], the general plane motion of a body with no fixed point [Sample Probs. 16.11 and 16.12], and the plane motion of various types of linkages [Sample Prob. 16.13].

Review Problems

- 16.153** A cyclist is riding a bicycle at a speed of 20 mph on a horizontal road. The distance between the axles is 42 in., and the mass center of the cyclist and the bicycle is located 26 in. behind the front axle and 40 in. above the ground. If the cyclist applies the brakes only on the front wheel, determine the shortest distance in which he can stop without being thrown over the front wheel.
- 16.154** The forklift truck shown weighs 2250 lb and is used to lift a crate of weight $W = 2500$ lb. The truck is moving to the left at a speed of 10 ft/s when the brakes are applied on all four wheels. Knowing that the coefficient of static friction between the crate and the fork lift is 0.30, determine the smallest distance in which the truck can be brought to a stop if the crate is not to slide and if the truck is not to tip forward.

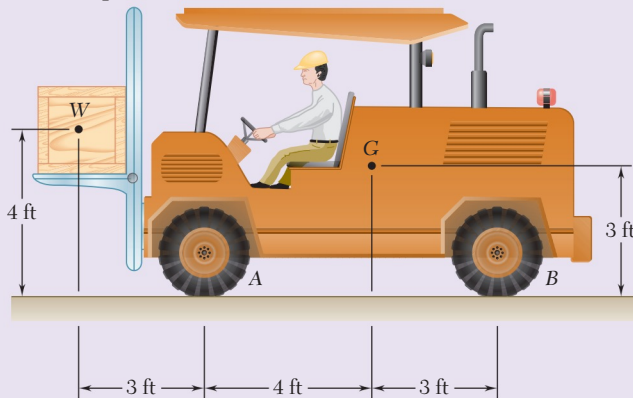


Fig. P16.154

- 16.155** The total mass of the Baja car and driver, including the wheels, is 250 kg. Each pair of 58-cm radius wheels and the axle has a total mass of 20 kg and a mass moment of inertia of $2.9 \text{ kg}\cdot\text{m}^2$. The center of gravity of the driver and Baja body (not including the wheels) is located $x = 0.70$ m from the rear axle A and $y = 0.55$ m from the ground. The wheelbase is $L = 1.60$ m. If the engine exerts a torque of $500 \text{ N}\cdot\text{m}$ on the rear axle, what is the car's acceleration?
- 16.156** Identical cylinders of mass m and radius r are pushed by a series of moving arms. Assuming the coefficient of friction between all surfaces to be $\mu < 1$ and denoting by a the magnitude of the acceleration of the arms, derive an expression for (a) the maximum allowable value of a if each cylinder is to roll without sliding, (b) the minimum allowable value of a if each cylinder is to move to the right without rotating.

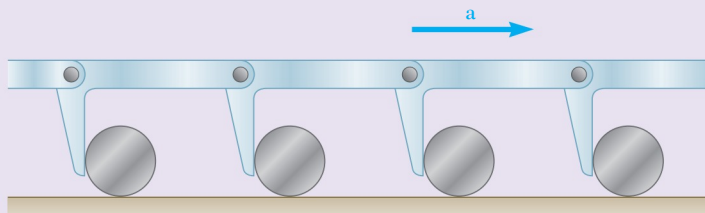


Fig. P16.156

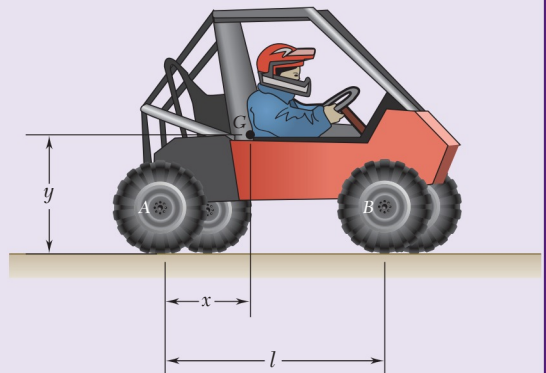


Fig. P16.155

- 16.157** The uniform rod AB of weight W is released from rest when $\beta = 70^\circ$. Assuming that the friction force between end A and the surface is large enough to prevent sliding, determine immediately after release (a) the angular acceleration of the rod, (b) the normal reaction at A , (c) the friction force at A .

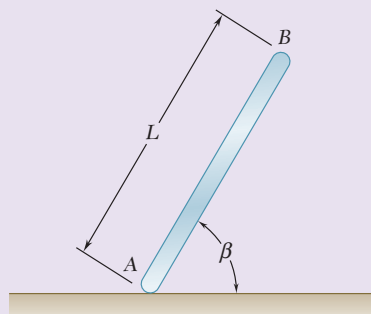


Fig. P16.157 and P16.158

- 16.158** The uniform rod AB of weight W is released from rest when $\beta = 70^\circ$. Assuming that the friction force is zero between end A and the surface, determine immediately after release (a) the angular acceleration of the rod, (b) the acceleration of the mass center of the rod, (c) the reaction at A .
- 16.159** A bar of mass $m = 5$ kg is held as shown between four disks, each of mass $m' = 2$ kg and radius $r = 75$ mm. Knowing that the normal forces on the disks are sufficient to prevent any slipping, for each of the cases shown determine the acceleration of the bar immediately after it has been released from rest.

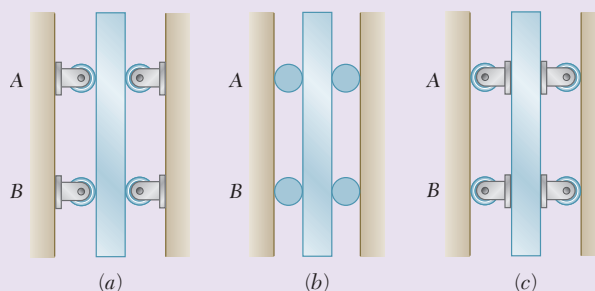


Fig. P16.159

- 16.160** A uniform plate of mass m is suspended in each of the ways shown. For each case determine immediately after the connection B has been released (a) the angular acceleration of the plate, (b) the acceleration of its mass center.

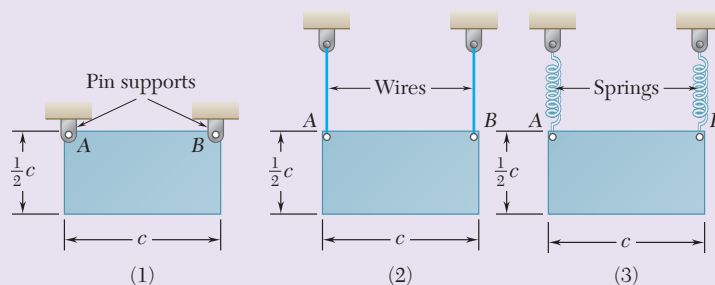


Fig. P16.160

- 16.161** A cylinder with a circular hole is rolling without slipping on a fixed curved surface as shown. The cylinder would have a weight of 16 lb without the hole, but with the hole it has a weight of 15 lb. Knowing that at the instant shown the disk has an angular velocity of 5 rad/s clockwise, determine (a) the angular acceleration of the disk, (b) the components of the reaction force between the cylinder and the ground at this instant.

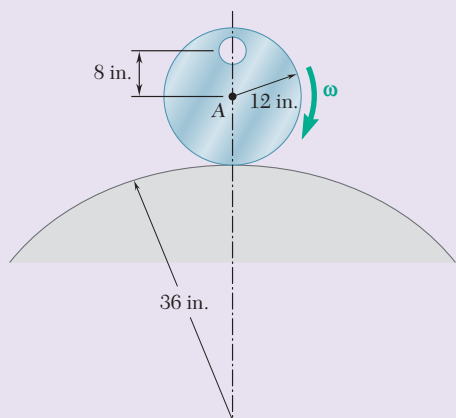


Fig. P16.161

- 16.162** The motion of a square plate of side 150 mm and mass 2.5 kg is guided by pins at corners A and B that slide in slots cut in a vertical wall. Immediately after the plate is released from rest in the position shown, determine (a) the angular acceleration of the plate, (b) the reaction at corner A.

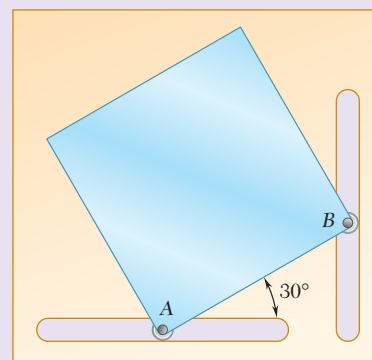


Fig. P16.162

- 16.163** The motion of a square plate of side 150 mm and mass 2.5 kg is guided by a pin at corner A that slides in a horizontal slot cut in a vertical wall. Immediately after the plate is released from rest in the position shown, determine (a) the angular acceleration of the plate, (b) the reaction at corner A.

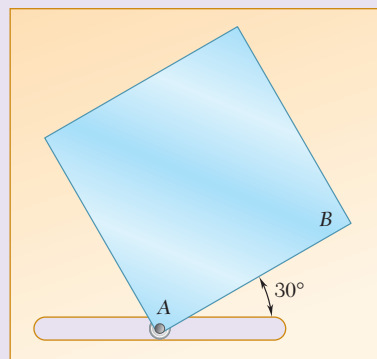


Fig. P16.163

16.164 The Geneva mechanism shown is used to provide an intermittent rotary motion of disk S . Disk D weighs 2 lb and has a radius of gyration of 0.9 in., and disk S weighs 6 lb and has a radius of gyration of 1.5 in. The motion of the system is controlled by a couple $\mathbf{M}$ applied to disk D . A pin P is attached to disk D and can slide in one of the six equally spaced slots cut in disk S . It is desirable that the angular velocity of disk S be zero as the pin enters and leaves each of the six slots; this will occur if the distance between the centers of the disks and the radii of the disks are related as shown. Knowing disk D rotates with a constant counterclockwise angular velocity of 8 rad/s and the friction between the slot and pin P is negligible, determine when $\phi = 150^\circ$ (a) the couple $\mathbf{M}$, (b) the magnitude of the force pin P applies to disk S .

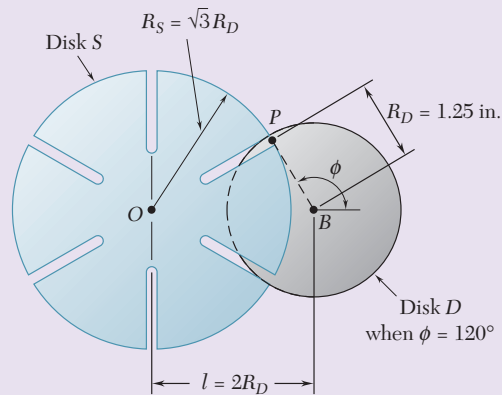


Fig. P16.164



17

Plane Motion of Rigid Bodies: Energy and Momentum Methods

In this chapter the energy and momentum methods will be added to the tools available for your study of the motion of rigid bodies. We can analyze the transfer between potential and kinetic energy as the gymnast goes from a high position to a lower one, and we can use conservation of angular momentum to examine how changes in the gymnast's body position affect his angular velocity.

Introduction

17.1 ENERGY METHODS FOR A RIGID BODY

- 17.1A. Principle of Work and Energy
- 17.1B. Work of Forces Acting on a Rigid Body
- 17.1C. Kinetic Energy of a Rigid Body in Plane Motion
- 17.1D. Systems of Rigid Bodies
- 17.1E. Conservation of Energy
- 17.1F. Power

17.2 MOMENTUM METHODS FOR A RIGID BODY

- 17.2A. Principle of Impulse and Momentum
- 17.2B. Systems of Rigid Bodies
- 17.2C. Conservation of Angular Momentum

17.3 ECCENTRIC IMPACT

Objectives

- **Calculate** the work done by a force or a moment on a rigid body.
- **Calculate** the kinetic energy of a rigid body in plane motion.
- **Solve** rigid body kinetics problems using the principle of work and energy.
- **Solve** rigid body kinetics problems using conservation of energy.
- **Calculate** the power of a mechanical system of rigid bodies.
- **Draw** complete and accurate impulse–momentum diagrams for problems involving rigid bodies.
- **Solve** rigid body kinetics problems using the principles of linear impulse and momentum and of angular impulse and momentum.
- **Solve** rigid body kinetics problems using conservation of angular momentum.
- **Solve** rigid body problems involving eccentric impact by using the principle of impulse and momentum and the coefficient of restitution.

Introduction

In this chapter, we return to the method of work and energy and the method of impulse and momentum that were introduced in Chapter 13 in the context of particle kinetics. Here we use them to analyze the plane motion of rigid bodies and of systems of rigid bodies.

We consider the method of work and energy first. We define the work of a force and of a couple, and we obtain an expression for the kinetic energy of a rigid body in plane motion. Then we use the principle of work and energy to solve problems involving displacements and velocities. We also apply the principle of conservation of energy to solve a variety of engineering problems.

In the second section, we apply the principle of impulse and momentum to solve problems involving velocities and time. We also discuss the concept of conservation of angular momentum for rigid bodies in plane motion.

In the last section of this chapter, we consider problems involving the eccentric impact of rigid bodies. As we did in Chap. 13, where we analyzed the impact of particles, we use the coefficient of restitution between colliding bodies, together with the principle of impulse and momentum, to solve impact problems. We will show that the method used is applicable not only when the colliding bodies move freely after the impact but also when the bodies are partially constrained in their motion.

17.1 ENERGY METHODS FOR A RIGID BODY

We now use the principle of work and energy to analyze the plane motion of rigid bodies. As we pointed out in Chap. 13, the method of work and energy is particularly well adapted to solving problems involving velocities and displacements. Its main advantage is that the work of forces and the kinetic energy of particles are scalar quantities.

17.1A Principle of Work and Energy

To apply the principle of work and energy to the motion of a rigid body, we again assume that the rigid body is made up of a large number n of particles of mass Δm_i . From Eq. (14.30) of Sec. 14.2B, we have

**Principle of work
and energy, rigid body**

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (17.1)$$

where T_1, T_2 = the initial and final values of total kinetic energy of particles forming the rigid body

$U_{1 \rightarrow 2}$ = work of all forces acting on various particles of the body

Just as we did in Chap. 13, we can express the work done by nonconservative forces as $U_{1 \rightarrow 2}^{\text{NC}}$, and we can define potential energy terms for conservative forces. Then we can express Eq. (17.1) as

$$T_1 + V_{g_1} + V_{e_1} + U_{1 \rightarrow 2}^{\text{NC}} = T_2 + V_{g_2} + V_{e_2} \quad (17.1')$$

where V_{g_1} and V_{g_2} are the initial and final gravitational potential energy of the center of mass of the rigid body with respect to a reference point or datum, and V_{e_1} and V_{e_2} are the initial and final values of the elastic energy associated with springs in the system.

We obtain the total kinetic energy

$$T = \frac{1}{2} \sum_{i=1}^n \Delta m_i v_i^2 \quad (17.2)$$

by adding positive scalar quantities, so it is itself a positive scalar quantity. You will see later how to determine T for various types of motion of a rigid body.

The expression $U_{1 \rightarrow 2}$ in Eq. (17.1) represents the work of all the forces acting on the various particles of the body, whether these forces are internal or external. However, the total work of the internal forces holding together the particles of a rigid body is zero. To see this, consider two particles A and B of a rigid body and the two equal and opposite forces $\mathbf{F}$ and $-\mathbf{F}$ they exert on each other (Fig. 17.1). Although, in general, small displacements $d\mathbf{r}$ and $d\mathbf{r}'$ of the two particles are different, the components of these displacements along AB must be equal; otherwise, the particles would not remain at the same distance from each other and the body would not be rigid. Therefore, the work of $\mathbf{F}$ is equal in magnitude and opposite



Photo 17.1 The work done by friction reduces the kinetic energy of the wheel.

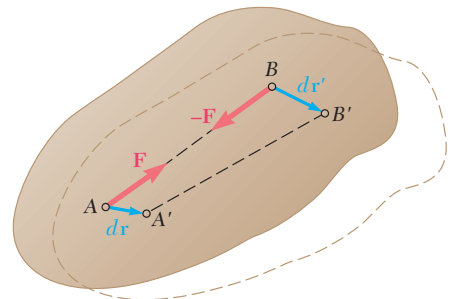


Fig. 17.1 The total work of the internal forces acting on the particles of a rigid body is zero.

in sign to the work of $-\mathbf{F}$, and their sum is zero. Thus, the total work of the internal forces acting on the particles of a rigid body is zero, and the expression $U_{1 \rightarrow 2}$ in Eq. (17.1) reduces to the work of the external forces acting on the body during the displacement considered.

17.1B Work of Forces Acting on a Rigid Body

We saw in Sec. 13.1A that the work of a force $\mathbf{F}$ during a displacement of its point of application from A_1 to A_2 is

Work of a force

$$U_{1 \rightarrow 2} = \int_{A_1}^{A_2} \mathbf{F} \cdot d\mathbf{r} \quad (17.3)$$

or

$$U_{1 \rightarrow 2} = \int_{s_1}^{s_2} (F \cos \alpha) ds \quad (17.3')$$

where F is the magnitude of the force, α is the angle it forms with the direction of motion of its point of application A , and s is the variable of integration that measures the distance traveled by A along its path.

In computing the work of the external forces acting on a rigid body, it is often convenient to determine the work of a couple without considering the work of each of the two forces forming the couple separately. Consider the two forces $\mathbf{F}$ and $-\mathbf{F}$ forming a couple of moment $\mathbf{M}$ and acting on a rigid body (Fig. 17.2). Any small displacement of the rigid body bringing A and B , respectively, into A' and B'' can be divided into two parts: in one part, points A and B undergo equal displacements $d\mathbf{r}_1$; in the other part, A' remains fixed, while B' moves into B'' through a displacement $d\mathbf{r}_2$ with a magnitude of $ds_2 = r d\theta$. In the first part of the motion, the work of $\mathbf{F}$ is equal in magnitude and opposite in sign to the work of $-\mathbf{F}$, and their sum is zero. In the second part of the motion, only force $\mathbf{F}$ works, and its work is $dU = F ds_2 = Fr d\theta$. But the product Fr is equal to the magnitude M of the moment of the couple. Thus, the work of a couple of moment $\mathbf{M}$ acting on a rigid body is

$$dU = M d\theta \quad (17.4)$$

where $d\theta$ is the small angle through which the body rotates and is expressed in radians. (We again note that work should be expressed in units obtained by multiplying units of force by units of length.) To obtain the work of the couple during a finite rotation of the rigid body, we integrate both members of Eq. (17.4) from the initial value θ_1 of the angle θ to its final value θ_2 .

$$U_{1 \rightarrow 2} = \int_{\theta_1}^{\theta_2} M d\theta \quad (17.5)$$

When the moment $\mathbf{M}$ of the couple is constant, formula (17.5) reduces to

$$U_{1 \rightarrow 2} = M(\theta_2 - \theta_1) \quad (17.6)$$

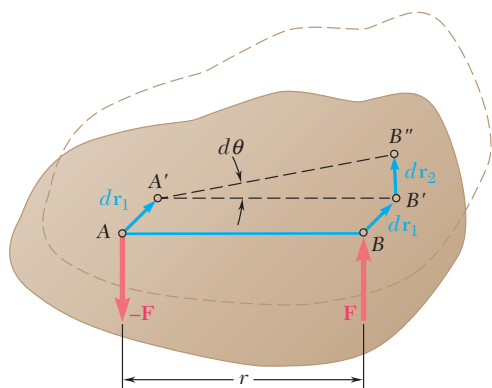


Fig. 17.2 The work of a couple acting on a rigid body equals the integral of the moment $\mathbf{M}$ of the couple with respect to the angular displacement of the body.

We pointed out in Sec. 13.1A that some forces encountered in problems of kinetics *do no work*. These include forces applied to fixed points or acting in a direction perpendicular to the displacement of their point of application. Among these forces are the reaction at a frictionless pin when the body rotates about the pin; the reaction at a frictionless surface when the body in contact moves along the surface; and the weight of a body when its center of gravity moves horizontally. We can now add that

When a rigid body rolls without sliding on a fixed surface, the friction force F at the point of contact C does no work.

The velocity $\mathbf{v}_C$ of the point of contact C is zero, and the work of the friction force $\mathbf{F}$ during a small displacement of the rigid body is

$$dU = F ds_C = F(v_C dt) = 0$$

17.1C Kinetic Energy of a Rigid Body in Plane Motion

Consider a rigid body with a mass m in plane motion. Recall from Sec. 14.2A that, if the absolute velocity $\mathbf{v}_i$ of each particle P_i of the body is expressed as the sum of the velocity $\bar{\mathbf{v}}$ of the mass center G of the body and of the velocity $\mathbf{v}'_i$ of the particle relative to a frame $Gx'y'$ attached to G and of fixed orientation (Fig. 17.3), we can express the kinetic energy of the system of particles forming the rigid body in the form

$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\sum_{i=1}^n \Delta m_i v_i'^2 \quad (17.7)$$

As you can see in Fig. 17.3, v'_i of particle P_i is equal to the product $r'_i\omega$, where r'_i is the distance from G to P_i and ω is the angular velocity of the body at the instant considered. Substituting into Eq. (17.7), we have

$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\left(\sum_{i=1}^n r_i'^2 \Delta m_i\right)\omega^2 \quad (17.8)$$

The sum represents the moment of inertia $\bar{I}$ of the body about the axis through G , so we have

Kinetic energy of a rigid body

$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\bar{I}\omega^2 \quad (17.9)$$

Note that, in the particular case of a body in translation ($\omega = 0$), this expression reduces to $\frac{1}{2}m\bar{v}^2$, whereas in the case of a centroidal rotation ($\bar{v} = 0$), it reduces to $\frac{1}{2}\bar{I}\omega^2$. We conclude that we can separate the kinetic energy of a rigid body in plane motion into two parts: (1) the kinetic energy $\frac{1}{2}m\bar{v}^2$ associated with the motion of the mass center G of the body and (2) the kinetic energy $\frac{1}{2}\bar{I}\omega^2$ associated with the rotation of the body about G .

Noncentroidal Rotation. The relation in Eq. (17.9) is valid for any type of plane motion, so we can use it to express the kinetic energy

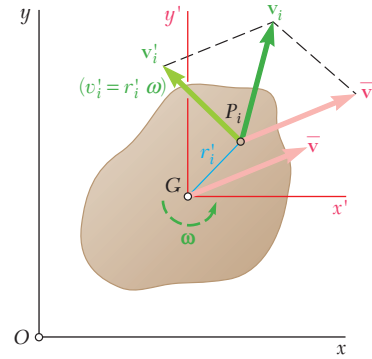


Fig. 17.3 The velocity of a particle P_i is the vector sum of the velocity of the mass center G and the tangential velocity $r'_i\omega$ due to rotation about G .

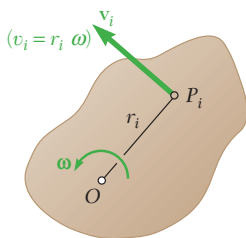


Fig. 17.4 For noncentroidal rotation, the velocity of a particle P_i is the tangential velocity $r_i\omega$ due to rotation about O .

of a rigid body rotating with an angular velocity ω about a fixed axis through O (Fig. 17.4). In that case, however, we can express the kinetic energy of the body more directly by noting that the speed v_i of particle P_i is equal to $r_i\omega$, where r_i is the distance from the fixed axis to P_i and ω is the angular velocity of the body at the instant considered. Substituting into Eq. (17.2), we have

$$T = \frac{1}{2} \sum_{i=1}^n \Delta m_i (r_i \omega)^2 = \frac{1}{2} \left(\sum_{i=1}^n r_i^2 \Delta m_i \right) \omega^2$$

The last sum represents the moment of inertia I_O of the body about the fixed axis through O , so this equation reduces to

$$T = \frac{1}{2} I_O \omega^2 \quad (17.10)$$

Note that these results are not limited to the motion of plane rigid bodies or to the motion of bodies that are symmetrical with respect to the reference plane—we can apply them to the study of the plane motion of any rigid body regardless of its shape. However, remember that Eq. (17.9) is applicable to any plane motion, whereas Eq. (17.10) is applicable only in cases involving rotating about a fixed axis.

17.1D Systems of Rigid Bodies

When a problem involves several rigid bodies, we usually analyze all of the bodies together as a system instead of analyzing each individual rigid body separately. Adding the kinetic energies of all the rigid bodies and considering the work of all the forces involved, we can write the equation of work and energy for the entire system. We have

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (17.11)$$

where T represents the arithmetic sum of the kinetic energies of the rigid bodies forming the system (all terms are positive) and $U_{1 \rightarrow 2}$ represents the work of all the forces acting on the various bodies—whether these forces are *internal* or *external* from the point of view of the system as a whole.

The method of work and energy is particularly useful in solving problems involving pin-connected members, blocks and pulleys connected by inextensible cords, and meshed gears. In all of these cases, the internal forces occur in pairs of equal and opposite forces, and the points of application of the forces in each pair *move through equal distances* during a small displacement of the system. As a result, the work of the internal forces is zero, and $U_{1 \rightarrow 2}$ reduces to the work of the *forces external to the system*.

17.1E Conservation of Energy

We saw in Sec. 13.2A that the work of conservative forces, such as the weight of a body or the force exerted by a spring, can be expressed as a change in potential energy. When a rigid body, or a system of rigid bodies, moves under the action of conservative forces, we can express the principle

of work and energy in a modified form. Substituting for $U_{1 \rightarrow 2}$ from Eq. (13.19') into Eq. (17.1), we have

Conservation of energy, rigid body

$$T_1 + V_1 = T_2 + V_2 \quad (17.12)$$

In Ch. 13, we discussed two types of potential energy: gravitational potential energy, V_g , and elastic potential energy, V_e . Therefore, another way to write Eq. (17.12) is

$$T_1 + V_{g1} + V_{e1} = T_2 + V_{g2} + V_{e2} \quad (17.12')$$

Formulas (17.12) and (17.12') indicate that when a rigid body, or a system of rigid bodies, moves under the action of conservative forces, **the sum of the kinetic energy and of the potential energy of the system remains constant**. Note that, in the case of the plane motion of a rigid body, the kinetic energy of the body should include both the *translational* term $\frac{1}{2}m\bar{v}^2$ and the *rotational* term $\frac{1}{2}\bar{I}\omega^2$.

As an example of applying the principle of conservation of energy, let us consider a slender rod AB with a length l and a mass m , whose ends are connected to blocks of negligible mass sliding along horizontal and vertical tracks. We assume that the rod is released with no initial velocity from a horizontal position (Fig. 17.5a), and we wish to determine its angular velocity after it has rotated through an angle θ (Fig. 17.5b).

Since the initial velocity is zero, we have $T_1 = 0$. Measuring the potential energy from the level of the horizontal track, we have $V_1 = 0$. After the rod has rotated through θ , the center of gravity G of the rod is at a distance $\frac{1}{2}l \sin \theta$ below the reference level, and we have

$$V_2 = -\frac{1}{2}Wl \sin \theta = -\frac{1}{2}mgl \sin \theta$$

In this position, the instantaneous center of the rod is located at C and $CG = \frac{1}{2}l$, so $\bar{v}_2 = \frac{1}{2}l\omega$, and we obtain

$$\begin{aligned} T_2 &= \frac{1}{2}m\bar{v}_2^2 + \frac{1}{2}\bar{I}\omega_2^2 = \frac{1}{2}m\left(\frac{1}{2}l\omega\right)^2 + \frac{1}{2}\left(\frac{1}{12}ml^2\right)\omega^2 \\ &= \frac{1}{2} \frac{ml^2}{3} \omega^2 \end{aligned}$$

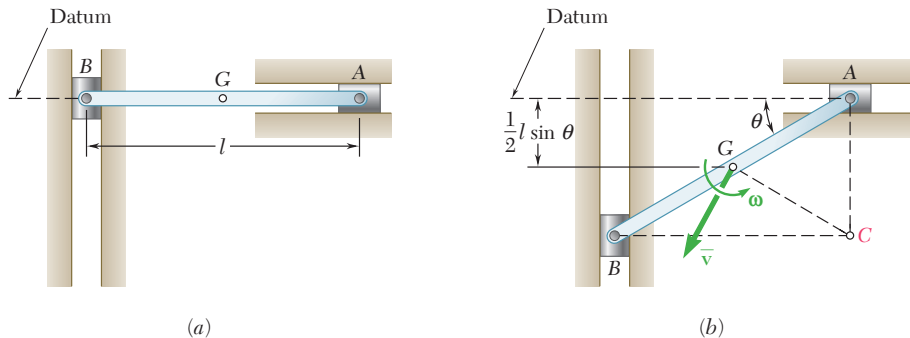


Fig. 17.5 (a) Rod AB in position 1 with the datum defined as shown. (b) Rod AB in position 2 with an instantaneous center C .

Applying the principle of conservation of energy gives

$$\begin{aligned}
 T_1 + V_1 &= T_2 + V_2 \\
 0 &= \frac{1}{2} \frac{ml^2}{3} \omega^2 - \frac{1}{2} mgl \sin \theta \\
 \omega &= \left(\frac{3g}{l} \sin \theta \right)^{1/2}
 \end{aligned}$$

The advantages of the method of work and energy, as well as its shortcomings, were indicated in Sec. 13.1C. Here we should add that we need to supplement the method of work and energy by the application of Newton's second law when we need to determine reactions at fixed axles, rollers, or sliding blocks. For example, in order to compute the reactions at the ends *A* and *B* of the rod of Fig. 17.5*b*, we need to draw a free-body diagram and a kinetic diagram to show that the system of the external forces applied to the rod is equivalent to both the vector $m\bar{\mathbf{a}}$ and the couple $\bar{I}\alpha$. However, we first need to determine the angular velocity ω of the rod using the method of work and energy before solving the equations of motion for the reactions. The complete analysis of the motion of the rod and of the forces exerted on the rod requires, therefore, the combined use of the method of work and energy and of the principle of equivalence of the external forces and moments and inertial terms.

17.1F Power

We defined **power** in Sec. 13.1D as the time rate at which work is done. In the case of a body acted upon by a force $\mathbf{F}$ and moving with a velocity $\mathbf{v}$, we expressed the power as

$$\text{Power} = \frac{dU}{dt} = \mathbf{F} \cdot \mathbf{v} \quad (13.13)$$

In the case of a rigid body rotating with an angular velocity ω and acted upon by a couple of moment $\mathbf{M}$ parallel to the axis of rotation, we have, by Eq. (17.4),

$$\text{Power} = \frac{dU}{dt} = \frac{Md\theta}{dt} = M\omega \quad (17.13)$$

The various units used to measure power, such as the watt and the horsepower, were defined in Sec. 13.1D.

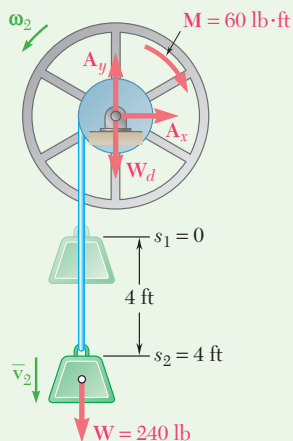
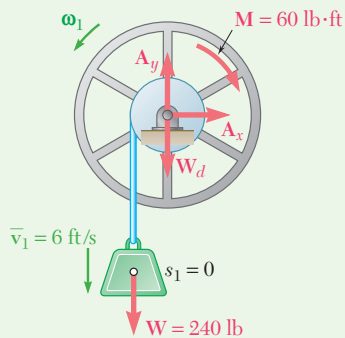
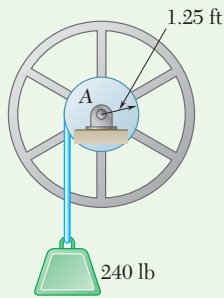


Fig. 1 Free body diagram of the system in positions 1 and 2.

Sample Problem 17.1

A 240-lb block is suspended from an inextensible cable that is wrapped around a drum with a 1.25-ft radius that is rigidly attached to a flywheel. The drum and flywheel have a combined centroidal moment of inertia of $\bar{I} = 10.5 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$. At the instant shown, the velocity of the block is 6 ft/s directed downward. Knowing that the bearing at A is poorly lubricated so that the bearing friction is equivalent to a couple $\mathbf{M}$ of magnitude 60 lb·ft, determine the velocity of the block after it has moved 4 ft downward.

STRATEGY: Since you have two positions and are interested in determining the velocity of the block, use the principle of work and energy.

MODELING: Consider the system formed by the flywheel and the block. Since the cable is inextensible, the work done by the internal forces exerted by the cable cancels out to zero. The initial and final positions of the system and the external forces acting on the system are shown in Fig. 1.

ANALYSIS: Apply the principle of work and energy

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (1)$$

Kinetic Energy. You need to calculate the initial and final kinetic energy and the work.

Position 1.

Block: $\bar{v}_1 = 6 \text{ ft/s}$

Flywheel: $\omega_1 = \frac{\bar{v}_1}{r} = \frac{6 \text{ ft/s}}{1.25 \text{ ft}} = 4.80 \text{ rad/s}$

$$\begin{aligned} T_1 &= \frac{1}{2} m \bar{v}_1^2 + \frac{1}{2} \bar{I} \omega_1^2 \\ &= \frac{1}{2} \frac{240 \text{ lb}}{32.2 \text{ ft/s}^2} (6 \text{ ft/s})^2 + \frac{1}{2} (10.5 \text{ lb}\cdot\text{ft}\cdot\text{s}^2) (4.80 \text{ rad/s})^2 \\ &= 255 \text{ ft}\cdot\text{lb} \end{aligned}$$

Position 2. Noting that $\omega_2 = \bar{v}_2/1.25$, you have

$$\begin{aligned} T_2 &= \frac{1}{2} m \bar{v}_2^2 + \frac{1}{2} \bar{I} \omega_2^2 \\ &= \frac{1}{2} \frac{240}{32.2} (\bar{v}_2)^2 + \left(\frac{1}{2}\right) (10.5) \left(\frac{\bar{v}_2}{1.25}\right)^2 = 7.09 \bar{v}_2^2 \end{aligned}$$

(continued)

Work. During the motion, only the weight $\mathbf{W}$ of the block and the friction couple $\mathbf{M}$ do work. Note that $\mathbf{W}$ does positive work, and the friction couple $\mathbf{M}$ does negative work. The total work done is

$$\begin{aligned}s_1 &= 0 & s_2 &= 4 \text{ ft} \\ \theta_1 &= 0 & \theta_2 &= \frac{s_2}{r} = \frac{4 \text{ ft}}{1.25 \text{ ft}} = 3.20 \text{ rad} \\ U_{1 \rightarrow 2} &= W(s_2 - s_1) - M(\theta_2 - \theta_1) \\ &= (240 \text{ lb})(4 \text{ ft}) - (60 \text{ lb}\cdot\text{ft})(3.20 \text{ rad}) \\ &= 768 \text{ ft}\cdot\text{lb}\end{aligned}$$

Substituting these expressions into Eq. (1) gives

$$\begin{aligned}T_1 + U_{1 \rightarrow 2} &= T_2 \\ 255 \text{ ft}\cdot\text{lb} + 768 \text{ ft}\cdot\text{lb} &= 7.09 \bar{v}_2^2 \\ \bar{v}_2 &= 12.01 \text{ ft/s} & \bar{v}_2 &= 12.01 \text{ ft/s} \downarrow \quad \blacktriangleleft\end{aligned}$$

REFLECT and THINK: The speed of the block increases as it falls, but much more slowly than if it were in free fall. This seems like a reasonable result. Rather than calculating the work done by gravity, you could have also treated the effect of the weight using gravitational potential energy, V_g .

Sample Problem 17.2

Gear A has a mass of 10 kg and a radius of gyration of 200 mm; gear B has a mass of 3 kg and a radius of gyration of 80 mm. The system is at rest when a couple $\mathbf{M}$ of magnitude 6 N·m is applied to gear B. Neglecting friction, determine (a) the number of revolutions executed by gear B before its angular velocity reaches 600 rpm, (b) the tangential force that gear B exerts on gear A.

STRATEGY: You are given a couple and are asked to determine the position at a given angular velocity, so use the principle of work and energy.

MODELING: For part (a), choose the system to be both gears and model each as a rigid body. In part (b), you are asked to determine an internal force, so you need to choose gear A as your system.

ANALYSIS:

Kinematics. The velocity of the point of contact, P , is the same for both gears (Fig. 1), so you have

$$v_P = r_A \omega_A = r_B \omega_B \quad \omega_A = \omega_B \frac{r_B}{r_A} = \omega_B \frac{100 \text{ mm}}{250 \text{ mm}} = 0.40 \omega_B$$

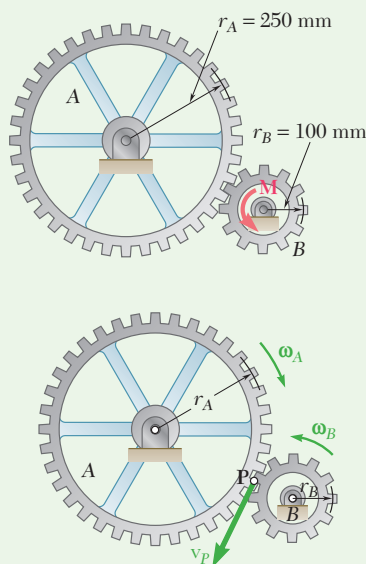


Fig. 1 The point of contact has the same velocity on each gear.

Calculations. For $\omega_B = 600$ rpm, you have

$$\begin{aligned}\omega_B &= 62.8 \text{ rad/s} & \omega_A &= 0.40\omega_B = 25.1 \text{ rad/s} \\ \bar{I}_A &= m_A \bar{k}_A^2 = (10 \text{ kg})(0.200 \text{ m})^2 = 0.400 \text{ kg}\cdot\text{m}^2 \\ \bar{I}_B &= m_B \bar{k}_B^2 = (3 \text{ kg})(0.080 \text{ m})^2 = 0.0192 \text{ kg}\cdot\text{m}^2\end{aligned}$$

Principle of Work and Energy: Apply the principle of work and energy

$$T_1 + U_{1\rightarrow 2} = T_2 \quad (1)$$

You need to calculate the initial and final kinetic energy and the work.

Kinetic Energy. The system is initially at rest, so $T_1 = 0$. Adding the kinetic energies of the two gears when $\omega_B = 600$ rpm gives

$$\begin{aligned}T_2 &= \frac{1}{2}\bar{I}_A\omega_A^2 + \frac{1}{2}\bar{I}_B\omega_B^2 \\ &= \frac{1}{2}(0.400 \text{ kg}\cdot\text{m}^2)(25.1 \text{ rad/s})^2 + \frac{1}{2}(0.0192 \text{ kg}\cdot\text{m}^2)(62.8 \text{ rad/s})^2 \\ &= 163.9 \text{ J}\end{aligned}$$

Work. Denote the angular displacement of gear B by θ_B . Then

$$U_{1\rightarrow 2} = M\theta_B = (6 \text{ N}\cdot\text{m})(\theta_B \text{ rad}) = (6\theta_B) \text{ J}$$

Substituting these terms into Eq. (1) gives you

$$\begin{aligned}0 + (6\theta_B) \text{ J} &= 163.9 \text{ J} \\ \theta_B &= 27.32 \text{ rad} & \theta_B &= 4.35 \text{ rev} \quad \blacktriangleleft\end{aligned}$$

Motion of Gear A.

Kinetic Energy. Initially, gear A is at rest, so $T_1 = 0$. When $\omega_B = 600$ rpm, the kinetic energy of gear A is

$$T_2 = \frac{1}{2}\bar{I}_A\omega_A^2 = \frac{1}{2}(0.400 \text{ kg}\cdot\text{m}^2)(25.1 \text{ rad/s})^2 = 126.0 \text{ J}$$

Work. The forces acting on gear A are shown in Fig. 2. The tangential force $\mathbf{F}$ does work equal to the product of its magnitude and of the length $\theta_A r_A$ of the arc described by the point of contact. Since $\theta_A r_A = \theta_B r_B$, you have

$$U_{1\rightarrow 2} = F(\theta_B r_B) = F(27.3 \text{ rad})(0.100 \text{ m}) = F(2.73 \text{ m})$$

Substituting these values into work and energy gives

$$\begin{aligned}T_1 + U_{1\rightarrow 2} &= T_2 \\ 0 + F(2.73 \text{ m}) &= 126.0 \text{ J} \\ F &= +46.2 \text{ N} & \mathbf{F} &= 46.2 \text{ N} \quad \blacktriangleleft\end{aligned}$$

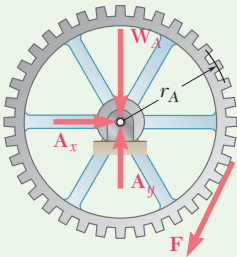


Fig. 2 Free-body diagram for gear A.

REFLECT and THINK: When the system was both gears, the tangential force between the gears did not appear in the work and energy equation, since it was internal to the system and therefore did no work. If you want to determine an internal force, you need to define a system where the force of interest is an external force. This problem, like most problems, also could have been solved using Newton's second law and kinematic relationships.

Sample Problem 17.3

A sphere, a cylinder, and a hoop, each having the same mass and the same radius, are released from rest on an incline. Determine the velocity of each body after it has rolled through a distance corresponding to a change in elevation h .

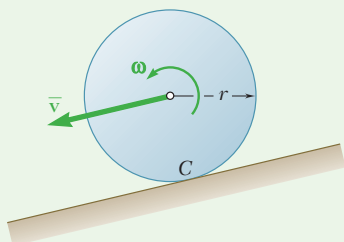


Fig. 1 Angular velocity and the velocity of the center of mass of the rolling object.

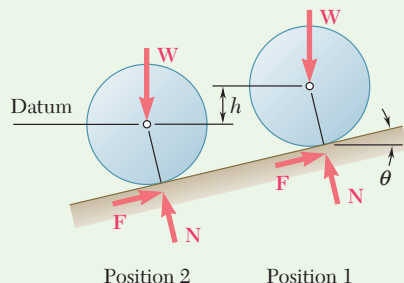


Fig. 2 Free-body diagrams of the system in positions 1 and 2.

STRATEGY: You are given two positions, want to find the velocities, and the friction force $\mathbf{F}$ in rolling motion does no work, so use the conservation of energy. First solve the problem in general terms, and then find the results for each body. Denote the mass by m , the centroidal moment of inertia by $\bar{I}$, and the radius by r .

MODELING: Choose the rolling object as your system and model it as a rigid body. Since each body rolls, the instantaneous center of rotation is located at C (Fig. 1). Free-body diagrams of at the two locations are shown in Fig. 2.

ANALYSIS:

Conservation of Energy.

$$T_1 + V_{g1} + V_{e1} = T_2 + V_{g2} + V_{e2} \quad (1)$$

Potential Energy.

Because there is no spring in the system, $V_{e1} = V_{e2} = 0$. If you place your datum at the center of mass of the system when it is at position 2, you have $V_{g2} = 0$ and $V_{g1} = mgh$.

Kinetic Energy.

$$\begin{aligned} T_1 &= 0 \\ T_2 &= \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\bar{I}\omega^2 \end{aligned}$$

Kinematics. You need to relate $\bar{v}$ and ω using kinematics. Since each body rolls, the instantaneous center of rotation is located at C (Fig. 1), which gives

$$\omega = \frac{\bar{v}}{r}$$

Substituting this into T_2 gives

$$T_2 = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\bar{I}\left(\frac{\bar{v}}{r}\right)^2 = \frac{1}{2}\left(m + \frac{\bar{I}}{r^2}\right)\bar{v}^2$$

Substituting these energy expressions into Eq. (1) gives

$$0 + mgh + 0 = \frac{1}{2}\left(m + \frac{\bar{I}}{r^2}\right)\bar{v}^2 + 0 + 0$$

Solving for the speed at position 2, you find

$$\bar{v}^2 = \frac{2gh}{1 + \bar{I}/mr^2}$$

Velocities of Sphere, Cylinder, and Hoop. Introducing the particular expressions for $\bar{I}$, you obtain

Sphere: $\bar{I} = \frac{2}{5}mr^2$ $\bar{v} = 0.845\sqrt{2gh}$ ◀

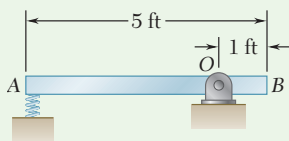
Cylinder: $\bar{I} = \frac{1}{2}mr^2$ $\bar{v} = 0.816\sqrt{2gh}$ ◀

Hoop: $\bar{I} = mr^2$ $\bar{v} = 0.707\sqrt{2gh}$ ◀

REFLECT and THINK: Comparing the results, we note that the velocity of the body is independent of both its mass and radius. However, the velocity does depend upon the quotient of $\bar{I}/mr^2 = \bar{k}^2/r^2$, which measures the ratio of the rotational kinetic energy to the translational kinetic energy. Thus the hoop, which has the largest $\bar{k}$ for a given radius r , attains the smallest velocity, whereas the sliding block, which does not rotate, attains the largest velocity.

Let us compare the results with the velocity attained by a frictionless block sliding through the same distance. The solution is identical to the previous solution except that $\omega = 0$; we find $\bar{v} = \sqrt{2gh}$. So, all the rolling objects are slower than one moving down a frictionless surface.

Sample Problem 17.4



A 30-lb slender rod AB is 5 ft long and is pivoted about a point O that is 1 ft from end B . The other end is pressed against a vertical spring with a constant of $k = 1800$ lb/in. until the spring is compressed 1 in. The rod is then in a horizontal position. If the rod is released from this position, determine its angular velocity and the reaction at the pivot O as the rod passes through a vertical position.

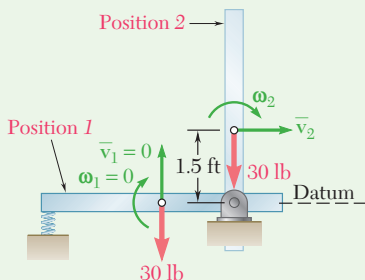


Fig. 1 The rod in positions 1 and 2.

STRATEGY: Since you are given two positions, want to find the velocities, and no external forces do work, use the conservation of energy. To determine the reactions at position 2, use a free-body diagram and a kinetic diagram.

MODELING: Choose the rod and the spring as your system and model the rod as a rigid body. Denote the initial position as position 1 and the vertical position as position 2 (Fig. 1). Choose your datum to be at position 1.

ANALYSIS:

Conservation of Energy.

$$T_1 + V_{g_1} + V_{e_1} = T_2 + V_{g_2} + V_{e_2} \quad (1)$$

You need to calculate the energy at position 1 and position 2.

(continued)

Position 1.

Potential Energy. The spring is compressed 1 in., so you have $x_1 = 1$ in. The elastic potential energy is

$$V_{e_1} = \frac{1}{2}kx_1^2 = \frac{1}{2}(1800 \text{ lb/in.})(1 \text{ in.})^2 = 900 \text{ in}\cdot\text{lb} = 75 \text{ ft}\cdot\text{lb}$$

Since the datum is at position 1, you have $V_{g_1} = 0$.

Kinetic Energy. The velocity in position 1 is zero, so you have $T_1 = 0$.

Position 2.

Potential Energy. The elongation of the spring is zero, so you have $V_{e_2} = 0$. Since the center of gravity of the rod is now 1.5 ft above the datum, you have

$$V_{g_2} = mgy = (30 \text{ lb})(1.5 \text{ ft}) = 45 \text{ ft}\cdot\text{lb}$$

Kinetic Energy. Denote the angular velocity of the rod in position 2 by ω_2 . The rod rotates about O , so you have $\bar{v}_2 = \bar{r}\omega_2 = 1.5\omega_2$ and

$$\bar{I} = \frac{1}{12}ml^2 = \frac{1}{12} \frac{30 \text{ lb}}{32.2 \text{ ft/s}^2} (5 \text{ ft})^2 = 1.941 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$$

$$T_2 = \frac{1}{2}m\bar{v}_2^2 + \frac{1}{2}\bar{I}\omega_2^2 = \frac{1}{2} \frac{30}{32.2} (1.5\omega_2)^2 + \frac{1}{2}(1.941)\omega_2^2 = 2.019\omega_2^2$$

Substituting these expressions into Eq. (1) give

$$0 + 0 + 75 \text{ ft}\cdot\text{lb} = 2.019\omega_2^2 + 45 \text{ ft}\cdot\text{lb} + 0 \quad \omega_2 = 3.86 \text{ rad/s} \quad \blacktriangleleft$$

Reaction. Since $\omega_2 = 3.86 \text{ rad/s}$, the components of the acceleration of G as the rod passes through position 2 are (Fig. 2)

$$\begin{aligned} \bar{a}_n &= \bar{r}\omega_2^2 = (1.5 \text{ ft})(3.86 \text{ rad/s})^2 = 22.3 \text{ ft/s}^2 & \bar{a}_n &= 22.3 \text{ ft/s}^2 \downarrow \\ \bar{a}_t &= \bar{r}\alpha & \bar{a}_t &= \bar{r}\alpha \rightarrow \end{aligned}$$

Draw free-body and kinetic diagrams (Fig. 3) to express that the system of external forces is equivalent to the vector of components $m\bar{a}_t$ and $m\bar{a}_n$ attached at G and the couple $\bar{I}\alpha$.

$$+\downarrow \Sigma M_O = \bar{I}\alpha + m\bar{a}_n\bar{r}: \quad 0 = \bar{I}\alpha + m(\bar{r}\alpha)\bar{r} \quad \alpha = 0$$

$$+\rightarrow \Sigma F_x = m\bar{a}_t: \quad R_x = m(\bar{r}\alpha) \quad R_x = 0$$

$$+\uparrow \Sigma F_y = m\bar{a}_y: \quad R_y - 30 \text{ lb} = -m\bar{a}_n$$

$$R_y - 30 \text{ lb} = -\frac{30 \text{ lb}}{32.2 \text{ ft/s}^2} (22.3 \text{ ft/s}^2)$$

$$R_y = +9.22 \text{ lb} \quad \mathbf{R} = 9.22 \text{ lb} \uparrow \quad \blacktriangleleft$$

REFLECT and THINK: This problem illustrates how you might need to supplement the conservation of energy with Newton's second law. What if the spring constant had been smaller, say 180 lb/in.? You would have found $V_{e_1} = 7.5 \text{ ft}\cdot\text{lb}$ and then solved Eq. (1) to obtain $\omega_2^2 = -18.57$. This is clearly impossible and means that the rod would not make it to position 2 as assumed.

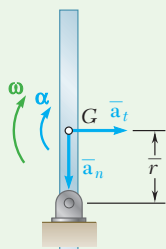


Fig. 2 The acceleration of the center of mass and the angular velocity and acceleration of the rod.

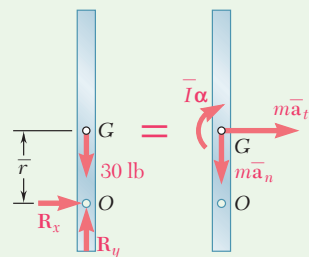


Fig. 3 Free-body diagram and kinetic diagram for the rod.

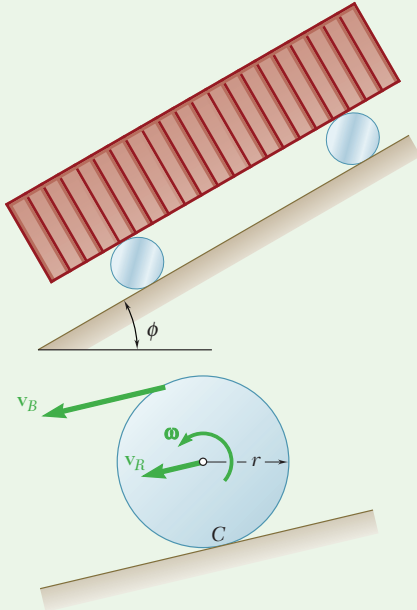


Fig. 1 Velocity of various points on the roller.

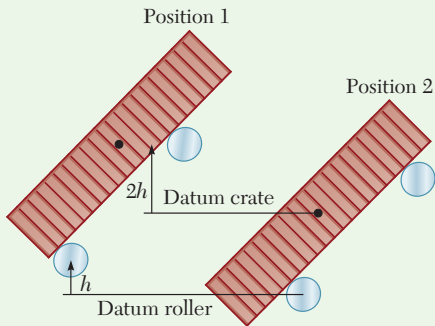


Fig. 2 The system in positions 1 and 2.

Sample Problem 17.5

A large box with a mass m and a flat bottom rests on two identical homogeneous cylindrical rollers, where each has radius r and a mass half that of the crate. The system is released from rest on a plane that is inclined at angle ϕ to the horizontal. Determine the speed of the box at the instant when the rollers have turned through an angle θ . Neglect rolling resistance and assume that the rollers do not slide.

STRATEGY: You are interested in the velocity after the rollers have moved a specified distance, $r\theta$, and the friction force in rolling motion does no work, so use the conservation of energy.

MODELING: Choose the box and the two cylindrical rollers as your system and model them as rigid bodies. In order for you to draw the system in its initial and final positions, you need to know how far each mass travels. You can determine this by using the instantaneous center of rotation. The rollers do not slide, so the instantaneous center of rotation for each roller is located at the point of contact, C , between the roller and the ground (Fig. 1). Using this instantaneous center of velocity, you know $v_B = 2\omega r$ and $v_R = \omega r$. Therefore, the box moves down a distance $2h$ when the rollers move a distance h (Fig. 2). Since you have three masses in the system (the two rollers and the box), you may define an individual datum for each mass to simplify the calculation of the gravitational potential energy.

ANALYSIS:

Conservation of Energy.

$$T_1 + V_{g1} + V_{e1} = T_2 + V_{g2} + V_{e2} \quad (1)$$

You need to calculate the energy at position 1 and position 2.

Potential Energy. Because there is no spring in the system, $V_{e1} = V_{e2} = 0$. If you place your datum at the center of mass of each object when the system is at position 2, you have $V_{g2} = 0$. The vertical distance a roller moves is $h = r\theta\sin\phi$, so

$$V_{g1} = mg(2h) + 2\left(\frac{m}{2}\right)g(h) = 3mgh = 3mgr\theta\sin\phi$$

Kinetic Energy. The velocity in position 1 is zero, so $T_1 = 0$.

At position 2,

$$T_2 = 2\left(\frac{1}{2}m_R v_R^2 + \frac{1}{2}\bar{I}\omega^2\right) + \frac{1}{2}mv_B^2$$

where m_R is the mass of the roller and $\bar{I}$ is the mass moment of inertia of the roller about its center of gravity. Substituting $v_B = 2\omega r$, $v_R = \omega r$, and $\bar{I} = \frac{1}{2}m_R r^2 = \frac{1}{2}\left(\frac{m}{2}\right)r^2 = \frac{1}{4}mr^2$ into T_2 gives

$$T_2 = \frac{11}{4}mr^2\omega^2$$

(continued)

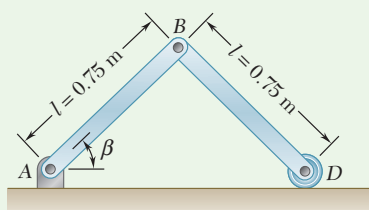
Substituting these expressions into Eq. (1), you find

$$0 + 3mgr\theta\sin\phi + 0 = \frac{11}{4}mr^2\omega^2 + 0 + 0$$

Solving for the angular velocity, $\omega = \sqrt{\frac{12g\theta\sin\phi}{11r}}$, so the velocity of the box at $v_B = 2\omega r$ is

$$v_B = 4\sqrt{\frac{3}{11}gr\theta\sin\phi} \quad \blacktriangleleft$$

REFLECT and THINK: If the rollers had been attached to the box by brackets, they would have traveled the same vertical distance as the box and the change in height of the centers of gravity of rollers and of the box would have been equal.



Sample Problem 17.6

Each of the two slender rods shown is 0.75 m long and has a mass of 6 kg. If the system is released from rest with $\beta = 60^\circ$, determine (a) the angular velocity of rod AB when $\beta = 20^\circ$, (b) the velocity of point D at the same instant.

STRATEGY: You have two positions and are interested in velocities, so use the conservation of energy. You will also need to use kinematics to relate the velocity terms in the kinetic energy expression.

MODELING: Choose the system to be both bars and model them as rigid bodies.

ANALYSIS: To illustrate that the order in which you solve a problem doesn't matter, let's start with kinematics.

Kinematics of Motion When $\beta = 20^\circ$. Since $\mathbf{v}_B$ is perpendicular to the rod AB and $\mathbf{v}_D$ is horizontal, the instantaneous center of rotation of rod BD is located at C (Fig. 1). From the geometry of the figure, you obtain

$$BC = 0.75 \text{ m} \quad CD = 2(0.75 \text{ m}) \sin 20^\circ = 0.513 \text{ m}$$

Apply the law of cosines to triangle CDE, where E is located at the mass center of rod BD. You find $EC = 0.522 \text{ m}$. Denoting the angular velocity of rod AB by ω , you have (Fig. 2)

$$\begin{aligned} \bar{v}_{AB} &= (0.375 \text{ m})\omega & \bar{v}_{AB} &= 0.375\omega \searrow \\ v_B &= (0.75 \text{ m})\omega & v_B &= 0.75\omega \searrow \end{aligned}$$

Since rod BD seems to rotate about point C, you have

$$\begin{aligned} v_B &= (BC)\omega_{BD} & (0.75 \text{ m})\omega &= (0.75 \text{ m})\omega_{BD} & \omega_{BD} &= \omega \nearrow \\ \bar{v}_{BD} &= (EC)\omega_{BD} & (0.522 \text{ m})\omega & & \bar{v}_{BD} &= 0.522\omega \searrow \end{aligned}$$

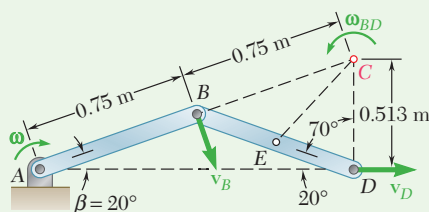


Fig. 1 Instantaneous center of rotation C for bar BD.

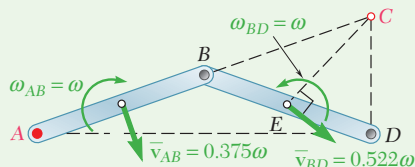


Fig. 2 Velocities of the center of masses of AB and BD in terms of ω .

Conservation of Energy. Since there are no springs in the system

$$T_1 + V_{g_1} = T_2 + V_{g_2}$$

You first need to determine the energy at the two positions.

Position 1.

Potential Energy. Choose the datum as shown in Fig. 3, and observe that $W = (6 \text{ kg})(9.81 \text{ m/s}^2) = 58.86 \text{ N}$. Then you have

$$V_{g_1} = 2W\bar{y}_1 = 2(58.86 \text{ N})(0.325 \text{ m}) = 38.26 \text{ J}$$

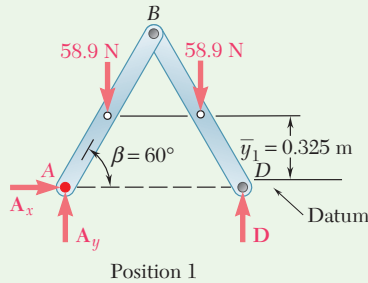


Fig. 3 Free-body diagram and distance from the datum in position 1.

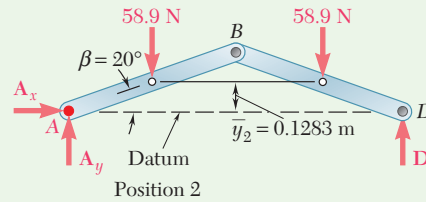


Fig. 4 Free-body diagram and distance from the datum in position 2.

Kinetic Energy. Initially, the system is at rest, so $T_1 = 0$.

Position 2.

Potential Energy. Compute the new height of the mass centers of the rods to be $\bar{y}_2 = 0.75 \sin(20) = 0.1283 \text{ m}$ (Fig. 4).

$$V_{g_2} = 2W\bar{y}_2 = 2(58.86 \text{ N})(0.1283 \text{ m}) = 15.10 \text{ J}$$

Kinetic Energy.

$$\begin{aligned} I_{AB} &= \bar{I}_{BD} = \frac{1}{12} ml^2 = \frac{1}{12} (6 \text{ kg})(0.75 \text{ m})^2 = 0.281 \text{ kg}\cdot\text{m}^2 \\ T_2 &= \frac{1}{2} m\bar{v}_{AB}^2 + \frac{1}{2} \bar{I}_{AB} \omega_{AB}^2 + \frac{1}{2} m\bar{v}_{BD}^2 + \frac{1}{2} \bar{I}_{BD} \omega_{BD}^2 \\ &= \frac{1}{2} (6)(0.375\omega)^2 + \frac{1}{2} (0.281)\omega^2 + \frac{1}{2} (6)(0.522\omega)^2 + \frac{1}{2} (0.281)\omega^2 \\ &= 1.520\omega^2 \end{aligned}$$

Conservation of Energy. Now you can write

$$\begin{aligned} T_1 + V_{g_1} &= T_2 + V_{g_2} \\ 0 + 38.26 \text{ J} &= 1.520\omega^2 + 15.10 \text{ J} \\ \omega &= 3.90 \text{ rad/s} \quad \omega_{AB} = 3.90 \text{ rad/s} \downarrow \end{aligned}$$

Velocity of Point D.

$$\begin{aligned} v_D &= (CD)\omega = (0.513 \text{ m})(3.90 \text{ rad/s}) = 2.00 \text{ m/s} \\ v_D &= 2.00 \text{ m/s} \rightarrow \end{aligned}$$

REFLECT and THINK: The only step in which you need to use forces is when calculating the gravitational potential energy in each position. However, it is good engineering practice to show the complete free-body diagram in each case to identify which, if any, forces do work. Rather than use the instantaneous center of rotation, you could have also used vector algebra to relate the velocities of the various objects.

SOLVING PROBLEMS ON YOUR OWN

In this section, we introduced energy methods to determine the velocity of rigid bodies for various positions during their motion. As you saw previously in Chap. 13, energy methods are particularly useful for problems involving displacements and velocities.

1. The method of work and energy, when applied to all of the particles forming a rigid body, yields the equation

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (17.1)$$

where T_1 and T_2 are, respectively, the initial and final values of the total kinetic energy of the particles forming the body and $U_{1 \rightarrow 2}$ is the work done by the external forces exerted on the rigid body. If we express the work done by nonconservative forces as $U_{1 \rightarrow 2}^{\text{NC}}$ and define the potential energy terms for conservative forces, we can express Eq. (17.1) as

$$T_1 + V_{g_1} + V_{e_1} + U_{1 \rightarrow 2}^{\text{NC}} = T_2 + V_{g_2} + V_{e_2} \quad (17.1')$$

where V_{g_1} and V_{g_2} are the initial and final gravitational potential energy of the center of mass of the rigid body and V_{e_1} and V_{e_2} are the initial and final values of the elastic energy associated with springs in the system. Recall that, for a linear spring, $V_e = \frac{1}{2}kx^2$, where x is the deflection of the spring from its unstretched length. For a single rigid body, $V_g = mgy$, where y is the elevation of the center of mass from a reference plane or datum.

a. Work of forces and couples. To the expression for the work of a force (Chap. 13), we added the expression for the work of a couple and wrote

$$U_{1 \rightarrow 2} = \int_{A_1}^{A_2} \mathbf{F} \cdot d\mathbf{r} \quad U_{1 \rightarrow 2} = \int_{\theta_1}^{\theta_2} M d\theta \quad (17.3, 17.5)$$

When the moment of a couple is constant, the work of the couple is

$$U_{1 \rightarrow 2} = M(\theta_2 - \theta_1) \quad (17.6)$$

where θ_1 and θ_2 are expressed in radians [Sample Probs. 17.1 and 17.2].

b. The kinetic energy of a rigid body in plane motion was found by considering the motion of the body as the sum of a translation with its mass center and a rotation about the mass center. So

$$T = \frac{1}{2} m \bar{v}^2 + \frac{1}{2} \bar{I} \omega^2 \quad (17.9)$$

where $\bar{v}$ is the velocity of the mass center and ω is the angular velocity of the body [Sample Probs. 17.3 and 17.4]. You will generally need to use kinematics to relate $\bar{v}$ and ω .

2. For a system of rigid bodies we again used the equation

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (17.1)$$

where T is the sum of the kinetic energies of the bodies forming the system and U is the work done by *all the forces acting on the bodies*—internal as well as external. Your computations will be simplified if you keep the following ideas in mind.

a. The forces exerted on each other by pin-connected members or by meshed gears are equal and opposite, and since they have the same point of application, they undergo equal small displacements. Therefore, *their total work is zero* and can be omitted from your calculations [Sample Prob. 17.2].

b. The forces exerted by an inextensible cord on the two bodies it connects have the same magnitude and their points of application move through equal distances, but the work of one force is positive and the work of the other is negative. Therefore, *their total work is zero* and again can be omitted from your calculations [Sample Prob. 17.1].

c. The forces exerted by a spring on the two bodies it connects also have the same magnitude, but their points of application generally move through different distances. Therefore, *their total work is usually not zero* and should be taken into account in your calculations. The easiest way to handle springs, therefore, is to use elastic potential energy.

3. The principle of conservation of energy can be expressed as

$$T_1 + V_1 = T_2 + V_2 \quad (17.12)$$

where V represents the potential energy of the system. If you prefer to write this equation in terms of gravitational potential energy, V_g , and elastic potential energy, V_e , you get

$$T_1 + V_{g1} + V_{e1} = T_2 + V_{g2} + V_{e2} \quad (17.12')$$

You can use this principle when a body or a system of bodies is acted upon by conservative forces, such as the force exerted by a spring or the force of gravity [Sample Probs. 17.4 through 17.6].

4. The last part of this section was devoted to power, which is the time rate at which work is done. For a body acted upon by a couple of moment $\mathbf{M}$, the power can be expressed as

$$\text{Power} = M\omega \quad (17.13)$$

where ω is the angular velocity of the body, expressed in rad/s. As you did in Chap. 13, you should express power either in watts or in horsepower (1 hp = 550 ft·lb/s).

Problems

CONCEPT QUESTIONS

- 17.CQ1** A round object of mass m and radius r is released from rest at the top of a curved surface and rolls without slipping until it leaves the surface with a horizontal velocity as shown. Will a solid sphere, a solid cylinder, or a hoop travel the greatest distance x ?
- Solid sphere
 - Solid cylinder
 - Hoop
 - They will all travel the same distance.

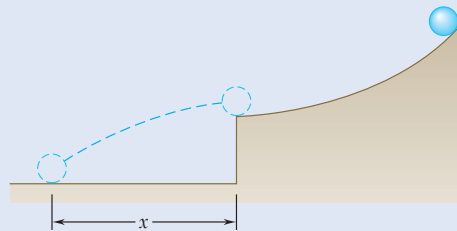


Fig. P17.CQ1

- 17.CQ2** A solid steel sphere A of radius r and mass m is released from rest and rolls without slipping down an incline as shown. After traveling a distance d , the sphere has a speed v . If a solid steel sphere of radius $2r$ is released from rest on the same incline, what will its speed be after rolling a distance d ?
- $0.25v$
 - $0.5v$
 - v
 - $2v$
 - $4v$

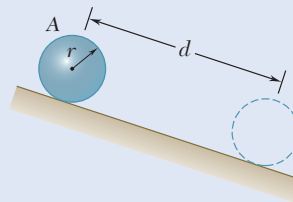


Fig. P17.CQ2

- 17.CQ3** Slender bar A is rigidly connected to a massless rod BC in Case 1 and two massless cords in Case 2 as shown. The vertical thickness of bar A is negligible compared to L . In both cases A is released from rest at an angle $\theta = \theta_0$. When $\theta = 0^\circ$, which system will have the larger kinetic energy?
- Case 1
 - Case 2
 - The kinetic energy will be the same.

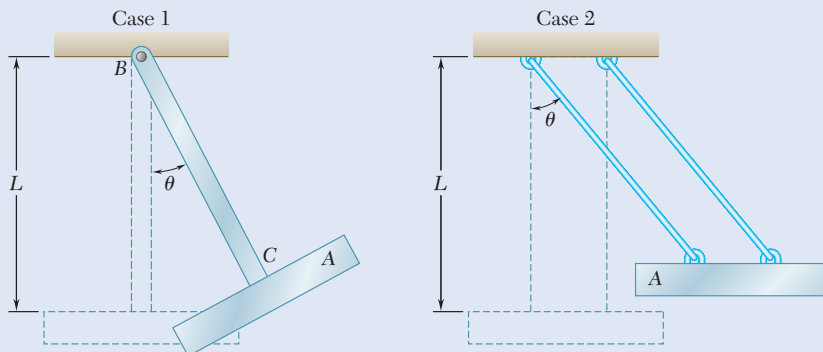


Fig. P17.CQ3 and P17.CQ5

- 17.CQ4** In Prob. 17.CQ3, how will the speeds of the centers of gravity compare for the two cases when $\theta = 0^\circ$?
- Case 1 will be larger.
 - Case 2 will be larger.
 - The speeds will be the same.
- 17.CQ5** Slender bar A is rigidly connected to a massless rod BC in Case 1 and two massless cords in Case 2 as shown. The vertical thickness of bar A is not negligible compared to L . In both cases A is released from rest at an angle $\theta = \theta_0$. When $\theta = 0^\circ$, which system will have the larger kinetic energy?
- Case 1
 - Case 2
 - The kinetic energy will be the same.

END-OF-SECTION PROBLEMS

- 17.1** A 200-kg flywheel is at rest when a constant 300 N·m couple is applied. After executing 560 revolutions, the flywheel reaches its rated speed of 2400 rpm. Knowing that the radius of gyration of the flywheel is 400 mm, determine the average magnitude of the couple due to kinetic friction in the bearing.
- 17.2** The rotor of an electric motor has an angular velocity of 3600 rpm when the load and power are cut off. The 110-lb rotor, which has a centroidal radius of gyration of 9 in., then coasts to rest. Knowing that the kinetic friction of the rotor produces a couple with a magnitude of 2.5 lb·ft, determine the number of revolutions that the rotor executes before coming to rest.

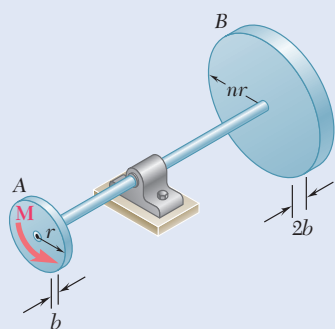


Fig. P17.3 and P17.4

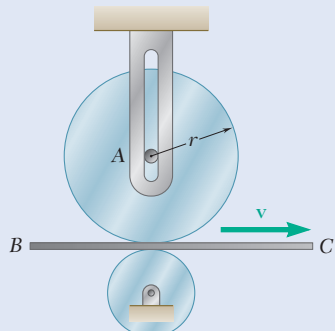


Fig. P17.7

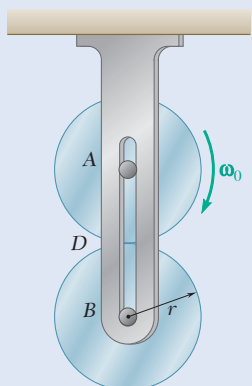


Fig. P17.8

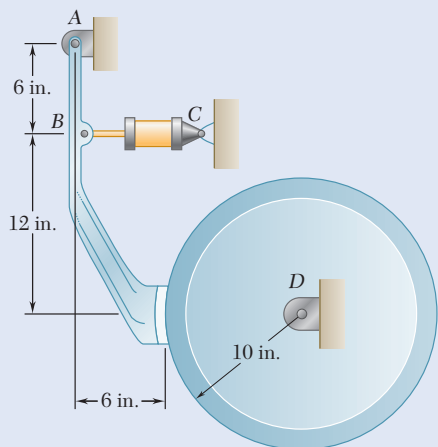


Fig. P17.9

17.3 Two uniform disks of the same material are attached to a shaft as shown. Disk A has a weight of 10 lb and a radius of $r = 6$ in. Disk B is twice as thick as disk A. Knowing that a couple $\mathbf{M}$ with a magnitude of 22 lb·ft is applied to disk A when the system is at rest, determine the radius nr of disk B if the angular velocity of the system is to be 480 rpm after five revolutions.

17.4 Two disks of the same material are attached to a shaft as shown. Disk A has a radius r and a thickness b , while disk B has a radius nr and a thickness $2b$. A couple $\mathbf{M}$ with a constant magnitude is applied when the system is at rest and is removed after the system has executed two revolutions. Determine the value of n that results in the largest final speed for a point on the rim of disk B.

17.5 The flywheel of a small punch rotates at 300 rpm. It is known that 1800 ft·lb of work must be done each time a hole is punched. It is desired that the speed of the flywheel after one punching be not less than 90 percent of the original speed of 300 rpm. (a) Determine the required moment of inertia of the flywheel. (b) If a constant 25-lb·ft couple is applied to the shaft of the flywheel, determine the number of revolutions that must occur between each punching, knowing that the initial velocity is to be 300 rpm at the start of each punching.

17.6 The flywheel of a punching machine has a mass of 300 kg and a radius of gyration of 600 mm. Each punching operation requires 2500 J of work. (a) Knowing that the speed of the flywheel is 300 rpm just before a punching, determine the speed immediately after the punching. (b) If a constant 25-N·m couple is applied to the shaft of the flywheel, determine the number of revolutions executed before the speed is again 300 rpm.

17.7 Disk A, of weight 10 lb and radius $r = 6$ in., is at rest when it is placed in contact with belt BC, which moves to the right with a constant speed $v = 40$ ft/s. Knowing that $\mu_k = 0.20$ between the disk and the belt, determine the number of revolutions executed by the disk before it attains a constant angular velocity.

17.8 The uniform 4-kg cylinder A with a radius of $r = 150$ mm has an angular velocity of $\omega_0 = 50$ rad/s when it is brought into contact with an identical cylinder B that is at rest. The coefficient of kinetic friction at the contact point D is μ_k . After a period of slipping, the cylinders attain constant angular velocities of equal magnitude and opposite direction at the same time. Knowing that cylinder A executes three revolutions before it attains a constant angular velocity and cylinder B executes one revolution before it attains a constant angular velocity, determine (a) the final angular velocity of each cylinder, (b) the coefficient of kinetic friction μ_k .

17.9 The 10-in.-radius brake drum is attached to a larger flywheel which is not shown. The total mass moment of inertia of the flywheel and drum is 16 lb·ft·s² and the coefficient of kinetic friction between the drum and the brake shoe is 0.40. Knowing that the initial angular velocity is 240 rpm clockwise, determine the force that must be exerted by the hydraulic cylinder if the system is to stop in 75 revolutions.

17.10 Solve Prob. 17.9, assuming that the initial angular velocity of the flywheel is 240 rpm counterclockwise.

17.11 Each of the gears *A* and *B* has a mass of 2.4 kg and a radius of gyration of 60 mm, while gear *C* has a mass of 12 kg and a radius of gyration of 150 mm. A couple $\mathbf{M}$ of constant magnitude 10 N·m is applied to gear *C*. Determine (a) the number of revolutions of gear *C* required for its angular velocity to increase from 100 to 450 rpm, (b) the corresponding tangential force acting on gear *A*.

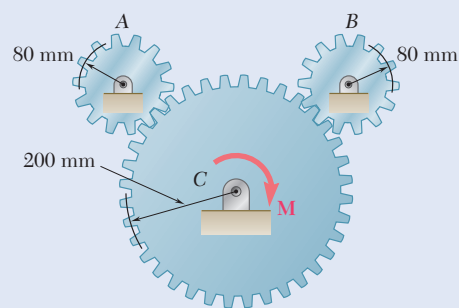


Fig. P17.11

17.12 Solve Prob. 17.11, assuming that the 10-N·m couple is applied to gear *B*.

17.13 The gear train shown consists of four gears of the same thickness and of the same material; two gears are of radius r , and the other two are of radius nr . The system is at rest when the couple $\mathbf{M}_0$ is applied to shaft *C*. Denoting by I_0 the moment of inertia of a gear of radius r , determine the angular velocity of shaft *A* if the couple $\mathbf{M}_0$ is applied for one revolution of shaft *C*.

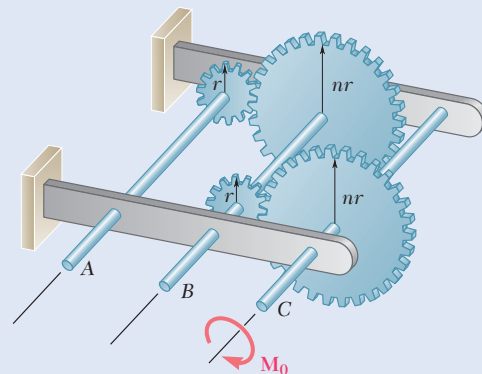


Fig. P17.13

17.14 The double pulley shown has a mass of 15 kg and a centroidal radius of gyration of 160 mm. Cylinder *A* and block *B* are attached to cords that are wrapped on the pulleys as shown. The coefficient of kinetic friction between block *B* and the surface is 0.2. Knowing that the system is at rest in the position shown when a constant force $\mathbf{P} = 200$ N is applied to cylinder *A*, determine (a) the velocity of cylinder *A* as it strikes the ground, (b) the total distance that block *B* moves before coming to rest.

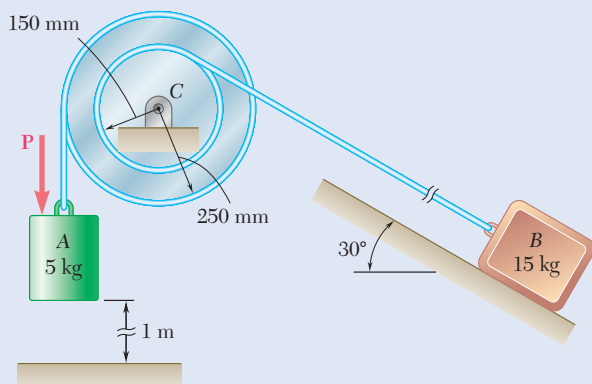


Fig. P17.14

17.15 Gear *A* has a mass of 1 kg and a radius of gyration of 30 mm; gear *B* has a mass of 4 kg and a radius of gyration of 75 mm; gear *C* has a mass of 9 kg and a radius of gyration of 100 mm. The system is at rest when a couple $\mathbf{M}_0$ of constant magnitude 4 N·m is applied to gear *C*. Assuming that no slipping occurs between the gears, determine the number of revolutions required for disk *A* to reach an angular velocity of 300 rpm.

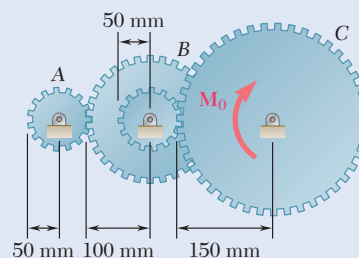


Fig. P17.15

17.16 A slender rod of length l and weight W is pivoted at one end as shown. It is released from rest in a horizontal position and swings freely. (a) Determine the angular velocity of the rod as it passes through a vertical position and determine the corresponding reaction at the pivot. (b) Solve part *a* for $W = 1.8$ lb and $l = 3$ ft.

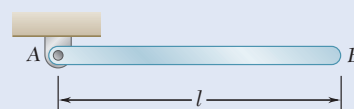


Fig. P17.16

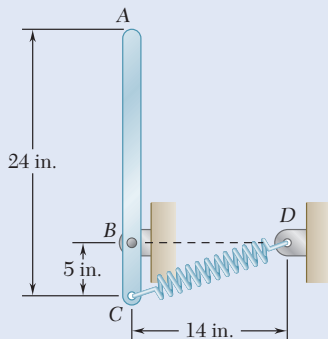


Fig. P17.18

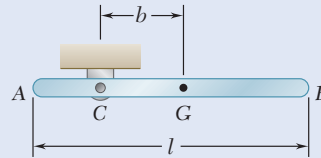


Fig. P17.17

- 17.17** A slender rod of length l is pivoted about a point C located at a distance b from its center G . It is released from rest in a horizontal position and swings freely. Determine (a) the distance b for which the angular velocity of the rod as it passes through a vertical position is maximum, (b) the corresponding values of its angular velocity and of the reaction at C .

- 17.18** A slender 9-lb rod can rotate in a vertical plane about a pivot at B . A spring of constant $k = 30$ lb/ft and of unstretched length 6 in. is attached to the rod as shown. Knowing that the rod is released from rest in the position shown, determine its angular velocity after it has rotated through 90° .

- 17.19** An adapted golf device attaches to a wheelchair to help people with mobility impairments play putt-putt. The stationary frame OD is attached to the wheelchair, and a club holder OB is attached to the pin at O . Holder OB is 6 in. long and weighs 8 oz, and the distance between O and D is $x = 1$ ft. The putter shaft has a length of $L = 36$ in. and weighs 10 oz, while the putter head at A weighs 12 oz. Knowing that the 1-lb/in. spring between D and B is unstretched when $\theta = 90^\circ$ and that the putter is released from rest at $\theta = 0$, determine the putter head speed when it hits the golf ball.

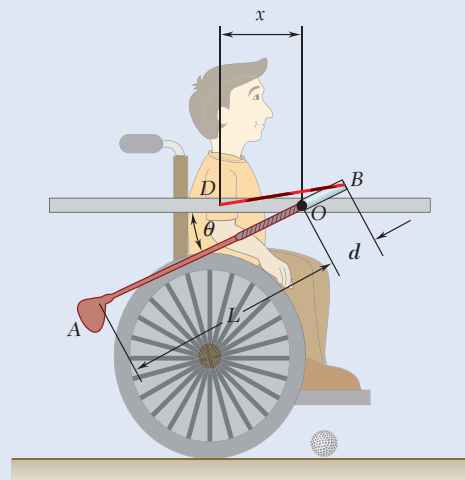


Fig. P17.19

- 17.20** A 160-lb gymnast is executing a series of full-circle swings on the horizontal bar. In the position shown, he has a small and negligible clockwise angular velocity and will maintain his body straight and rigid as he swings downward. Assuming that during the swing the centroidal radius of gyration of his body is 1.5 ft, determine his angular velocity and the force exerted on his hands after he has rotated through (a) 90° , (b) 180° .

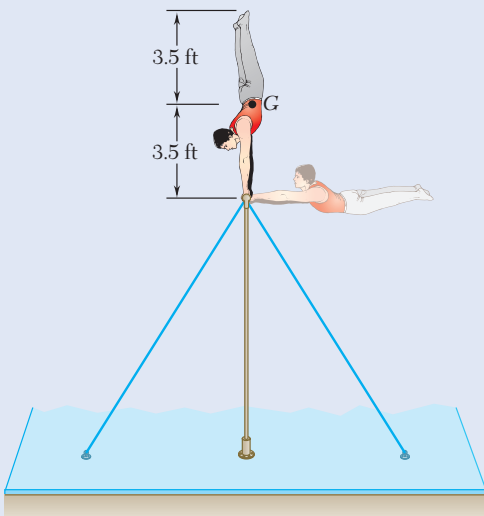


Fig. P17.20

- 17.21** A collar with a mass of 1 kg is rigidly attached at a distance $d = 300$ mm from the end of a uniform slender rod AB . The rod has a mass of 3 kg and is of length $L = 600$ mm. Knowing that the rod is released from rest in the position shown, determine the angular velocity of the rod after it has rotated through 90° .

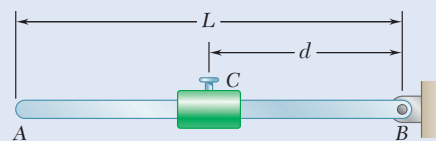


Fig. P17.21 and P17.22

- 17.22** A collar with a mass of 1 kg is rigidly attached to a slender rod AB of mass 3 kg and length $L = 600$ mm. The rod is released from rest in the position shown. Determine the distance d for which the angular velocity of the rod is maximum after it has rotated through 90° .

- 17.23** Two identical slender rods AB and BC are welded together to form an L-shaped assembly. The assembly is pressed against a spring at D and released from the position shown. Knowing that the maximum angle of rotation of the assembly in its subsequent motion is 90° counterclockwise, determine the magnitude of the angular velocity of the assembly as it passes through the position where rod AB forms an angle of 30° with the horizontal.

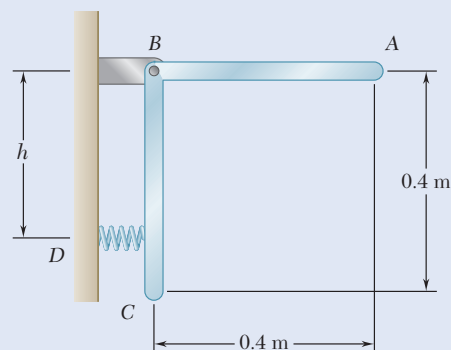


Fig. P17.23

- 17.24** The 30-kg turbine disk has a centroidal radius of gyration of 175 mm and is rotating clockwise at a constant rate of 60 rpm when a small blade of weight 0.5 N at point A becomes loose and is thrown off. Neglecting friction, determine the change in the angular velocity of the turbine disk after it has rotated through (a) 90° , (b) 270° .

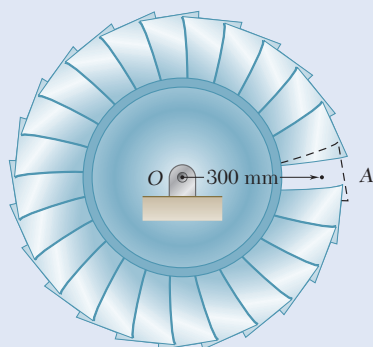


Fig. P17.24

- 17.25** A rope is wrapped around a cylinder of radius r and mass m as shown. Knowing that the cylinder is released from rest, determine the velocity of the center of the cylinder after it has moved downward a distance s .

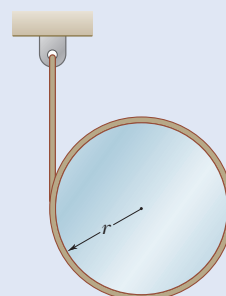


Fig. P17.25

- 17.26** Solve Prob. 17.25, assuming that the cylinder is replaced by a thin-walled pipe of radius r and mass m .

- 17.27** Greek engineers had the unenviable task of moving large columns from the quarries to the city. One engineer, Chersiphron, tried several different techniques to do this. One method was to cut pivot holes into the ends of the stone and then use oxen to pull the column. The 4-ft diameter column weighs 12,000 lbs, and the team of oxen generates a constant pull force of 1500 lbs on the center of the cylinder G . Knowing that the column starts from rest and rolls without slipping, determine (a) the velocity of its center G after it has moved 5 ft, (b) the minimum static coefficient of friction that will keep it from slipping.

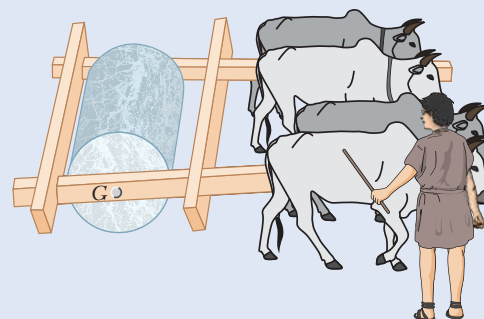


Fig. P17.27

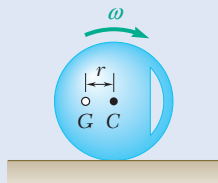


Fig. P17.29

- 17.28** A small sphere of mass m and radius r is released from rest at A and rolls without sliding on the curved surface to point B where it leaves the surface with a horizontal velocity. Knowing that $a = 1.5$ m and $b = 1.2$ m, determine (a) the speed of the sphere as it strikes the ground at C , (b) the corresponding distance c .

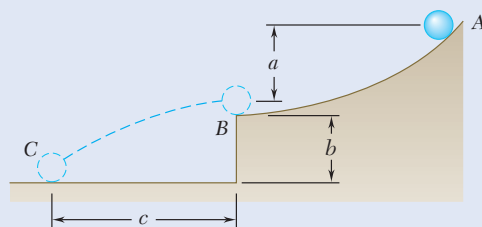


Fig. P17.28

- 17.29** The mass center G of a 3-kg wheel of radius $R = 180$ mm is located at a distance $r = 60$ mm from its geometric center C . The centroidal radius of gyration of the wheel is $\bar{k} = 90$ mm. As the wheel rolls without sliding, its angular velocity is observed to vary. Knowing that $\omega = 8$ rad/s in the position shown, determine (a) the angular velocity of the wheel when the mass center G is directly above the geometric center C , (b) the reaction at the horizontal surface at the same instant.
- 17.30** A half-cylinder with mass m and radius r is released from rest in the position shown. Knowing that the half-cylinder rolls without sliding, determine (a) its angular velocity after it has rolled through 90° , (b) the reaction at the horizontal surface at the same instant. [Hint: Note that $GO = 4r/3\pi$ and that, by the parallel-axis theorem, $\bar{I} = \frac{1}{2}mr^2 - m(GO)^2$.]

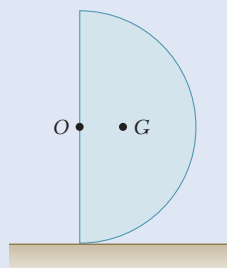


Fig. P17.30

- 17.31** A sphere of mass m and radius r rolls without slipping inside a curved surface of radius R . Knowing that the sphere is released from rest in the position shown, derive an expression for (a) the linear velocity of the sphere as it passes through B , (b) the magnitude of the vertical reaction at that instant.

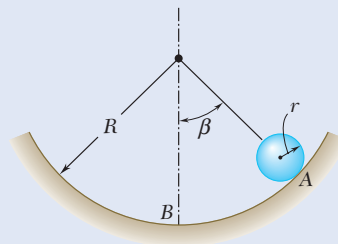


Fig. P17.31

17.32 Two uniform cylinders, each of weight $W = 14$ lb and radius $r = 5$ in., are connected by a belt as shown. Knowing that at the instant shown the angular velocity of cylinder B is 30 rad/s clockwise, determine (a) the distance through which cylinder A will rise before the angular velocity of cylinder B is reduced to 5 rad/s, (b) the tension in the portion of belt connecting the two cylinders.

17.33 Two uniform cylinders, each of weight $W = 14$ lb and radius $r = 5$ in., are connected by a belt as shown. If the system is released from rest, determine (a) the velocity of the center of cylinder A after it has moved through 3 ft, (b) the tension in the portion of belt connecting the two cylinders.

17.34 A bar of mass $m = 5$ kg is held as shown between four disks each of mass $m' = 2$ kg and radius $r = 75$ mm. Knowing that the forces exerted on the disks are sufficient to prevent slipping and that the bar is released from rest, for each of the cases shown, determine the velocity of the bar after it has moved through the distance h .

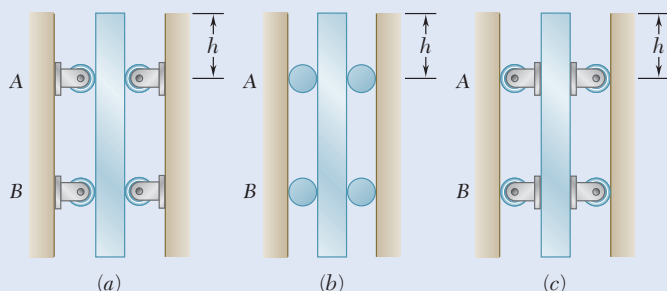


Fig. P17.34

17.35 The 1.5 -kg uniform slender bar AB is connected to the 3 -kg gear B that meshes with the stationary outer gear C . The centroidal radius of gyration of gear B is 30 mm. Knowing that the system is released from rest in the position shown, determine (a) the angular velocity of the bar as it passes through the vertical position, (b) the corresponding angular velocity of gear B .

17.36 The motion of the uniform rod AB is guided by small wheels of negligible mass that roll on the surface shown. If the rod is released from rest when $\theta = 0$, determine the velocities of A and B when $\theta = 30^\circ$.

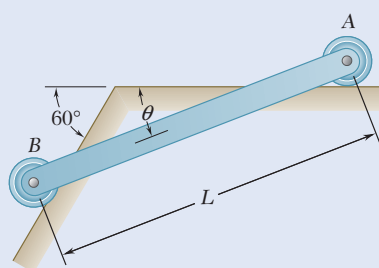


Fig. P17.36

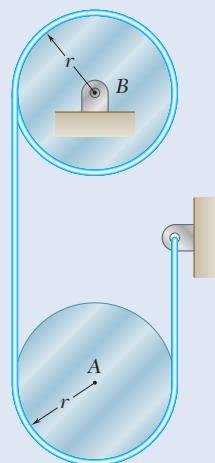


Fig. P17.32 and P17.33

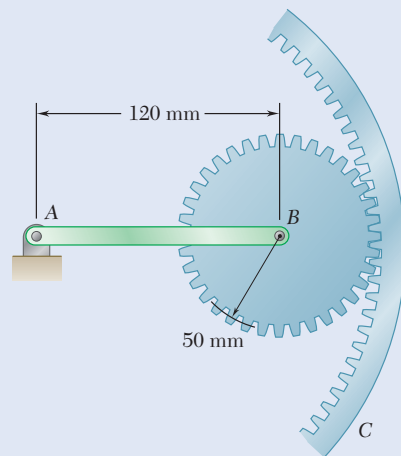


Fig. P17.35

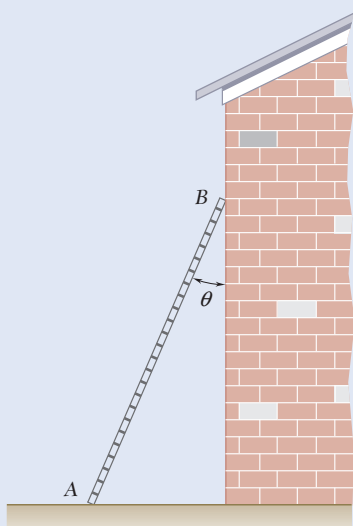


Fig. P17.37 and P17.38

- 17.37** A 5-m-long ladder has a mass of 15 kg and is placed against a house at an angle $\theta = 20^\circ$. Knowing that the ladder is released from rest, determine the angular velocity of the ladder and the velocity of end A when $\theta = 45^\circ$. Assume the ladder can slide freely on the horizontal ground and on the vertical wall.
- 17.38** A long ladder of length l , mass m , and centroidal mass moment of inertia $\bar{I}$ is placed against a house at an angle $\theta = \theta_0$. Knowing that the ladder is released from rest, determine the angular velocity of the ladder when $\theta = \theta_2$. Assume the ladder can slide freely on the horizontal ground and on the vertical wall.
- 17.39** The ends of a 9-lb rod AB are constrained to move along slots cut in a vertical plate as shown. A spring of constant $k = 3$ lb/in. is attached to end A in such a way that its tension is zero when $\theta = 0$. If the rod is released from rest when $\theta = 50^\circ$, determine the angular velocity of the rod and the velocity of end B when $\theta = 0$.

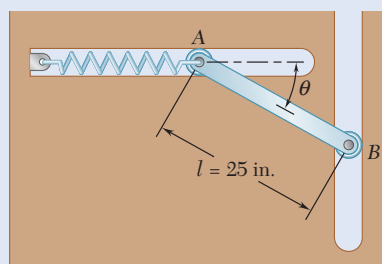


Fig. P17.39

- 17.40** The mechanism shown is one of two identical mechanisms attached to the two sides of a 200-lb uniform rectangular door. Edge ABC of the door is guided by wheels of negligible mass that roll in horizontal and vertical tracks. A spring with a constant of $k = 40$ lb/ft is attached to wheel B. Knowing that the door is released from rest in the position $\theta = 30^\circ$ with the spring unstretched, determine the velocity of wheel A just as the door reaches the vertical position.
- 17.41** The mechanism shown is one of two identical mechanisms attached to the two sides of a 200-lb uniform rectangular door. Edge ABC of the door is guided by wheels of negligible mass that roll in horizontal and vertical tracks. A spring with a constant k is attached to wheel B in such a way that its tension is zero when $\theta = 30^\circ$. Knowing that the door is released from rest in the position $\theta = 45^\circ$ and reaches the vertical position with an angular velocity of 0.6 rad/s, determine the spring constant k .

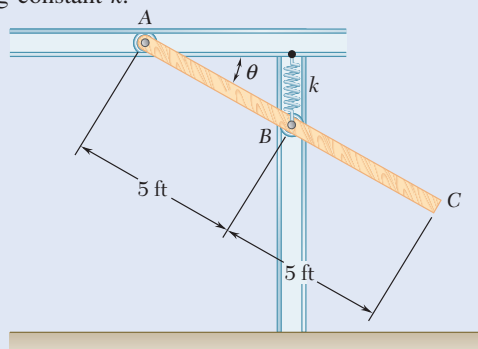


Fig. P17.40 and P17.41

17.42 Each of the two rods shown is of length $L = 1$ m and has a mass of 5 kg. Point D is connected to a spring of constant $k = 20$ N/m and is constrained to move along a vertical slot. Knowing that the system is released from rest when rod BD is horizontal and the spring connected to point D is initially unstretched, determine the velocity of point D when it is directly to the right of point A .

17.43 The 4-kg rod AB is attached to a collar of negligible mass at A and to a flywheel at B . The flywheel has a mass of 16 kg and a radius of gyration of 180 mm. Knowing that in the position shown the angular velocity of the flywheel is 60 rpm clockwise, determine the velocity of the flywheel when point B is directly below C .

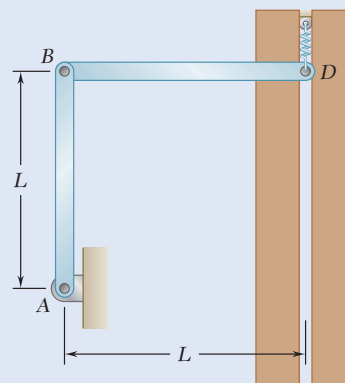


Fig. P17.42

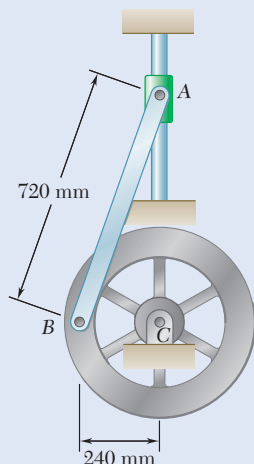


Fig. P17.43 and P17.44

17.44 If in Prob. 17.43 the angular velocity of the flywheel is to be the same in the position shown and when point B is directly above C , determine the required value of its angular velocity in the position shown.

17.45 The uniform rods AB and BC weigh 2.4 kg and 4 kg, respectively, and the small wheel at C is of negligible weight. If the wheel is moved slightly to the right and then released, determine the velocity of pin B after rod AB has rotated through 90° .

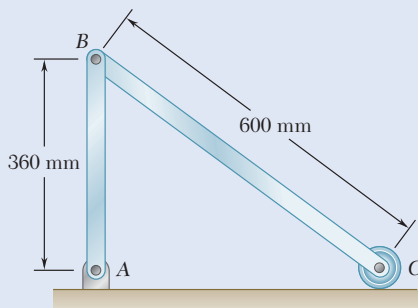


Fig. P17.45 and P17.46

17.46 The uniform rods AB and BC weigh 2.4 kg and 4 kg, respectively, and the small wheel at C is of negligible weight. Knowing that in the position shown the velocity of wheel C is 2 m/s to the right, determine the velocity of pin B after rod AB has rotated through 90° .

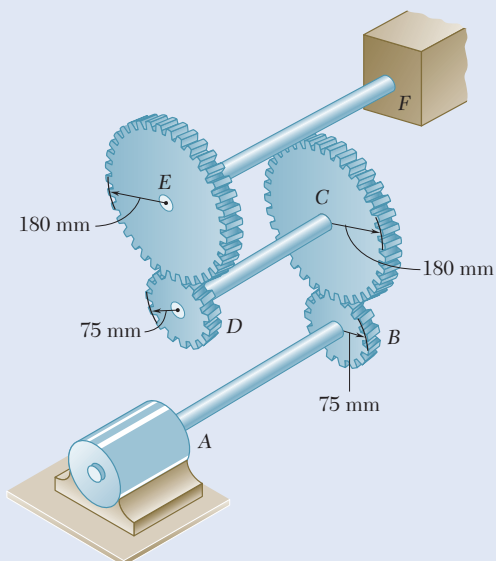


Fig. P17.49

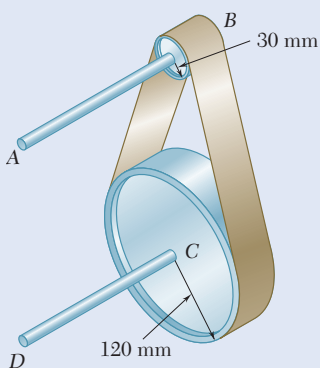


Fig. P17.50

- 17.47** The 80-mm-radius gear shown has a mass of 5 kg and a centroidal radius of gyration of 60 mm. The 4-kg rod AB is attached to the center of the gear and to a pin at B that slides freely in a vertical slot. Knowing that the system is released from rest when $\theta = 60^\circ$, determine the velocity of the center of the gear when $\theta = 20^\circ$.

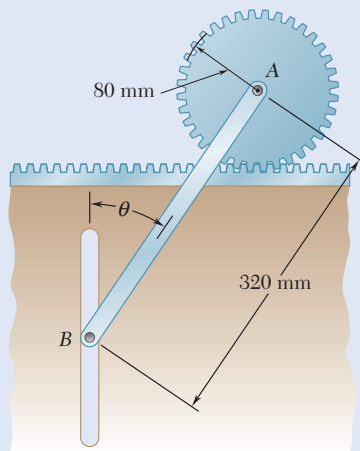


Fig. P17.47

- 17.48** Knowing that the maximum allowable couple that can be applied to a shaft is 15.5 kip·in., determine the maximum horsepower that can be transmitted by the shaft at (a) 180 rpm, (b) 480 rpm.
- 17.49** Three shafts and four gears are used to form a gear train which will transmit 7.5 kW from the motor at A to a machine tool at F . (Bearings for the shafts are omitted from the sketch.) Knowing that the frequency of the motor is 30 Hz, determine the magnitude of the couple that is applied to shaft (a) AB , (b) CD , (c) EF .
- 17.50** The shaft-disk-belt arrangement shown is used to transmit 2.4 kW from point A to point D . Knowing that the maximum allowable couples that can be applied to shafts AB and CD are 25 N·m and 80 N·m, respectively, determine the required minimum speed of shaft AB .
- 17.51** The drive belt on a vintage sander transmits $\frac{1}{2}$ hp to a pulley that has a diameter of $d = 4$ in. Knowing that the pulley rotates at 1450 rpm, determine the tension difference $T_1 - T_2$ between the tight and slack sides of the belt.

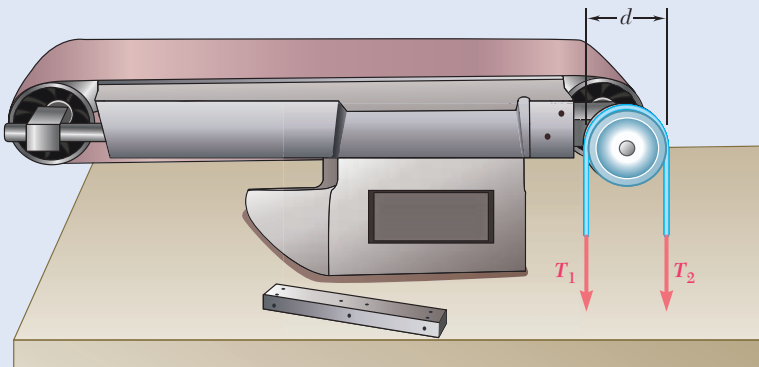


Fig. P17.51

17.2 MOMENTUM METHODS FOR A RIGID BODY

We now apply the principle of impulse and momentum to the plane motion of rigid bodies and of systems of rigid bodies. As we pointed out in Chap. 13, the method of impulse and momentum is particularly well adapted to the solution of problems involving time and velocities. Moreover, the principle of impulse and momentum provides the only practicable method for the solution of problems involving impulsive motion or impact (Sec. 17.3).

17.2A Principle of Impulse and Momentum

Consider again a rigid body made of a large number of particles P_i . Recall from Sec. 14.2C that impulse–momentum diagrams are a pictorial representation of the principle of impulse and momentum. They show (a) the system formed by the momenta of the particles at time t_1 and (b) the system of the impulses of the external forces applied from t_1 to t_2 are together equipollent to (c) the system formed by the momenta of the particles at time t_2 (Fig. 17.6). We can consider the vectors associated with a rigid body to be sliding vectors, so it follows (*Statics*, Sec. 3.4B) that the systems of vectors shown in Fig. 17.6 are not only equipollent, but they are truly *equivalent*. In other words, the vectors on the left-hand side of the equal sign can be transformed into the vectors on the right-hand side through the use of the fundamental operations listed in Sec. 3.3B. We therefore have

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2 \quad (17.14)$$

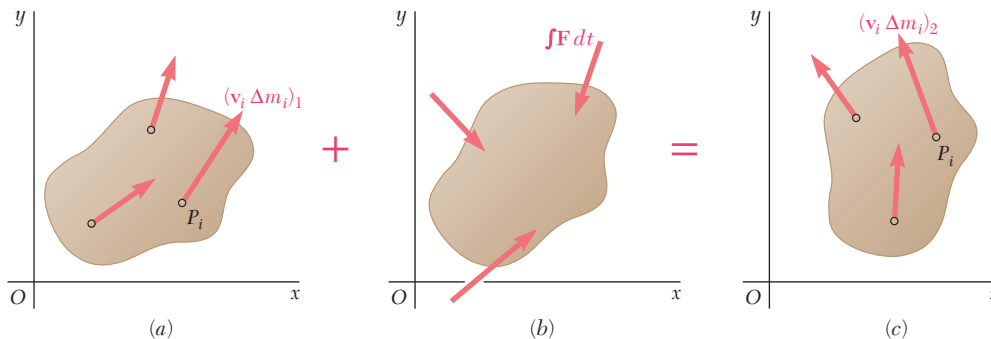


Fig. 17.6 For a rigid body in plane motion: (a) the system of particle momenta at time t_1 plus (b) the system of impulses of the external forces from time t_1 to t_2 is equivalent to (c) the system of particle momenta at time t_2 .

The momenta $\mathbf{v}_i \Delta m_i$ of the particles can be reduced to a vector attached at G that is equal to their sum

$$\mathbf{L} = \sum_{i=1}^n \mathbf{v}_i \Delta m_i$$



Photo 17.2 A Charpy impact test is used to determine the amount of energy absorbed by a material during impact. To determine the amount of energy absorbed, the final gravitational potential energy of the arm is subtracted from its initial gravitational potential energy.

and a couple of moment equal to the sum of their moments about G , as

$$\mathbf{H}_G = \sum_{i=1}^n \mathbf{r}'_i \times \mathbf{v}_i \Delta m_i$$

Recall from Sec. 14.1B that $\mathbf{L}$ and $\mathbf{H}_G$ define, respectively, the linear momentum and the angular momentum about G of the system of particles forming the rigid body. Also note from Eq. (14.14) that $\mathbf{L} = m\bar{\mathbf{v}}$. On the other hand, by restricting the present analysis to the plane motion of a rigid body or of a rigid body symmetrical with respect to the reference plane, we recall from Eq. (16.4) that $\mathbf{H}_G = \bar{I}\boldsymbol{\omega}$. We thus conclude that the system of the momenta $\mathbf{v}_i \Delta m_i$ is equivalent to the **linear momentum vector** $m\bar{\mathbf{v}}$ attached at G and to the **angular momentum couple** $\bar{I}\boldsymbol{\omega}$ (Fig. 17.7).

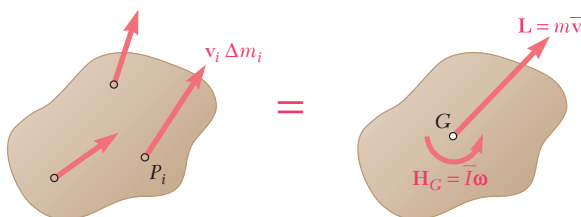


Fig. 17.7 The system of momenta of a rigid body is equivalent to a linear momentum vector attached at G and an angular momentum couple.

The system of momenta reduces to the vector $m\bar{\mathbf{v}}$ in the particular case of a translation ($\boldsymbol{\omega} = 0$) and to the couple $\bar{I}\boldsymbol{\omega}$ in the particular case of a centroidal rotation ($\bar{\mathbf{v}} = 0$). Thus, we verify once more that the plane motion of a rigid body that is symmetrical with respect to the reference plane can be resolved into a translation with the mass center G and a rotation about G .

Replacing the system of momenta in Fig. 17.6a and c by the equivalent linear momentum vector and angular momentum couple, we obtain the three diagrams shown in Fig. 17.8. This impulse-momentum diagram is a visual representation of the fundamental relation in Eq. (17.14) in the case of the plane motion of a rigid body or of a rigid body symmetrical with respect to the reference plane.

We can derive three equations of motion from Fig. 17.8. Two equations come from summing and equating the x and y components of the momenta

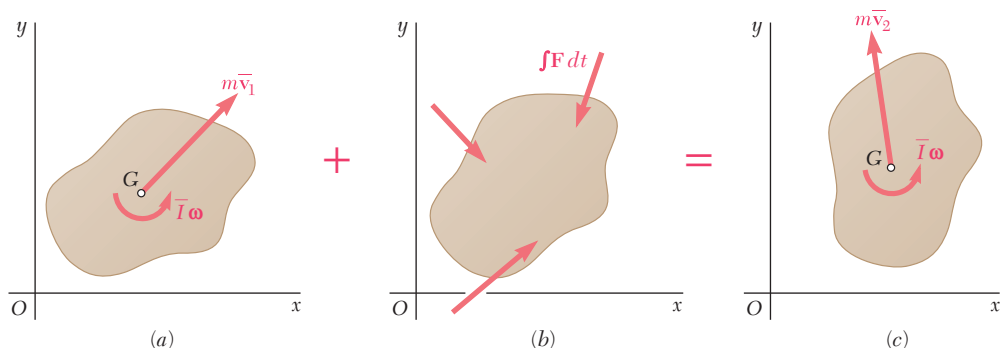


Fig. 17.8 An impulse-momentum diagram is used for applying the principle of impulse and momentum.

and impulses. The third equation is obtained by summing and equating the *moments* of these vectors *about any given point*. We can choose the coordinate axes to be fixed in space or allowed to move with the mass center of the body while maintaining a fixed direction. In either case, the point about which moments are taken should keep the same position relative to the coordinate axes during the interval of time considered. If you choose to sum moments about a point P , Eq. (17.14) can be expressed as

$$\bar{I}\omega_1 + m\bar{v}_1d_\perp + \sum \int_{t_1}^{t_2} M_P dt = \bar{I}\omega_2 + m\bar{v}_2d_\perp \quad (17.14')$$

where $d_\perp$ is the perpendicular distance from point P to the line of action of the linear velocity of G . If you choose to sum moments about the center of gravity of the body, then Eq. (17.14') reduces to

$$\bar{I}\omega_1 + \sum \int_{t_1}^{t_2} M_G dt = \bar{I}\omega_2 \quad (17.14'')$$

In deriving the three equations of motion for a rigid body, you should take care to avoid adding linear and angular momenta indiscriminately. Remember that $m\bar{v}_x$ and $m\bar{v}_y$ represent the *components of a vector*, namely, the linear momentum vector $m\bar{\mathbf{v}}$, whereas $\bar{I}\omega$ represents the *magnitude of a couple*, namely, the angular momentum couple $\bar{\mathbf{I}}\boldsymbol{\omega}$. Thus, you should add the quantity $\bar{I}\omega$ only to the *moment* of the linear momentum $m\bar{\mathbf{v}}$ —never to this vector itself nor to its components. All angular momentum quantities involved then will be expressed in the same units, namely $\text{N}\cdot\text{m}\cdot\text{s}$ or $\text{lb}\cdot\text{ft}\cdot\text{s}$.

Noncentroidal Rotation. In this particular case of plane motion, the magnitude of the velocity of the mass center of the body is $\bar{v} = \bar{r}\omega$, where $\bar{r}$ represents the distance from the mass center to the fixed axis of rotation and ω represents the angular velocity of the body at the instant considered. The magnitude of the momentum vector attached at G is thus $m\bar{v} = m\bar{r}\omega$. Summing the moments about O of the momentum vector and momentum couple (Fig. 17.9) and using the parallel-axis theorem for moments of inertia, we find that the angular momentum $\mathbf{H}_O$ of the body about fixed axis O has the magnitude[†]

$$\bar{I}\omega + (m\bar{r}\omega)\bar{r} = (\bar{I} + m\bar{r}^2)\omega = I_O\omega \quad (17.15)$$

Equating the moments about O of the momenta and impulses in Eq. (17.14), we have

$$I_O\omega_1 + \sum \int_{t_1}^{t_2} M_O dt = I_O\omega_2 \quad (17.16)$$

In the general case of plane motion of a rigid body symmetrical with respect to the reference plane, you can use Eq. (17.16) with respect to the instantaneous axis of rotation under certain conditions. We recommend, however, that all problems of plane motion be solved by the general method described earlier in this section.

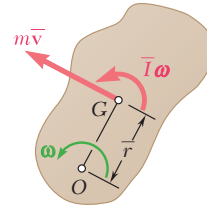


Fig. 17.9 The linear and angular momenta for a noncentroidal rotation.

[†]Note that the sum $\mathbf{H}_P$ of the moments about an arbitrary point P of the momenta of the particles of a rigid body is, in general, not equal to $I_P\boldsymbol{\omega}$ (see Prob. 17.67).

17.2B Systems of Rigid Bodies

We can analyze the motion of several rigid bodies by applying the principle of impulse and momentum to each body separately (Sample Prob. 17.7). However, in solving problems involving no more than three unknowns (including the impulses of unknown reactions), it is often convenient to apply the principle of impulse and momentum to the system as a whole.

To do this, first draw impulse–momentum diagrams for the entire system of bodies. For each moving part of the system, the diagrams of momenta should include a linear momentum vector and a momentum couple. You can omit impulses of forces internal to the system from the diagram showing the impulses, since they occur in pairs of equal and opposite vectors. Summing and equating successively the x components, y components, and moments of all vectors involved, you obtain three relations expressing that the momenta at time t_1 and the impulses of the external forces form a system equipollent to the system of the momenta at time t_2 . Again, you should take care not to add linear and angular momenta indiscriminately; check each equation to make sure that consistent units are used. This approach has been used in Sample Probs. 17.9 through 17.13.

17.2C Conservation of Angular Momentum



Photo 17.3 A figure skater at the beginning and at the end of a spin. By using the principle of conservation of angular momentum you will find that her angular velocity is much higher at the end of the spin.

When no external force acts on a rigid body or a system of rigid bodies, the impulses of the external forces are zero and the system of the momenta at time t_1 is equipollent to the system of the momenta at time t_2 . Summing and equating successively the x components, y components, and moments of the momenta at times t_1 and t_2 , we conclude that the total linear momentum of the system is conserved in any direction and that its total angular momentum is conserved about any point.

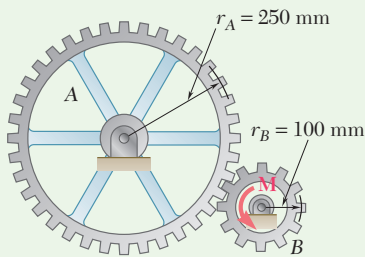
In many engineering applications, however, the linear momentum is not conserved, yet the angular momentum $\mathbf{H}_P$ of the system about a given point P is conserved. That is,

$$(\mathbf{H}_P)_1 = (\mathbf{H}_P)_2 \quad (17.17)$$

Such cases occur when the lines of action of all external forces pass through P or, more generally, when the sum of the angular impulses of the external forces about P is zero.

You can solve problems involving the **conservation of angular momentum** about a point P using the general method of impulse and momentum, i.e., by drawing impulse-momentum diagrams as described earlier. You then obtain Eq. (17.17) by summing and equating moments about P (Sample Prob. 17.9). As you will see in Sample Prob. 17.11, you can obtain two additional equations by summing and equating the x and y components of the linear momentum; then you can use these equations to determine two unknown linear impulses, such as the impulses of the reaction components at a fixed point.

Sample Problem 17.7



Gear *A* has a mass of 10 kg and a radius of gyration of 200 mm, and gear *B* has a mass of 3 kg and a radius of gyration of 80 mm. The system is at rest when a couple **M** with a magnitude of 6 N·m is applied to gear *B*. (These gears were considered in Sample Prob. 17.2.) Neglecting friction, determine (a) the time required for the angular velocity of gear *B* to reach 600 rpm, (b) the tangential force that gear *B* exerts on gear *A*.

STRATEGY: Since you are given an angular velocity and are asked for time, use the principle of impulse and momentum.

MODELING: You are asked to find the internal tangential force, so you need two systems for this problem; that is, gear *A* and gear *B*. Model the gears as rigid bodies. Since all forces and couples are constant, you can obtain the impulses by multiplying the forces and moments by the unknown time *t*.

ANALYSIS: Recall from Sample Prob. 17.2 that the centroidal moments of inertia and the final angular velocities are

$$\begin{aligned}\bar{I}_A &= 0.400 \text{ kg}\cdot\text{m}^2 & \bar{I}_B &= 0.0192 \text{ kg}\cdot\text{m}^2 \\ (\omega_A)_2 &= 25.1 \text{ rad/s} & (\omega_B)_2 &= 62.8 \text{ rad/s}\end{aligned}$$

Principle of Impulse and Momentum for Gear A. The impulse-momentum diagram (Fig. 1) for gear *A* shows the initial momenta, impulses, and final momenta.

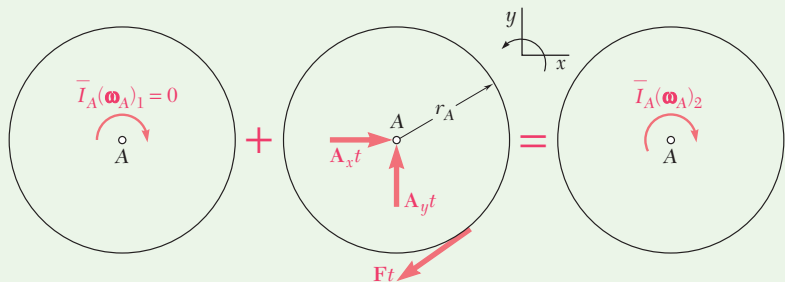


Fig. 1 Impulse-momentum diagram for gear *A*.

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2$$

$$+\curvearrowleft \text{moments about A:} \quad 0 - Ftr_A = -\bar{I}_A(\omega_A)_2$$

$$Ft(0.250 \text{ m}) = (0.400 \text{ kg}\cdot\text{m}^2)(25.1 \text{ rad/s})$$

$$Ft = 40.2 \text{ N}\cdot\text{s}$$

(continued)

Principle of Impulse and Momentum for Gear B. Draw a separate impulse–momentum diagram for gear B (Fig. 2).

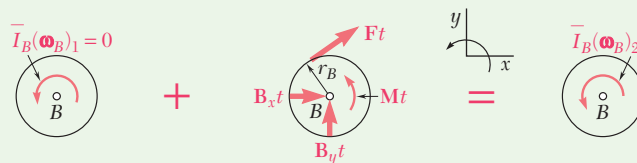


Fig. 2 Impulse–momentum diagram for gear B.

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2$$

$$\begin{aligned} + \uparrow \text{moments about } B: \quad & 0 + Mt - Ftr_B = \bar{I}_B(\omega_B)_2 \\ + (6 \text{ N}\cdot\text{m})t - (40.2 \text{ N}\cdot\text{s})(0.100 \text{ m}) &= (0.0192 \text{ kg}\cdot\text{m}^2)(62.8 \text{ rad/s}) \\ & t = 0.871 \text{ s} \quad \blacktriangleleft \end{aligned}$$

Recall that $Ft = 40.2 \text{ N}\cdot\text{s}$, so you have

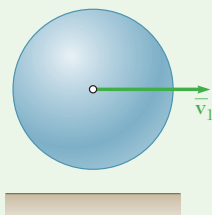
$$F(0.871 \text{ s}) = 40.2 \text{ N}\cdot\text{s} \quad F = +46.2 \text{ N}$$

Thus, the force exerted by gear B on gear A is

$$\mathbf{F} = 46.2 \text{ N} \quad \blacktriangleleft$$

REFLECT and THINK: This is the same answer obtained in Sample Prob. 17.2 by the method of work and energy, as you would expect. The difference is that in Sample Prob. 17.2, you were asked to find the number of revolutions, and in this problem, you were asked to find the time. What you are asked to find will often determine the best approach to use when solving a problem.

Sample Problem 17.8



A uniform sphere with a mass m and a radius r is projected along a rough horizontal surface with a linear velocity $\bar{v}_1$ and no angular velocity. Denote the coefficient of kinetic friction between the sphere and the surface by μ_k . Determine (a) the time t_2 at which the sphere starts rolling without sliding, (b) the linear and angular velocities of the sphere at time t_2 .

STRATEGY: You are asked to find the time, so use the principle of impulse and momentum. You can apply this principle to the sphere from the time $t_1 = 0$ when it is placed on the surface until the time $t_2 = t$ when it starts rolling without sliding.

MODELING: Choose the sphere as your system and model it as a rigid body. While the sphere is sliding relative to the surface, it is acted upon by the normal force $\mathbf{N}$, the friction force $\mathbf{F}$, and its weight $\mathbf{W}$ with a magnitude of $W = mg$. An impulse–momentum diagram for this system is shown in Fig. 1.

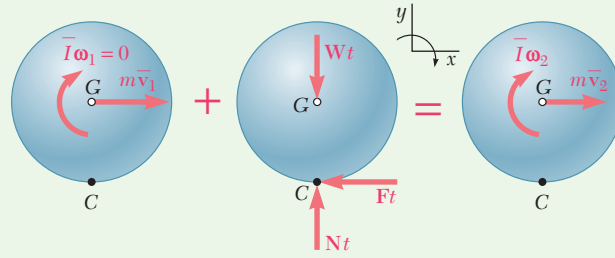


Fig. 1 Impulse-momentum diagram for the sphere.

ANALYSIS:

Principle of Impulse and Momentum. Apply the principle of impulse and momentum for this system between time t_1 and t_2

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2$$

$$+\uparrow y \text{ components:} \quad Nt - Wt = 0 \quad (1)$$

$$+\rightarrow x \text{ components:} \quad m\bar{v}_1 - Ft = m\bar{v}_2 \quad (2)$$

$$+\downarrow \text{ moments about } G: \quad Ftr = \bar{I}\omega_2 \quad (3)$$

From Eq. (1) you obtain $N = W = mg$. During the entire time interval considered, sliding occurs at point C , and $F = \mu_k N = \mu_k mg$. Substituting this expression for F into Eq. (2), you have

$$m\bar{v}_1 - \mu_k mgt = m\bar{v}_2 \quad \bar{v}_2 = \bar{v}_1 - \mu_k gt \quad (4)$$

Substituting $F = \mu_k mg$ and $\bar{I} = \frac{2}{5}mr^2$ into Eq. (3) gives

$$\mu_k mgtr = \frac{2}{5}mr^2\omega_2 \quad \omega_2 = \frac{5}{2} \frac{\mu_k g}{r} t \quad (5)$$

The sphere starts rolling without sliding when the velocity $\mathbf{v}_C$ of the point of contact is zero. At that time, point C becomes the instantaneous center of rotation, and you have $\bar{v}_2 = r\omega_2$. Substituting Eqs. (4) and (5) into this equation, you obtain

$$\bar{v}_1 - \mu_k gt = r \left(\frac{5}{2} \frac{\mu_k g}{r} t \right) \quad t = \frac{2}{7} \frac{\bar{v}_1}{\mu_k g} \quad \blacktriangleleft$$

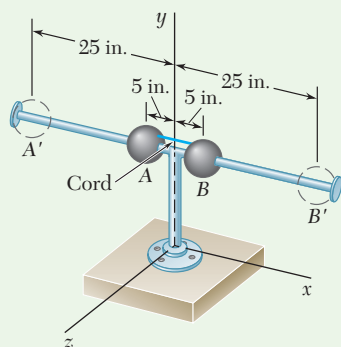
Substituting this expression for t into Eq. (5), you have

$$\omega_2 = \frac{5}{2} \frac{\mu_k g}{r} \left(\frac{2}{7} \frac{\bar{v}_1}{\mu_k g} \right) \quad \omega_2 = \frac{5}{7} \frac{\bar{v}_1}{r} \quad \omega_2 = \frac{5}{7} \frac{\bar{v}_1}{r} \quad \blacktriangleleft$$

$$\bar{v}_2 = r\omega_2 \quad \bar{v}_2 = r \left(\frac{5}{7} \frac{\bar{v}_1}{r} \right) \quad \bar{v}_2 = \frac{5}{7} \bar{v}_1 \rightarrow \quad \blacktriangleleft$$

REFLECT and THINK: This is the same answer obtained in Sample Prob. 16.6 by first dealing directly with force and acceleration and then applying kinematic relationships.

Sample Problem 17.9



Two solid spheres with a radius 3 in. and weighing 2 lb each are mounted at A and B on the horizontal rod $A'B'$ that rotates freely about a vertical axis with a counterclockwise angular velocity of 6 rad/s. The spheres are held in position by a cord, which is suddenly cut. The centroidal moment of inertia of the rod and pivot is $\bar{I}_R = 0.25 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$. Determine (a) the angular velocity of the rod after the spheres have moved to positions A' and B' , (b) the energy lost due to the plastic impact of the spheres and the stops at A' and B' .

STRATEGY: You can first use the principle of impulse and momentum to find the angular velocity of the rod and then use the definition of kinetic energy to determine the change in energy.

MODELING: Choose the two solid spheres and the horizontal rod as your system and model these as rigid bodies. The impulse–momentum diagram for this system is shown in Fig. 1.

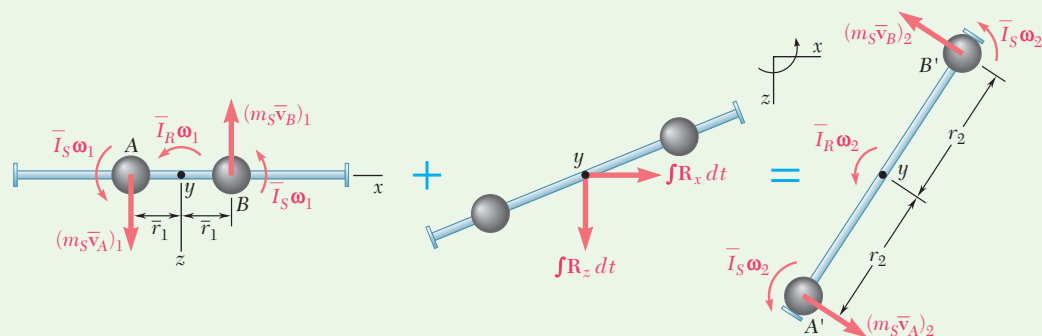


Fig. 1 Impulse–momentum diagram for the system.

ANALYSIS:

a. Principle of Impulse and Momentum. Apply the principle of impulse and momentum for this system between time t_1 (when the spheres are at r_1) and t_2 (when the spheres are at r_2)

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2$$

The external forces consist of the weights and the reaction at the pivot, which have no moment about the y axis. Noting that the rod is undergoing centroidal rotation and $\bar{v}_A = \bar{v}_B = \bar{r}\omega$, you can equate moments about the y axis as

$$\begin{aligned} 2(m_S \bar{r}_1 \omega_1) \bar{r}_1 + 2\bar{I}_S \omega_1 + \bar{I}_R \omega_1 &= 2(m_S \bar{r}_2 \omega_2) \bar{r}_2 + 2\bar{I}_S \omega_2 + \bar{I}_R \omega_2 \\ (2m_S \bar{r}_1^2 + 2\bar{I}_S + \bar{I}_R) \omega_1 &= (2m_S \bar{r}_2^2 + 2\bar{I}_S + \bar{I}_R) \omega_2 \end{aligned} \quad (1)$$

This states that the angular momentum of the system about the y axis is conserved. You can now compute

$$\begin{aligned} \bar{I}_S &= \frac{2}{5} m_S a^2 = \frac{2}{5} (2 \text{ lb}/32.2 \text{ ft/s}^2) \left(\frac{3}{12} \text{ ft}\right)^2 = 0.00155 \text{ lb}\cdot\text{ft}\cdot\text{s}^2 \\ m_S \bar{r}_1^2 &= (2/32.2) \left(\frac{5}{12}\right)^2 = 0.0108 & m_S \bar{r}_2^2 &= (2/32.2) \left(\frac{25}{12}\right)^2 = 0.2696 \end{aligned}$$

Substituting these values, along with $\bar{I}_R = 0.25 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$ and $\omega_1 = 6 \text{ rad/s}$, into Eq. (1) gives

$$0.275(6 \text{ rad/s}) = 0.792\omega_2 \quad \omega_2 = 2.08 \text{ rad/s} \quad \blacktriangleleft$$

b. Energy Lost. The kinetic energy of the system at any instant is

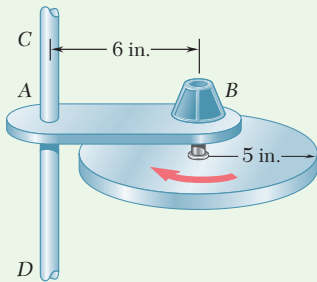
$$T = 2\left(\frac{1}{2}m_S\bar{v}^2 + \frac{1}{2}\bar{I}_S\omega^2\right) + \frac{1}{2}\bar{I}_R\omega^2 = \frac{1}{2}(2m_S\bar{r}^2 + 2\bar{I}_S + \bar{I}_R)\omega^2$$

Using the numerical values found here, you have

$$T_1 = \frac{1}{2}(0.275)(6)^2 = 4.95 \text{ ft}\cdot\text{lb} \quad T_2 = \frac{1}{2}(0.792)(2.08)^2 = 1.713 \text{ ft}\cdot\text{lb}$$

$$\Delta T = T_2 - T_1 = 1.71 - 4.95 \quad \Delta T = -3.24 \text{ ft}\cdot\text{lb} \quad \blacktriangleleft$$

REFLECT and THINK: As expected, when the spheres move outward, the angular velocity of the system decreases. This is similar to an ice skater who throws her arms outward to reduce her angular speed.



Sample Problem 17.10

A 10-lb uniform disk is attached to the shaft of a motor mounted on arm AB that is free to rotate about the vertical axle CD . The arm-and-motor unit has a moment of inertia of $0.032 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$ about axle CD . Knowing that the system is initially at rest, determine the angular velocities of the arm and of the disk when the motor reaches a speed of 360 rpm.

STRATEGY: Since you have two times—when the system starts from rest and when the motor has reached a speed of 360 rpm—use the conservation of angular momentum. You cannot use the conservation of energy because the motor converts electrical energy into mechanical energy.

MODELING: Choose the arm AB , the motor, and the disk to be your system and model them as rigid bodies. The impulse-momentum diagram for this system is shown in Fig. 1.

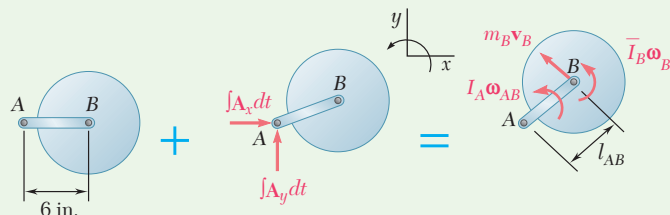


Fig. 1 Impulse-momentum diagram for the system.

(continued)

Moments of Inertia. The mass moment of inertia of the arm and motor about the axle is $I_A = 0.032 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$, and the mass moment of inertia of disk B about its center of mass is

$$\bar{I}_B = \frac{1}{2} \frac{W}{g} r^2 = \frac{1}{2} \left(\frac{10}{32.2} \right) \left(\frac{5}{12} \right)^2 = 0.02696 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$$

ANALYSIS:

Principle of Impulse and Momentum. Apply the principle of impulse and momentum for this system between time t_1 (when the system is at rest) and t_2 (when the motor has an angular velocity of 360 rpm)

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2$$

Taking moments about A gives

$$+\curvearrowright \text{ moments about } A: \quad 0 + 0 = (m_B v_B) l_{AB} + I_A \omega_{AB} + \bar{I}_B \omega_B \quad (1)$$

Kinematics. You can relate the velocity of B to the angular velocity of AB using

$$v_B = l_{AB} \omega_{AB} = \frac{6}{12} \omega_{AB} \quad (2)$$

The velocity of the motor is $\omega_M = 360 \text{ rpm} = 12\pi \text{ rad/s}$, which is the angular velocity of the disk relative to the arm. Thus,

$$\omega_B = \omega_{AB} + \omega_M \quad (3)$$

Substituting Eqs. (2) and (3) into Eq. (1) and solving for ω_{AB} gives

$$\begin{aligned} (m_B l_{AB}^2 + I_A) \omega_{AB} + \bar{I}_B (\omega_{AB} + \omega_M) &= 0 \\ \left[\left(\frac{10}{32.2} \right) \left(\frac{6}{12} \right)^2 + 0.032 \right] \omega_{AB} + 0.02696 (\omega_{AB} + 12\pi) &= 0 \\ \omega_{AB} &= -7.44 \text{ rad/s} \end{aligned}$$

$$\omega_{AB} = 71.0 \text{ rpm} \downarrow \quad \blacktriangleleft$$

The angular velocity of the disk is

$$\omega_B = -7.44 + 12\pi = 30.26 \text{ rad/s}$$

$$\omega_B = 289 \text{ rpm} \uparrow \quad \blacktriangleleft$$

REFLECT and THINK: When the motor spins the disk counterclockwise (as viewed from above), the arm AB rotates in a clockwise direction. One key to solving this problem is recognizing that the angular velocity of the motor is the relative angular velocity of the disk with respect to the bar.

SOLVING PROBLEMS ON YOUR OWN

In this section, we described how to use the method of impulse and momentum to solve problems involving the plane motion of rigid bodies. As you found out previously in Chap. 13, this method is most effective when used in the solution of problems involving velocities and time.

1. The principle of impulse and momentum for the plane motion of a rigid body is expressed by the vector equation:

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2 \quad (17.14)$$

where **Syst Momenta** represents the system of the momenta of the particles forming the rigid body and **Syst Ext Imp** represents the system of all the external impulses exerted during the motion.

a. The system of the momenta of a rigid body is equivalent to a linear momentum vector $m\bar{v}$ attached at the mass center of the body and an angular momentum couple about the center of mass $\bar{I}\omega$ (Fig. 17.7).

b. You should draw an impulse–momentum diagram for the rigid body to express the vector equation (17.14) graphically. Your diagram should consist of three sketches of the body representing, respectively, the initial momenta, the impulses of the external forces, and the final momenta. This shows that the system of the initial momenta and the system of the impulses of the external forces are together equivalent to the system of the final momenta (Fig. 17.8).

c. By using the impulse–momentum diagram, you can sum components in any direction and sum moments about any point. For a single rigid body, if you choose to sum moments about an arbitrary point P , you can write Eq. (17.14) as

$$\bar{I}\omega_1 + m\bar{v}_1 d_{\perp} + \sum \int_{t_1}^{t_2} M_P dt = \bar{I}\omega_2 + m\bar{v}_2 d_{\perp} \quad (17.14')$$

where $d_{\perp}$ is the perpendicular distance from point P to the line of action of the linear velocity of G . If you choose to sum moments about the center of gravity of the body, Eq. (17.14') reduces to

$$\bar{I}\omega_1 + \sum \int_{t_1}^{t_2} M_G dt = \bar{I}\omega_2 \quad (17.14'')$$

If you choose to sum moments about a fixed point O , Eq. (17.14') reduces to

$$I_O\omega_1 + \sum \int_{t_1}^{t_2} M_O dt = I_O\omega_2 \quad (17.16)$$

where I_O is the mass moment of inertia about point O . In most cases, you will be able to select and solve an equation that involves only one unknown.

2. In problems involving a system of rigid bodies, you can apply the principle of impulse and momentum to the system as a whole. Since internal forces occur in equal and opposite pairs, they should not be part of your solution [Sample Probs. 17.9 and 17.10].

3. Conservation of angular momentum about a given axis occurs when, for a system of rigid bodies, *the sum of the moments of the external impulses about that axis is zero*. You can indeed easily observe from the impulse–momentum diagram that the initial and final angular momenta of the system about that axis are equal and, thus, that the angular momentum of the system about the given axis is conserved. You can then sum the angular momenta of the various bodies of the system and the moments of their linear momenta about that axis to obtain an equation that you can solve for one unknown [Sample Prob. 17.9 and 17.10].

Problems

CONCEPT QUESTIONS

17.CQ6 Slender bar A is rigidly connected to a massless rod BC in Case 1 and two massless cords in Case 2 as shown. The vertical thickness of bar A is negligible compared to L . If bullet D strikes A with a speed v_0 and becomes embedded in it, how will the speeds of the center of gravity of A immediately after the impact compare for the two cases?

- Case 1 will be larger.
- Case 2 will be larger.
- The speeds will be the same.

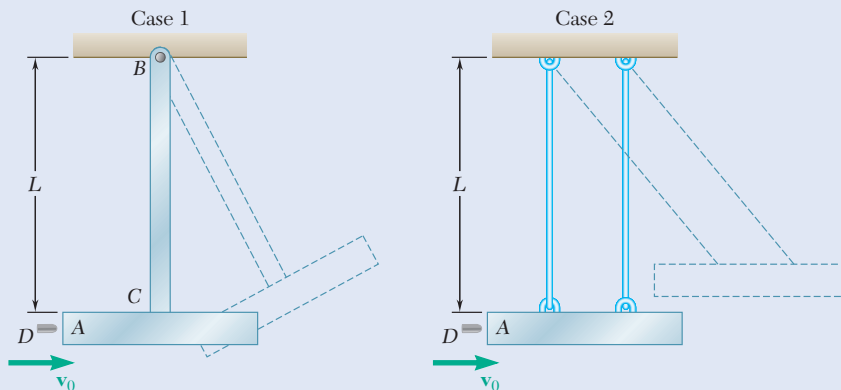


Fig. P17.CQ6

17.CQ7 A 1-m-long uniform slender bar AB has an angular velocity of 12 rad/s and its center of gravity has a velocity of 2 m/s as shown. About which point is the angular momentum of A smallest at this instant?

- P_1
- P_2
- P_3
- P_4
- It is the same about all the points.

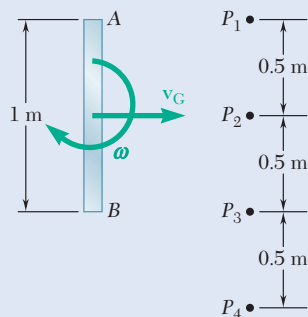


Fig. P17.CQ7

IMPULSE–MOMENTUM DIAGRAM PRACTICE PROBLEMS

- 17.F1** The 350-kg flywheel of a small hoisting engine has a radius of gyration of 600 mm. If the power is cut off when the angular velocity of the flywheel is 100 rpm clockwise, draw an impulse–momentum diagram that can be used to determine the time required for the system to come to rest.

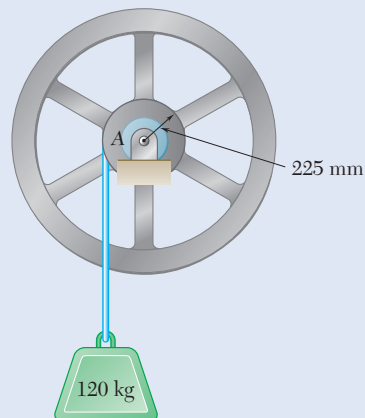


Fig. P17.F1

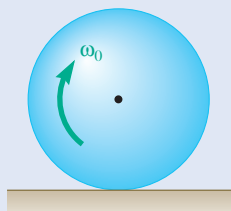


Fig. P17.F2

- 17.F2** A sphere of radius r and mass m is placed on a horizontal floor with no linear velocity but with a clockwise angular velocity ω_0 . Denoting by μ_k the coefficient of kinetic friction between the sphere and the floor, draw the impulse–momentum diagram that can be used to determine the time t_1 at which the sphere will start rolling without sliding.

- 17.F3** Two panels A and B are attached with hinges to a rectangular plate and held by a wire as shown. The plate and the panels are made of the same material and have the same thickness. The entire assembly is rotating with an angular velocity ω_0 when the wire breaks. Draw the impulse–momentum diagram that is needed to determine the angular velocity of the assembly after the panels have come to rest against the plate.

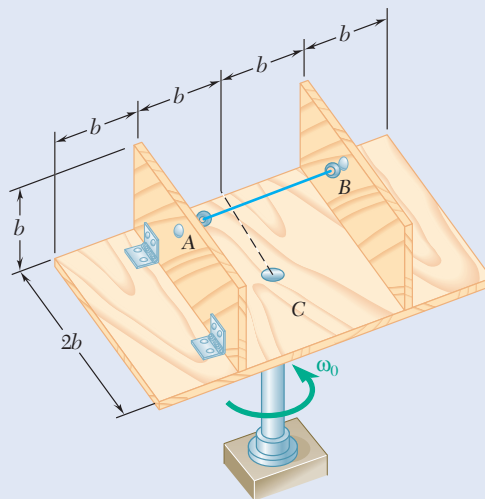


Fig. P17.F3

END-OF-SECTION PROBLEMS

17.52 The rotor of an electric motor has a mass of 25 kg, and it is observed that 4.2 min is required for the rotor to coast to rest from an angular velocity of 3600 rpm. Knowing that kinetic friction produces a couple of magnitude 1.2 N·m, determine the centroidal radius of gyration for the rotor.

17.53 A small grinding wheel is attached to the shaft of an electric motor that has a rated speed of 3600 rpm. When the power is turned off, the unit coasts to rest in 70 s. The grinding wheel and rotor have a combined weight of 6 lb and a combined radius of gyration of 2 in. Determine the average magnitude of the couple due to kinetic friction in the bearings of the motor.

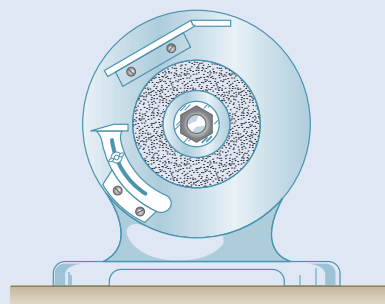


Fig. P17.53

17.54 A bolt located 50 mm from the center of an automobile wheel is tightened by applying the couple shown for 0.10 s. Assuming that the wheel is free to rotate and is initially at rest, determine the resulting angular velocity of the wheel. The wheel has a mass of 19 kg and has a radius of gyration of 250 mm.

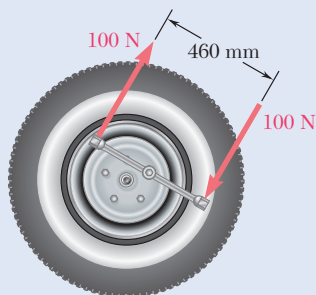


Fig. P17.54

17.55 A uniform 144-lb cube is attached to a uniform 136-lb circular shaft as shown, and a couple $\mathbf{M}$ with a constant magnitude is applied to the shaft when the system is at rest. Knowing that $r = 4$ in., $L = 12$ in., and the angular velocity of the system is 960 rpm after 4 s, determine the magnitude of the couple $\mathbf{M}$.

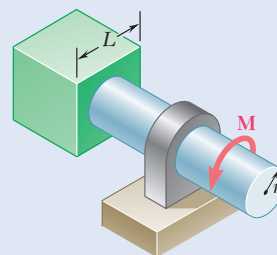


Fig. P17.55 and P17.56

17.56 A uniform 75-kg cube is attached to a uniform 70-kg circular shaft as shown, and a couple $\mathbf{M}$ with a constant magnitude of 20 N·m is applied to the shaft. Knowing that $r = 100$ mm and $L = 300$ mm, determine the time required for the angular velocity of the system to increase from 1000 rpm to 2000 rpm.

17.57 A disk of constant thickness, initially at rest, is placed in contact with a belt that moves with a constant velocity $\mathbf{v}$. Denoting by μ_k the coefficient of kinetic friction between the disk and the belt, derive an expression for the time required for the disk to reach a constant angular velocity.

17.58 Disk A, of weight 5 lb and radius $r = 3$ in., is at rest when it is placed in contact with a belt that moves at a constant speed $v = 50$ ft/s. Knowing that $\mu_k = 0.20$ between the disk and the belt, determine the time required for the disk to reach a constant angular velocity.

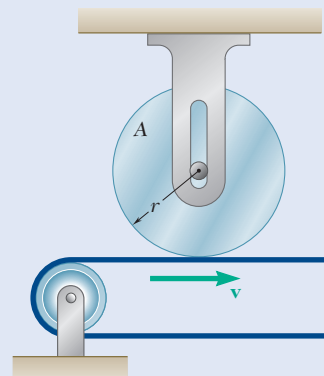


Fig. P17.57 and P17.58

- 17.59** A cylinder of radius r and weight W with an initial counterclockwise angular velocity ω_0 is placed in the corner formed by the floor and a vertical wall. Denoting by μ_k the coefficient of kinetic friction between the cylinder and the wall and the floor, derive an expression for the time required for the cylinder to come to rest.

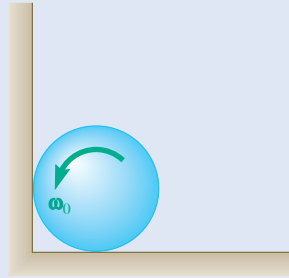


Fig. P17.59

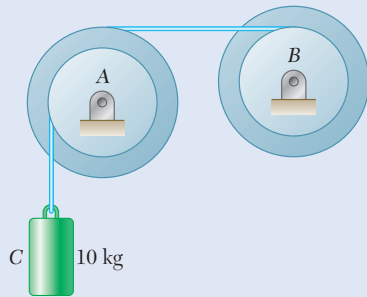


Fig. P17.60

- 17.60** Each of the double pulleys shown has a centroidal mass moment of inertia of $0.25 \text{ kg}\cdot\text{m}^2$, an inner radius of 100 mm, and an outer radius of 150 mm. Neglecting bearing friction, determine (a) the velocity of the cylinder 3 s after the system is released from rest, (b) the tension in the cord connecting the pulleys.

- 17.61** Each of the gears A and B has a mass of 675 g and a radius of gyration of 40 mm, while gear C has a mass of 3.6 kg and a radius of gyration of 100 mm. Assume that kinetic friction in the bearings of gears A , B , and C produces couples of constant magnitude $0.15 \text{ N}\cdot\text{m}$, $0.15 \text{ N}\cdot\text{m}$, and $0.3 \text{ N}\cdot\text{m}$, respectively. Knowing that the initial angular velocity of gear C is 2000 rpm, determine the time required for the system to come to rest.

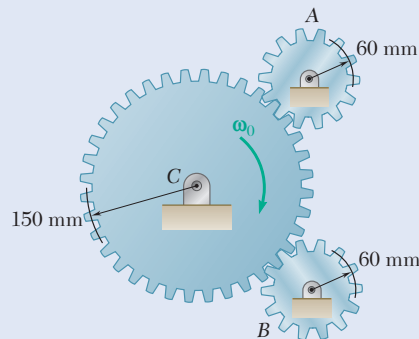


Fig. P17.61

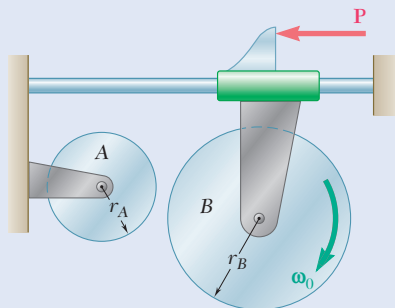


Fig. P17.62 and P17.63

- 17.62** Disk B has an initial angular velocity ω_0 when it is brought into contact with disk A , which is at rest. Show that the final angular velocity of disk B depends only on ω_0 and the ratio of the masses m_A and m_B of the two disks.

- 17.63** The 7.5-lb disk A has a radius $r_A = 6 \text{ in.}$ and is initially at rest. The 10-lb disk B has a radius $r_B = 8 \text{ in.}$ and an angular velocity ω_0 of 900 rpm when it is brought into contact with disk A . Neglecting friction in the bearings, determine (a) the final angular velocity of each disk, (b) the total impulse of the friction force exerted on disk A .

- 17.64** A tape moves over the two drums shown. Drum A weighs 1.4 lb and has a radius of gyration of 0.75 in., while drum B weighs 3.5 lb and has a radius of gyration of 1.25 in. In the lower portion of the tape the tension is constant and equal to $T_A = 0.75$ lb. Knowing that the tape is initially at rest, determine (a) the required constant tension T_B if the velocity of the tape is to be $v = 10$ ft/s after 0.24 s, (b) the corresponding tension in the portion of the tape between the drums.

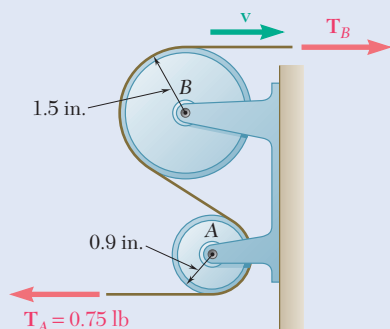


Fig. P17.64

- 17.65** Show that the system of momenta for a rigid body in plane motion reduces to a single vector, and express the distance from the mass center G to the line of action of this vector in terms of the centroidal radius of gyration $\bar{k}$ of the body, the magnitude $\bar{v}$ of the velocity of G , and the angular velocity ω .

- 17.66** Show that, when a rigid body rotates about a fixed axis through O perpendicular to the body, the system of the momenta of its particles is equivalent to a single vector of magnitude $m\bar{r}\omega$, perpendicular to the line OG , and applied to a point P on this line, called the *center of percussion*, at a distance $GP = \bar{k}^2/\bar{r}$ from the mass center of the body.

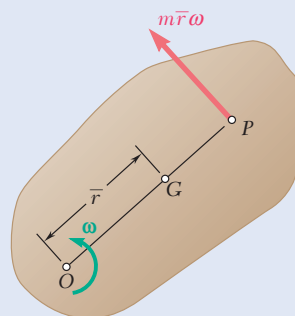


Fig. P17.66

- 17.67** Show that the sum $\mathbf{H}_A$ of the moments about a point A of the momenta of the particles of a rigid body in plane motion is equal to $I_A\omega$, where ω is the angular velocity of the body at the instant considered and I_A the moment of inertia of the body about A, if and only if one of the following conditions is satisfied: (a) A is the mass center of the body, (b) A is the instantaneous center of rotation, (c) the velocity of A is directed along a line joining point A and the mass center G .

- 17.68** Consider a rigid body initially at rest and subjected to an impulsive force $\mathbf{F}$ contained in the plane of the body. We define the *center of percussion* P as the point of intersection of the line of action of $\mathbf{F}$ with the perpendicular drawn from G . (a) Show that the instantaneous center of rotation C of the body is located on line GP at a distance $GC = \bar{k}^2/GP$ on the opposite side of G . (b) Show that if the center of percussion were located at C , the instantaneous center of rotation would be located at P .

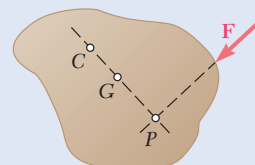


Fig. P17.68

- 17.69** A flywheel is rigidly attached to a 1.5-in.-radius shaft that rolls without sliding along parallel rails. Knowing that after being released from rest the system attains a speed of 6 in./s in 30 s, determine the centroidal radius of gyration of the system.

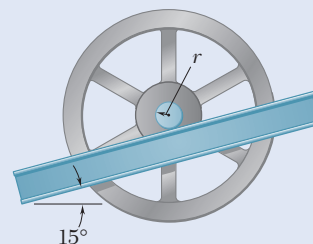


Fig. P17.69

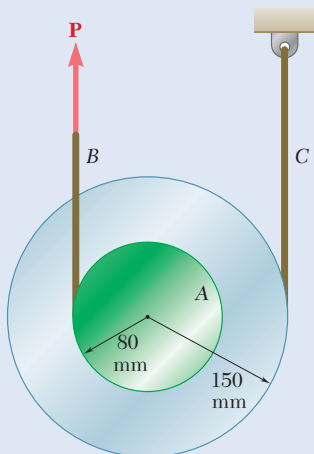


Fig. P17.71

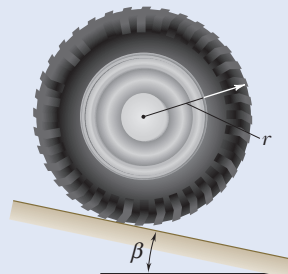


Fig. P17.70

- 17.70** A wheel of radius r and centroidal radius of gyration $\bar{k}$ is released from rest on the incline shown at time $t = 0$. Assuming that the wheel rolls without sliding, determine (a) the velocity of its center at time t , (b) the coefficient of static friction required to prevent slipping.

- 17.71** The double pulley shown has a mass of 3 kg and a radius of gyration of 100 mm. Knowing that when the pulley is at rest, a force $\mathbf{P}$ of magnitude 24 N is applied to cord B , determine (a) the velocity of the center of the pulley after 1.5 s, (b) the tension in cord C .

- 17.72 and 17.73** A 9-in.-radius cylinder of weight 18 lb rests on a 6-lb carriage. The system is at rest when a force $\mathbf{P}$ of magnitude 2.5 lb is applied as shown for 1.2 s. Knowing that the cylinder rolls without sliding on the carriage and neglecting the mass of the wheels of the carriage, determine the resulting velocity of (a) the carriage, (b) the center of the cylinder.

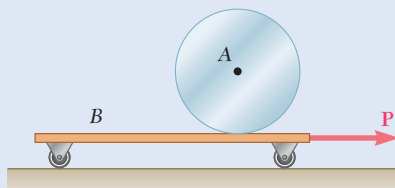


Fig. P17.72

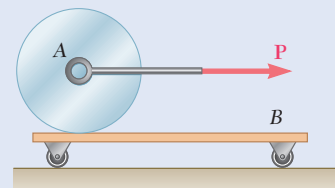


Fig. P17.73

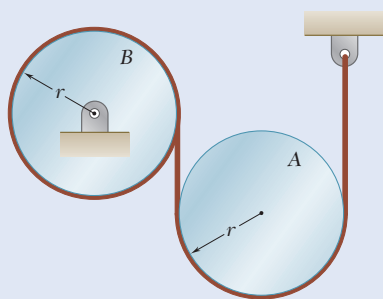


Fig. P17.74 and P17.75

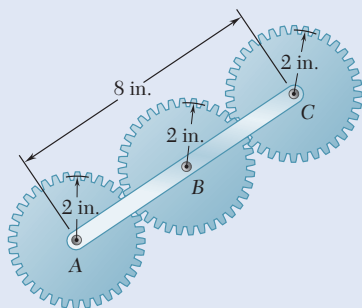


Fig. P17.76

- 17.74** Two uniform cylinders, each of mass $m = 6$ kg and radius $r = 125$ mm, are connected by a belt as shown. If the system is released from rest when $t = 0$, determine (a) the velocity of the center of cylinder B at $t = 3$ s, (b) the tension in the portion of belt connecting the two cylinders.

- 17.75** Two uniform cylinders, each of mass $m = 6$ kg and radius $r = 125$ mm, are connected by a belt as shown. Knowing that at the instant shown the angular velocity of cylinder A is 30 rad/s counter-clockwise, determine (a) the time required for the angular velocity of cylinder A to be reduced to 5 rad/s, (b) the tension in the portion of belt connecting the two cylinders.

- 17.76** In the gear arrangement shown, gears A and C are attached to rod ABC , that is free to rotate about B , while the inner gear B is fixed. Knowing that the system is at rest, determine the magnitude of the couple $\mathbf{M}$ that must be applied to rod ABC , if 2.5 s later the angular velocity of the rod is to be 240 rpm clockwise. Gears A and C weigh 2.5 lb each and may be considered as disks of radius 2 in.; rod ABC weighs 4 lb.

- 17.77** A sphere of radius r and mass m is projected along a rough horizontal surface with the initial velocities shown. If the final velocity of the sphere is to be zero, express (a) the required magnitude of ω_0 in terms of v_0 and r , (b) the time required for the sphere to come to rest in terms of v_0 and the coefficient of kinetic friction μ_k .

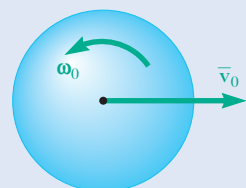


Fig. P17.77

- 17.78** A bowler projects an 8.5-in.-diameter ball weighing 16 lb along an alley with a forward velocity $\mathbf{v}_0$ of 25 ft/s and a backspin ω_0 of 9 rad/s. Knowing that the coefficient of kinetic friction between the ball and the alley is 0.10, determine (a) the time t_1 at which the ball will start rolling without sliding, (b) the speed of the ball at time t_1 .



Fig. P17.78

- 17.79** A semicircular panel with a radius r is attached with hinges to a circular plate with a radius r and initially is held in the vertical position as shown. The plate and the panel are made of the same material and have the same thickness. Knowing that the entire assembly is rotating freely with an initial angular velocity of ω_0 , determine the angular velocity of the assembly after the panel has been released and comes to rest against the plate.

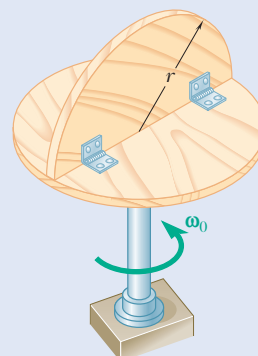


Fig. P17.79

- 17.80.** A satellite has a total weight (on Earth) of 250 lbs, and each of the solar panels weighs 15 lbs. The body of the satellite has mass moment of inertia about the z -axis of 6 slug-ft², and the panels can be modeled as flat plates. The satellite spins with a rate of 10 rpm about the z -axis when the solar panels are positioned in the xy plane. Determine the spin rate about z after a motor on the satellite has rotated both panels to be positioned in the yz plane (as shown in the figure).

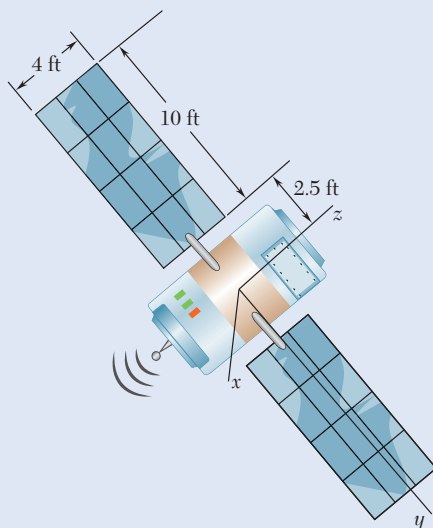


Fig. P17.80

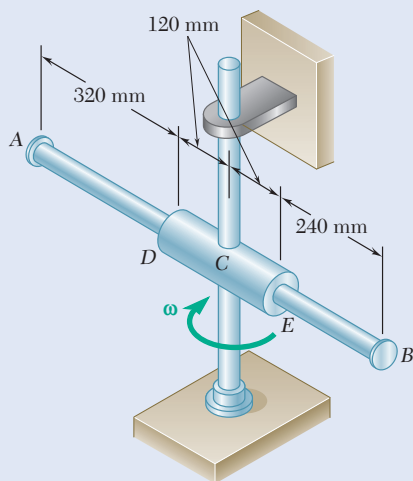


Fig. P17.82

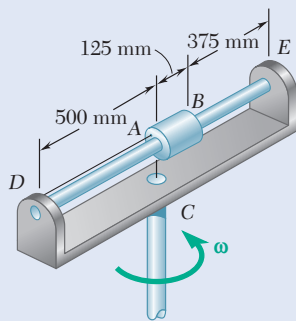


Fig. P17.83

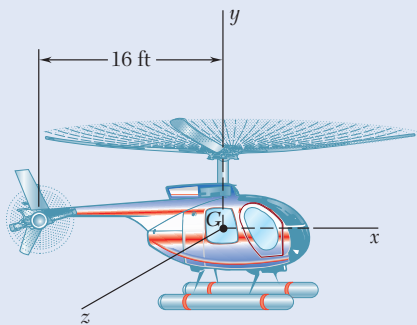


Fig. P17.84

- 17.81** Two 10-lb disks and a small motor are mounted on a 15-lb rectangular platform that is free to rotate about a central vertical spindle. The normal operating speed of the motor is 180 rpm. If the motor is started when the system is at rest, determine the angular velocity of all elements of the system after the motor has attained its normal operating speed. Neglect the mass of the motor and of the belt.

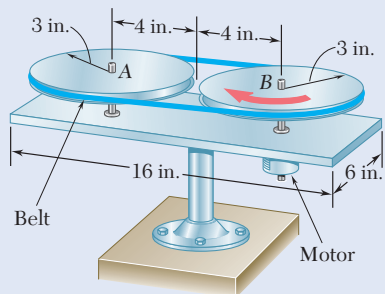


Fig. P17.81

- 17.82** A 3-kg rod of length 800 mm can slide freely in the 240-mm cylinder *DE*, which in turn can rotate freely in a horizontal plane. In the position shown, the assembly is rotating with an angular velocity of magnitude $\omega = 40$ rad/s and end *B* of the rod is moving toward the cylinder at a speed of 75 mm/s relative to the cylinder. Knowing that the centroidal mass moment of inertia of the cylinder about a vertical axis is $0.025 \text{ kg}\cdot\text{m}^2$ and neglecting the effect of friction, determine the angular velocity of the assembly as end *B* of the rod strikes end *E* of the cylinder.

- 17.83** A 1.6-kg tube *AB* can slide freely on rod *DE*, which in turn can rotate freely in a horizontal plane. Initially the assembly is rotating with an angular velocity of magnitude $\omega = 5$ rad/s and the tube is held in position by a cord. The moment of inertia of the rod and bracket about the vertical axis of rotation is $0.30 \text{ kg}\cdot\text{m}^2$ and the centroidal moment of inertia of the tube about a vertical axis is $0.0025 \text{ kg}\cdot\text{m}^2$. If the cord suddenly breaks, determine (a) the angular velocity of the assembly after the tube has moved to end *E*, (b) the energy lost during the plastic impact at *E*.

- 17.84** In the helicopter shown, a vertical tail propeller is used to prevent rotation of the cab as the speed of the main blades is changed. Assuming that the tail propeller is not operating, determine the final angular velocity of the cab after the speed of the main blades has been changed from 180 to 240 rpm. (The speed of the main blades is measured relative to the cab, and the cab has a centroidal moment of inertia of $650 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$. Each of the four main blades is assumed to be a slender 14-ft rod weighing 55 lb.)

- 17.85** Assuming that the tail propeller in Prob. 17.84 is operating and that the angular velocity of the cab remains zero, determine the final horizontal velocity of the cab when the speed of the main blades is changed from 180 to 240 rpm. The cab weighs 1250 lb and is initially at rest. Also, determine the force exerted by the tail propeller if the change in speed takes place uniformly in 12 s.

- 17.86** The circular platform *A* is fitted with a rim of 200-mm inner radius and can rotate freely about the vertical shaft. It is known that the platform-rim unit has a mass of 5 kg and a radius of gyration of 175 mm with respect to the shaft. At a time when the platform is rotating with an angular velocity of 50 rpm, a 3-kg disk *B* of radius 80 mm is placed on the platform with no velocity. Knowing that disk *B* then slides until it comes to rest relative to the platform against the rim, determine the final angular velocity of the platform.

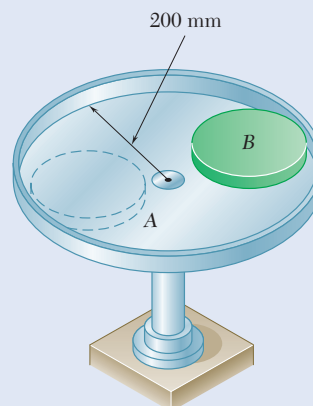


Fig. P17.86

- 17.87** The 30-kg uniform disk *A* and the bar *BC* are at rest and the 5-kg uniform disk *D* has an initial angular velocity of ω_1 with a magnitude of 440 rpm when the compressed spring is released and disk *D* contacts disk *A*. The system rotates freely about the vertical spindle *BE*. After a period of slippage, disk *D* rolls without slipping. Knowing that the magnitude of the final angular velocity of disk *D* is 176 rpm, determine the final angular velocities of bar *BC* and disk *A*. Neglect the mass of bar *BC*.

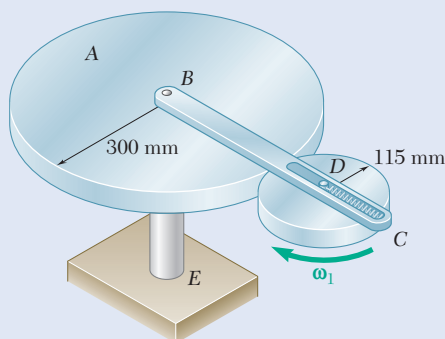


Fig. P17.87

- 17.88** The 4-kg rod *AB* can slide freely inside the 6-kg tube. The rod was entirely within the tube ($x = 0$) and released with no initial velocity relative to the tube when the angular velocity of the assembly was 5 rad/s. Neglecting the effect of friction, determine the speed of the rod relative to the tube when $x = 400$ mm.

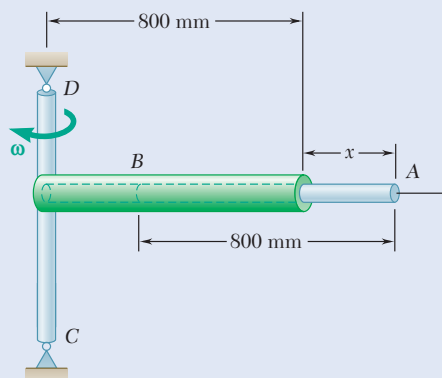


Fig. P17.88

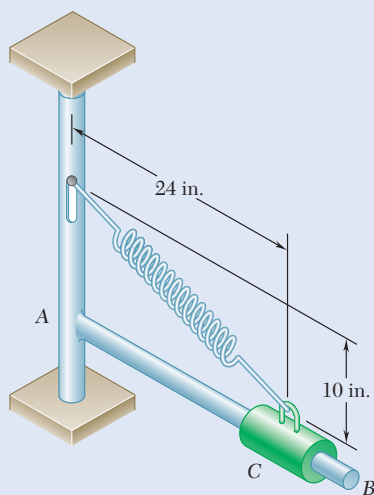


Fig. P17.90

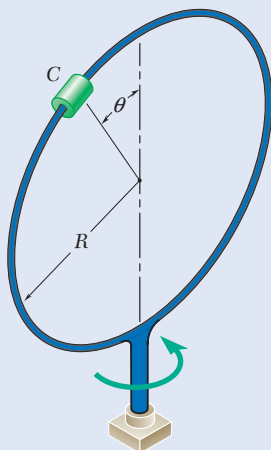


Fig. P17.91

- 17.89** A 1.8-kg collar *A* and a 0.7-kg collar *B* can slide without friction on a frame, consisting of the horizontal rod *OE* and the vertical rod *CD*, which is free to rotate about its vertical axis of symmetry. The two collars are connected by a cord running over a pulley that is attached to the frame at *O*. At the instant shown, the velocity $\mathbf{v}_A$ of collar *A* has a magnitude of 2.1 m/s and a stop prevents collar *B* from moving. The stop is suddenly removed and collar *A* moves toward *E*. As it reaches a distance of 0.12 m from *O*, the magnitude of its velocity is observed to be 2.5 m/s. Determine at that instant the magnitude of the angular velocity of the frame and the moment of inertia of the frame and pulley system about *CD*.

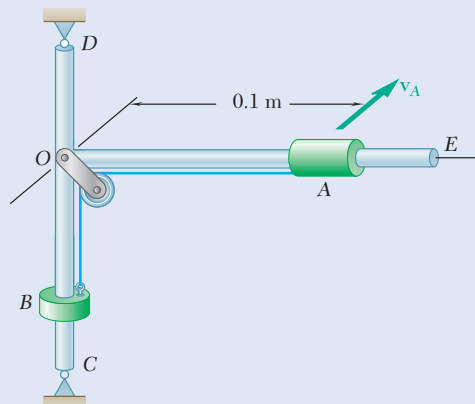


Fig. P17.89

- 17.90** A 6-lb collar *C* is attached to a spring and can slide on rod *AB*, which in turn can rotate in a horizontal plane. The mass moment of inertia of rod *AB* with respect to end *A* is $0.35 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$. The spring has a constant $k = 15 \text{ lb/in.}$ and an undeformed length of 10 in. At the instant shown, the velocity of the collar relative to the rod is zero and the assembly is rotating with an angular velocity of 12 rad/s. Neglecting the effect of friction, determine (a) the angular velocity of the assembly as the collar passes through a point located 7.5 in. from end *A* of the rod, (b) the corresponding velocity of the collar relative to the rod.
- 17.91** A small 4-lb collar *C* can slide freely on a thin ring of weight 6 lb and radius 10 in. The ring is welded to a short vertical shaft, which can rotate freely in a fixed bearing. Initially, the ring has an angular velocity of 35 rad/s and the collar is at the top of the ring ($\theta = 0$) when it is given a slight nudge. Neglecting the effect of friction, determine (a) the angular velocity of the ring as the collar passes through the position $\theta = 90^\circ$, (b) the corresponding velocity of the collar relative to the ring.

- 17.92** Rod AB has a weight of 6 lb and is attached to a 10-lb cart C . Knowing that the system is released from rest in the position shown and neglecting friction, determine (a) the velocity of point B as rod AB passes through a vertical position, (b) the corresponding velocity of the cart C .

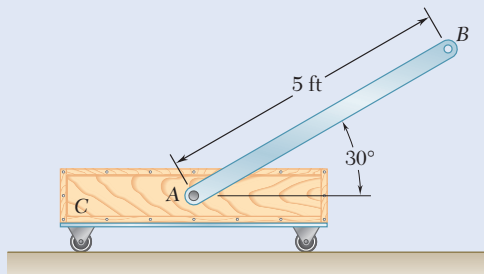


Fig. P17.92

- 17.93** In Prob. 17.82, determine the velocity of rod AB relative to cylinder DE as end B of the rod strikes end E of the cylinder.
- 17.94** In Prob. 17.83, determine the velocity of the tube relative to the rod as the tube strikes end E of the assembly.
- 17.95** The 6-lb steel cylinder A of radius r and the 10-lb wooden cart B are at rest in the position shown when the cylinder is given a slight nudge, causing it to roll without sliding along the top surface of the cart. Neglecting friction between the cart and the ground, determine the velocity of the cart as the cylinder passes through the lowest point of the surface at C .

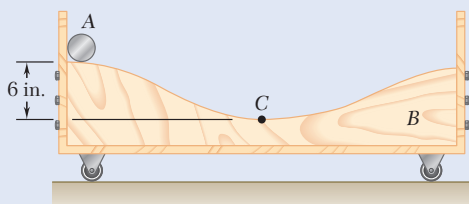


Fig. P17.95

17.3 ECCENTRIC IMPACT

You saw in Chap. 13 that the method of impulse and momentum is the only practicable method for solving problems involving the impulsive motion of a particle. Now you will see that problems involving the impulsive motion of a rigid body are particularly well suited to a solution using the method of impulse and momentum. Since the time interval considered in the computation of linear impulses and angular impulses is very short, we can assume the bodies involved occupy the same position during that time interval, making the computation quite simple.

In Sec. 13.4, we described how to solve problems of **central impact**, i.e., problems in which the mass centers of the two colliding bodies are located on the line of impact. We now analyze the **eccentric impact** of two rigid bodies.

Consider two colliding bodies and denote the velocities of the two points of contact A and B before impact by $\mathbf{v}_A$ and $\mathbf{v}_B$ (Fig. 17.10a). Under

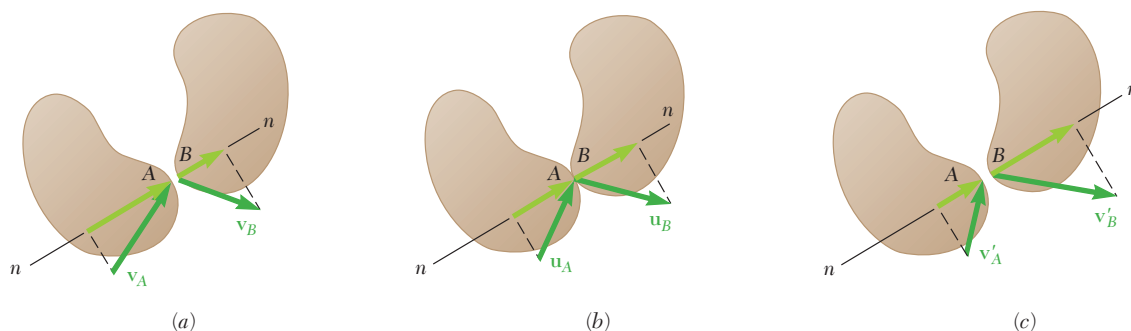


Fig. 17.10 When two rigid bodies collide, (a) the velocities of the points of contact before impact (b) change during the period of deformation and (c) change again during the period of restitution.



Photo 17.4 A swinging bat applies an impulsive force on contact with the ball. You can use the principle of impulse and momentum to determine the final velocities of the ball and bat.

the impact, the two bodies *deform*, and at the end of the period of deformation, the velocities $\mathbf{u}_A$ and $\mathbf{u}_B$ of A and B have equal components along the line of impact nn (Fig. 17.10b). A period of *restitution* then takes place, at the end of which points A and B have velocities of $\mathbf{v}'_A$ and $\mathbf{v}'_B$ (Fig. 17.10c). Assuming that the bodies are frictionless, we find that the forces they exert on each other are directed along the line of impact. We denote the magnitude of the impulse of one of these forces during the period of deformation by $\int P dt$ and the magnitude of its impulse during the period of restitution by $\int R dt$. Recall that we define the coefficient of restitution e as the ratio of

$$e = \frac{\int R dt}{\int P dt} \quad (17.18)$$

We propose to show that the relation established in Sec. 13.4 between the relative velocities of two particles before and after impact also holds between the components along the line of impact of the relative velocities of the two points of contact A and B . That is, we want to show that

$$(v'_B)_n - (v'_A)_n = e[(v_A)_n - (v_B)_n] \quad (17.19)$$

First, we assume that the motion of each of the two colliding bodies of Fig. 17.10 is unconstrained. Thus, the only impulsive forces exerted on the bodies during the impact are applied at A and B , respectively. Consider the body to which point A belongs and draw the impulse–momentum diagram corresponding to the period of deformation (Fig. 17.11). We

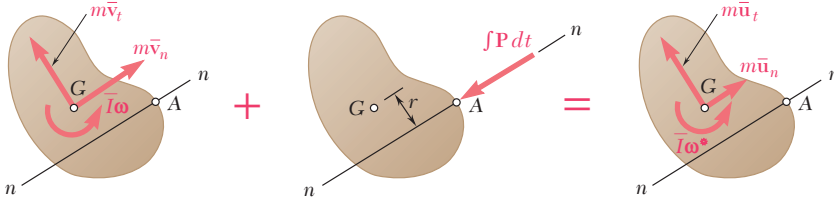


Fig. 17.11 An impulse–momentum diagram for a body undergoing an eccentric impact during the period of deformation.

denote the velocity of the mass center at the beginning and at the end of the period of deformation by $\bar{\mathbf{v}}$ and $\bar{\mathbf{u}}$, respectively, and we denote the angular velocity of the body at the same instants by $\bar{\omega}$ and $\bar{\omega}^*$. Summing and equating the components of the momenta and impulses along the line of impact nn , we have

$$m\bar{v}_n - \int P dt = m\bar{u}_n \quad (17.20)$$

Summing and equating the moments about G of the momenta and impulses, we also have

$$\bar{I}\bar{\omega} - r\int P dt = \bar{I}\bar{\omega}^* \quad (17.21)$$

where r represents the perpendicular distance from G to the line of impact. Considering now the period of restitution, we obtain in a similar way

$$m\bar{u}_n - \int R dt = m\bar{v}'_n \quad (17.22)$$

$$\bar{I}\bar{\omega}^* - r\int R dt = \bar{I}\bar{\omega}' \quad (17.23)$$

where $\bar{\mathbf{v}}'$ and $\bar{\omega}'$ represent, respectively, the velocity of the mass center and the angular velocity of the body after impact. First solving Eqs. (17.20) and (17.22) for the two impulses and substituting into Eq. (17.18) and then solving Eqs. (17.21) and (17.23) for the same two impulses and substituting again into Eq. (17.18), we obtain the two alternative expressions for the coefficient of restitution as

$$e = \frac{\bar{u}_n - \bar{v}'_n}{\bar{v}_n - \bar{u}_n} \quad e = \frac{\bar{\omega}^* - \bar{\omega}'}{\bar{\omega} - \bar{\omega}^*} \quad (17.24)$$

Multiplying the numerator and denominator of the second expression for e by r and adding them, respectively, to the numerator and denominator of the first expression, we have

$$e = \frac{\bar{u}_n + r\bar{\omega}^* - (\bar{v}'_n + r\bar{\omega}')}{\bar{v}_n + r\bar{\omega} - (\bar{u}_n + r\bar{\omega}^*)} \quad (17.25)$$

Observe that $\bar{v}_n + r\bar{\omega}$ represents the component $(v_A)_n$ along nn of the velocity of the point of contact A and that, similarly, $\bar{u}_n + r\bar{\omega}^*$ and $\bar{v}'_n + r\bar{\omega}'$ represent, respectively, the components $(u_A)_n$ and $(v'_A)_n$. Thus, we have

$$e = \frac{(u_A)_n - (v'_A)_n}{(v_A)_n - (u_A)_n} \quad (17.26)$$

The analysis of the motion of the second body leads to a similar expression for e in terms of the components along nn of the successive velocities of point B . Recalling that $(u_A)_n = (u_B)_n$, and eliminating these two velocity components by a manipulation similar to the one used in Sec. 13.4, we obtain the relation in Eq. (17.19).

If one or both of the colliding bodies is constrained to rotate about a fixed point O —as in the case of a compound pendulum (Fig. 17.12a)—an impulsive reaction is exerted at O (Fig. 17.12b). Let us verify that, although

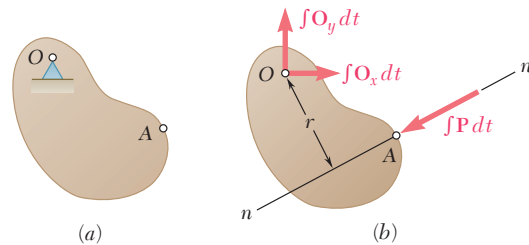


Fig. 17.12 (a) A rigid body constrained to rotate about a fixed point O ; (b) impulsive reaction at O resulting from an eccentric impact.

their derivation must be modified, Eqs. (17.26) and (17.19) remain valid. Applying formula (17.16) to the period of deformation and to the period of restitution, we have

$$I_O \omega - r \int P dt = I_O \omega^* \quad (17.27)$$

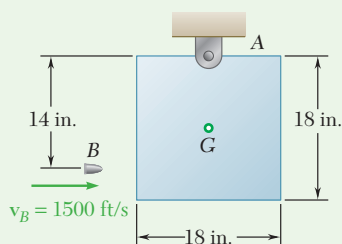
$$I_O \omega^* - r \int R dt = I_O \omega' \quad (17.28)$$

where r represents the perpendicular distance from the fixed point O to the line of impact. We solve Eqs. (17.27) and (17.28) for the two impulses and substitute them into Eq. (17.18). Noting that $r\omega$, $r\omega^*$, and $r\omega'$ represent the components along nn of the successive velocities of point A , we obtain

$$e = \frac{\omega^* - \omega'}{\omega - \omega^*} = \frac{r\omega^* - r\omega'}{r\omega - r\omega^*} = \frac{(u_A)_n - (v'_A)_n}{(v_A)_n - (u_A)_n}$$

This verifies that Eq. (17.26) still holds. Thus, Eq. (17.19) remains valid when one or both of the colliding bodies is constrained to rotate about a fixed point O .

In order to determine the velocities of the two colliding bodies after impact, we need to use the relation in Eq. (17.19) in conjunction with one or several other equations obtained by applying the principle of impulse and momentum (Sample Prob. 17.11 and 17.13).



Sample Problem 17.11

A 0.05-lb bullet B is fired with a horizontal velocity of 1500 ft/s into the side of a 20-lb square panel suspended from a hinge at A . Knowing that the panel is initially at rest, determine (a) the angular velocity of the panel immediately after the bullet becomes embedded, (b) the impulsive reaction at A , assuming that the bullet becomes embedded in 0.0006 s.

STRATEGY: Since you have an impact, use the principle of impulse and momentum.

MODELING: Choose your system to be the bullet and the panel, where you model the bullet as a particle and the panel as a rigid body. The impulse-momentum diagram for this system is shown in Fig. 1. Since the time interval $\Delta t = 0.0006$ s is very short, you can neglect all nonimpulsive forces and consider only the external impulses $A_x \Delta t$ and $A_y \Delta t$.

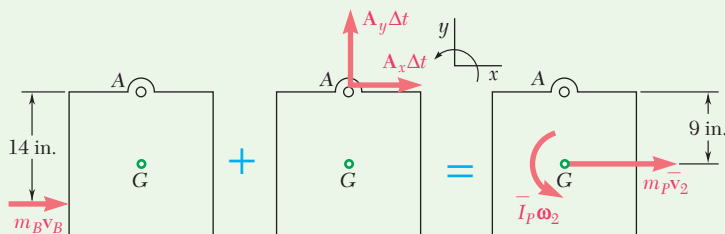


Fig. 1 Impulse-momentum diagram for the system. The bullet is neglected at time 2.

ANALYSIS:

Principle of Impulse and Momentum.

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2$$

$$+\circlearrowleft \text{moments about } A: \quad m_B v_B \left(\frac{14}{12} \text{ ft} \right) + 0 = m_P \bar{v}_2 \left(\frac{9}{12} \text{ ft} \right) + \bar{I}_P \omega_2 \quad (1)$$

$$\rightarrow x \text{ components:} \quad m_B v_B + A_x \Delta t = m_P \bar{v}_2 \quad (2)$$

$$+\uparrow y \text{ components:} \quad 0 + A_y \Delta t = 0 \quad (3)$$

Note that the weight of the bullet is negligible compared to the weight of the panel, so we did not include it on the right-hand side of Eq. (1). The centroidal mass moment of inertia of the square panel is

$$\bar{I}_P = \frac{1}{6} m_P b^2 = \frac{1}{6} \left(\frac{20 \text{ lb}}{32.2} \right) \left(\frac{18}{12} \text{ ft} \right)^2 = 0.2329 \text{ lb} \cdot \text{ft} \cdot \text{s}^2$$

Substituting this value as well as the given data into Eq. (1) and noting that from kinematics, you know

$$\bar{v}_2 = \left(\frac{9}{12} \text{ ft} \right) \omega_2$$

(continued)

So you have

$$\left(\frac{0.05}{32.2}\right)(1500)\left(\frac{14}{12}\right) = 0.2329\omega_2 + \left(\frac{20}{32.2}\right)\left(\frac{9}{12}\omega_2\right)\left(\frac{9}{12}\right)$$

$$\omega_2 = 4.67 \text{ rad/s} \quad \omega_2 = 4.67 \text{ rad/s} \quad \blacktriangleleft$$

$$\bar{v}_2 = \left(\frac{9}{12} \text{ ft}\right)\omega_2 = \left(\frac{9}{12} \text{ ft}\right)(4.67 \text{ rad/s}) = 3.50 \text{ ft/s}$$

Substituting $\bar{v}_2 = 3.50 \text{ ft/s}$, $\Delta t = 0.0006 \text{ s}$, and the given data into Eq. (2) gives you

$$\left(\frac{0.05}{32.2}\right)(1500) + A_x(0.0006) = \left(\frac{20}{32.2}\right)(3.50)$$

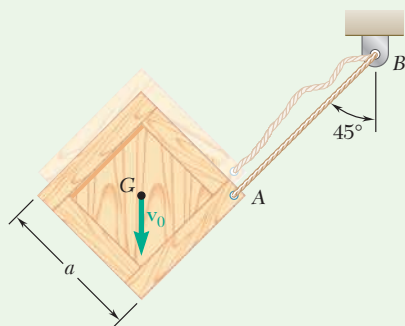
$$A_x = -259 \text{ lb} \quad A_x = 259 \text{ lb} \leftarrow \quad \blacktriangleleft$$

From Eq. (3), you find $A_y = 0$.

$$A_y = 0 \quad \blacktriangleleft$$

REFLECT and THINK: The speed of the bullet is in the range of a modern high-performance rifle. Notice that the reaction at A is over 5000 times the weight of the bullet and over 10 times the weight of the plate.

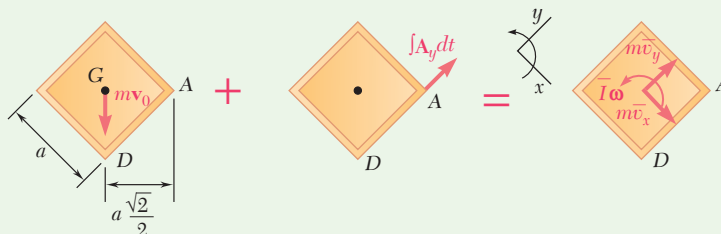
Sample Problem 17.12



A uniformly loaded square crate is falling freely with a velocity v_0 when cable AB suddenly becomes taut. Assuming that the impact is perfectly plastic, determine the angular velocity of the crate and the velocity of its mass center immediately after the cable becomes taut.

STRATEGY: Since impact occurs, use the principle of impulse and momentum.

MODELING: Choose the crate as your system and model it as a rigid body. The impulse-momentum diagram for this system is shown in Fig. 1. The mass moment of inertia of the plate about G is $\bar{I} = \frac{1}{6}ma^2$.



$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2$$

Fig. 1 Impulse-momentum diagram for the crate.

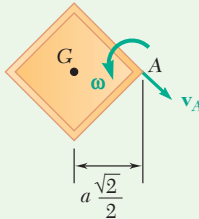


Fig. 2 Velocity of point A.

ANALYSIS:

Principle of Impulse and Momentum. Applying the impulse–momentum principle in the x -direction and taking moments about A gives

$$+\curvearrowright \text{ moments about A: } mv_0 a \frac{\sqrt{2}}{2} + 0 = \bar{I}\omega + m\bar{v}_x \frac{a}{2} - m\bar{v}_y \frac{a}{2} \quad (1)$$

$$+\rightarrow x \text{ components: } mv_0 \frac{\sqrt{2}}{2} + 0 = m\bar{v}_x \quad (2)$$

There are three unknowns in these two equations, ω , $\bar{v}_x$, and $\bar{v}_y$. For additional equations, you can use kinematics. Since you are told the impact is perfectly plastic, point A has a velocity perpendicular to the rope (Fig. 2). Therefore, you can relate the acceleration of A to that of G, as

$$\begin{aligned} \bar{\mathbf{v}} &= \mathbf{v}_G = \mathbf{v}_A + \mathbf{v}_{G/A} \\ &= [v_A \searrow 45^\circ] + \left[a \frac{\sqrt{2}}{2} \omega \downarrow \right] \end{aligned}$$

Equating components in the x and y directions, you find

$$+\rightarrow x\text{-components: } \bar{v}_x = v_A + a \frac{\sqrt{2}}{2} \omega \frac{\sqrt{2}}{2} = v_A + \frac{a\omega}{2} \quad (3)$$

$$+\uparrow y\text{-components: } \bar{v}_y = -a \frac{\sqrt{2}}{2} \omega \frac{\sqrt{2}}{2} = -\frac{a\omega}{2} \quad (4)$$

You now have four equations and four unknowns. Solving these gives

$$\omega = \frac{3\sqrt{2}}{5} \frac{v_0}{a} \quad \bar{v}_x = \frac{\sqrt{2}}{2} v_0 \quad \bar{v}_y = -\frac{3\sqrt{2}}{10} v_0 \quad v_A = \frac{\sqrt{2}}{5} v_0$$

So

$$\omega = 0.849 \frac{v_0}{d} \searrow \blacktriangleleft$$

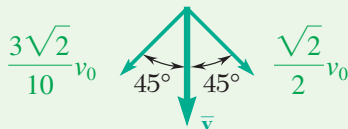


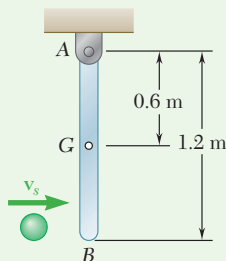
Fig. 3 Diagram to determine the magnitude and direction of $\bar{\mathbf{v}}$

Resolving the velocity of the center of mass into a magnitude and direction using Fig. 3 gives you

$$\bar{\mathbf{v}} = 0.825 v_0 \searrow 76.0^\circ \blacktriangleleft$$

REFLECT and THINK: If the impact had not been plastic, point A would have rebounded and the rope would have become slack. To solve the problem in this case, you would have needed to use the equation for the coefficient of restitution.

Sample Problem 17.13



A 2-kg sphere moving horizontally to the right with an initial velocity of 5 m/s strikes the lower end of an 8-kg rigid rod AB . The rod is suspended from a hinge at A and is initially at rest. Knowing that the coefficient of restitution between the rod and the sphere is 0.80, determine the angular velocity of the rod and the velocity of the sphere immediately after the impact.

STRATEGY: Since you have an impact, use the principle of impulse and momentum.

MODELING: Choose the sphere and the rod as your system; model the sphere as a particle and the rod as a rigid body. You also need to use the coefficient of restitution equation. The impulse–momentum diagram for this system is shown in Fig. 1. Note that the only impulsive force external to the system is the impulsive reaction at A .

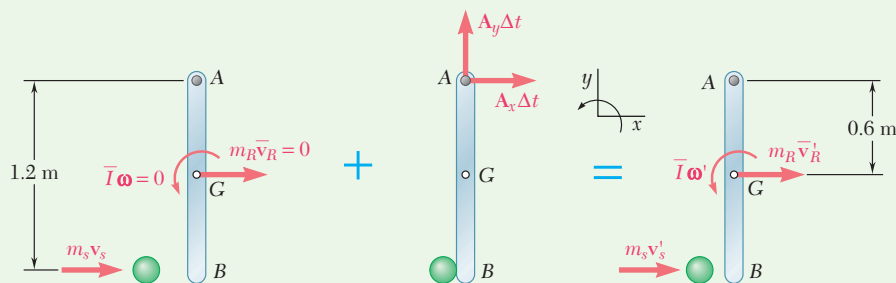


Fig. 1 Impulse–momentum diagram for the system.

ANALYSIS:

Principle of Impulse and Momentum.

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2$$

+ $\curvearrowright$ moments about A :

$$m_s v_s (1.2 \text{ m}) = m_s v'_s (1.2 \text{ m}) + m_R \bar{v}'_R (0.6 \text{ m}) + \bar{I} \omega' \quad (1)$$

In this case, the mass of the sphere is not negligible compared to the rod, so we must include it on the right-hand side of Eq. (1). Since the rod rotates about A , from kinematics, you know $\bar{v}'_R = \bar{r} \omega' = (0.6 \text{ m}) \omega'$. Also,

$$\bar{I} = \frac{1}{12} m L^2 = \frac{1}{12} (8 \text{ kg}) (1.2 \text{ m})^2 = 0.96 \text{ kg} \cdot \text{m}^2$$

Substituting these values and the given data into Eq. (1), you obtain

$$\begin{aligned} (2 \text{ kg})(5 \text{ m/s})(1.2 \text{ m}) &= (2 \text{ kg})v'_s(1.2 \text{ m}) + (8 \text{ kg})(0.6 \text{ m})\omega'(0.6 \text{ m}) \\ &\quad + (0.96 \text{ kg} \cdot \text{m}^2)\omega' \\ 12 &= 2.4v'_s + 3.84\omega' \end{aligned} \quad (2)$$

Coefficient of Restitution. Choosing positive to the right, you have

$$v'_B - v'_s = e(v_s - v_B)$$

Substituting $v_s = 5 \text{ m/s}$, $v_B = 0$, and $e = 0.80$ gives

$$v'_B - v'_s = 0.8(5 \text{ m/s} - 0) \quad (3)$$

Again noting that the rod rotates about A, you have

$$v'_B = (1.2 \text{ m}) \omega' \quad (4)$$

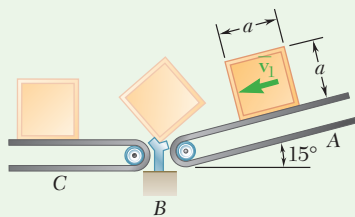
Solving Eqs. (2) to (4) simultaneously, you obtain

$$\omega' = 3.21 \text{ rad/s} \quad \omega' = 3.21 \text{ rad/s} \nearrow$$

$$v'_s = -0.143 \text{ m/s} \quad v'_s = 0.143 \text{ m/s} \leftarrow$$

REFLECT and THINK: The negative value for the velocity of the sphere after impact means that it bounces back to the left. Given the masses of the sphere and the rod, this seems reasonable.

Sample Problem 17.14



A square package of side a and mass m moves down a conveyor belt A with a constant velocity $\bar{v}_1$. At the end of the conveyor belt, the corner of the package strikes a rigid support at B . Assuming that the impact at B is perfectly plastic, derive an expression for the smallest magnitude of the velocity $\bar{v}_1$ for which the package will rotate about B and reach conveyor belt C .

STRATEGY: Because you have an impact, use the principle of impulse and momentum for when the package strikes the rigid support at B , and then apply the conservation of energy for the rotation of the package about the support B after the impact.

MODELING: Choose the package to be your system and model it as a rigid body. The impulse–momentum diagram for this system is shown in Fig. 1. Note that the only impulsive force external to the package is the impulsive reaction at B .

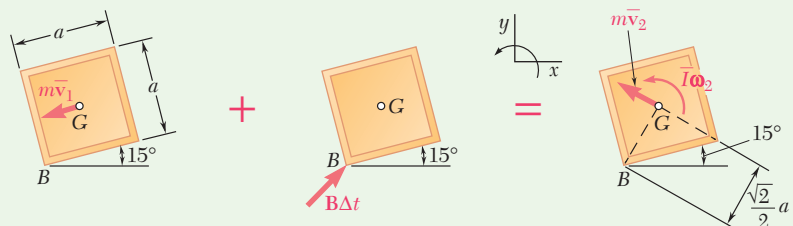
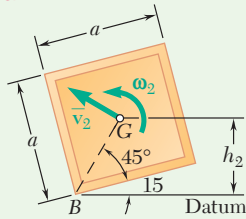


Fig. 1 Impulse–momentum diagram for the crate.

(continued)

Position 2

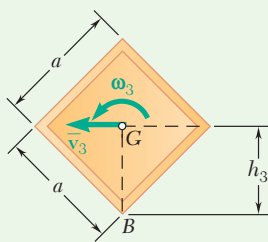


$$GB = \frac{1}{2} \sqrt{2}a = 0.707a$$

$$h_2 = GB \sin(45 + 15)$$

$$= 0.612a$$

Position 3



$$h_3 = GB = 0.707a$$

Fig. 2 The crate in positions 2 and 3.

ANALYSIS:

Principle of Impulse and Momentum.

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2$$

$$+\uparrow \text{ moments about } B: (m\bar{v}_1)(\tfrac{1}{2}a) + 0 = (m\bar{v}_2)(\tfrac{1}{2}\sqrt{2}a) + \bar{I}\omega_2 \quad (1)$$

Since the package rotates about B , from kinematics you have $\bar{v}_2 = (GB)\omega_2 = \tfrac{1}{2}\sqrt{2}a\omega_2$.

Substitute this expression, together with $\bar{I} = \tfrac{1}{6}ma^2$, into Eq. (1) for

$$(m\bar{v}_1)(\tfrac{1}{2}a) = m(\tfrac{1}{2}\sqrt{2}a\omega_2)(\tfrac{1}{2}\sqrt{2}a) + \tfrac{1}{6}ma^2\omega_2 \quad \bar{v}_1 = \tfrac{4}{3}a\omega_2 \quad (2)$$

Conservation of Energy. Apply the principle of conservation of energy between position 2 and position 3 (Fig. 2) as

$$T_2 + V_2 = T_3 + V_3 \quad (3)$$

You need to determine the energy at these two positions.

Position 2. $V_2 = Wh_2$. Since $\bar{v}_2 = \tfrac{1}{2}\sqrt{2}a\omega_2$, you have

$$T_2 = \tfrac{1}{2}m\bar{v}_2^2 + \tfrac{1}{2}\bar{I}\omega_2^2 = \tfrac{1}{2}m(\tfrac{1}{2}\sqrt{2}a\omega_2)^2 + \tfrac{1}{2}(\tfrac{1}{6}ma^2)\omega_2^2 = \tfrac{1}{3}ma^2\omega_2^2$$

Position 3. The package must reach conveyor belt C , so it must pass through position 3 where G is directly above B . Also, since you wish to determine the smallest velocity for which the package will reach this position, choose $\bar{v}_3 = \omega_3 = 0$. Therefore, $T_3 = 0$ and $V_3 = Wh_3$.

Substituting these into Eq. (3)

$$\tfrac{1}{3}ma^2\omega_2^2 + Wh_2 = 0 + Wh_3$$

$$\omega_2^2 = \frac{3W}{ma^2}(h_3 - h_2) = \frac{3g}{a^2}(h_3 - h_2) \quad (4)$$

Substituting the computed values of h_2 and h_3 into Eq. (4), you obtain

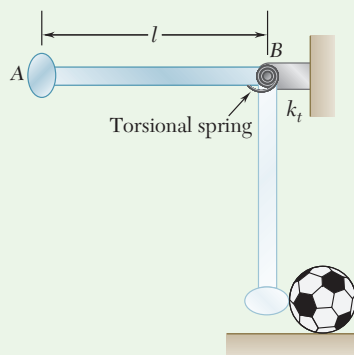
$$\omega_2^2 = \frac{3g}{a^2}(0.707a - 0.612a) = \frac{3g}{a^2}(0.095a) \quad \omega_2 = \sqrt{0.285g/a}$$

$$\bar{v}_1 = \tfrac{4}{3}a\omega_2 = \tfrac{4}{3}a\sqrt{0.285g/a} \quad \bar{v}_1 = 0.712\sqrt{ga} \quad \blacktriangleleft$$

REFLECT and THINK: The combination of energy and momentum methods is typical of many design analyses. If you had been interested in determining the reaction at B immediately after the impact or at some other point in the motion, you would have needed to draw a free-body diagram and a kinetic diagram and apply Newton's second law.

Sample Problem 17.15

A soccer ball tester consists of a 15-kg slender rod AB with a 1.1-kg simulated foot located at A and a torsional spring located at pin B . The torsional spring has a spring constant of $k_t = 910 \text{ N}\cdot\text{m}$ and is unstretched when AB is vertical. The length of AB is 0.9 m, and you can assume that the foot can be modeled as a point mass. Knowing that the velocity of the 0.45-kg soccer ball is 30 ft/s after impact, determine (a) the coefficient of restitution between the simulated foot and the ball, (b) the impulse at B during the impact.



STRATEGY: This problem can be broken into two distinct stages of motion. In stage 1, the arm moves downward under the influence of gravity and the torsional spring. You can use the conservation of energy for this stage. In stage 2, the foot hits the ball, and you need to use both the principle of impulse and momentum and the coefficient of restitution.

MODELING: Each stage requires a different system. For stage 1, your system is rod AB , foot B , and the torsional spring. In stage 2, your system is rod AB , foot B , and the soccer ball. The appropriate diagrams are drawn in the analysis section. You can model AB as a slender rod, so its mass moment of inertia is

$$\bar{I}_{AB} = \frac{1}{12}m_{AB}l^2 = \frac{1}{12}(15 \text{ kg})(0.9 \text{ m})^2 = 1.0125 \text{ kg}\cdot\text{m}^2$$

ANALYSIS:

Rod AB Moves Down. Apply the principle of conservation of energy

$$T_1 + V_{g_1} + V_{e_1} = T_2 + V_{g_2} + V_{e_2} \quad (1)$$

(continued)

Position 1. The system starts from rest, so $T_1 = 0$. Using the datum defined in Fig. 1, you know $V_{g_1} = 0$, and since the spring is unstretched at position 2, you find

$$V_{e_1} = \frac{1}{2}k_t\theta^2 = \frac{1}{2}(910 \text{ N}\cdot\text{m})\left(\frac{\pi}{2}\right)^2 = 1123 \text{ J}$$

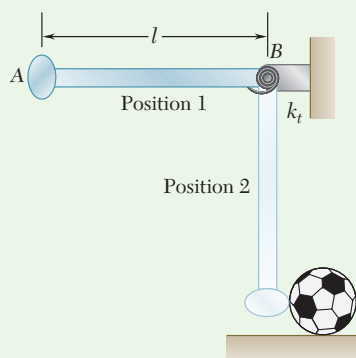


Fig. 1 The rod in positions 1 and 2.

Position 2. The elastic potential energy is $V_{e_2} = 0$, and the gravitational potential energy is

$$\begin{aligned} V_{g_2} &= -m_{AB}g\frac{l}{2} - m_Agl = -(15 \text{ kg})(9.81 \text{ m/s}^2)(0.45 \text{ m}) - (1.1 \text{ kg})(9.81 \text{ m/s}^2)(0.9 \text{ m}) \\ &= -75.93 \text{ J} \end{aligned}$$

The kinetic energy is

$$T_2 = \frac{1}{2}m_A v_A^2 + \frac{1}{2}m_{AB} v_G^2 + \frac{1}{2}\bar{I}_{AB}\omega^2$$

You can relate the velocity of the foot and the velocity of the center of gravity of the rod to the angular velocity of AB by recognizing that AB is undergoing fixed-axis rotation. Therefore, $v_G = \omega\frac{l}{2}$ and $v_A = \omega l$. Substituting these into the expression for T_2 and putting in values gives

$$T_2 = \frac{1}{2}\left(m_A l^2 + m_{AB}\left(\frac{l}{2}\right)^2 + \bar{I}_{AB}\right)\omega^2 = 2.4705\omega^2$$

Substituting these energy terms into Eq. (1) gives

$$0 + 0 + 1123 = 2.4705\omega^2 - 75.93 + 0$$

Solving for the angular velocity, you find $\omega = 22.03 \text{ rad/s}$. Knowing ω , you can calculate the velocities $v_G = 9.912 \text{ m/s}$ and $v_A = 19.824 \text{ m/s}$.

Foot A Impacts the Soccer Ball. Impulse–momentum diagrams for the impact on the ball are shown in Fig. 2.

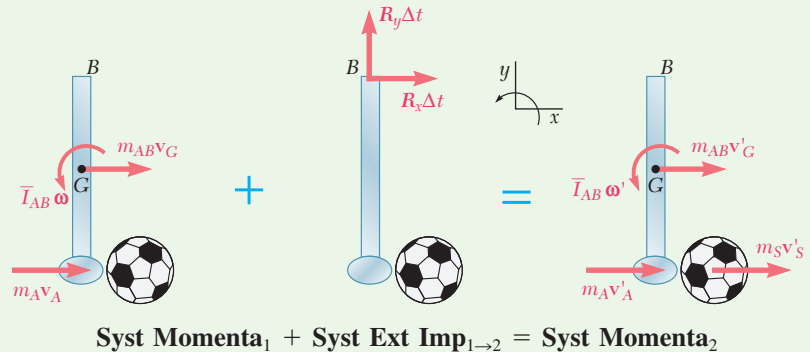


Fig. 2

Taking moments about B gives you

+ ↺ moments about B :

$$m_A v_A l + m_{AB} v_G \frac{l}{2} + I_{AB} \omega + 0 = m_A v'_A l + m_{AB} v'_G \frac{l}{2} + \bar{I}_{AB} \omega' + m_S v'_S l \quad (2)$$

The equation for the coefficient of restitution is

$$v'_S - v'_A = e(v_A - 0) \quad (3)$$

where $v'_S = 30$ m/s. From kinematics, you know $v'_A = \omega' l$ and $v'_G = \omega' (l/2)$. Using these kinematic equations and Eqs. (2) and (3), you can solve for the unknown quantities

$$v'_A = 17.61 \text{ m/s} \quad v'_G = 8.81 \text{ m/s} \quad \omega' = 19.57 \text{ rad/s} \quad e = 0.625$$

$$e = 0.625 \quad \blacktriangleleft$$

Impulses During Impact. Applying impulse–momentum in the x - and y -directions gives

$$\rightarrow x\text{-components: } m_{AB} v_G + m_A v_A + R_x \Delta t = m_{AB} v'_G + m_A v'_A + m_S v'_S \quad (4)$$

$$+\uparrow y\text{-components: } 0 + R_y \Delta t = 0 \quad (5)$$

Solving these equations, you find $R_x \Delta t = -5.53$ N and $R_y \Delta t = 0$.

$$R \Delta t = 5.53 \text{ N} \quad \blacktriangleleft$$

REFLECT and THINK: This coefficient of restitution seems reasonable. As you decrease the pressure in the ball, you would expect the coefficient of restitution to decrease; therefore, the distance the ball travels will decrease. If you had been asked to determine the reactions at B after the impact, you would need to draw a free-body diagram and kinetic diagram for your system and apply Newton's second law.

SOLVING PROBLEMS ON YOUR OWN

This section was devoted to **impulsive motion** and to the **eccentric impact** of rigid bodies.

1. Impulsive motion occurs when a rigid body is subjected to a very large force $\mathbf{F}$ for a very short interval of time Δt ; the resulting impulse $\mathbf{F}_{\text{avg}} \Delta t$ is both finite and different from zero. Such forces are referred to as **impulsive forces** and arise whenever an impact occurs between two rigid bodies. Forces for which the impulse is negligible are referred to as **nonimpulsive forces**. As discussed in Chap. 13, you can assume the following forces to be nonimpulsive: the weight of a body, the force exerted by a spring, and any other force that is known to be small by comparison with the impulsive forces. Unknown reactions, however, cannot be assumed to be nonimpulsive.

2. Eccentric impact of rigid bodies. When two bodies collide, the velocity components along the line of impact of the points of contact A and B before and after impact satisfy

$$(v'_B)_n - (v'_A)_n = e[(v_A)_n - (v_B)_n] \quad (17.19)$$

where the left-hand side is the *relative velocity after the impact* and the right-hand side is the product of the coefficient of restitution and the *relative velocity before the impact*.

This equation expresses the same relation between the velocity components of the points of contact before and after an impact that you used for particles in Chap. 13.

3. To solve a problem involving an impact you should use the *method of impulse and momentum* and take the following steps.

a. Draw an impulse–momentum diagram of the system showing the momenta immediately before impact plus the impulses of the external forces acting during the impact; this sum is equivalent to the momenta immediately after impact.

b. Write the governing equations for the angular momentum about some point. Depending on the problem type (especially when you want to find support impulsive reactions), you may also need to write the equations for linear momentum [Sample Prob. 17.11].

c. In the case of an impact in which $e > 0$, the number of unknowns will be greater than the number of equations that you can write by summing components and moments. You should supplement the equations obtained from the impulse–momentum diagram with the coefficient of restitution from Eq. (17.19) that relates the relative velocities of the points of contact before and after impact [Sample Prob. 17.13 and 17.15].

d. During an impact, you must use the method of impulse and momentum. However, *before and after the impact* you can, if necessary, use some of the other methods of solution that you have learned, such as the conservation of energy [Sample Prob. 17.14 and 17.15] or Newton's second law.

Problems

IMPULSE–MOMENTUM DIAGRAM PRACTICE PROBLEMS

- 17.F4** A uniform slender rod AB of mass m is at rest on a frictionless horizontal surface when hook C engages a small pin at A . Knowing that the hook is pulled upward with a constant velocity $\mathbf{v}_0$, draw the impulse–momentum diagram that is needed to determine the impulse exerted on the rod at A and B . Assume that the velocity of the hook is unchanged and that the impact is perfectly plastic.

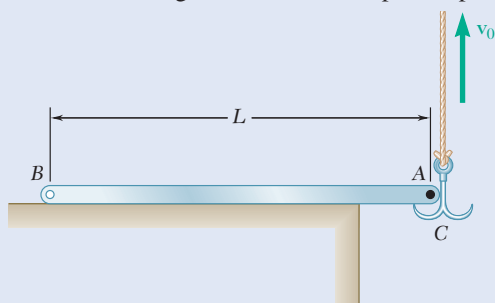


Fig. P17.F4

- 17.F5** A uniform slender rod AB of length L is falling freely with a velocity $\mathbf{v}_0$ when cord AC suddenly becomes taut. Assuming that the impact is perfectly plastic, draw the impulse–momentum diagram that is needed to determine the angular velocity of the rod and the velocity of its mass center immediately after the cord becomes taut.

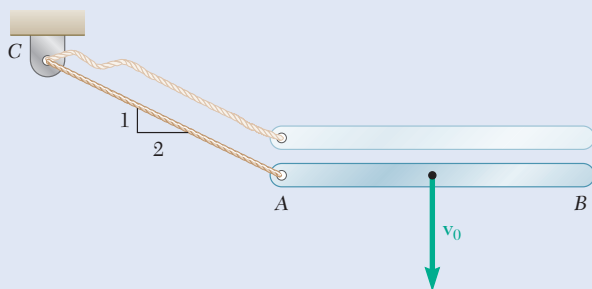


Fig. P17.F5

- 17.F6** A slender rod CDE of length L and mass m is attached to a pin support at its midpoint D . A second and identical rod AB is rotating about a pin support at A with an angular velocity ω_1 when its end B strikes end C of rod CDE . The coefficient of restitution between the rods is e . Draw the impulse–momentum diagrams that are needed to determine the angular velocity of each rod immediately after the impact.

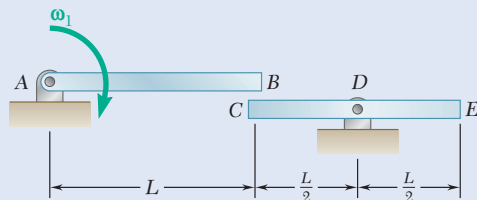


Fig. P17.F6

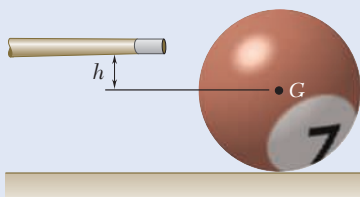


Fig. P17.96

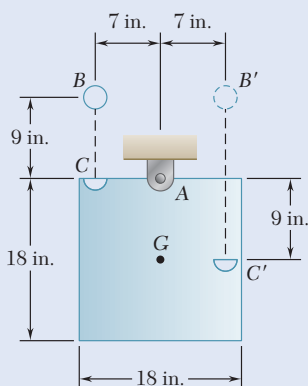


Fig. P17.99 and P17.100

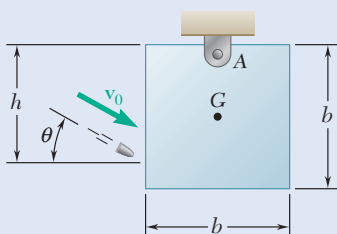


Fig. P17.101 and P17.102

END-OF-SECTION PROBLEMS

17.96 At what height h above its center G should a billiard ball of radius r be struck horizontally by a cue if the ball is to start rolling without sliding?

17.97 A bullet weighing 0.08 lb is fired with a horizontal velocity of 1800 ft/s into the lower end of a slender 15-lb bar of length $L = 30$ in. Knowing that $h = 12$ in. and that the bar is initially at rest, determine (a) the angular velocity of the bar immediately after the bullet becomes embedded, (b) the impulsive reaction at C , assuming that the bullet becomes embedded in 0.001 s.

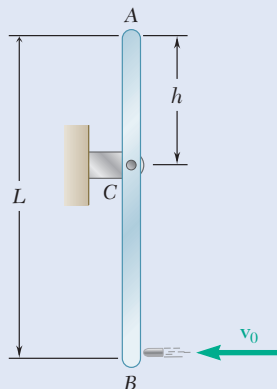


Fig. P17.97

17.98 In Prob. 17.97, determine (a) the required distance h if the impulsive reaction at C is to be zero, (b) the corresponding angular velocity of the bar immediately after the bullet becomes embedded.

17.99 A 16-lb wooden panel is suspended from a pin support at A and is initially at rest. A 4-lb metal sphere is released from rest at B and falls into a hemispherical cup C attached to the panel at a point located on its top edge. Assuming that the impact is perfectly plastic, determine the velocity of the mass center G of the panel immediately after the impact.

17.100 A 16-lb wooden panel is suspended from a pin support at A and is initially at rest. A 4-lb metal sphere is released from rest at B' and falls into a hemispherical cup C' attached to the panel at the same level as the mass center G . Assuming that the impact is perfectly plastic, determine the velocity of the mass center G of the panel immediately after the impact.

17.101 A 45-g bullet is fired with a velocity of 400 m/s at $\theta = 30^\circ$ into a 9-kg square panel of side $b = 200$ mm. Knowing that $h = 150$ mm and that the panel is initially at rest, determine (a) the velocity of the center of the panel immediately after the bullet becomes embedded, (b) the impulsive reaction at A , assuming that the bullet becomes embedded in 2 ms.

17.102 A 45-g bullet is fired with a velocity of 400 m/s at $\theta = 5^\circ$ into a 9-kg square panel of side $b = 200$ mm. Knowing that the panel is initially at rest, determine (a) the required distance h if the horizontal component of the impulsive reaction at A is to be zero, (b) the corresponding velocity of the center of the panel immediately after the bullet becomes embedded.

- 17.103** Two uniform rods, each of mass m , form the L-shaped rigid body ABC , which is initially at rest on the frictionless horizontal surface when hook D of the carriage E engages a small pin at C . Knowing that the carriage is pulled to the right with a constant velocity v_0 , determine immediately after the impact (a) the angular velocity of the body, (b) the velocity of corner B . Assume that the velocity of the carriage is unchanged and that the impact is perfectly plastic.

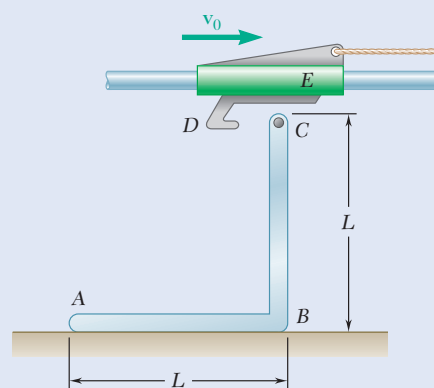


Fig. P17.103

- 17.104** The uniform slender rod AB of weight 5 lb and length 30 in. forms an angle $\beta = 30^\circ$ with the vertical as it strikes the smooth corner shown with a vertical velocity v_1 of magnitude 8 ft/s and no angular velocity. Assuming that the impact is perfectly plastic, determine the angular velocity of the rod immediately after the impact.

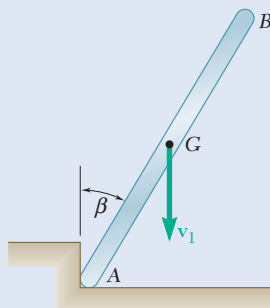


Fig. P17.104

- 17.105** A bullet weighing 0.08 lb is fired with a horizontal velocity of 1800 ft/s into the 15-lb wooden rod AB of length $L = 30$ in. The rod, which is initially at rest, is suspended by a cord of length $L = 30$ in. Determine the distance h for which, immediately after the bullet becomes embedded, the instantaneous center of rotation of the rod is point C .

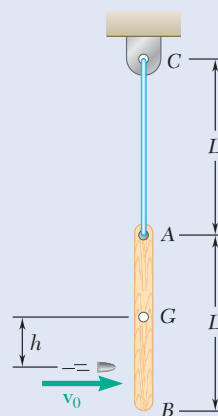


Fig. P17.105

- 17.106** A prototype of an adapted bowling device is a simple ramp that attaches to a wheelchair. The bowling ball has a mass moment of inertia about its center of gravity of cmr^2 , where c is a unitless constant, r is the radius, and m is its mass. The athlete nudges the ball slightly from a height of h , and the ball rolls down the ramp without slipping. It hits the bowling lane, and after slipping for a short distance, it begins to roll again. Assuming that the ball does not bounce as it hits the lane, determine the angular velocity and velocity of the mass center of the ball after it has resumed rolling.

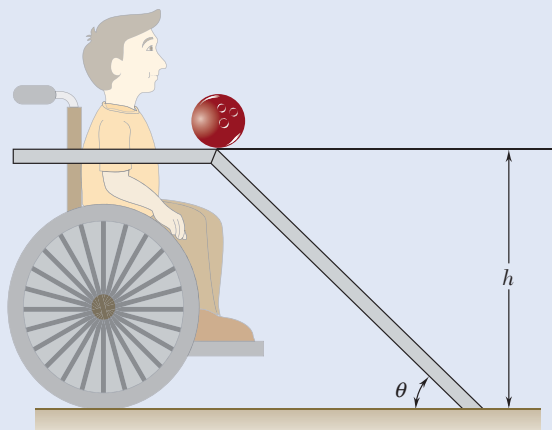


Fig. P17.106

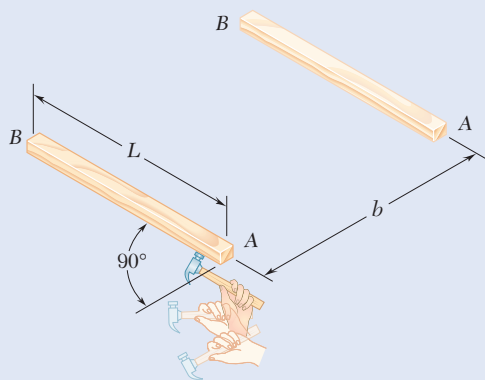


Fig. P17.107

17.107 A uniform slender rod AB is at rest on a frictionless horizontal table when end A of the rod is struck by a hammer that delivers an impulse that is perpendicular to the rod. In the subsequent motion, determine the distance b through which the rod will move each time it completes a full revolution.

17.108 A bullet of mass m is fired with a horizontal velocity v_0 and at a height $h = \frac{1}{2}R$ into a wooden disk of much larger mass M and radius R . The disk rests on a horizontal plane and the coefficient of friction between the disk and the plane is finite. (a) Determine the linear velocity $\bar{v}_1$ and the angular velocity ω_1 of the disk immediately after the bullet has penetrated the disk. (b) Describe the ensuing motion of the disk and determine its linear velocity after the motion has become uniform.

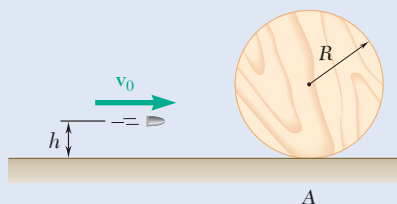


Fig. P17.108 and P17.109

17.109 Determine the height h at which the bullet of Prob. 17.108 should be fired (a) if the disk is to roll without sliding immediately after impact, (b) if the disk is to slide without rolling immediately after impact.

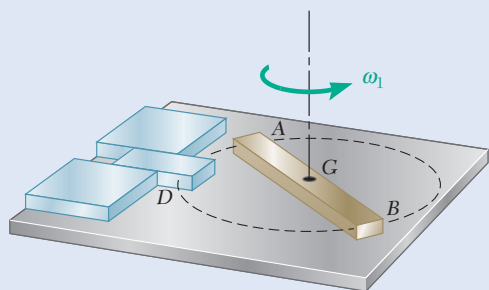


Fig. P17.110

17.110 A uniform slender bar of length $L = 200$ mm and mass $m = 0.5$ kg is supported by a frictionless horizontal table. Initially the bar is spinning about its mass center G with a constant angular speed $\omega_1 = 6$ rad/s. Suddenly latch D is moved to the right and is struck by end A of the bar. Knowing that the coefficient of restitution between A and D is $e = 0.6$, determine the angular velocity of the bar and the velocity of its mass center immediately after the impact.

17.111 A uniform slender rod of length L is dropped onto rigid supports at A and B . Since support B is slightly lower than support A , the rod strikes A with a velocity $\bar{v}_1$ before it strikes B . Assuming perfectly elastic impact at both A and B , determine the angular velocity of the rod and the velocity of its mass center immediately after the rod (a) strikes support A , (b) strikes support B , (c) again strikes support A .

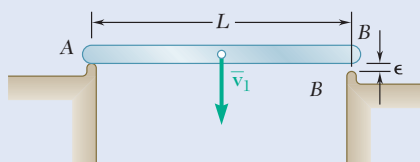


Fig. P17.111

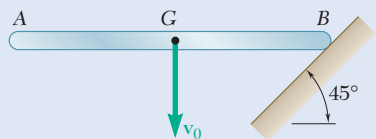


Fig. P17.112

17.112 A uniform slender rod AB has a mass m , a length L , and is falling freely with a velocity v_0 when end B strikes a smooth inclined surface as shown. Assuming that the impact is perfectly elastic, determine the angular velocity of the rod and the velocity of its mass center immediately after the impact.

17.113 The slender rod AB of length $L = 1$ m forms an angle $\beta = 30^\circ$ with the vertical as it strikes the frictionless surface shown with a vertical velocity $\bar{v}_1 = 2$ m/s and no angular velocity. Knowing that the coefficient of restitution between the rod and the ground is $e = 0.8$, determine the angular velocity of the rod immediately after the impact.

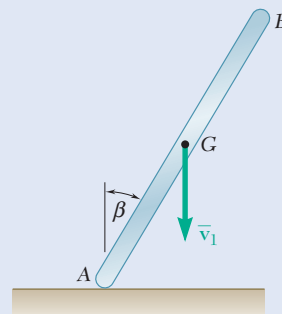
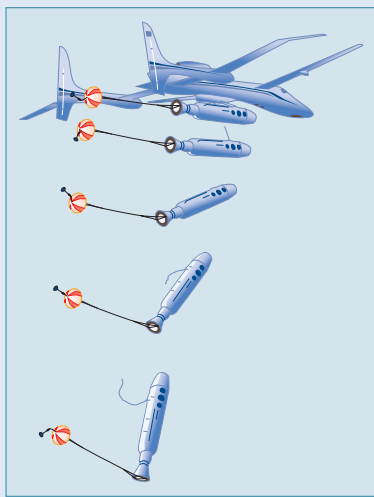
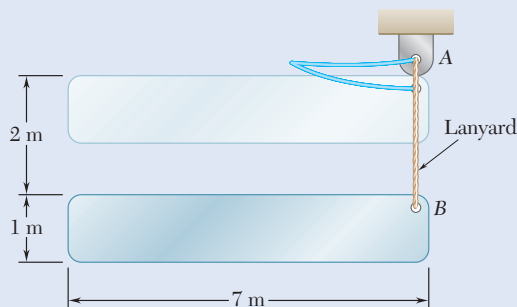


Fig. P17.113

17.114 The trapeze/lanyard air drop (t/LAD) launch is a proposed innovative method for airborne launch of a payload-carrying rocket. The release sequence involves several steps as shown in (1) where the payload rocket is shown at various instances during the launch. To investigate the first step of this process, where the rocket body drops freely from the carrier aircraft until the 2-m lanyard stops the vertical motion of B , a trial rocket is tested as shown in (2). The rocket can be considered a uniform 1×7 -m rectangle with a mass of 4000 kg. Knowing that the rocket is released from rest and falls vertically 2 m before the lanyard becomes taut, determine the angular velocity of the rocket immediately after the lanyard is taut.



(1)



(2)

Fig. P17.114

17.115 The uniform rectangular block shown is moving along a frictionless surface with a velocity $\bar{v}_1$ when it strikes a small obstruction at B . Assuming that the impact between corner A and obstruction B is perfectly plastic, determine the magnitude of the velocity $\bar{v}_1$ for which the maximum angle θ through which the block will rotate will be 30° .

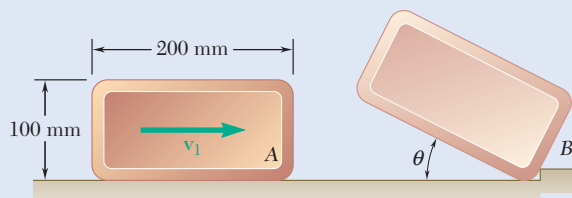


Fig. P17.115

- 17.116** The 40-kg gymnast drops from her maximum height of $h = 0.5$ m straight down to the bar as shown. Her hands hit the bar and clasp onto it, and her body remains straight in the position shown. Her center of mass is 0.75 meters away from her hands, and her mass moment of inertia about her center of mass is $7.5 \text{ kg}\cdot\text{m}^2$. Assuming that friction between the bar and her hands is negligible and that she remains in the same position throughout the swing, determine her angular velocity when she swings around to $\theta = 135^\circ$.

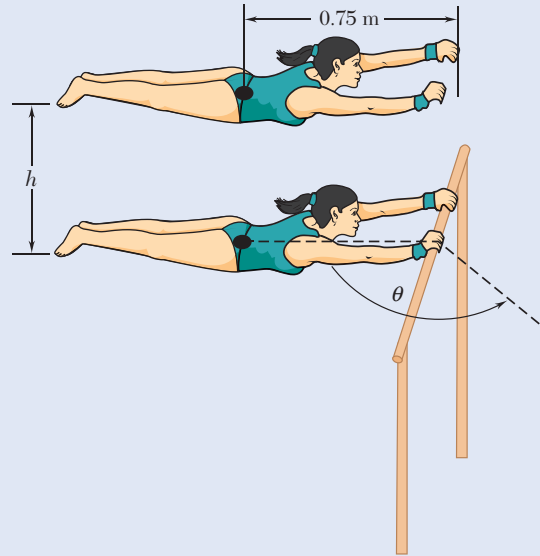


Fig. P17.116

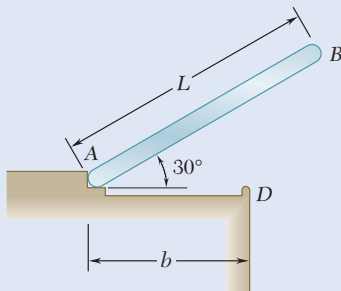


Fig. P17.117

- 17.117** A slender rod of mass m and length L is released from rest in the position shown and hits edge D . Assuming perfectly plastic impact at D , determine for $b = 0.6L$, (a) the angular velocity of the rod immediately after the impact, (b) the maximum angle through which the rod will rotate after the impact.

- 17.118** A uniformly loaded square crate is released from rest with its corner D directly above A ; it rotates about A until its corner B strikes the floor, and then rotates about B . The floor is sufficiently rough to prevent slipping and the impact at B is perfectly plastic. Denoting by ω_0 the angular velocity of the crate immediately before B strikes the floor, determine (a) the angular velocity of the crate immediately after B strikes the floor, (b) the fraction of the kinetic energy of the crate lost during the impact, (c) the angle θ through which the crate will rotate after B strikes the floor.

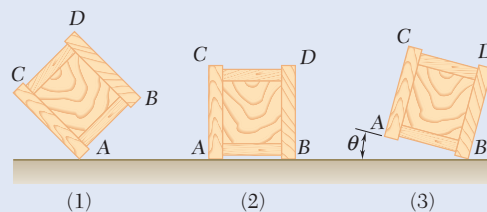


Fig. P17.118

- 17.119** A 1-oz bullet is fired with a horizontal velocity of 750 mi/h into the 18-lb wooden beam AB . The beam is suspended from a collar of negligible mass that can slide along a horizontal rod. Neglecting friction between the collar and the rod, determine the maximum angle of rotation of the beam during its subsequent motion.

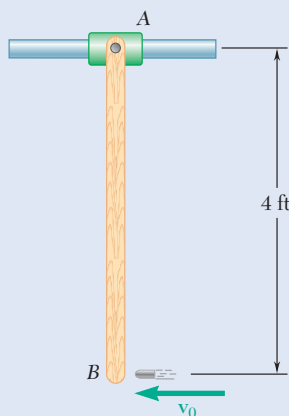


Fig. P17.119

- 17.120** For the beam of Prob. 17.119, determine the velocity of the 1-oz bullet for which the maximum angle of rotation of the beam will be 90° .
- 17.121** The plank CDE has a mass of 15 kg and rests on a small pivot at D . The 55-kg gymnast A is standing on the plank at C when the 70-kg gymnast B jumps from a height of 2.5 m and strikes the plank at E . Assuming perfectly plastic impact and that gymnast A is standing absolutely straight, determine the height to which gymnast A will rise.
- 17.122** Solve Prob. 17.121, assuming that the gymnasts change places so that gymnast A jumps onto the plank while gymnast B stands at C .
- 17.123** A slender rod AB is released from rest in the position shown. It swings down to a vertical position and strikes a second and identical rod CD that is resting on a frictionless surface. Assuming that the coefficient of restitution between the rods is 0.4, determine the velocity of rod CD immediately after the impact.
- 17.124** A slender rod AB is released from rest in the position shown. It swings down to a vertical position and strikes a second and identical rod CD that is resting on a frictionless surface. Assuming that the impact between the rods is perfectly elastic, determine the velocity of rod CD immediately after the impact.

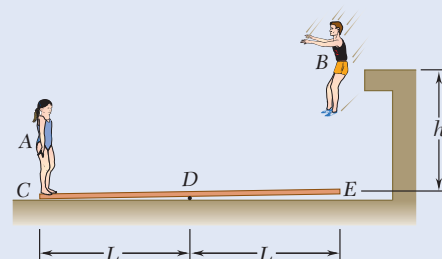


Fig. P17.121

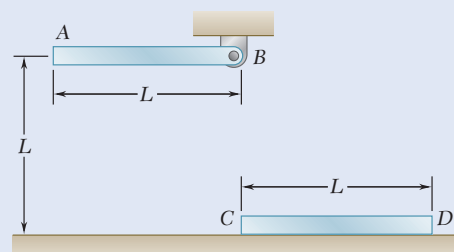


Fig. P17.123 and P17.124

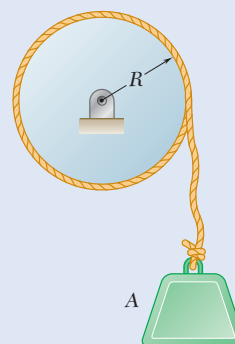


Fig. P17.125

- 17.125** Block A has a mass m and is attached to a cord that is wrapped around a uniform disk with a mass M . The block is released from rest and falls through a distance h before the cord becomes taut. Derive expressions for the velocity of the block and the angular velocity of the disk immediately after the impact. Assume that the impact is (a) perfectly plastic, (b) perfectly elastic.

- 17.126** A 2-kg solid sphere of radius $r = 40$ mm is dropped from a height $h = 200$ mm and lands on a uniform slender plank AB of mass 4 kg and length $L = 500$ mm that is held by two inextensible cords. Knowing that the impact is perfectly plastic and that the sphere remains attached to the plank at a distance $a = 40$ mm from the left end, determine the velocity of the sphere immediately after impact. Neglect the thickness of the plank.

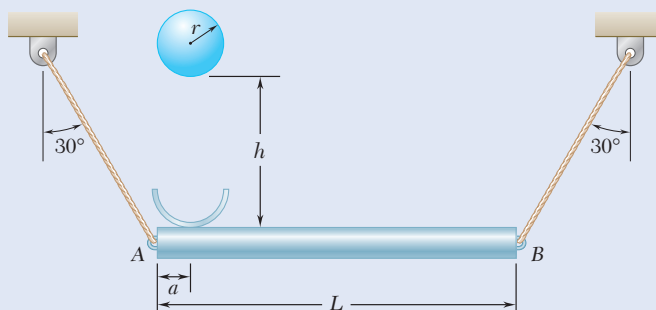


Fig. P17.126

- 17.127 and 17.128** Member ABC has a mass of 2.4 kg and is attached to a pin support at B . An 800-g sphere D strikes the end of member ABC with a vertical velocity v_1 of 3 m/s. Knowing that $L = 750$ mm and that the coefficient of restitution between the sphere and member ABC is 0.5, determine immediately after the impact (a) the angular velocity of member ABC , (b) the velocity of the sphere.

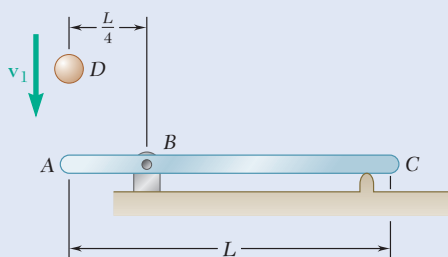


Fig. P17.127

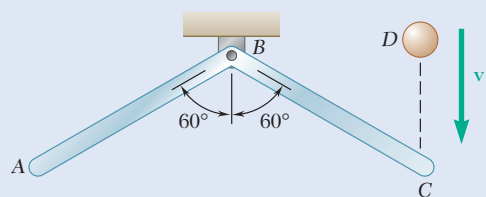


Fig. P17.128

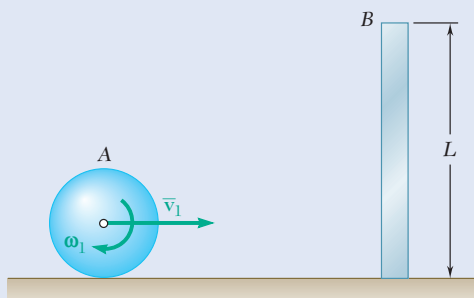


Fig. P17.129

- 17.129** Sphere A of mass $m_A = 2$ kg and radius $r = 40$ mm rolls without slipping with a velocity $\bar{v}_1 = 2$ m/s on a horizontal surface when it hits squarely a uniform slender bar B of mass $m_B = 0.5$ kg and length $L = 100$ mm that is standing on end and is at rest. Denoting by μ_k the coefficient of kinetic friction between the sphere and the horizontal surface, neglecting friction between the sphere and the bar, and knowing the coefficient of restitution between A and B is 0.1, determine the angular velocities of the sphere and the bar immediately after the impact.

- 17.130** A large 3-lb sphere with a radius $r = 3$ in. is thrown into a light basket at the end of a thin, uniform rod weighing 2 lb and length $L = 10$ in. as shown. Immediately before the impact, the angular velocity of the rod is 3 rad/s counterclockwise and the velocity of the sphere is 2 ft/s down. Assume the sphere sticks in the basket. Determine after the impact (a) the angular velocity of the bar and sphere, (b) the components of the reactions at A.

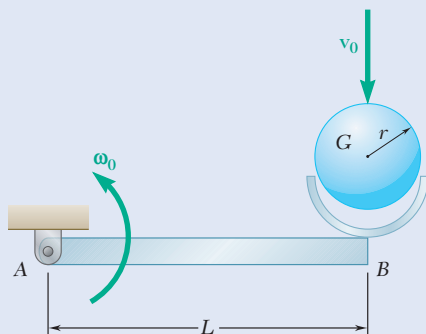


Fig. P17.130

- 17.131** A small rubber ball of radius r is thrown against a rough floor with a velocity $\bar{v}_A$ of magnitude v_0 and a backspin ω_A of magnitude ω_0 . It is observed that the ball bounces from A to B, then from B to A, then from A to B, etc. Assuming perfectly elastic impact, determine the required magnitude ω_0 of the backspin in terms of $\bar{v}_0$ and r .

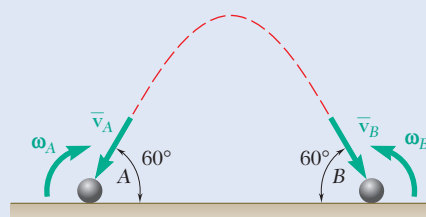


Fig. P17.131

- 17.132** Sphere A of mass m and radius r rolls without slipping with a velocity $\bar{v}_1$ on a horizontal surface when it hits squarely an identical sphere B that is at rest. Denoting by μ_k the coefficient of kinetic friction between the spheres and the surface, neglecting friction between the spheres, and assuming perfectly elastic impact, determine (a) the linear and angular velocities of each sphere immediately after the impact, (b) the velocity of each sphere after it has started rolling uniformly.

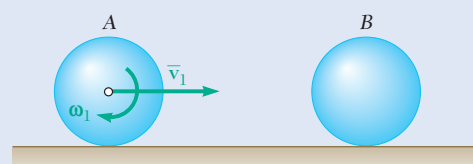


Fig. P17.132

- 17.133** In a game of pool, ball A is rolling without slipping with a velocity $\bar{v}_0$ as it hits obliquely ball B, which is at rest. Denoting by r the radius of each ball and by μ_k the coefficient of kinetic friction between a ball and the table, and assuming perfectly elastic impact, determine (a) the linear and angular velocity of each ball immediately after the impact, (b) the velocity of ball B after it has started rolling uniformly.

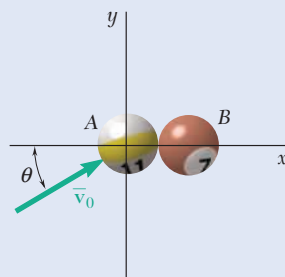


Fig. P17.133

- 17.134** Each of the bars AB and BC is of length $L = 400$ mm and mass $m = 1.2$ kg. Determine the angular velocity of each bar immediately after the impulse $\mathbf{Q}\Delta t = (1.5 \text{ N}\cdot\text{s})\mathbf{i}$ is applied at C.

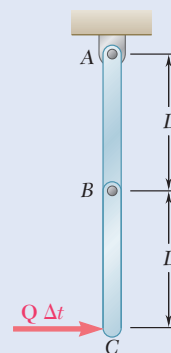


Fig. P17.134

Review and Summary

In this chapter, we again considered the method of work and energy and the method of impulse and momentum. In the first section, we applied the method of work and energy to the analysis of the motion of rigid bodies and systems of rigid bodies.

The second section was devoted to the method of impulse and momentum and its application to the solution of various types of problems involving the plane motion of rigid bodies and rigid bodies symmetrical with respect to the reference plane.

Principle of Work and Energy for a Rigid Body

In Sec. 17.1, we first expressed the principle of work and energy for a rigid body in the form

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (17.1)$$

where T_1 and T_2 represent the initial and final values of the kinetic energy of the rigid body and $U_{1 \rightarrow 2}$ represents the work of the external forces acting on it. If we express the work done by nonconservative forces as $U_{1 \rightarrow 2}^{\text{NC}}$ and define potential energy terms for conservative forces, we can express Eq. (17.1) as

$$T_1 + V_{g_1} + V_{e_1} + U_{1 \rightarrow 2}^{\text{NC}} = T_2 + V_{g_2} + V_{e_2} \quad (17.1')$$

where V_{g_1} and V_{g_2} are the initial and final gravitational potential energy of the center of mass of the rigid body and V_{e_1} and V_{e_2} are the initial and final values of the elastic energy associated with springs in the system, respectively.

Work of a Force or a Couple

In Sec. 17.1B, we recalled the expression found in Chap. 13 for the work of a force $\mathbf{F}$ applied at a point A , namely

$$U_{1 \rightarrow 2} = \int_{A_1}^{A_2} \mathbf{F} \cdot d\mathbf{r} \quad (17.3)$$

or

$$U_{1 \rightarrow 2} = \int_{s_1}^{s_2} (F \cos \alpha) ds \quad (17.3')$$

where F is the magnitude of the force, α is the angle it forms with the direction of motion of A , and s is the variable of integration measuring the distance traveled by A along its path. We also derived the expression for the work of a couple of moment $\mathbf{M}$ applied to a rigid body during a rotation in θ of the rigid body as

$$U_{1 \rightarrow 2} = \int_{\theta_1}^{\theta_2} M d\theta \quad (17.5)$$

Kinetic Energy in Plane Motion

We then derived an expression for the kinetic energy of a rigid body in plane motion [Sec. 17.1C]:

$$T = \frac{1}{2} m \bar{v}^2 + \frac{1}{2} I \bar{\omega}^2 \quad (17.9)$$

where $\bar{v}$ is the speed of the mass center G of the body, ω is the angular speed of the body, and $\bar{I}$ is its moment of inertia about an axis through G perpendicular to the plane of reference (Fig. 17.13) [Sample Prob. 17.3]. We noted that the kinetic energy of a rigid body in plane motion can be separated into two parts: (1) the kinetic energy $\frac{1}{2}m\bar{v}^2$ associated with the motion of the mass center G of the body and (2) the kinetic energy $\frac{1}{2}\bar{I}\omega^2$ associated with the rotation of the body about G . You will generally need to use kinematics to relate $\bar{v}$ and ω .

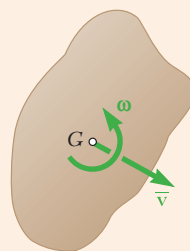


Fig. 17.13

Kinetic Energy in Rotation About a Fixed Axis

For a rigid body rotating about a fixed axis through O with an angular velocity ω , we had

$$T = \frac{1}{2}I_O\omega^2 \quad (17.10)$$

where I_O is the moment of inertia of the body about the fixed axis. We noted that this result is not limited to the rotation of plane rigid bodies or of bodies symmetrical with respect to the reference plane, but it also is valid regardless of the shape of the body or of the location of the axis of rotation.

Systems of Rigid Bodies

Equation (17.1) can be applied to the motion of systems of rigid bodies [Sec. 17.1D] as long as all the forces acting on the various bodies involved—internal as well as external to the system—are included in the computation of $U_{1 \rightarrow 2}$. However, in the case of systems consisting of pin-connected members or blocks and pulleys connected by inextensible cords or meshed gears, the points of application of the internal forces move through equal distances and the work of these forces cancels out [Sample Probs. 17.1, 17.2, and 17.6].

Conservation of Energy

When a rigid body or a system of rigid bodies moves under the action of conservative forces, the principle of work and energy can be expressed in the form

$$T_1 + V_1 = T_2 + V_2 \quad (17.12)$$

or

$$T_1 + V_{g_1} + V_{e_1} = T_2 + V_{g_2} + V_{e_2} \quad (17.12')$$

This is referred to as the *principle of conservation of energy* [Sec. 17.1E]. We can use this principle to solve problems involving conservative forces such as the force of gravity or the force exerted by a spring [Sample Probs. 17.4 through 17.6]. However, if we need to determine a reaction, we must supplement the principle of conservation of energy by using Newton's second law [Sample Prob. 17.4].

Power

In Sec. 17.1F, we extended the concept of power to a rotating body subjected to a couple as

$$\text{Power} = \frac{dU}{dt} = \frac{Md\theta}{dt} = M\omega \quad (17.13)$$

where M is the magnitude of the couple and ω is the magnitude of the angular velocity of the body.

Principle of Impulse and Momentum for a Rigid Body

In Sec. 17.2, we applied the principle of impulse and momentum as had been derived in Sec. 14.2C for a system of particles to the motion of a rigid body [Sec. 17.2A]. We have

$$\text{Syst Momenta}_1 + \text{Syst Ext Imp}_{1 \rightarrow 2} = \text{Syst Momenta}_2 \quad (17.14)$$

Next we showed that, for a rigid body symmetrical with respect to the reference plane, the system of the momenta of the particles forming the body is equivalent to a vector $m\bar{v}$ attached at the mass center G of the body and a couple $\bar{I}\omega$ (Fig. 17.14). The vector $m\bar{v}$ is associated with the translation of the body with G and represents the *linear momentum* of the body, whereas the couple $\bar{I}\omega$ corresponds to the rotation of the body about G and represents the *angular momentum* of the body about an axis through G .

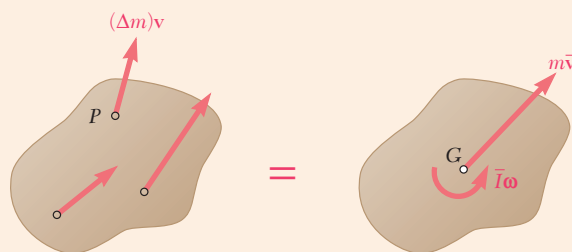


Fig. 17.14

We can express Eq. (17.14) graphically using an impulse–momentum diagram, as shown in Fig. 17.15. This diagram represents the system of the

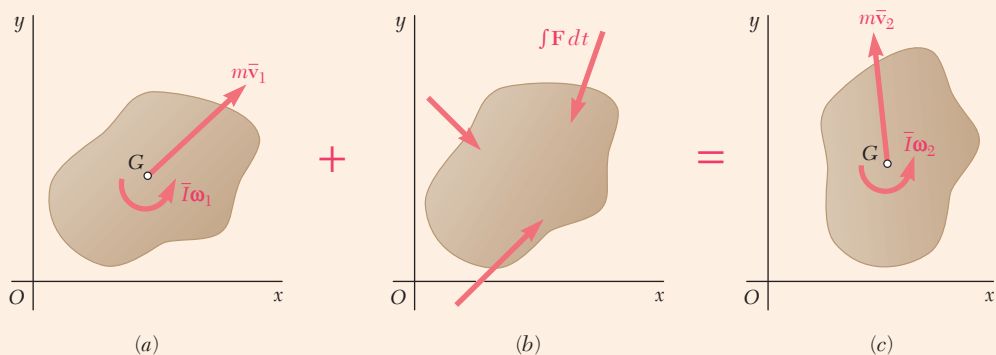


Fig. 17.15

initial momenta of the body, the impulses of the external forces acting on the body, and the system of the final momenta of the body, respectively. We can choose to sum moments about an arbitrary point P using

$$\bar{I}\omega_1 + m\bar{v}_1 d_{\perp} + \sum \int_{t_1}^{t_2} M_P dt = \bar{I}\omega_2 + m\bar{v}_2 d_{\perp} \quad (17.14')$$

the center of mass G using

$$\bar{I}\omega_1 + \sum \int_{t_1}^{t_2} M_G dt = \bar{I}\omega_2 \quad (17.14'')$$

or a fixed axis of rotation O using

$$I_O\omega_1 + \sum \int_{t_1}^{t_2} M_O dt = I_O\omega_2 \quad (17.16)$$

Using one of these expressions and the x and y components of the linear impulse–momentum equation, we obtain three equations of motion that we can solve for the desired unknowns [Sample Probs. 17.7 and 17.8].

In problems dealing with several connected rigid bodies [Sec. 17.2B], we can consider each body separately [Sample Prob. 17.7], or if no more than three unknowns are involved, we can apply the principle of impulse and momentum to the entire system, considering the impulses of the external forces only [Sample Prob. 17.9].

Conservation of Angular Momentum

When the lines of action of all the external forces acting on a system of rigid bodies pass through a given point O , the angular momentum of the system about O is conserved [Sec. 17.2C]. We suggested that problems involving conservation of angular momentum be solved by the general method described previously [Sample Prob. 17.9 and 17.10].

Impulsive Motion

Section 17.3 was devoted to the **impulsive motion** and the **eccentric impact** of rigid bodies. We recalled that the method of impulse and momentum is the only practicable method for the solution of problems involving impulsive motion and that the computation of impulses in such problems is particularly simple [Sample Prob. 17.11 and 17.12].

Eccentric Impact

We also recalled that the eccentric impact of two rigid bodies is defined as an impact in which the mass centers of the colliding bodies are *not* located on the line of impact. We showed that, in such a situation, a relation similar to that derived in Chap. 13 for the central impact of two particles and involving the coefficient of restitution e still holds, but *the velocities of points A and B where contact occurs during the impact should be used*. We have

$$(v'_B)_n - (v'_A)_n = e[(v_A)_n - (v_B)_n] \quad (17.19)$$

where $(v_A)_n$ and $(v_B)_n$ are the components along the line of impact of the velocities of A and B before the impact, and $(v'_A)_n$ and $(v'_B)_n$ are their components after the impact (Fig. 17.16). Equation (17.19) applies not only when the colliding bodies move freely after the impact but also when the bodies are partially constrained in their motion. You should use it in conjunction with one or several other equations obtained by applying the principle of impulse and momentum [Sample Prob. 17.13]. We also considered problems where the method of impulse and momentum and the method of work and energy can be combined [Sample Prob. 17.14].

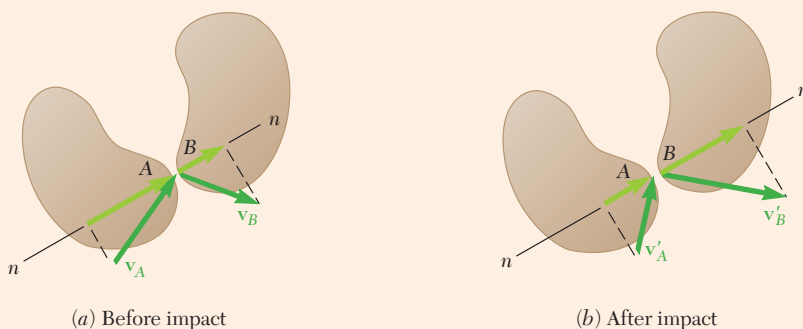


Fig. 17.16

Review Problems

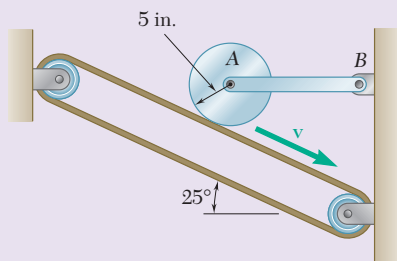


Fig. P17.135

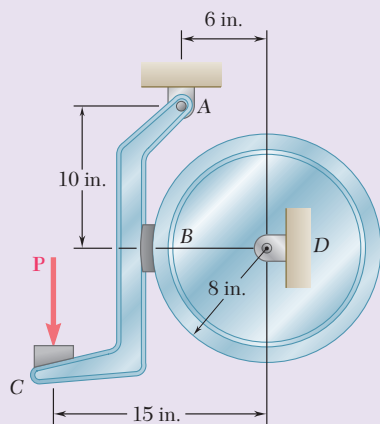


Fig. P17.136

17.135 A uniform disk of constant thickness and initially at rest is placed in contact with the belt shown, which moves at a constant speed $v = 80$ ft/s. Knowing that the coefficient of kinetic friction between the disk and the belt is 0.15, determine (a) the number of revolutions executed by the disk before it reaches a constant angular velocity, (b) the time required for the disk to reach that constant angular velocity.

17.136 The 8-in.-radius brake drum is attached to a larger flywheel that is not shown. The total mass moment of inertia of the flywheel and drum is $14 \text{ lb}\cdot\text{ft}\cdot\text{s}^2$ and the coefficient of kinetic friction between the drum and the brake shoe is 0.35. Knowing that the initial angular velocity of the flywheel is 360 rpm counterclockwise, determine the vertical force $\mathbf{P}$ that must be applied to the pedal C if the system is to stop in 100 revolutions.

17.137 Charpy impact test pendulums are used to determine the amount of energy a test specimen absorbs during an impact (see ASTM Standard E23). The hammer weighs 71.2 lbs and has a mass moment of inertia about its center of gravity G_H of $20.9 \text{ slug}\cdot\text{in}^2$. The arm weighs 19.5 lbs and has a mass moment of inertia about its own center of gravity G_A of $47.1 \text{ slug}\cdot\text{in}^2$. The pendulum is released from rest from an initial position of $\theta = 39^\circ$. Knowing that the friction at pin O is negligible, determine (a) the impact speed when the hammer hits the test specimen, (b) the force on the pin O just before the hammer hits the test specimen, (c) the amount of energy that the test specimen absorbs if the hammer swings up to a maximum of $\phi = 70^\circ$ after the impact.

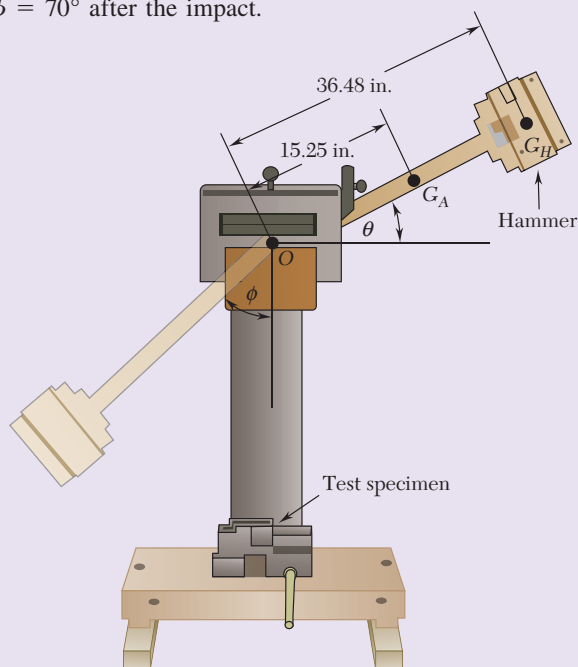


Fig. P17.137

- 17.138** The gear shown has a radius $R = 150$ mm and a radius of gyration $\bar{k} = 125$ mm. The gear is rolling without sliding with a velocity $\bar{v}_1$ of magnitude 3 m/s when it strikes a step of height $h = 75$ mm. Because the edge of the step engages the gear teeth, no slipping occurs between the gear and the step. Assuming perfectly plastic impact, determine (a) the angular velocity of the gear immediately after the impact, (b) the angular velocity of the gear after it has rotated to the top of the step.

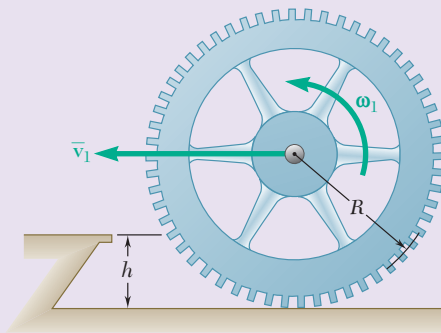


Fig. P17.138

- 17.139** A uniform slender rod is placed at corner B and is given a slight clockwise motion. Assuming that the corner is sharp and becomes slightly embedded in the end of the rod so that the coefficient of static friction at B is very large, determine (a) the angle β through which the rod will have rotated when it loses contact with the corner, (b) the corresponding velocity of end A .

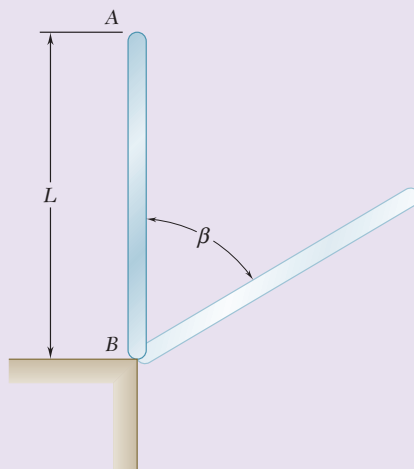


Fig. P17.139

- 17.140** The motion of the slender 250-mm rod AB is guided by pins at A and B that slide freely in slots cut in a vertical plate as shown. Knowing that the rod has a mass of 2 kg and is released from rest when $\theta = 0$, determine the reactions at A and B when $\theta = 90^\circ$.

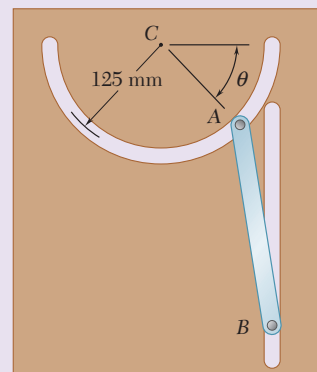


Fig. P17.140

- 17.141** A baseball attachment that helps people with mobility impairments play T-ball and baseball is powered by a spring that is unstretched at position 2. The spring is attached to a cord that is fastened to point B on the 75-mm radius pulley. The pulley is fixed at point O , rotates backwards to the cocked position at θ , and the rope wraps around the pulley and stretches the spring with a stiffness of $k = 2000 \text{ N/m}$. The combined mass moment of inertia of all the rotating components about point O is $0.40 \text{ kg}\cdot\text{m}^2$. The swing is timed perfectly to strike a 145-gram baseball travelling with a speed of $v_0 = 10 \text{ m/s}$ at a distance of $h = 0.7 \text{ m}$ away from point O . Knowing that the coefficient of restitution between the bat and ball is 0.59, determine the velocity of the baseball immediately after the impact. Assume that the ball is travelling primarily in the horizontal plane and that its spin is negligible.

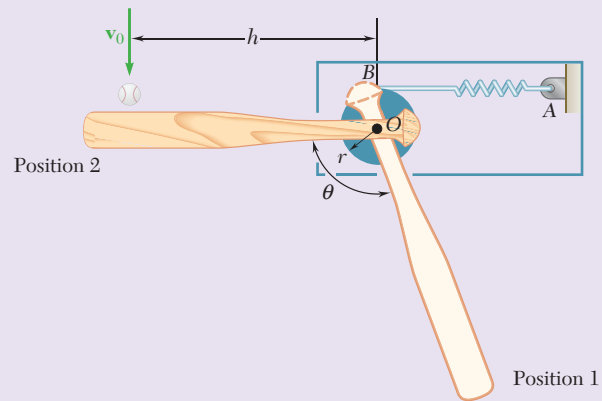


Fig. P17.141

- 17.142** Two panels A and B are attached with hinges to a rectangular plate and held by a wire as shown. The plate and the panels are made of the same material and have the same thickness. The entire assembly is rotating with an angular velocity ω_0 when the wire breaks. Determine the angular velocity of the assembly after the panels have come to rest against the plate.

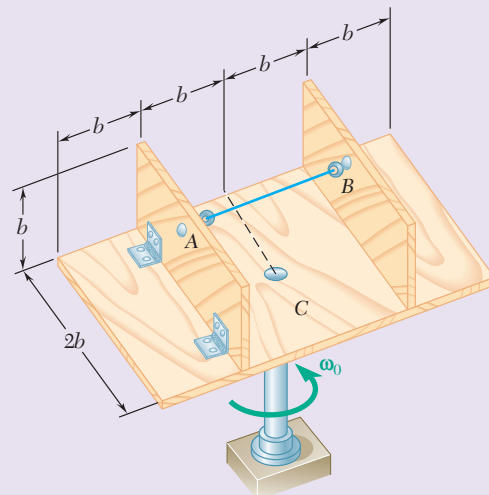


Fig. P17.142

17.143 Disks *A* and *B* are made of the same material, are of the same thickness, and can rotate freely about the vertical shaft. Disk *B* is at rest when it is dropped onto disk *A*, which is rotating with an angular velocity of 500 rpm. Knowing that disk *A* has a mass of 8 kg, determine (a) the final angular velocity of the disks, (b) the change in kinetic energy of the system.

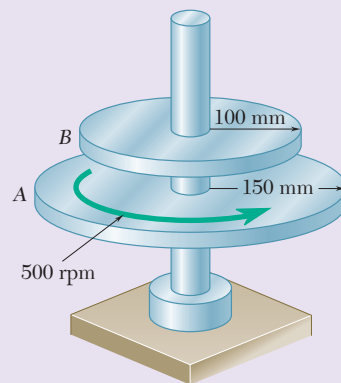


Fig. P17.143

17.144 A square block of mass m is falling with a velocity $\bar{v}_1$ when it strikes a small obstruction at *B*. Knowing that the coefficient of restitution for the impact between corner *A* and the obstruction *B* is $e = 0.5$, determine immediately after the impact (a) the angular velocity of the block, (b) the velocity of its mass center *G*.

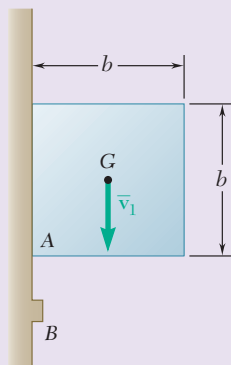


Fig. P17.144

17.145 A 3-kg bar *AB* is attached by a pin at *D* to a 4-kg square plate, which can rotate freely about a vertical axis. Knowing that the angular velocity of the plate is 120 rpm when the bar is vertical, determine (a) the angular velocity of the plate after the bar has swung into a horizontal position and has come to rest against pin *C*, (b) the energy lost during the plastic impact at *C*.

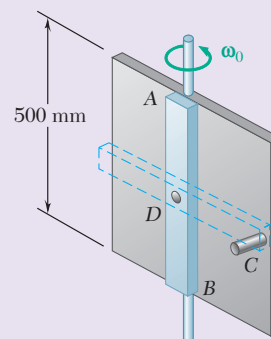


Fig. P17.145

17.146 A 1.8-lb javelin *DE* impacts a 10-lb slender rod *ABC* with a horizontal velocity of $v_0 = 30$ ft/s as shown. Knowing that the javelin becomes embedded into the end of the rod at point *C* and does not penetrate very far into it, determine immediately after the impact (a) the angular velocity of the rod *ABC*, (b) the components of the reaction at *B*. Assume the javelin and the rod move as a single rigid body after the impact.

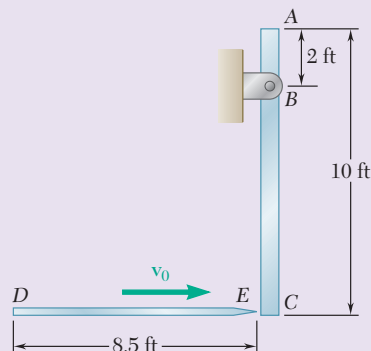
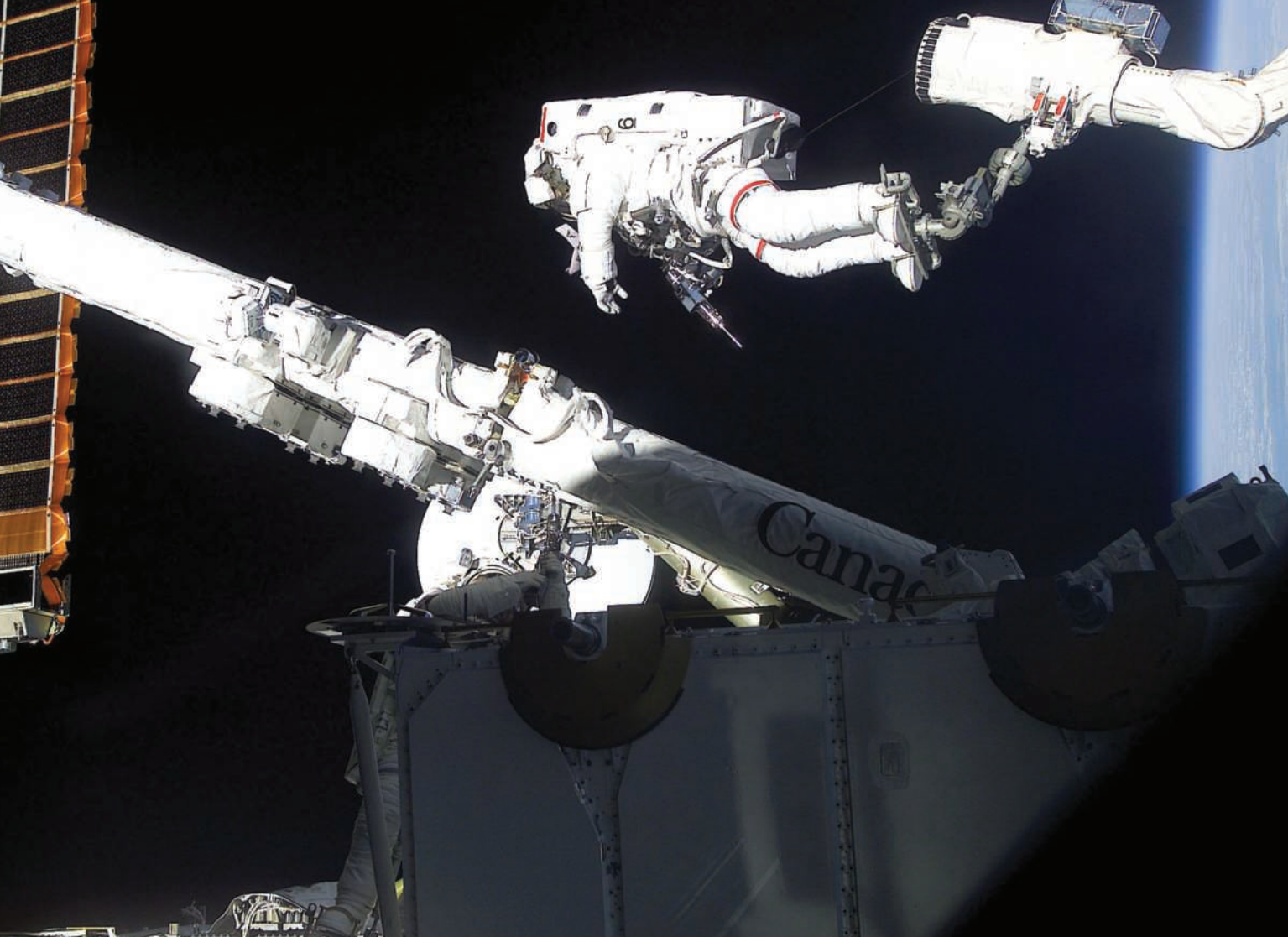


Fig. P17.146



18

Kinetics of Rigid Bodies in Three Dimensions

While the general principles that you learned in earlier chapters can be used again to solve problems involving the three-dimensional motion of rigid bodies, the solution of these problems requires a new approach and is considerably more involved than the solution of two-dimensional problems. One example is the determination of the forces acting on the robotic arm of the spacecraft.

Introduction

18.1 ENERGY AND MOMENTUM OF A RIGID BODY

18.1A Angular Momentum of a Rigid Body in Three Dimensions

18.1B Applying the Principle of Impulse and Momentum to the Three-Dimensional Motion of a Rigid Body

18.1C Kinetic Energy of a Rigid Body in Three Dimensions

18.2 MOTION OF A RIGID BODY IN THREE DIMENSIONS

18.2A Rate of Change of Angular Momentum

18.2B Euler's Equations of Motion

18.2C Motion of a Rigid Body About a Fixed Point

18.2D Rotation of a Rigid Body About a Fixed Axis

18.3 MOTION OF A GYROSCOPE

18.3A Eulerian Angles

18.3B Steady Precession of a Gyroscope

18.3C Motion of an Axisymmetrical Body Under No Force

Objectives

- **Calculate** the angular momentum and kinetic energy of a rigid body undergoing general three-dimensional motion.
- **Define** the inertia tensor, products of inertia, and principal axes of inertia.
- **Apply** the principle of impulse and momentum to solve three-dimensional rigid body kinetics problems.
- **Solve** three-dimensional rigid body kinetics problems, including fixed point rotation, fixed axis rotation, and gyroscopic motion.
- **Describe** the relationship between applied moment, precession, and spin of a gyroscope undergoing steady precession.
- **Analyze** the motion of a rotating axisymmetric body under no external forces.

Introduction

In Chaps. 16 and 17, we were concerned with the plane motion of rigid bodies and of systems of rigid bodies. In Chap. 16 and in the second half of Chap. 17 (impulse and momentum), our study was further restricted to the motion of plane rigid bodies and of bodies symmetrical with respect to the reference plane. However, many of the fundamental results obtained in these two chapters remain valid in the case of the motion of a rigid body in three dimensions. For example, the two fundamental equations

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}} \quad (18.1)$$

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G \quad (18.2)$$

on which we based the analysis of the plane motion of a rigid body remain valid in the most general case of motion of a rigid body. As indicated in Sec. 16.1, these equations express that the system of external forces is equipollent to the system consisting of the vector $m\bar{\mathbf{a}}$ attached at G and the couple of moment $\dot{\mathbf{H}}_G$ (Fig. 18.1).

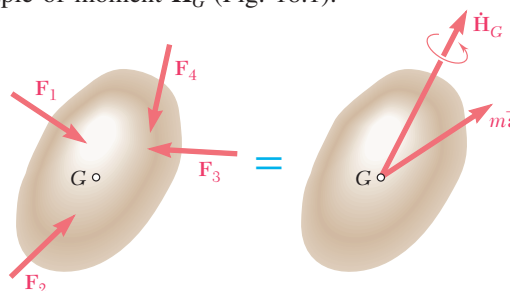


Fig. 18.1 The external forces acting on the rigid body are equipollent to a vector $m\bar{\mathbf{a}}$ attached to the mass center G and a rotational inertia vector $\dot{\mathbf{H}}_G$.

The relation $\mathbf{H}_G = \bar{I}\boldsymbol{\omega}$ enabled us to determine the angular momentum of a rigid body and played an important part in the solution of problems involving the plane motion of rigid bodies and bodies symmetrical with respect to the reference plane. However, this equation ceases to be valid in the case of nonsymmetrical bodies or three-dimensional motion. Thus, we need to develop a more general method for computing the angular momentum $\mathbf{H}_G$ of a rigid body in three dimensions.

Similarly, the main feature of the impulse–momentum method discussed in Sec. 17.2A is the reduction of the momenta of the particles of a rigid body to a linear momentum vector $m\bar{\mathbf{v}}$ attached at the mass center G of the body and an angular momentum couple $\mathbf{H}_G$. This method remains valid in the more general case, but we must discard the relation $\mathbf{H}_G = \bar{I}\boldsymbol{\omega}$ and replace it with a more general relation before we can apply this method to the three-dimensional motion of a rigid body (Sec. 18.1B).

Also note that the work–energy principle and the principle of conservation of energy still apply in the case of the motion of a rigid body in three dimensions. However, we need to replace the expression obtained in Sec. 17.1C for the kinetic energy of a rigid body in plane motion with a new expression for a rigid body in three-dimensional motion.

In the second part of this chapter, you will learn to determine the rate of change $\dot{\mathbf{H}}_G$ of the angular momentum $\mathbf{H}_G$ of a three-dimensional rigid body using a rotating frame of reference where the moments and products of inertia of the body remain constant. Then you can express Eqs. (18.1) and (18.2) in the form of free-body and kinetic diagrams that you can use to solve various problems involving the three-dimensional motion of rigid bodies (Sec. 18.2).

The last part of this chapter (Sec. 18.3) is devoted to the study of the motion of gyroscopes or, more generally, of an axisymmetric body with a fixed point located on its axis of symmetry. We first consider the particular case of the steady precession of a gyroscope and then analyze the motion of an axisymmetric body subjected to no force except its own weight.

18.1 ENERGY AND MOMENTUM OF A RIGID BODY

All of the methods you studied in earlier chapters for analyzing the plane motion of a rigid body have corresponding versions for motion in three dimensions. However, some of the formulas for determining kinetic quantities such as energy and angular momentum need to be replaced by more general equations. In this section, we examine some of the basic quantities and equations needed for the study of motion in space.

*18.1A Angular Momentum of a Rigid Body in Three Dimensions

In this section you will see how to determine the angular momentum $\mathbf{H}_G$ of a body about its mass center G from the angular velocity $\boldsymbol{\omega}$ of the body in the case of three-dimensional motion.

According to Eq. (14.24), we can express the angular momentum of the body about G as

$$\mathbf{H}_G = \sum_{i=1}^n (\mathbf{r}'_i \times \mathbf{v}'_i \Delta m_i) \quad (18.3)$$

where $\mathbf{r}'_i$ and $\mathbf{v}'_i$ denote, respectively, the position vector and the velocity of the particle P_i with a mass Δm_i that is relative to the centroidal frame $Gxyz$ (Fig. 18.2). However, $\mathbf{v}'_i = \boldsymbol{\omega} \times \mathbf{r}'_i$, where $\boldsymbol{\omega}$ is the angular velocity of the body at the instant considered. Substituting into Eq. (18.3), we have

$$\mathbf{H}_G = \sum_{i=1}^n [\mathbf{r}'_i \times (\boldsymbol{\omega} \times \mathbf{r}'_i) \Delta m_i]$$

From the rule for determining the rectangular components of a vector product (*Statics*, Sec. 3.1D, or Appendix A), we obtain the following expression for the x -component of the angular momentum as

$$\begin{aligned} H_x &= \sum_{i=1}^n [y_i(\boldsymbol{\omega} \times \mathbf{r}'_i)_z - z_i(\boldsymbol{\omega} \times \mathbf{r}'_i)_y] \Delta m_i \\ &= \sum_{i=1}^n [y_i(\omega_x y_i - \omega_y x_i) - z_i(\omega_z x_i - \omega_x z_i)] \Delta m_i \\ &= \omega_x \sum_i (y_i^2 + z_i^2) \Delta m_i - \omega_y \sum_i x_i y_i \Delta m_i - \omega_z \sum_i z_i x_i \Delta m_i \end{aligned}$$

Replacing the sums by integrals in this expression and in the two similar expressions obtained for H_y and H_z , we have

$$\begin{aligned} H_x &= \omega_x \int (y^2 + z^2) dm - \omega_y \int xy dm - \omega_z \int zx dm \\ H_y &= -\omega_x \int xy dm + \omega_y \int (z^2 + x^2) dm - \omega_z \int yz dm \\ H_z &= -\omega_x \int zx dm - \omega_y \int yz dm + \omega_z \int (x^2 + y^2) dm \end{aligned} \quad (18.4)$$

Note that the integrals containing squares represent the *centroidal mass moments of inertia* of the body about the x , y , and z axes, respectively (*Statics*, Sec. 9.5A, or Appendix B). That is,

$$\begin{aligned} \bar{I}_x &= \int (y^2 + z^2) dm & \bar{I}_y &= \int (z^2 + x^2) dm \\ \bar{I}_z &= \int (x^2 + y^2) dm \end{aligned} \quad (18.5)$$

Similarly, the integrals containing products of coordinates represent the *centroidal mass products of inertia* of the body (Sec. 9.6A); we have

$$\bar{I}_{xy} = \int xy dm \quad \bar{I}_{yz} = \int yz dm \quad \bar{I}_{zx} = \int zx dm \quad (18.6)$$

Substituting from Eqs. (18.5) and (18.6) into Eq. (18.4), we obtain the components of the angular momentum $\mathbf{H}_G$ of the body about its mass center as

Angular momentum about mass center

$$\begin{aligned} H_x &= +\bar{I}_x \omega_x - \bar{I}_{xy} \omega_y - \bar{I}_{xz} \omega_z \\ H_y &= -\bar{I}_{yx} \omega_x + \bar{I}_y \omega_y - \bar{I}_{yz} \omega_z \\ H_z &= -\bar{I}_{zx} \omega_x - \bar{I}_{zy} \omega_y + \bar{I}_z \omega_z \end{aligned} \quad (18.7)$$

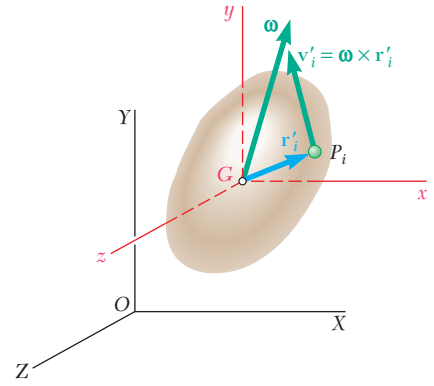


Fig. 18.2 The velocity of particle P_i is needed to derive the angular momentum of a rigid body in three dimensions.